Diebold 10XX/478X Device Support Manual

BASE24-atm®

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Contents

ATM Configuration *continued*

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What’s New

The following table highlights the major changes that have been made in the most recent update to the ***Diebold 10XX/478X Device Support Manual.*** The first column of the table lists the sections and appendixes in which major changes have been made. The second column of the table describes the major changes for each section or appendix.

September 2012

Section/ Appendix	Major Changes
G	Clarifies the description of the MOBILE-OPERATOR-ID-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type. This param specifies a mobile operator six-digit IIN, two-digit digit product ID, and the mobile operator name (up to 20 characters) associated with the mobile telephone operator.

ACI Worldwide, Inc.

Preface

The ***BASE24-atm Diebold 10XX/478X Device Support Manual*** provides an overview of the Diebold 10XX/478X ATM Device Handler process and Diebold-related software options required by the BASE24-atm product. This manual discusses the interaction between BASE24 and the Diebold 10XX/478X Modular Delivery System (MDS), *i*-series, *ix*-series, and Opteva-series ATMs or ATMs emulating the Diebold 10XX/478X-series ATMs. It also examines the role of the BASE24 product in the features, specifications, and functions of the Diebold 10XX/478X ATMs.

Audience

ACI prepared this publication for distribution to institution personnel responsible for configuring and maintaining Diebold 10XX/478X ATMs or ATMs emulating the Diebold 10XX/478X ATMs for use with the BASE24-atm product.

Prerequisites

This manual discusses a number of general topics relating to Diebold 10XX/478X ATMs. These discussions have been included primarily to provide a stronger insight into the way the Device Handler process handles Diebold 10XX/478X ATMs. This manual is not intended as a substitute for the vendor literature relating to Diebold 10XX/478X ATMs. It is recommended that individuals making decisions on BASE24-atm Diebold 10XX/478X ATM processing be familiar with the vendor literature available for the type of Diebold 10XX/478X ATM they are using.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24-atm Diebold 10XX/478X SSB Manual*** describes the self-service banking (SSB) modules that can be added to the BASE24-atm product, field descriptions for SSB-specific screens, receipt templates for new and modified transactions, and SSB-specific configuration.
- The ***BASE24 Device Control Manual*** describes operational control of the Diebold ATMs by BASE24. This includes instructions for such actions as downloading information and performing status inquiries. It also describes the EMT Control Commands screen used to reallocate extended memory tables.
- The ***BASE24 Base Files Maintenance Manual*** describes screen images and field descriptions of the Acquirer Processing Code File (APCF), Institution Definition File (IDF), the External Message File (EMF), and the Surcharge File (SURF).
- The ***BASE24-atm Files Maintenance Manual*** describes screen images and field descriptions of the BASE24-atm Terminal Data files (ATD), Receipt File (RCPT), Terminal Receipt File (TRF), and the Transaction Log File (TLF).
- The ***BASE24-atm EMV Support Manual*** describes screen images and field images of the BASE24-atm Terminal Data files (ATD) screen specific to the EMV product.
- The ***BASE24-atm Settlement and Reporting Manual*** describes the settlement and cutover procedures, and contains samples and field descriptions of the reports produced by the BASE24-atm product.
- The ***BASE24-atm Transaction Processing Manual*** describes the supported transactions and response codes, and the standard internal message (STM) used by BASE24.
- The ***BASE24-atm Multiple Currency Support Manual*** describes the authorization and currency conversion portions of the BASE24-atm multiple currency functions.
- The ***BASE24 Logical Network Configuration File Manual*** describes the Logical Network Configuration File (LCONF) and its assigns and params.
- The ***BASE24 Transaction Security Manual*** describes message authentication.

- The ***BASE24 Text Command Reference Manual*** describes commands that can be sent to the Device Handler process.
- The ***BASE24 Event Message Reference Manual*** describes event messages and how to access documentation for event messages.
- The ***BASE24 User Security Manual*** describes logging on to the BASE24 user access system, using the default function keys, and creating screen access security.
- The ***BASE24 External Message Manual*** describes the standard external message used by ISO hosts.
- The ***BASE24 ISO Host Interface Manual*** describes message processing between BASE24 and ISO hosts.
- The ***BASE24 ANSI Host Interface Manual*** describes the external message used to route information between BASE24 and an ANSI host.
- The ***BASE24 Tokens Manual*** describes the tokens that are added to the standard internal message to carry additional information.
- The ***BASE24 Transaction Security Services Processing Guide*** describes the Transaction Security Services process requirements for implementing the Automated Key Distribution System add-on product. This manual also describes the first phases of automated key distribution and the Transaction Security Services user interface.

In addition to BASE24 publications, the following Diebold manuals are referenced in this manual:

- The Diebold ***Data Security Procedures and Reference Manual*** describes setting up the configuration file to support local PIN verification.
- The Diebold ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** describes Diebold terminals features, formats, status information, load information, and configuration options.
- The Diebold ***Agilis 91x Terminal Programming Manual*** describes terminal features, formats, status information, load information and configuration options for Opteva-series terminals.
- The ***Diebold i series, ix Series, and MDS Maintenance Manual*** describes the Diebold 10XX/478X firmware and the device installation table (DIT).
- The ***Visa II Dial Protocol Manual*** (TP-820321-001A) describes the Visa II Dial Application File (CONFIG.DAT) and the Application Dial Interface File (ADI.DAT).

The following Wincor-Nixdorf manuals are referenced in this manual:

- The ***Wincor-Nixdorf NDC/Diebold D91x Message Format Extension for Cheque In, June 13, 2007, Release 1.34*** contains information about the Wincor Enhanced Check In state.
- The ***Wincor-Nixdorf NDC_Diebold D91x Message Format Extension for CashIn V1.44*** contains information about the Wincor Enhanced Cash In state.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual:

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, **“Introduction,”** introduces the reader to the Diebold 10XX/478X ATM models that are available and describes how BASE24-atm provides support for Diebold 10XX/478X ATMs.

Section 2, **“ATM Configuration,”** contains information about Diebold 10XX/478X ATM configuration requirements, Diebold 10XX/478X configuration files, and the BASE24 files used in configuring Diebold 10XX/478X ATMs.

Section 3, **“BASE24-atm Device Handler Process,”** contains an overview of the BASE24-atm Diebold 10XX/478X Device Handler process, the processing steps for the Device Handler process when it is first started, an overview of the network commands that can be sent to the Device Handler process, and information on how to access the event messages generated by the Device Handler process.

Section 4, **“Configuration File Maintenance,”** describes the Diebold Configuration File (CNFGIX) maintenance that can be performed using BASE24-atm. It includes the different screens used to change information in the configuration file.

Section 5, **"Host-Controlled Solenoid-Activated Lock Feature,"** discusses the tasks required to establish operator access to the ATM chest door when the host-controlled solenoid-activated lock feature is supported. It discusses the Operator Identification File (OIF) and how it is set up to control operator access to the ATM chest door.

Section 6, **"ATM Printer Support,"** describes standard printer support for Diebold 10XX/478X ATMs. Examples of default receipts generated at a Diebold 10XX/478X ATM are shown as they would appear to the cardholder or administrative personnel. This section also describes how you can establish receipt formats that are unique to the institution.

Section 7, "Diebold Merchant Banking Center Support," describes BASE24-atm operations used to support the Diebold Merchant Banking Center. The Diebold 10XX/478X Merchant Banking Center Device Handler extension supports money exchange and bag deposit transactions and provides receipts and settlement and reporting information.

Appendix A, **"Statement Print Data Compression,"** describes data compression techniques used by BASE24 to send and receive messages from the host system and to send messages to an ATM. The statement print data compression feature is used to increase efficiency of information transfer in order to save time and expenses associated with transmission of repeated information such as blanks used to format and position printed text.

Appendix B, "Dispensing Bills and Noncurrency Items," describes the bill dispensing methods used by the Diebold 10XX/478X Device Handler process, and describes the BASE24-atm Non–Currency Dispense add-on product.

Appendix C, "Diebold 10XX/478X Dial-Up Device Handler Extension," describes the Diebold 10XX/478X Dial-Up Device Handler extension.

Appendix D, "Multiple Currency Dispense Add-on Feature," describes the BASE24-atm Multiple Currency Dispense add-on feature. This feature, part of the BASE24-atm Multiple Currency add-on product, enables you to dispense multiple currencies from a single ATM.

Appendix E, "Distributing Master Keys and Transaction Keys," describes the Automated Key Distribution System add-on product and the Dynamic Key Management add-on product. The Automated Key Distribution System add-on product enables you to remotely load master keys to an ATM. The Dynamic Key Management add-on product enables you to automatically load PIN and message authentication code (MAC) keys to an ATM.

Appendix F, "BASE24-atm Mobile Top-Up Extension," describes the BASE24-atm Mobile Top-Up Device Handler Extension. This extension, which is used in conjunction with the BASE24-atm Non–Currency Dispense add-on product, enables an institution to offer mobile telephone top-up minutes as an option at an ATM.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-DA150-01 for the ***BASE24-atm Diebold 10XX/478X Device Support Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Sep-2012 for September 2012) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how command syntax and fields are documented in this manual.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
lowercase italics	Identify variables that you must provide.
Brackets [ ]	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar (|).
Vertical bar |	Separates alternatives in a list of items.
Ellipsis …	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Field Descriptions

Each field appearing on a files maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the Operator Identification File (OIF).

Example

OPERATOR GROUP — The group number of the operator servicing the ATM. The operator group is defined at the ATM. The OPERATOR GROUP field, along with the OPERATOR NUMBER and TERM ID fields, are the primary keys to the OIF. Valid values are 00 through 05.

Field Length:	2 numeric characters
Required Field:	Yes
Default Value:	No default value
Data Name:	OIF.PRIKEY.GRP

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Demonstrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term *alphanumeric* includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term *hexadecimal* includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is *System Protected*.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the **F8** key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

1: Introduction

This section provides an introduction to the support that the BASE24-atm product provides for Diebold 10XX/478X ATMs. It contains information about the following items:

- Diebold 10XX/478X ATM models supported by the BASE24-atm product
- BASE24-atm support capabilities
- Diebold 10XX/478X ATM features supported by the BASE24-atm product
- Protocols supported by the BASE24-atm product for Diebold 10XX/478X ATMs
- Transaction security
- Cutover and settlement
- Financial and administrative transactions supported by the BASE24-atm product for the Diebold 10XX/478X ATMs
- Device configuration and control functions supported by the BASE24-atm product

Diebold 10XX/478X Models

The BASE24-atm product supports the Diebold 10XX/478X ATMs described below. These are full-service terminals that allow cardholders to perform routine banking transactions without assistance from bank personnel.

All Diebold ATMs meet the applicable requirements of the Bank Protection Act and Regulation P, as well as the applicable UL Standard 291 (Business Hour Service) for ATMs.

The BASE24-atm product supports Modular Delivery System (MDS) 10XX/478X terminals, *ix*-series, and Opteva-series terminals. The *ix*-series indicates 10XX/478X terminals that have hardware and software upgrades for additional functionality. An intermediate level of hardware upgrades, the *i*-series, is also supported by the BASE24-atm product.

BASE24-atm Support

The BASE24-atm product provides the following support specifically tailored to the Diebold 10XX/478X terminals:

- Supported features
- Supported protocols
- Transaction security

Supported Features

The BASE24-atm Diebold 10XX/478X Device Handler process is compatible with Diebold TCS releases through release 5.3 and TCS Plus release 1.3 or higher.

The BASE24-atm product supports Diebold terminals that are equipped with one of the following operating systems: STP releases up to and including 4.0 (firmware-based), BTP (software-based), ATP (PC-based), and CTP (advanced PC-based).

The following Diebold 10XX/478X ATM features are supported by the BASE24-atm product:

- Receipt printer. Standard and alternate character sets are supported. Statement receipts up to 80 columns wide can be printed.
- Journal printer. The receipt width is configurable by the terminal owner.
- Depository printer. The BASE24-atm product overrides the default information.
- Statement printer. The BASE24-atm product supports a statement printer on *ix*-series and Opteva-series only.
- Data compression on statement print transaction receipts.
- 48-column thermal consumer printer. Use of this feature requires *ix*-series TCS 4.0.
- Space compression on receipts other than statement print transaction receipts.
- DES PIN encryption.
- ANSI X9.8 standard PIN block format.
- Local PIN verification.

- Withdrawal door. The BASE24-atm product supports no door, manual door, or motorized door.
- Withdrawal area sensor check.
- Host-controlled solenoid-activated lock.
- Intelligent Depository Module (IDM), IDM II, IDM III, and IDM IV.
- Display grid, with an enhanced display size of 40 x 20.
- Touch screen. This feature is supported on *ix*-series only.
- Enhanced Monochrome Graphics (EMG). The BASE24-atm product supports repositioning or restructuring of Diebold prepackaged graphics and downloading to the ATM using the DCT. Standard and Chinese character sets are supported.
- Color graphics.
- Super VGA mode.
- Consumer interface panel, with four standard operation keys and four optional operations keys. The BASE24-atm product supports the international key pad.
- Lead-through lamps. Use of this feature requires the *ix*-series TCS 3.1.
- Decimal format entry.
- Downloading of configuration files.
- Remote and rear balancing. Rear balancing initiated by continuous availability does not result in the terminal being closed (with the exception of the CPLF Print administrative transaction which closes the terminal).
- Multiple languages on screen and on receipt. This feature allows eight languages if the ATM is using STP release 4.0, ATP, BTP, or CTP; otherwise it allows two languages.
- Character sets. The BASE24-atm product supports the following character sets for the Consumer Display (CRT): standard, alternate, mosaic, and icon. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** or the ***Agilis 91x Terminal Programming Manual*** for a detailed description of the supported character sets.
- Card reader. The BASE24-atm product supports the use of either the motorized or the swipe style card readers. The motorized card reader receives cards in a right-handed, face up presentation and can be programmed to capture cards for security purposes.
- ICC chip card reader.

- Card return. This feature indicates whether to return the card to the cardholder before dispensing cash on financial transactions. This feature is set up through the use of configuration file load groups.
- Media Player. Use of this feature requires the *ix*-series TCS 4.5.
- Transaction Verification Status processing. Use of this feature requires the *ix*-series TCS 4.5.
- Phase I, II, and III Voice. Use of this feature requires the *ix*-series TCS 4.5.
- Expanded range of state numbers to allow 000–250, 256–999, and A00–ZZZ. Use of this feature require the *ix*-series TCS 4.5.
- Dial-up protocol.
- Diebold Merchant Banking Center.
- Printing an entire scanned check image or scanned check amount to a printer. Use of this feature requires TCS release 5.3.
- Selecting a specific character set for printing. Use of this feature requires TCS release 5.3.
- Two-color printing to thermal receipt printers. This feature is supported by the Opteva-series only.
- Two-color printing to thermal statement printers This feature is supported by the Opteva-series only.
- Envelope dispenser solicited and unsolicited status messages.
- LCD monitor status messages.
- Automatic envelope dispenser.
- Text to speech. This feature is supported by the *ix*-series and Opteva-series only.
- Enhanced voice prompting and voice keypad commands. This feature is supported by the *ix*-series and Opteva-series only.
- Bulk note acceptor. This feature is supported by the Agilis 91x/Opteva-series only.
- Enhanced note acceptor. This feature is supported by the Agilis 91x/Opteva-series only.

Supported Protocols

The BASE24-atm product can support communications with Diebold terminals using the following protocols:

- Burroughs Multipoint Supervisor
- IBM 3275 Bisynchronous ASCII
- IBM 3275 Bisynchronous EBCDIC
- IBM SDLC/SNA
- TCP/IP
- X.25 SVC
- X.25 PVC

BASE24-atm Self-Service Banking

The BASE24-atm self-service banking (SSB) package is an add-on product for use with the Diebold 10XX/478X ATMs. SSB supports cash deposits, coin dispensing, check imaging, check bin sorting, and check transactions. Configuration and maintenance requirements of the BASE24-atm SSB product for Diebold 10XX/478X ATMs are described in the ***BASE24-atm Diebold 10XX/ 478X SSB Manual***.

BASE24-atm Non–Currency Dispense

The BASE24-atm Non–Currency Dispense add-on product enables an institution to dispense noncurrency items from its ATMs. Noncurrency items can include cash-value items such as travelers checks, postage stamps, or event tickets and nonvalue items such as coupons and advertisements. Refer to appendix B for more information about the Non–Currency Dispense add-on product.

Transaction Security

The BASE24-atm product offers transaction security options to protect an institution and its cardholders from fraud and includes such functions as PIN encryption, PIN verification, and message authentication.

The BASE24-atm product supports the PIN Change transaction for the Diebold 10XX/478X ATMs, which enables the cardholder to perform a PIN change at the ATM. The BASE24-atm product also supports the EMV PIN Unblock transaction for the Diebold ATMs, which enables the cardholder to reset the bad PIN tries accumulator for the EMV card's offline PIN at the ATM. If local PIN verification is used, the PIN Change transaction and the EMV PIN unblock transaction are not supported.

The BASE24-atm product supports transaction security using both hardware and software. Hardware security options include interfaces to various models of Atalla and Thales e-Security (Racal) security hardware. Software security options include several methods of PIN encryption and PIN verification.

The BASE24-atm Diebold 10XX/478X Device Handler process is responsible for the interface between the BASE24 system and the Diebold 10XX/478X ATMs attached to it. In this capacity, the Diebold 10XX/478X Device Handler process is responsible for PIN encryption between the BASE24-atm system and the Diebold 10XX/478X terminals. Based on a combination of device capabilities and your preferences, the Device Handler process passes on or decrypts PINs upon entry into the system.

For complete information on BASE24 transaction security options, refer to the ***BASE24 Transaction Security Manual.***

Local PIN Verification

The BASE24-atm product supports local PIN verification in which Diebold 10XX/478X devices that have PIN verification capability perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network.

To perform local PIN verification, the Diebold 10XX/478X terminal must be equipped with the proper hardware. Only those PINs with an offset encoded on the card can be verified at the ATM.

Institutions must consider which cards can be verified at the ATM because some interchanges do not support a preverified PIN.

The BASE24-atm product does not keep track of the number of bad PIN attempts when the ATM performs local PIN verification.

Local PIN verification requires a setting in the BASE24-atm Terminal Data files (ATD) as well as updates to the Financial Institution Table (FIT), the PIN Entry state, and the Transaction Request state in the Diebold Configuration File (CNFGIX). The CNFGIX must be downloaded after these changes are made to implement local PIN verification. Refer to section 4 for complete information on how to access the CNFGIX. Refer to Diebold's ***Data Security Procedures and Reference Manual*** for complete information on setting up the CNFGIX to support local PIN verification.

Message Authentication Code (MAC)

Message authentication ensures transaction messages are received exactly as created by the legitimate originator. Message authentication protects financial transaction messages against accidental or deliberate alteration.

The message authentication code (MAC) is generated by the originator of the message, based upon message elements identified in advance by the originator and recipient, and included with the message.

The BASE24-atm system supports the generation and verification of MAC values in hardware. Full message authentication is supported for the following message types:

- Consumer Request (transaction request)
- Function Command (transaction response)
- Write Command I (download of state data)
- Write Command VI (download of FIT data)

Device Handler Process Keys

There are two types of keys that are used in PIN encryption and message authentication between terminals and their BASE24-atm Device Handler processes: transaction keys and master keys.

Transaction Keys

A transaction key is distributed electronically by the BASE24 system to its terminals using the same lines over which all other communications travel. There are separate transaction keys for PIN encryption and message authentication.

Master Keys

Since transaction keys are distributed to BASE24-atm terminals using the same lines over which all other communications travel, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key known to both the terminal and the BASE24 system. The master key is used to encrypt transaction keys for movement over the communication lines. The same master key is used to encrypt the PIN encryption transaction key and message authentication transaction key.

Master keys are by nature *long-term* and, once in place, usually remain unchanged unless they are compromised. Unlike transaction keys, master keys cannot be transmitted over communications lines. The master key of each terminal must be physically entered at the ATM.

Transmitting Transaction Keys to the Terminal

There are several instances in which Device Handler processes transmit new transaction keys to their terminals. They are identified in the following list:

- During load sequences of the ATMs.
- When commands are issued from the BASE24 Device Control Terminal (DCT) or from a network control facility.

Each time a new transaction key is sent to a terminal, the Device Handler process checks the MAC SUPPORT TYPE field on ATD NCR 5XXX screen 9. The Device Handler process provides software message authentication and security module message authentication.

Software Message Authentication. If the MAC SUPPORT TYPE field on ATD screen 9 is set to a value of 1, the Device Handler process generates a new transaction key, encrypts the key under the hard-coded master key, and sends the key to the terminal.

Security Module Message Authentication. If the MAC SUPPORT TYPE field on ATD screen 9 is set to a value of 3, the Device Handler process requests a new transaction key from the security module. The Device Handler process stores the encrypted transaction key in the ATD so that the key can be used to generate MACs for messages outbound for the terminal. In addition, MACs in messages received from the terminal can be verified.

Cutover and Settlement

The role of the BASE24-atm product in settlement consists of providing institutions with detailed transaction reports and reflecting cardholder and terminal activity on both a daily and monthly basis. The BASE24-atm system defines the business day, captures and keeps track of each day's transaction totals, provides a daily file of transactions for processing by a host, and reports the funds due from and owed to the various participants in the logical network for balancing purposes.

A term used in connection with the BASE24-atm settlement process is cutover. Cutover refers to closing the current posting day and opening the next. It refers to the point in time when transactions stop being logged to the current business day and start being logged to the next business day by that terminal.

During institution cutover, the Settlement Initiator process reads the BASE24-atm Terminal Data files (ATD) to find ATMs that are not yet cut over and performs a force cutover on those terminals, and logs the totals to the Transaction Log File (TLF). An institution can purchase the Forced Cutover Audit module to write the terminal totals to the journal receipt at the ATM.

For a complete description of BASE24-atm settlement and cutover and the Forced Cutover Audit module, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Transaction Support

The BASE24-atm product supports a number of financial and administrative transactions for Diebold 10XX/478X ATMs. They are listed below. For complete descriptions of supported transactions, refer to the ***BASE24-atm Transaction Processing Manual***. Additional transactions are provided with the BASE24-atm self-service banking (SSB) add-on product and are described in the ***BASE24-atm Diebold 10XX/478X SSB Manual.***

Financial Transactions

The following financial transactions are supported by BASE24-atm for the Diebold 10XX/478X ATMs:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash
- Fast cash from checking account
- Fast cash from savings account
- Fast cash from credit account
- Fast cash for dollars
- Fast cash from checking account for dollars
- Fast cash from savings account for dollars
- Fast cash from credit account for dollars
- Cash advance from credit account
- Deposit to checking account
- Deposit to savings account
- Transfer from checking to savings
- Transfer from savings to checking
- Transfer from credit account to checking
- Transfer from checking to checking
- Transfer from savings to savings
- Payment enclosed

- Payment from checking to credit account
- Payment from savings to credit account
- PIN change
- EMV PIN unblock
- Balance inquiry from checking
- Balance inquiry from savings
- Inquiry into credit account availability
- Message enclosed from cardholder to financial institution
- Statement print request
- Log only transactions (four user-defined possibilities available)
- Non–Currency Dispense transaction
- Mobile top-up from checking
- Mobile top-up from savings
- Mobile top-up from credit

Transaction Receipt Printing

All of the transactions listed above can have receipts printed for the cardholder's records. There are several ways that you can determine whether the receipt prints, and if so, how the receipt looks. Refer to section 6 for additional discussion about these options.

Optional Receipts

You can determine whether a receipt is printed by transaction type. You can also allow the cardholder to determine if a receipt is printed on all transactions except for the statement print transaction and administrative transactions.

Configurable Receipts

The BASE24-atm product comes with standard formats, or templates, that determine how financial and administrative receipts look. You can use these templates, or modify them to print the information that you want the cardholder to receive. Header and trailer templates allow you to customize the receipt to print your institution's name and marketing information. You can even print your graphic logo, if the printer is capable.

PAN Truncation

You can determine if the account numbers on a receipt are printed in full or are truncated so only some digits of the number appear. This option allows you to reduce the amount of fraud caused by carelessly discarded receipts falling into the wrong hands.

Data Compression

The BASE24-atm product supports data compression on statement print transactions. Data compression involves reducing the amount of transaction data that is carried in a statement print transaction message. Reducing the amount of transmitted data results in a more efficient use of processing resources. For more information on the statement print data compression feature, refer to appendix A.

Transaction Surcharge

An optional surcharge can be added to certain transactions performed at your terminals. This surcharge is in addition to any fees that the cardholder's institution imposes; in other words, surcharges are for transaction acquirers, not issuers. A surcharge can be positive or negative. A positive surcharge amount debits the cardholder account, while a negative surcharge amount (i.e., an incentive) credits the cardholder account.

You determine whether the surcharge is added, and the amount added, depending on such variables as whether the transaction is on-us or not-on-us, the account type of the transaction, the transaction amount, the card prefix, and the card type. Terminals can also be grouped for surcharging purposes; for example, all of your terminals in airports can have the same surcharge profile. You also determine whether or not a surcharge is assessed when a partial amount is dispensed and the rest of the transaction is reversed.

Surcharging is customized through the Card Prefix File (CPF) and the Surcharge File (SURF), both of which are documented in the ***BASE24 Base Files Maintenance Manual***, and the ATD, which is documented in the ***BASE24-atm***

Files Maintenance Manual. The surcharge amount is carried in the Surcharge Data token, which is appended to the BASE24-atm Standard Internal Message (STM). Tokens are described in the ***BASE24 Tokens Manual***.

The following cardholder transactions can have a surcharge added by the BASE24-atm system:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash withdrawal
- Cash advance from credit account
- Noncurrency dispense
- Deposit
- Deposit with cash back
- Split deposit

Interactive Request Notification

The standard surcharging feature uses an interactive request notification sequence to notify the cardholder of a transaction surcharge. That is, the notification and acceptance of the surcharge is processed before the Authorization process sends the transaction to the issuer for authorization. The cardholder account is impacted only after the cardholder accepts the fee.

Interactive request notification is the default processing for the surcharging feature. To use this processing, the ATM-FEE-INTRACTV-RESP-NTFY param must be set to a value of N. Interactive request notification also is used if the ATM-FEE-INTRACTV-RESP-NTFY param is not present.

Interactive Response Notification

Surcharge interactive response notification processing routes the transaction request, including fees, to the issuer for authorization before notifying the cardholder of a fee. The cardholder account is impacted before the cardholder accepts the fee. If the cardholder does not accept the fee, the transaction is reversed.

To use surcharge interactive response notification processing, the ATM-FEE-INTRACTV-RESP-NTFY param must be set to a value of Y.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) for acquirer fees only; it does not maintain information in its database for issuer fees. The setting of issuer fees must be done at a host system or an interchange. Issuer fee information is carried in the Issuer Fee/Rebate token. This token is described in the ***BASE24 Tokens Manual.***

Administrative Transactions

You can perform administrative transactions in one of the following ways:

- You can use a machine-readable administrative card at the terminal.
- You can use the rear balancing feature of the terminal to initiate administrative transactions from the rear of the ATM without the use of an administrative card.

  Note: ATM customers are strongly discouraged from entering PINs into rear keypads, based on Payment Card Industry (PCI) security restrictions. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information on configuring the ATM-REAR-ADMN-TRAN-PIN-ALWD param to decline rear balancing administrative transaction requests containing PINs.
- You can perform administrative services using the Device Control Terminal (DCT).

The administrative transactions available through the BASE24-atm product include the following transactions:

- Balance with current cash
- Balance with standard replenishment
- Terminal subtotals (includes rolled coin subtotals for the Diebold Merchant Banking Center)
- Increase cash (includes rolled coin increase for the Diebold Merchant Banking Center)
- Decrease cash (includes rolled coin decrease for the Diebold Merchant Banking Center)

The terminal balancing transactions (that is, balance with current cash and balance with standard replenishment) allow terminals to be balanced using either the standard replenishment balancing method or the current cash balancing method. Totals stored in the BASE24-atm Terminal Data files (ATD) are impacted and cleared as a result of these transactions.

The terminal subtotals transaction allows activity totals accumulated since the last balancing transaction (and up to the time the terminal subtotal transaction is initiated) to be printed on journal and receipt printers. Terminal totals are not cleared by this transaction.

The increase and decrease cash transactions (for example, the midpoint cash adjustment transaction) allow the amount of cash contained in a cassette to either be replenished or decreased without balancing the terminal.

For more information on administrative transactions, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Device Configuration and Control

The Diebold 10XX/478X ATMs are configured using the device installation table (DIT), which is a series of parameters stored in the memory of the terminal. The terminal can also be modified by the Diebold Configuration File (CNFGIX) that is downloaded to the ATM using the BASE24 network. Refer to section 4 for information on the CNFGIX.

Changes to the configuration using the DIT settings can be made only by service personnel physically present at the ATM. For more information on the Diebold 10XX/478X firmware or the DIT, refer to the ***Diebold i Series, ix Series, and MDS Maintenance Manual.***

BASE24-atm support provides the ability to customize and control the CNFGIX. If you want to change the configuration file, the changes can be made using the Diebold entry program (ENTIBOLD) and downloaded to the ATM using the DCT functions.

Terminal Balancing

A terminal must be balanced on a regular basis once it is operational. This can be accomplished using the DCT functions or using administrative transactions performed at the ATM. The DCT allows a terminal to be closed, balanced, and opened again and cash totals to be increased or decreased. Other functions such as adding to the cash supply and servicing the terminal require administrative transactions to be performed at the terminal site.

Event Messages

The Device Handler process generates event messages periodically and as needed to notify an institution of the state of an ATM. The messages range in severity from informational to critical and can be printed or displayed using the EMSperus utility.

The Diebold 10XX/478X Device Handler process also can capture native ATM status messages and display them in their native format to the EMSperus utility. If the ATM-DH-DISPLAY-NATIVE-STATUS-MSG param is present and set to the value Y, the Diebold 10XX/478X Device Handler process appends the native status message it received from the ATM to existing status event messages sent to the EMSperus utility. The Device Handler process does this for all Diebold unsolicited status messages with a status descriptor of 8 or J and any Diebold solicited status messages it receives. All fields starting with the Solicited/ Unsolicited Message field through the Status field are displayed to the EMS system. If the param is set to a value other than Y or N, EMS raises an event stating that the value was invalid and defaults to the value N. If the param is set to the value N, the EMS system does not capture native ATM messages.

The format of the data appended to the existing event message is [NATIVE]D91X*native-message,* where *native-message* is the message received from the ATM.

Terminal Control

The BASE24-atm product provides several ways for you to control Diebold 10XX/ 478X terminals. One of these ways is using the DCT. The DCT allows you to perform all of the functions necessary to control a terminal including starting and stopping, downloading, balancing, and displaying the status of the terminal.

Other ways to impact terminals within the BASE24 system include issuing commands using a network control facility. Refer to the ***BASE24 Text Command Reference Manual*** for more information about network control commands. However, the DCT is the heart of the control mechanism for terminals within the BASE24-atm system. The DCT is the one module that contains all of the functions necessary to control the terminal. Anyone who is responsible for controlling terminals should have access to the DCT. Refer to the ***BASE24 User Security Manual*** and the ***BASE24 Device Control Manual*** for more information about accessing and using the DCT.

Basic Device Control Functions

The basic device control functions common to all types of terminals include the following:

- Balancing terminals
- Displaying the status of the terminals
- Opening and closing the terminals
- Starting and stopping the terminals
- Displaying subtotals
- Increasing and decreasing cash totals in the terminal hoppers

Each of the functions listed above is associated with the DCT, since each is critical to the operation of ATMs. This central location allows these functions to be used more efficiently. For example, when you need to balance your terminal, you can close the ATM from the same screen from which you execute the balancing command.

BASE24-atm processes also can be reinitialized or warmbooted from the DCT. This is important because Device Handler processes must be reinitialized when changes are made that affect the Device Handler processes. Refer to the ***BASE24 Device Control Manual*** for more information about device control functions.

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2: ATM Configuration

The support of Diebold 10XX/478X ATMs in the BASE24-atm system includes flexible configuration options through the configuration file. Diebold ATMs can be configured in numerous ways, depending on the institution's preferences. For example, an institution can allow some ATMs to offer certain transactions, while prohibiting ATMs in other locations from offering the same transactions. Each ATM can have its own transaction set, based on selections made by the institution. Institutions can also change the wording on screen displays for ATMs in different locations.

This section contains information about the following items:

- The Diebold Configuration File (CNFGIX)
- BASE24 files used by the Device Handler process during transaction processing

Diebold 10XX/478X ATM Configuration

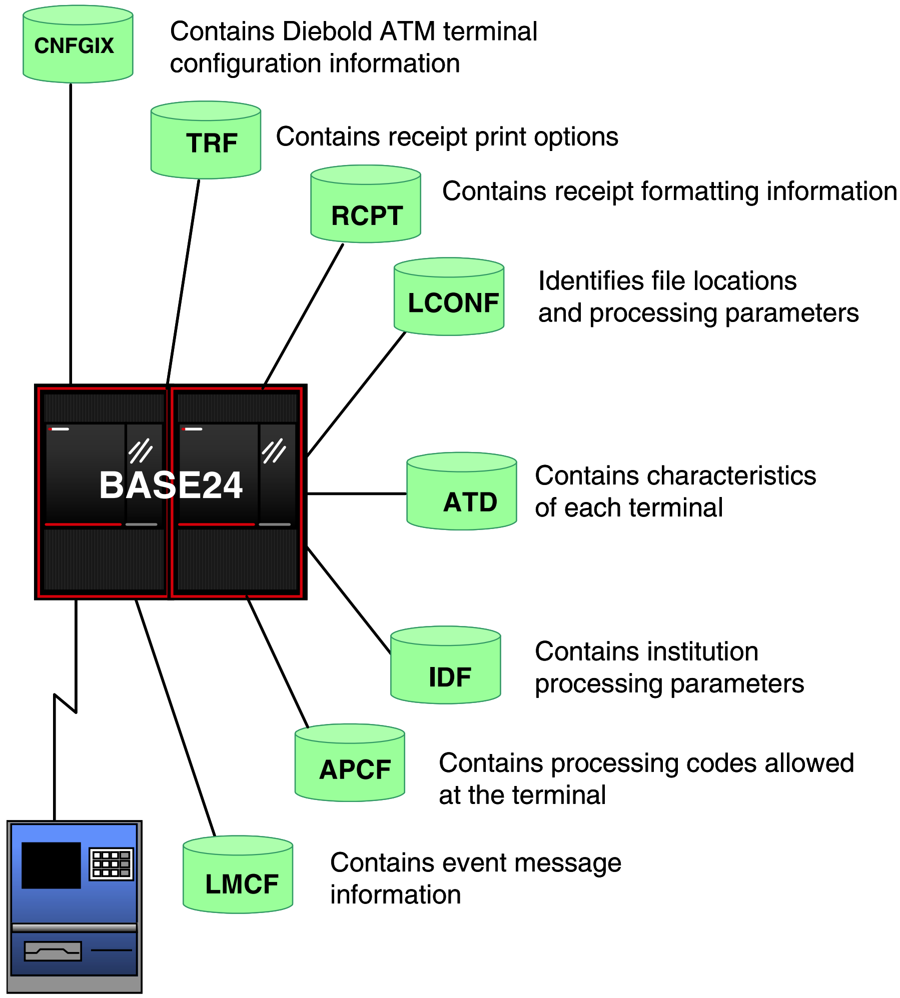
The Diebold 10XX/478X ATM is controlled by software contained in the ATM itself. The Diebold Configuration File (CNFGIX) provides a convenient method of configuring the control logic and associated parameters that determine how transaction processing is configured at the ATM level (for example, screen data and timer data). The CNFGIX data is downloaded to the ATM.

In addition to the Diebold 10XX/478X internal software, a number of BASE24 files are used to configure transaction processing at the network level to support the Diebold 10XX/478X ATMs. BASE24 files provide indirect support to the ATM by defining the interaction between the Device Handler process and the ATM.

Because of the many combinations of options that are available during configuration, the configuration file and BASE24 files are discussed in this section one at a time. The various combinations that can be derived from these individual options are left up to the institutions. The following files are used by Diebold 10XX/478X ATMs:

- Diebold Configuration File (CNFGIX)
- Logical Network Configuration File (LCONF)
- BASE24-atm Terminal Data files (ATD)
- Receipt File (RCPT)
- Terminal Receipt File (TRF)
- Institution Definition File (IDF)
- Log Message Configuration File (LMCF)

The following illustration shows the components required to configure processing at a Diebold terminal using a BASE24-atm system.



In addition, the Diebold 10XX/478X ATMs can use the Operator Identification File (OIF) to control access to the solenoid lock of the ATM chest door. Refer to section 5 for a description of the OIF. When the BASE24-atm Multiple Currency Dispense add-on feature is used, the institutions use the Exchange Rate File (ERF) to configure exchange rates at the network level. Refer to appendix D for a description of the ERF.

Diebold Configuration File

Changes can be made to the Diebold Configuration File (CNFGIX) on the BASE24-atm system using the Diebold entry program and downloaded to the ATM from the BASE24 Device Control Terminal (DCT). Refer to the ***BASE24 Device Control Manual*** for information on the procedures used when downloading files. For further information on how to modify the configuration file, refer to section 4.

All of the information in the Diebold Configuration File is combined into groups. When a change is made, you can download a data group to the ATM without physically going to the terminal.

Groups 300 through 303 and 311 through 313 are described in section 6 of this manual, and the following groups are described in more detail in the ***BASE24-atm Diebold 10XX/478X SSB Manual***: 12, 52, 54, 56, and 58.

Receipt File (RCPT)

The Receipt File (RCPT) is used to support the configurable receipt option, which provides you with the ability to establish ATM receipt formats that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, and multiple languages. The configurable receipt option includes approved and denied financial transaction receipts as well as administrative receipts.

Receipt configuration is supported by device type, receipt group, transaction type, language code, card issuer FIID, *from* and *to* account type, and printer type. Receipt groups allow the use of a single receipt format for a group of ATMs rather than configuring receipts for an individual terminal. Receipt groups are specified for each terminal on screen 8 of the BASE24-atm Terminal Data files (ATD). For further information on establishing receipt groups in the ATD, refer to the ***BASE24-atm Files Maintenance Manual***.

The information on a receipt can also be tailored for specific card issuers; for example, you can have different marketing information for on-us and not-on-us transaction receipts. Having different templates for different printers allows you to truncate the PAN on a cardholder receipt to reduce the chance of fraud, while allowing the entire PAN to be printed on the journal so that you have a complete record of the transaction.

The receipt also can be customized to allow separate templates for different *from* and *to* account types; for example, you can truncate savings account numbers differently than credit card numbers.

Note that there are several ways you can configure the BASE24-atm product to require receipts or allow transactions without a printed receipt. The first option allows you to decide by transaction type whether receipts are required, optional (that is, allow the transaction if the receipt printer is not operational), or not printed. These settings are contained in the Terminal Receipt File (TRF) using the terminal receipt profile configured for the ATM in the TERMINAL RECEIPT PROFILE field on ATD screen 8. The second option allows the cardholder to decide if he or she wants a receipt for a particular transaction or set of chained transactions, which overrides your settings on the ATD.

For a description of how the Device Handler process uses the RCPT in determining receipt wording and format as well as examples of default receipt templates for various transactions using the TRF, refer to section 6.

Terminal Receipt File (TRF)

The Terminal Receipt File (TRF) contains receipt print options for approved and denied transactions. The TRF contains one record for each terminal receipt profile, message category (the second digit of the message type), and ISO processing code. The terminal receipt profile defines the receipt requirements for various transaction types and in defined in the BASE24-atm Terminal Data files (ATD).

Refer to section 6 for more information about how the TRF is used in receipt processing. For more information about TRF screens, refer to the ***BASE24-atm Files Maintenance Manual.***

Logical Network Configuration File (LCONF)

The Logical Network Configuration File (LCONF) contains the records that link physical file names to the file names as they are referenced in BASE24 code and the records that set processing values for a logical network.

The records that link physical file names to the file names as they are referenced in code are known as assigns. The records that set processing values are known as params.

The LCONF assigns and params used by Diebold devices are listed below, including a short description of each. The following add-on products, features, and extensions use additional assigns and params that are described only in appendixes of this manual:

Add-On Product, Feature, or Extension	Appendix
BASE24-atm Non–Currency Dispense	Appendix B
Diebold 10XX/478X Dial-Up Device Handler Extension	Appendix C
Multiple Currency Dispense	Appendix D
Automated Key Distribution System	Appendix E
Dynamic Key Management	Appendix E

The BASE24-atm self-service banking (SSB) product uses additional assigns and params that are described in the ***BASE24-atm Diebold 10XX/478X SSB Manual.*** Refer to the ***BASE24 Logical Network Configuration File Manual*** for more detailed information on LCONF assigns and params.

Assigns

The following assigns are used by the Diebold Device Handler processes to locate the appropriate files. Add-on products, features and extensions use additional assigns that are described only in appendixes in this manual.

APCFEMT — The fully qualified file name of the Acquirer Processing Code File extended memory table (APCFEMT), which contains APCF records.

ATM-ATD-DYN-GNRL — The fully qualified file name of the BASE24-atm Terminal Data Dynamic File—general data (ATDD1). This file contains dynamic core terminal data (i.e., data common to all types of devices that is subject to change during transaction processing), device information (i.e., data specific to a device type), and device-dependent data (i.e., data specific to each terminal defined). Key STM fields for the last transaction processed for a device are also stored in this file. The Diebold 10XX/478X Device Handler process uses this assign to read and update dynamic terminal data as transactions are processed.

ATM-ATD-DYN-SCRATCH-PAD — The fully qualified file name of the BASE24-atm Terminal Data Dynamic File—scratch pad (ATDD2). This file contains dynamic token data for reversal processing. This assign is generally optional, but required for ATDD2 processing.

ATM-ATD-STATIC-GNRL — The fully qualified file name of the BASE24-atm Terminal Data Static File—general data (ATDS1). This file contains static data, which is information that does not change during the course of transaction processing. The Diebold 10XX/478X Device Handler process uses this assign to retrieve static terminal data, which is maintained in extended memory.

ATM-DH-ATD-EXTMEM-SWAPVOL — The fully qualified name of the BASE24-atm Terminal Data files extended memory swap volume. This assign is optional.

Example: \B24.$DATA

ATM-DH-CNF1000 — The fully qualified file name of the configuration file for Diebold 10XX/478X ATMs (for example, \B24.$DATA.PRO1DATA.CNFGIX). This assign is used by Diebold 10XX/478X Device Handler processes when downloading data to terminals. This value can be overridden by a corresponding value in the device-dependent information in the individual ATD record for the ATM.

ATM-DH-OIF1000 — The fully qualified file name of the Operator Identification File (OIF) (for example, \B24.$DATA.PRO1DATA.OIF1000). This assign is used by Diebold 10XX/478X Device Handler processes when the host-controlled solenoid-activated lock feature is supported. This feature is available with Diebold 2.1 firmware.

ATM-NIDFEMT — The fully qualified file name of the Note ID File (NIDF) extended memory table, which contains NIDF records.

ATM-TRFEMT — The fully qualified file name of the Terminal Receipt File extended memory table (TRFEMT), which contains TRF records.

ERFEMT — The fully qualified file name of the Exchange Rate File (ERF) extended memory table. This assign is used by the Currency Conversion utilities.

IDF — The fully qualified file name of the Institution Definition File (IDF).

IDFEMT — The fully qualified file name of the Institution Definition File (IDF) extended memory table (IDFEMT), which contains IDF records.

LMCF — The fully qualified file name of the Log Message Configuration File (LMCF). This file is used to specify event message formats.

RCPTEMT — The fully qualified file name of the Receipt File extended memory table (RCPTEMT), which contains Receipt File (RCPT) records.

SECURE-DEV1 — The name of the primary Transaction Security Services process used for PIN management, message authentication processing, and security module key management.

SECURE-DEV2 — The name of the secondary Transaction Security Services process used for PIN management, message authentication processing, and security module key management.

SECURE-DEV-*xx* — The name of one of 20 Transaction Security Services processes accessed on a rotating basis, where *xx* is any two characters.

Params

The following params are used for setting processing values at the logical network level. Add-on products, features and extensions use additional params that are described only in appendixes in this manual.

ATM-DH-ADDENDS — The number used as an offset to determine which set of screens to display if a secondary language has been selected at the ATM. The Device Handler process adds the value of this param to the screen number of the primary language to determine the screen number of the secondary language.

ATM-DH-AUTH-TIMER — The number of seconds the Device Handler process waits for a reply from the BASE24-atm Authorization process before denying a request.

ATM-DH-CONS-MAC-ERR-LMT — A code indicating the number of consecutive message authentication code (MAC) errors that can occur before the Device Handler process sends a shutdown message to the ATM. If the institution has purchased the Dynamic Key Management add-on product, the Device Handler process shuts down the ATM, loads new PIN and MAC keys, and attempts to reopen the ATM. Refer to appendix E for more information about the Dynamic Key Management add-on product.

ATM-DH-DISPLAY-NATIVE-STATUS-MSG — A value indicating whether the native status message data is displayed in the event message. If this parameter does not exist or has a value other than Y, the native status message data is not displayed in the event message.

ATM-DH-DYN-TBL-RETN-PRD-LEASE — The number of seconds the Device Handler process waits for the next leg of a multiple-leg transaction before the dynamic data for a terminal connected to a leased line can be removed from the dynamic data table. A value of 0 disables this processing.

ATM-DH-MAX-TERMS — The maximum number of BASE24-atm Terminal Data files (ATD) records that each BASE24-atm Diebold 10XX/478X Device Handler process can hold in extended memory at one time. Thus, this param indicates the maximum number of terminals each Diebold 10XX/478X Device Handler process can service.

ATM-DH-MAX-TERMS-IN-CPLF-TBL — The maximum number of unique terminals that can exist at one time in the CPLF extended memory table. A symbolic name (for example, P1A^D912) is required in this field. A value of 1/5 of the value specified in the ATM-DH-MAX-TERMS param is the default.

ATM-DH-MAX-TERMS-IN-DYN-TBL — The maximum number of BASE24-atm Terminal Data Dynamic File—general data (ATDD1) records that each BASE24-atm Diebold 10XX/478X Device Handler process can store in extended memory at one time. Thus, this param indicates the amount of extended memory each Device Handler process must reserve for ATDD1 records and the maximum number of transactions each Device Handler process can process at one time.

ATM-DH-MAX-TERMS-IN-STATIC-TBL — The maximum number of BASE24-atm Terminal Data Static File—general data (ATDS1) records that each BASE24-atm Diebold 10XX/478X Device Handler process can store in extended memory at one time. Thus, this param indicates the amount of extended memory each Device Handler process must reserve for ATDS1 records.

ATM-DH-OIF1000-CHK — A flag indicating whether the Device Handler process checks the Operator Identification File (OIF) to determine access to the solenoid lock on the ATM chest door. This param is used by the Device Handler process for ATMs that support the host-controlled solenoid-activated lock feature.

ATM-DH-OPT-LVL — The number of times the Device Handler process writes to the ATD for normal transactions. This param indicates whether the Device Handler process uses one write or two writes to the ATD for normal transactions.

ATM-DH-STALE-RQST-TIMEOUT — The number of seconds an incoming transaction can remain in the queue of the Device Handler process before being discarded by the Device Handler process as too old to process.

ATM-DH-TERM-IN-CPLF-RETN-PRD — The number of minutes that the BASE24-atm Check Proof List files (CPLF) records and its terminal are retained in extended memory. Valid values are 1 through 60. A value of 5 is the default.

ATM-DISCR-DATA-STM-SUPPRS — A flag indicating whether discretionary data carried in the STM should be suppressed on the ATD and in the STM held in memory when the transaction has been completed or cancelled. If this parameter does not exist or has a value other than Y, discretionary data is not suppressed in the STM.

ATM-FCA-CUTOVER-AUDIT — A flag indicating whether the Forced Cutover Audit function is turned on. This param specifies the use of the Forced Cutover Audit function for the entire logical network. It is used only if the Forced Cutover Audit add-on product has been purchased and installed.

ATM-FEE-INTRACTV-RESP-NTFY — A flag indicating whether surcharge interactive response notification processing requirements (i.e., performing interactive notification sequences for acquirer and issuer fees and rebates) is being used.

ATM-REAR-ADMN-TRAN-PIN-ALWD — A flag indicating whether rear administrative transaction requests containing PIN data should be allowed.

DATE-DISPLAY-TYPE — A code indicating the format of the month and day portion of the dates on reports and BASE24-atm Device Handler receipts (that is, DDMM or MMDD). The DATE-DISPLAY-TYPE param does not affect the year portion of dates that include the year, such as the format YYMMDD.

REVERSE-BAL-INQ — A code indicating whether or not balance inquiry transactions can be reversed.

SECURE-DEV-FAIL-NOTIFY — The interval at which critical messages are logged to EMS due to the failure of a process' primary or secondary security modules.

SECURE-DEV-TIM-LMT — The maximum amount of time a BASE24 process waits for a response from the security device before the message times out.

SECURE-DEV-TYP — A code indicating the type of security module that is being used by the logical network.

BASE24-atm Terminal Data Files (ATD)

The BASE24-atm Terminal Data file (ATD) records for an ATM define much of its functionality. Whenever the BASE24-atm system receives a message from an ATM, or a message is sent to an ATM, the ATD records for the ATM is accessed. Separate ATD records must exist for each ATM controlled by the BASE24-atm product.

BASE24-atm uses the BASE24-atm Terminal Data Dynamic File—general data (ATDD1) to help maintain ATM control. Each record in this file contains information reflecting the current status of an ATM. The status of every terminal is monitored continuously so that the BASE24-atm system operator is notified when a problem is detected. The system operator, in turn, can notify the terminal-owning institution about any errors at its ATMs that may require service intervention. The institution can determine from the type of error reported whether the problem is of a nature that the ATM vendor should be called or if it can be corrected by the institution staff.

The BASE24-atm Terminal Data Static File—general data (ATDS1) for an ATM contains the ATM static data. This is data that does not change during transaction processing. Whenever BASE24 receives a message from an ATM, or a message is sent by BASE24 to an ATM, the static ATD record for that ATM is accessed. A separate ATDS1 record must exist for each ATM controlled by BASE24-atm.

You can access ATD records using BASE24-atm files maintenance functions. ATD information is accessed by 15 screens. To view these screens for Diebold 10XX/478X ATMs, set the DEVICE TYPE field on ATD screen 1 to 30. The first six screens contain general information that is common to all ATMs, regardless of manufacturer. The next three screens contain information specific to Diebold 10XX/478X ATMs. The following three screens contain information specific to the BASE24-atm self-service banking (SSB) product, and are only used if you have installed the SSB product. Screen 13 contains configurable dispense information specific to Diebold 10XX/478X ATMs. Screen 21 contains EMV information. Screen 22 contains AKDS (Automated Key Distribution System) information that is common to all ATMs regardless of the manufacturer. For information on the values of the individual fields in the ATD and how the ATD can be maintained, refer to the ***BASE24-atm Files Maintenance Manual.*** Refer to the ***BASE24-atm Diebold 10XX/478X SSB Manual*** for information about the SSB screens. Refer to the ***BASE24-atm EMV Support Manual*** for information about the EMV screen.

General ATD Information

The basic information that is contained in ATD records for a Diebold 10XX/478X ATM includes the following:

- Reference information
- Transaction processing information
- Status information
- Transaction security information
- AKDS (Automated Key Distribution System) configuration data

Reference information includes information such as the physical location of the terminal, the owner of the terminal, and the Device Handler and Authorization processes that are involved in the transaction processing of the specified ATM.

Transaction processing information includes information on the kind of not-on-us transactions allowed at the terminal for a specific type of terminal. The Authorization process determines if a match exists between the sharing groups indicated in the ATD record for the ATM and the sharing groups set in the Institution Definition File (IDF) for the institution that issued the card accepted by the ATM. This check determines the type of transaction agreement, if any, that has been made between the ATM owner and the institution owning the card.

Information on status changes to the ATM such as the state of the ATM and the amount of transaction activity for the ATM is also kept in the ATD.

Transaction security information relating to the encryption and decryption of data, local PIN verification, and message authentication is defined in the ATD.

AKDS configuration information relating to the name of the Host Public Key Owner (also known as the Key Signer), the ATM manufacturer, and the EPP format, is also defined in the ATD.

Specific ATD Information about the Diebold 10XX/478X

The ATD also contains information specific to Diebold ATMs. The Diebold-specific screens contain information such as the type of line protocol used with the terminal, cardholder balance information, cardholder receipt information, message authentication code (MAC) information, inactivity timer information, and the PIN encryption method used at the ATM.

Transaction activity can be monitored through a series of timer settings established in the ATD for peak and nonpeak periods and holiday schedules. Receipt groups for configurable receipts are defined in the ATD.

The Diebold 10XX/478X Device Handler process supports Diebold MDS, *i*-series, *ix*-series, and Opteva-series ATMs using either the 911 or 912 message format. The message format used affects the relationship between the Diebold ATM dispense cassettes and the hopper contents established in the ATD. The differences in these relationships are important to know when setting up the ATD.

911 Mode

When operating in 911 mode, the Diebold Device Handler process determines the denomination of a given cassette based on the ATD bill value associated with that cassette type. When a Diebold 10XX/478X ATM is operating in 911 mode, the only permissible cassette codes are C and D. Small denomination bills must be in the C cassette and large denomination bills must be in the D cassette. Only two different denominations can be used in 911 mode.

The Diebold 10XX/478X ATM always dispenses small denomination bills from the C cassette and large denomination bills from the D cassette. The Diebold Device Handler process determines the denomination of a given cassette based solely on the bill value of the hopper associated with the type of cassette established in the ATD. The ATM chooses from cassette C the number of bills specified in Hopper 1 (LO bills) of the ATD and chooses from cassette D the number of bills specified in Hopper 2 (HI bills). If only one cassette type is present in the ATM, only one hopper should be established in the ATD.

912 Mode

When a Diebold 10XX/478X ATM is operating in 912 mode, all eight cassette types (A through H) can be used in the ATM, and up to four different denominations can be used. Unlike 911 mode, the denomination present in a given cassette is determined more by the cassette type than by the bill value established in the ATD.

The denominations associated with different cassette types are established in the diebold_1000_curr_tbl table in the C100NAMS File used by the Diebold Device Handler process. You can modify this table.

Even though the cassette types are the major factor in determining the cassette denominations in 912 mode, the hoppers in the ATD must also reflect the cassette denominations. There must be one hopper established in the ATD for each cassette denomination and the ATD hoppers must be established in ascending order. The cassette positions do not need to be in ascending order.

For more information on Diebold 10XX/478X ATM dispensers, refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual.***

Device Type

The device type in the ATD must be set to 30 regardless of the Diebold 10XX/478X model or emulation mode.

Institution Definition File (IDF)

The Institution Definition File (IDF) is used to define the processing parameters for each institution participating in the logical network. The IDF is read by the Device Handler process during an administrative transaction in order to retrieve the dates the Device Handler process requires for cutting over its terminals. In addition, the Device Handler process reads the holiday table in the IDF extended memory table if terminals are configured, through the BASE24-atm Terminal Data files (ATD), to support inactivity timers. The IDF also contains routing tables used for transaction routing within the BASE24-atm system, date parameters, transaction sets, and ATM processing control parameters. For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Acquirer Processing Code File (APCF)

The Acquirer Processing Code File (APCF) contains the processing codes allowed at the terminal. The key to the APCF is the acquirer transaction profile, the message category (the second digit of the message type), and the ISO processing code (a six-character code containing the ISO transaction code, a *from* account type, and a *to* account type). The acquirer transaction profile, which is defined in the ATD or IDF, can be assigned to an individual terminal or to a group of terminals. Acquirer transaction profiles defined in the ATD override the default acquirer transaction profiles defined in the IDF. Each acquirer transaction profile defines a set of transactions supported for an individual acquiring terminal or group of terminals.

For more information on the APCF, refer to the ***BASE24 Base Files Maintenance Manual.***

Log Message Configuration File (LMCF)

The Log Message Configuration File (LMCF) contains one record for each event message that has been altered by a BASE24 network operator. Because LMCF records must be read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized.

An LMCF record is accessed using BASE24 operator functions. The file itself contains changes made to event messages in the form of records. The routing code, severity, and text of each event message can be modified using BASE24 operator functions.

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3: BASE24-atm Device Handler Process

Device Handler processes are the primary interfaces between ATMs and the BASE24-atm system. All messages that are sent by the BASE24-atm system to ATMs must first be received by the Device Handler process and then forwarded to the ATM. Likewise, all messages that are sent by an ATM to the BASE24-atm system must first be received by the Device Handler process.

Because each type of ATM has its own message format and way of processing transactions, separate Device Handler processes must exist for each type of terminal supported by the BASE24-atm product. For example, Diebold 910 ATMs must have their own type of Device Handler process, which differs from the Device Handler process used for Diebold 10XX/478X ATMs or NCR 5XXX ATMs.

This section provides the following information:

- An overview of the BASE24-atm Diebold 10XX/478X Device Handler process
- Processing steps taken by the Device Handler process when it is first started
- BASE24 commands received by the Device Handler process to perform specialized types of activities
- Information on how to access the event messages generated by the Diebold 10XX/478X Device Handler process

BASE24-atm Diebold 10XX/478X Device Handler Process Overview

Diebold 10XX/478X terminals, or other vendor's ATMs emulating the Diebold 10XX/478X-series terminals, are linked to BASE24-atm transaction processing by a Device Handler process tailored to the Diebold 10XX/478X terminals. Every transaction initiated at a Diebold 10XX/478X terminal connected to the BASE24-atm system is passed to this Device Handler process when it enters the BASE24-atm system. The message coming from the terminal is in the native mode of the device and the Device Handler process converts this message into the BASE24-atm Standard Internal Message (STM) for sending to a BASE24-atm Authorization process. The BASE24-atm product supports both financial and administrative transactions.

Transactions processed at different Diebold 10XX/478X terminals can be handled differently depending on how the individual terminal is configured in the BASE24-atm Terminal Data files (ATD). Each terminal has a record in the ATD. This allows each terminal to have different sharing arrangements and transaction sets. The Device Handler process accesses the ATD data whenever it needs information about a terminal. It uses the acquirer transaction profile specified in the ATD as part of the key to the APCF extended memory table. Values in the APCF indicate if and how the transaction is allowed.

The Institution Definition File (IDF) is read by the Device Handler process to obtain the dates the Device Handler process requires for administrative transaction processing. The IDF extended memory table is read by the Device Handler process to obtain the default acquirer transaction profile, if needed. It is also used to determine the holidays established by an institution for use in supporting the inactivity timer option. Inactivity timers enable an institution to monitor transaction activity during specified periods.

The Terminal Receipt File (TRF) extended memory table is read by the Device Handler process to determine whether a cardholder receipt and journal receipt are required for the transaction at the terminal.

The Receipt File (RCPT) extended memory table is read by the Device Handler process to determine the wording and format of financial and administrative receipts.

The Device Handler process also uses the Logical Network Configuration File (LCONF). The Device Handler process reads the LCONF to obtain the names of the various extended memory table files, ATD, and IDF as well as a number of other file names and processing params.

When problems occur with the terminal, the Device Handler process creates event messages and sends them to the Hewlett-Packard (HP) NonStop Event Management Service (EMS).

Initialization

The following steps are taken by the Device Handler process when it is first started or initialized:

1. Initializes its timer table.

   When a Device Handler process creates timers, the timers are placed in the Device Handler timer table. Device Handler processes maintain three types of timers:

   - The authorization response timer monitors each authorization response. If an authorization response is not returned to the Device Handler process within this time limit, the request is automatically denied. This timer is set using the ATM-DH-AUTH-TIMER param.
   - The download timer monitors each station download record. If no response is received from the station within this time limit, the Device Handler process automatically aborts the download. This timer is always set to 45 seconds.
   - The inactivity timer indicates a period of time during which some activity should occur at the terminal. If no activity is determined, the Device Handler process logs a message. This timer is set on ATD screen 7, with different settings for business day and weekend and holiday peak usage time periods. This Device Handler process allows a maximum number of inactivity timeouts according to the value specified in the TIMEOUTS ALLOWED field on ATD screen 7. Once the maximum is reached, terminal monitoring is discontinued.

2. Reads the name of the LCONF into memory.

   The startup message from the XPNET process contains the name of the LCONF, which the Device Handler process reads into memory. This name allows the Device Handler process to access the LCONF for the assigns and params it uses in processing.

3. Opens the LCONF.

4. Reads the following assigns from the LCONF:

   APCFEMT
   ATM-ATD-DYN-GNRL
   ATM-ATD-DYN-SCRATCH-PAD
   ATM-ATD-STATIC-GNRL
   ATM-DH-ATD-EXTMEM-SWAPVOL
   ATM-DH-CNF1000
   ATM-DH-MDS1000-CPLF

ATM-DH-OIF1000
ATM-NIDFEMT
ATM-TRFEMT
IDF
IDFEMT
LMCF
RCPTEMT
SECURE-DEV1
SECURE-DEV2
SECURE-DEV-xx
SSB-BSF

5. Retrieves ATM-ADCF assign for dial-up processing.

6. Retrieves the ERFEMT assign for the Multiple Currency Dispense add-on feature and the Shared BNA Support feature.

7. Initializes the network EMS for writing event messages.

8. Reads the LMCF for its modified event messages.

   The LMCF contains a record for each modified event message in the system. The Device Handler process opens the LMCF, retrieves the modified event messages, and closes the file.

9. Loads the following params from the LCONF:

   ATM-DH-ADDENDS
   ATM-DH-AUTH-TIMER
   ATM-DH-CHK-DISP-FAULT-ACT
   ATM-DH-CHK-BIN-THRHLD
   ATM-DH-CONS-MAC-ERR-LMT
   ATM-DH-DISPLAY-NATIVE-STATUS-MSG
   ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP
   ATM-DH-DYN-TBL-RETN-PRD-LEASE
   ATM-DH-MAX-TERMS
   ATM-DH-MAX-TERMS-IN-DYN-TBL
   ATM-DH-MAX-TERMS-IN-STATIC-TBL
   ATM-DH-MDS1000-CPLF-REQ
   ATM-DH-NON-VALUE-THRESHOLD
   ATM-DH-OIF1000-CHK
   ATM-DH-OPT-LVL
   ATM-DH-PEAK-LEAD-TIME
   ATM-DH-STALE-RQST-TIMEOUT
   ATM-DISCR-DATA-STM-SUPPRS
   ATM-FEE-INTRACTV-RESP-NTFY

ATM-REAR-ADMN-TRAN-PIN-ALWD
DATE-DISPLAY-TYPE
REVERSE-BAL-INQ

10. Loads the BASE-CURRENCY-CODE and CURRENCY-CODE params for the Multiple Currency Dispense add-on feature.
11. Loads the ATM-DH-MAX-QUEUED-CMD and ATM-DH-MAX-LOAD-CMD params for dial-up processing.
12. Loads the ATM-DH-AKDS-TSS-TIMER and ATM-DH-1000-AKDS-SERIAL-NUM-VRFY params for the Automated Key Distribution System add-on product.
13. Loads the following params for Dynamic Key Management add-on product processing. These params are loaded only if the Dynamic Key Management add-on product has been purchased and installed.

    ATM-DH-DKM-KEY-CHNG-INTRVL
    ATM-DH-DKM-MAC-ERR-LMT
    ATM-DH-DKM-MAC-TXN-LMT
    ATM-DH-DKM-PIN-ERR-LMT
    ATM-DH-DKM-PIN-TXN-LMT

14. Loads the ATM-FCA-CUTOVER-AUDIT param if the Forced Cutover Audit add-on feature has been purchased and installed.
15. Loads the ATM-DH-MAX-TERMS-IN-CPLF-TBL and ATM-DH-MAX-TERMS-IN-CPLF-TBL params if the Shared CPM Support add-on feature has been purchased and installed.
16. Retrieves the APCF extended memory table information using the name specified in the APCFEMT assign.
17. Retrieves the TRF extended memory table information using the name specified in the ATM-TRFEMT assign.
18. Retrieves the RCPT extended memory table information using the name specified in the RCPTEMT assign.
19. Retrieves the IDF extended memory table information using the name specified in the IDFEMT assign.
20. Retrieves the ERF extended memory table information using the name specified in the ERFEMT assign if the Multiple Currency Dispense add-on feature is being used.
21. Opens the ATD files using the names specified in the ATD file assigns.
22. Initializes an extended memory area for finding and keeping ATD records.

The Device Handler process uses the ATM-DH-MAX-TERMS param and allocates sufficient memory for the required number of terminals.

23. Opens the IDF with read access using the name specified in the IDF assign.
24. Opens the OIF, if configured to do so, using the ATM-DH-OIF1000-CHK param.
25. Calls a security utility procedure that reads either the SECURE-DEV1 and SECURE-DEV2 assigns or just the SECURE-DEV-*xx* assign from the LCONF and attempts to open the security device, if a security device is being used.

    If the security devices cannot be opened, the Device Handler process creates an event message and sends it to the EMS indicating that one or both of the security devices could not be opened.

    The SECURE-DEV-FAIL-NOTIFY, SECURE-DEV-TIM-LMT, and SECURE-DEV-TYP params are also accessed at this time by the security utility procedure.
26. Reads all ATD records for this Device Handler process name. Initializes station information in the ATD if this is a network cold start.
27. If you purchased the BASE24-atm self-service banking (SSB) Device Handler Extension, the Device Handler process initializes the Check Proof List File (CPLF).
28. If you purchased the SSB Enhanced Check Processing Application, the Device Handler process initializes the Bin Sort File.
29. Closes the LCONF.
30. Raises an event stating initialization is complete.

Warmboot Procedures

The following paragraphs describe how the Device Handler process handles warmboot commands.

WARMBOOT Command Procedures

The WARMBOOT command is a general use operator command sent using the Device Control Terminal (DCT) screen or a network control facility. General use operator commands enable the operator to control or change the way the Device Handler process is operating.

When Device Handler processes are warmbooted, they do not go through their entire initialization routine. Instead, they only perform the following steps:

1. Release the current extended memory used for the ATD.
2. Close and reopen the files they use.
3. Reread the LCONF and retrieve all of their assigns and params (including module-specific assigns and params).
4. Reinitialize the LMCF.
5. Release extended memory used for extended memory tables and reallocate new extended memory based on the current settings in the LCONF, including new extended memory tables.
6. Close and reopen the security module.
7. Reinitialize the ATD.
8. Raise a “Warmboot Complete” event.

For detailed information on the WARMBOOT command and other operator commands, refer to the ***BASE24 Text Command Reference Manual***. For more information on the DCT, refer to the ***BASE24 Device Control Manual***.

Warmboot ATD Procedures

When the Device Handler process receives a 9529ATD command, it warmboots static terminal data held in extended memory. The module rereads its ATM Terminal Data Static File—general data (ATDS1) into extended memory. Any time static data is changed for an Diebold 10XX/478X ATM on the BASE24-atm Terminal Data files (ATD) screens, it is not used in processing until the associated Device Handler process receives this command or is reinitialized.

Device Handler Text Commands

Text commands are used to send information from a network control facility to a process. These commands serve as a tool that you can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process.

The following text commands can be used with the BASE24-atm Diebold 10XX/478X Device Handler process:

- The 9500OPEN or 9500STARTUP command causes the Device Handler process to send the appropriate native-mode message to open the ATM.
- The 9500CLOSE or 9500SHUTDOWN command causes the Device Handler process to send the appropriate native-mode message to close the ATM.
- The 9500RESET command causes the Device Handler process to interrupt any current transaction or load in progress for the given ATM. You should not enter this command unless requested to do so by ACI personnel.
- The 9501 command directs the Device Handler process to warmboot.
- The 9529ATD command causes the Device Handler process to release static terminal data from extended memory.
- The ALTERATTRIBUTE command causes the Device Handler process to warmboot (reallocate) a specified shared extended memory table.
- The STATUSEXTMEM command causes the Device Handler process to display statistical information about its BASE24-atm Terminal Data files (ATD) extended memory segment on the network log.

In addition, load commands entered from the Device Control Terminal (DCT) direct the Device Handler process to download configuration information to the ATM.

For a list of the steps the Diebold 10XX/478X Device Handler process follows when a warmboot command is issued, refer to the "Warmboot Procedures" topic earlier in this section.

For detailed descriptions and the syntax of all BASE24 text commands, refer to the ***BASE24 Text Command Reference Manual.*** For more information about entering load commands from the DCT, refer to the ***BASE24 Device Control Manual.***

Event Messages

The Diebold 10XX/478X Device Handler process generates event messages to assist network operators in performing their daily tasks. Event messages provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

The Diebold 10XX/478X Device Handler process generates event messages using a standard routing code of one.

Event message documentation specific to the Diebold 10XX/478X Device Handler process can be found in the $vol.ba*xx*logm.at1000 file where *xx* is the current BASE24 release (for example, 60 for BASE24 Release 6.0, regardless of the version). Each file on the ba*xx*logm subvolume contains event messages for a specific BASE24 module. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 module.

Operators also can view the contents of ATM status messages received by the Device Handler process in their native format. The ATM-DH-DISPLAY-NATIVE-STATUS-MSG param in the LCONF indicates whether the Device Handler process sends this information to the EMS system so it can be reviewed.

For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual.***

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4: Configuration File Maintenance

The Diebold Configuration File (CNFGIX), which is resident in the BASE24-atm system, can be accessed using the Diebold entry program and downloaded using the DCT.

With access to the CNFGIX, you can make changes to ATM screen displays, state table entries, financial institution table (FIT) data, configuration parameters, timers, and printer configuration. You can then download these changes to the ATM. Information on downloading configuration file information to Diebold ATMs can be found in the ***BASE24 Device Control Manual***. Information about the Diebold terminals and configuration options can be found in Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual***.

This section details the Diebold entry program screens you can use to make changes to the CNFGIX.

Diebold Entry Programs

The Diebold entry programs access the Diebold Configuration File (CNFGIX) that controls Diebold ATM display screens, states, timers, financial institution table (FIT) data, and printer configuration. All of the data in the CNFGIX can be modified through the entry programs.

The Diebold 10XX/478X supports the following Diebold entry programs:

ENTIBOLD — Supports all Diebold states and cardholder screens and the FIT records, miscellaneous data, rear balance functions, consumer and statement printer configuration, and other information for Diebold and Wincor ATMs. The program supports operation codes 070–078 for state SC (smart card). New Diebold-specific enhancements should be added to this program.

ENTIBLD2 — Supports all Diebold states, Wincor EMV state e, Wincor EMV AID table, cardholder screens, FIT records, miscellaneous data, timers, and printer configuration. This program supports operation codes 010, 011, 095, 097, and 099 for state SC (smart card). This program is provided for backwards compatibility and should not be updated for new enhancements.

ENTWBOLD — Supports all Wincor states, Wincor tables and cardholder screens. This program supports Wincor-specific screens, states, and tables only. Any new Wincor-specific enhancements should be added to this program.

The Diebold entry programs provide the mechanism to configure Diebold ATMs or Wincor ATMs using the BASE24-atm product. For comprehensive documentation regarding the Diebold or Wincor ATMs, individuals should consult the available vendor literature.

The Diebold entry programs allow you to define the following attributes:

- ATM display screens
- State tables
- Miscellaneous data
- Financial institution table
- Rear balance function definition data
- Consumer printer-configured text definition
- Statement printer-configured text definition

- Merchant Banking Center buttons and amounts
- EMV application data
- EMV scheme data
- Wincor EMV AID table

Note that the ENTWBOLD program is used to configure Wincor ATM display screens, Wincor State screens and the Wincor EMV AID table. All other functions for Wincor devices must be configured through the ENTIBOLD program.

Accessing the Diebold Entry Programs

The Diebold entry programs can be used to inquire, add, delete, change, and print the CNFGIX. The process normally resides on the AT*xx*OBJ subvolume, where *xx* is the number of the current release. To access the Diebold entry program, the following command must be entered from the Hewlett-Packard (HP) NonStop TACL prompt.

Syntax

RUN {ENTIBOLD | ENTIBLD2 | ENTWBOLD} [/ [IN *configfile*] [,OUT *listfile*] /] [$*terminal*]

IN *configfile* — Specifies the file name of the configuration file to be accessed. This name must be entered using the HP NonStop file naming convention of $*volume.subvolume.filename*. If omitted, the default is CNFGIX on the current volume and subvolume.

OUT *listfile* — Specifies the device where print functions direct their output. If omitted, the default is $S.#HOLD.

$*terminal* — Specifies the terminal from which you perform the file maintenance. It must be a HP NonStop 6520/6530 terminal. If no terminal is specified, the default is the home terminal from which the program was started.

Example

```
RUN ENTIBOLD / IN $data.pro1data.CNFGIX,OUT $s.#cnfg /
RUN ENTIBLD2 / IN $data.pro1data.CNFGIX,OUT $s.#cnfg /
RUN ENTWBOLD / IN $data.pro1data.CNFGIX,OUT $s.#cnfg /
```

Configuration File Function Keys

The function keys specific to most of the screens associated with the CNFGIX are described below.

Throughout BASE24 manuals, references to these function keys include only BASE24 function keys. Specific keyboards may require the use of a combination of keys to achieve the functionality. The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these function keys.

Key	Description
F1	**Enter —** Checks the data that has been entered on the screen for errors and enters the data in the file.
F2	**Read Data —** Reads data in the file.
F3	**Add Data —** Adds data to the file. The data added must be unique within the file.
F4	**Delete Data —** Deletes data from the file.
F5	**Update Data —** Changes data already in the file.
F9	**Next —** Displays the next entry screen of the file in which you are working. If the last screen of the file is displayed when you press this key, an end of file error is displayed.
F10	**Print Screen —** Sends the screen currently being displayed to the spooler location specified on the Diebold Entry Program Menu in the Output Device For PRINT Function field as explained in this section.
F16	**Exit —** Exits the Diebold entry program or the configuration screen currently being accessed. This function key is operable on the Help screen to enable you to resume normal processing.
Shift-F16	**Exit —** Exits the Diebold entry program.

Diebold Entry Program Menus

When the Diebold entry program is run, the system displays the main menu for the Diebold Configuration File (CNFGIX). The menu is used to select the type of configuration file data to be accessed, to print all, or selected parts, of the CNFGIX, and to create a new CNFGIX.

ENTIBOLD Diebold Entry Program Menu

The Diebold Entry Program Menu for the ENTIBOLD program is shown below, followed by descriptions of its function codes. Refer to the topic “ENTIBOLD and ENTIBLD2 Menu Field Descriptions” for a description of the menu fields.

```
      INTERBOLD i Series/MDS CONFIGURATOR PROCESS - VERSION n.n.nn (DDMmmYY)

Code  Functional Description
----  ------------------------------------------------------------------------
  1.  ATM Display Screens  **> PRINT functions from this menu generate cross-
  2.  State Tables         **> references of Screen, Configured Text and States.
  3.  Miscellaneous Data
  4.  FIT (Financial Institution Table)
  5.  Rear Balance Function Definition Data (Screen 009)
  6.  Consumer Printer Configured Text Definition
  7.  Statement Printer Configured Text Definition
  8.  Merchant Banking Center Data
  9.  EMV Application Data
 10.  EMV Scheme Data


Enter Function Code (from above menu) and press appropriate function key ...

Interbold Configuration File Name ......... $YYYY.XXXX.CNFGIX
Output Device for PRINT function .......... $S.#HOLD

================================================================================
    F1=ENTER        F3=CREATE FILE           F10=PRINT          F16=EXIT
                                       SHIFT-F10=PRINT FILE
```

The Diebold Entry Program Menu for the ENTIBOLD program supports the following function codes:

1 = ATM Display Screens. This code allows you to access the Diebold ATM high-level and low-level display screens.

2 = State Tables. This code allows you to access the InterBold State Table screen.

3 = Miscellaneous Data. This code allows you to access the Miscellaneous Data screen.

4 = FIT (Financial Institution Table). This code allows you to access the Financial Institution Table screen.
5 = Rear Balance Function Definition Data (Screen 009). This code allows you to access the Rear Balance Function Definition Data screen.
6 = Consumer Printer Configured Text Definition. This code allows you to access the Consumer Printer Configured Text Definition screen.
7 = Statement Printer Configured Text Definition. This code allows you to access the Statement Printer Configured Text Definition screen.
8 = Merchant Banking Center Data. This code allows you to access the Merchant Banking Center Data screen.
9 = EMV Application Data. This code allows you to access the EMV Application Data screen.
10 = EMV Scheme Data. This code allows you to access the EMV Scheme Data screen.

ENTIBLD2 Diebold Entry Program Menu

The Diebold Entry Program Menu for the ENTIBLD2 program is shown below, followed by descriptions of its function codes. Refer to the topic “ENTIBOLD and ENTIBLD2 Menu Field Descriptions” for a description of the menu fields.

```
      INTERBOLD i Series/MDS CONFIGURATOR PROCESS - VERSION n.n.nn (DDMmmYY)

Code  Functional Description
----  ------------------------------------------------------------------------
  1.  ATM Display Screens  **> PRINT functions from this menu generate cross-
  2.  State Tables         **> references of Screen, Configured Text and States.
  3.  Miscellaneous Data
  4.  FIT (Financial Institution Table)
  5.  Rear Balance Function Definition Data (Screen 009)
  6.  Consumer Printer Configured Text Definition
  7.  Statement Printer Configured Text Definition
  8.  Merchant Banking Center Data
  9.  EMV Application Data
 10.  EMV Scheme Data
 11.  Wincor EMV AID table


Enter Function Code (from above menu) and press appropriate function key ...

Interbold Configuration File Name ......... $YYYY.XXXX.CNFGIX
Output Device for PRINT function .......... $S.#HOLD

================================================================================
    F1=ENTER        F3=CREATE FILE          F10=PRINT         F16=EXIT
```

The Diebold Entry Program Menu for the ENTIBLD2 program supports the following function codes:

1 = ATM Display Screens. This code allows you to access the Diebold ATM high-level and low-level display screens.
2 = State Tables. This code allows you to access the InterBold State Table screen.
3 = Miscellaneous Data. This code allows you to access the Miscellaneous Data screen.
4 = FIT (Financial Institution Table). This code allows you to access the Financial Institution Table screen.
5 = Rear Balance Function Definition Data (Screen 009). This code allows you to access the Rear Balance Function Definition Data screen.
6 = Consumer Printer Configured Text Definition. This code allows you to access the Consumer Printer Configured Text Definition screen.
7 = Statement Printer Configured Text Definition. This code allows you to access the Statement Printer Configured Text Definition screen.
8 = Merchant Banking Center Data. This code allows you to access the Merchant Banking Center Data screen.
9 = EMV Application Data. This code allows you to access the EMV Application Data screen.
10 = EMV Scheme Data. This code allows you to access the EMV Scheme Data screen.
11 = Wincor EMV AID Table. This code allows you to access the Wincor EMV AID Table screen.

ENTWBOLD Diebold Entry Program Menu

The Diebold Entry Program Menu for the ENTWBOLD program is shown below, followed by descriptions of its function codes. Refer to the topic “ENTWBOLD Menu Field Descriptions” for a description of the menu fields.

```
DIEBOLD ENTRY PROGRAM SHARED/WINCOR SUPPORT MAIN MENU (version 11.09)

Code  Functional Description

----  ------------------------------------------------------------------------

   1.  Customer ATM Display Screens
   2.  Wincor/Shared State Screens
   3.  Wincor EMV Application ID (AID) Table



Enter Function Code then press the appropriate function key ................

Diebold Configuration File Name ........... $BRIDGE.AT20IBLD.CNFGIX

Output Device for PRINT function .......... $S.#HOLD



================================================================================

    F1=ENTER        F3=CREATE FILE           F10=PRINT          F16=EXIT
                                        SHIFT-F10=PRINT FILE
```

The Diebold Entry Program Menu for the ENTWBOLD program supports the following function codes:

1 = Customer ATM Display Screens. This code allows you to access the Wincor ATM high-level and low-level display screens.
2 = Wincor State Screens. This code allows you to access the Wincor State Table screen.
3 = Wincor EMV Application ID (AID) Table. This code allows you to access the Wincor EMV AID Table screen.

ENTIBOLD and ENTIBLD2 Menu Field Descriptions

The ENTIBOLD and ENTIBLD2 Diebold Entry Program Menus contain the following fields:

Enter Function Code — A code identifying the configuration data to be modified. The list at the top of the screen identifies the valid values. The **F1** key must be pressed to enter the desired code. The system responds by displaying the appropriate screen for modifying the selected configuration data.

Field Length:	2 numeric characters
Required Field:	Yes, if the **F1** key is pressed.
Default Value:	No default value

Interbold Configuration File Name — The file name of the Diebold Configuration File. This name must be entered using the HP NonStop file naming convention of $*volume.subvolume.filename*. The default is CNFGIX on the current volume and subvolume.

When the file name entered does not exist, the system displays a message at the bottom of the screen advising that the file name entered does not exist. You then select another file or create a new configuration file.

To create a new configuration file, enter the new file name directly over the displayed file name and press the **F3** key. The system responds by creating the new file and displays a message at the bottom of the screen indicating that the new file was created.

At times, the file can be opened with read-only access due to a possible security violation or because another process has the CNFGIX open. The message "FILE CAN ONLY BE OPENED READ-ONLY" is displayed at the bottom of the screen. When this occurs, all functions can still be performed except for add, delete, or update.

If the file can only be opened with read-only access, you should check for a possible security violation. If another process has the file open, you must select another file or wait until the other user is finished.

Field Length:	35 alphanumeric characters
Required Field:	Yes
Default Value:	$*currvol.currsubv*.CNFGIX, where *currvol* is the current volume and *currsubv* is the current subvolume

Output Device for PRINT Function — The name of the output device for items to be printed.

This field is required and can be either a printer (or the spooler), an entry-sequenced file, or another terminal. It cannot be the same terminal where the Diebold entry program is running.

When the output device named is another terminal, the Diebold entry program performs paging and prompts you to press the **Return** key before starting a new page. To stop the print function, press the **Control-Y** keys.

Whenever a print function is in progress, the terminal keyboard is locked (all keyboard functions are disabled). Once the print function is completed, the keyboard is unlocked.

Example:	$S.#PLP
Field Length:	35 alphanumeric characters
Required Field:	Yes, if the **F10** key is pressed.
Default Value:	$S.#HOLD

ENTWBOLD Menu Field Descriptions

The ENTWBOLD Diebold Entry Program Menus contain the following fields:

Enter Function Code — A code identifying the configuration data to be modified. The list at the top of the screen identifies the valid values. The **F1** key must be pressed to enter the desired code. The system responds by displaying the appropriate screen for modifying the selected configuration data.

Field Length:	2 numeric characters
Required Field:	Yes, if the **F1** key is pressed.
Default Value:	No default value

Diebold Configuration File Name — The file name of the Diebold Configuration File. This name must be entered using the HP NonStop file naming convention of $*volume.subvolume.filename*. The default is CNFGIX on the current volume and subvolume.

When the file name entered does not exist, the system displays a message at the bottom of the screen advising that the file name entered does not exist. You then select another file or create a new configuration file.

To create a new configuration file, enter the new file name directly over the displayed file name and press the **F3** key. The system responds by creating the new file and displays a message at the bottom of the screen indicating that the new file was created.

At times, the file can be opened with read-only access due to a possible security violation or because another process has the CNFGIX open. The message “FILE CAN ONLY BE OPENED READ-ONLY” is displayed at the bottom of the screen. When this occurs, all functions can still be performed except for add, delete, or update.

If the file can only be opened with read-only access, you should check for a possible security violation. If another process has the file open, you must select another file or wait until the other user is finished.

Field Length:	35 alphanumeric characters
Required Field:	Yes
Default Value:	$*currvol.currsubv*.CNFGIX, where *currvol* is the current volume and *currsubv* is the current subvolume

Output Device for PRINT Function — The name of the output device for items to be printed.

This field is required and can be either a printer (or the spooler), an entry-sequenced file, or another terminal. It cannot be the same terminal where the Diebold entry program is running.

When the output device named is another terminal, the Diebold entry program performs paging and prompts you to press the **Return** key before starting a new page. To stop the print function, press the **Control-Y** keys.

Whenever a print function is in progress, the terminal keyboard is locked (all keyboard functions are disabled). Once the print function is completed, the keyboard is unlocked.

Example:	$S.#PLP
Field Length:	35 alphanumeric characters
Required Field:	Yes, if the **F10** key is pressed.
Default Value:	$S.#HOLD

Function Keys

The use of two function keys on the Diebold Entry Program Menu varies from the standard function keys explained earlier in this section. In addition, one key which was not assigned a function in the standard definitions is assigned a special function for this screen. The use of these function keys is explained below.

Key	Description
F3	**Create File —** Creates a new configuration file.
Shift-F10	**Print File —** Prints the contents of the configuration file to the specified output device.

Configuration File Data

The following are Diebold entry program screens that are used to modify the configuration file.

ATM Display Screens

ATM display screens relay information and ask for input from the cardholder at the ATM. Display screens can be modified using the entry program.

State Tables

The states define the actions and processing the ATM performs under predefined conditions. Refer to the topic “State Tables” for a complete list of states.

Miscellaneous Data

Five miscellaneous data fields and 16 internal timeout values can be altered using the entry program. For example, some of the timeout values include cardholder response time, cardholder timeout response, close screen display time, depository response time, and delivery door timeout.

Financial Institution Table Screen

The Financial Institution Table screen contains PIN verification information, institution IDs, routing information for multi-institution terminal use, and data on where to find specific information on a card.

Rear Balance Function Definition Data

The Rear Balance Function Definition Data screen defines special information needed when doing a rear balance administrative transaction.

Note: ATM customers are strongly discouraged from entering PINs into rear keypads, based on Payment Card Industry (PCI) security restrictions. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information on configuring the ATM-REAR-ADMN-TRAN-PIN-ALWD param to decline rear balancing administrative transaction requests containing PINs.

Consumer Printer Configured Text Definition

The Consumer Printer Configured Text Definition screen is used to define special codes that control how the receipts printed on the consumer printer look. These codes are then downloaded to the terminal.

Statement Printer Configured Text Definition

The Statement Printer Configured Text Definition screen is used to define special codes that control how the receipts printed on the statement printer look. These codes are then downloaded to the terminal. Note that the statement printer is only supported on Diebold *ix*-series terminals.

Merchant Banking Center Data

The Merchant Banking Center Data screen is used to define the x and y coordinates for placing buttons and amounts on the Merchant Banking Center change mix touch screen at the ATM. This screen is used by the Diebold Merchant Banking Center product only.

EMV Application Data

The EMV Application Data screen is used to define the EMV application IDs. Refer to the ***BASE24-atm EMV Support Manual*** for more information about this screen.

EMV Scheme Data

The EMV Scheme Data screen is used to define EMV card scheme data. Refer to the ***BASE24-atm EMV Support Manual*** for more information about this screen.

Wincor EMV AID Table

The Wincor EMV AID Table screen is used to add an entry to the application ID table for Wincor D91X EMV devices. This screen can be accessed from the ENTIBLD2 and ENTWBOLD programs only. Refer to the ***BASE24-atm EMV Support Manual*** for more information about this screen.

Local PIN Verification

The BASE24-atm product supports local PIN verification in which Diebold terminals, which have PIN verification capability, perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network.

Local PIN verification requires changes to the FIT, the PIN Entry state, and the Transaction Request state in the CNFGIX. Refer to the topic “Financial Institution Table” for a description of the changes to the FIT. The following changes must be made to the PIN Entry state and the Transaction Request state to support local PIN verification.

Note: Local PIN verification and PIN Change transactions cannot be used simultaneously. You can select one or the other—but not both.

PIN Entry State

The following fields must be set to specific values in the PIN Entry state to support local PIN verification:

- The GOOD PIN NEXT STATE field must be configured to specify the next state to which the terminal proceeds upon successful local PIN verification.
- The REMOTE PIN NEXT STATE field must contain the value 255 (not used).
- The MAX BAD PIN NEXT STATE field must be set appropriately.
- The ERROR SCREEN field must be set appropriately.

Transaction Request State

The SEND PIN BUFFER field in the Transaction Request state must be set to 000 (do not send PIN buffer).

For complete information on local PIN verification, refer to Diebold's ***Data Security Procedures and Reference Manual.***

ATM Display Screens

The ATM Display screens can be accessed by entering function code 1 for ATM Display screens on the Diebold Entry Program Menu. These screens allow modification of the screens displayed on the ATM.

ATM screens can be displayed in a low-level or in a high-level mode for changing screens. You can move between the two modes by pressing the **F1** key.

The Diebold standard display screens for high-level display and low-level display are described below and on the following pages.

In general, the text on a screen appears on the Diebold ATM just as it appears on the InterBold i series/MDS Display screen (also known as the high-level display screen). The InterBold Low-Level Display screen allows you to customize the screen for such things as cursor movement, voice response, and animation sequence. Refer to the topic "Screen Control Codes" that follows the description of the low-level standard display screen for a complete list of items that can be customized.

InterBold i Series/MDS Display Screen

The InterBold i Series/MDS Display screen displays the text as it appears on the ATM. Diebold display screens 4 through 9 are restricted to certain formats; the Diebold entry program issues a warning before allowing access to these screens. Screens 0 through 3 also have a special meaning to the Diebold ATM. The Diebold entry program issues a warning when attempting to add or update screens 0 through 3; each must contain a specific message.

Screen Number	Screen Message
0	The cardholder has taken too much time to make a selection.
1	The ATM is offline.
2	The ATM is out of service.
3	The ATM is out of service.

If the complete screen table is printed, a cross-reference listing is generated as well, showing a list of screens that are available. (You can print the screen table using either the **F10** or **Shift-F10** keys. To use the **F10** key, you must put 1 in the function code field. Using the **Shift-F10** keys prints the entire file.)

The InterBold i Series/MDS Display screen is shown below, followed by descriptions of its fields.

```
 Screen 000  Lang Bank 000        @ABCDEFGHIJKLMNO0123456789:;<=>?PQRSTUVW
 Group 0000  Clear Screen? Y    @ >                                        @
                                A >                                        A
 **************************     B >                                        B
 * INTERBOLD i Series/MDS *     C >                                        C
 *     DISPLAY SCREEN     *     D >                                        D
 **************************     E >                                        E
                                F >                                        F
   Overlays: 1.  N/A            G >                                        G
             2.                 H >             <text entry>               H
             3.              F >I >                                        I< A
             4.                 J >                                        J
             5.                 K >                                        K
                             G >L >                                        L< B
 +--------------------+         M >                                        M
 | NOTE:  use '^' to  |         N >                                        N
 | display    Track1  |      H >O >                                        O< C
 | Name.  Use '@' to  |         0 >                                        0
 | position   cursor  |         1 >                                        1
 | for data entry.    |      I >2 >                                        2< D
 +--------------------+         3 >                                        3
                                  @ABCDEFGHIJKLMNO0123456789:;<=>?PQRSTUVW

F1=LO  F2=RD  SF2=RD-OVRLAY  F3=ADD  F4=DEL  F5=UPDT  F9=NXT  F10=PRNT  F16=EXIT
```

Screen — A number that is associated with the desired screen. Valid values are 000 through 511, 900, 950, and 999 for STP/DX-based processors. Valid values are 000 through 009 (reserved for use by Diebold), 010 through 899, 900 through 950 (reserved for use by Diebold), and 999 for PC-based processors (BTP, ATP, and CTP).

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Lang Bank — A number associated with the language in which the screen is displayed. This key field allows the same screen number to be displayed in multiple languages. Valid values are 0 through 7.

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Group — The group number used to identify a particular load group such as a collection of states, screens, etc., that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field on Diebold 10XX/478X screen 7 of the BASE24-atm Terminal Data files (ATD).

Groups 300 through 303 and 311 through 313 are described in section 6 of this manual, and groups 305, 306, 310, 315, 354, 355, 364, and 365 are described in appendix C. The following groups are described in more detail in the ***BASE24-atm Diebold 10XX/478X SSB Manual***: 12, 52, 54, 56, 58.

The following groups of data are provided by ACI as defaults in the Diebold Configuration File.

0 = The basic set of states and screens that control all ATM transactions; this group is always downloaded to the terminal.
1 = Adds icons to some of the screens.
3 = Replaces dollar entry states with information entry states. Note that this group does not allow statement print transactions.
4 = Supports the Diebold *ix*-series statement printer running in native mode.
5 = Fast cash transactions.
6 = Supports the Diebold *ix*-series touch screen.
7 = Changes screen formats to support ATMs that have left-hand operation keys.
9 = Changes screen formats to support ATMs that have card swipes.
10 = Supports Diebold 10XX/478X terminals that are configured to use eight operation keys.
11 = Adds states and screens necessary to allow cardholders to conduct transactions in a second language. This group does not use the language bank feature. This group must be used if you want to use multiple language support, but the terminal is firmware-based with an STP release prior to 4.0.
12 = Adds states and screens necessary to support the Device Handler Extension of the BASE24-atm self-service banking (SSB) add-on product.

13 = The sample set of color graphics screens.
14 = Supports withdrawal area sensors.
15 = Adds states and screens necessary to allow cardholders to conduct transactions in up to eight languages through language banks. Note that this group requires STP release 4.0 or later, or a software-based terminal (ATP, BTP, or CTP).
16 = Supports rear balancing, on the group 0 transaction set only.
17 = Supports card before cash, on group 0 transaction set only.
19 = Supports card before cash on fast cash transactions. This group must be loaded after group 5.
20 = Supports ANSI PIN blocks.
21 = Supports withdrawals and inquiries on default account.
22 = Enhanced Monochrome Graphics (EMG) control sequences.
24 = Supports the PIN change transaction with encrypted new keys.
52 = Supports the Split Deposit transaction with the Base Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0 and 12.
54 = Supports transaction time windows with the Base Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0 and 12.
56 = Supports the Super Teller card with the Base Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0 and 12.
58 = Supports the check-cashing card with the Enhanced Check Processing Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0 and 12.
100 = Supports local PIN verification.
200 = Supports multiple account selection by qualifier.
212 = Supports multiple account selection by qualifier plus the Device Handler Extension of the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, and 200.
220 = Supports multiple account selection by qualifier plus the Non–Currency Dispense add-on product. This group must be merged with groups 0, 200, and 500.
232 = Supports multiple account selection by qualifier plus the Non–Currency Dispense add-on product and the Device Handler Extension of the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, 200, 212, and 500.
252 = Supports MAC multiple account selection plus the split deposit transaction with the Base Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0 and 12.
300 = The basic group for the optional receipts function. The states and screens from the appropriate Printer Operable Group (group 301, 303, 311, 313, 351, 353, 361, or 363) should be copied into this group.

301 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, on all approved or approved and denied transactions.
302 = Optional receipts. The printer currently is inoperable. The cardholder selects optional receipts once for all chained transactions.
303 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, only if the printer becomes inoperable.
304 = Surcharging group.
305 = The group for dial-up ATM noninteractive surcharge processing for withdrawal and deposit transactions.
306 = The group for dial-up ATM noninteractive surcharge processing for fast cash transactions.
310 = The group for surcharging with surcharge interactive response notification processing.
311 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts before each transaction, on all approved or approved and denied transactions.
312 = Optional receipts. The printer currently is inoperable. The cardholder selects optional receipts before each transaction.
313 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts before each transaction, only if the printer becomes inoperable.
314 = The group for surcharge processing and logging canceled transactions to the journal.
315 = The group for dial-up noninteractive surcharging and logging canceled transactions to the journal.
320 = The group for basic DCC (Dynamic Currency Conversion) support.
321 = The group providing DCC support for interactive response notification surcharging. This group must be loaded after groups 310 and 320.
351 = Optional receipts for the Device Handler Extension of the BASE24-atm SSB add-on product. Same functionality as group 301 above.
352 = Optional receipts for the Device Handler Extension of the BASE24-atm SSB add-on product. Same functionality as group 302 above.
353 = Optional receipts for the Device Handler Extension of the BASE24-atm SSB add-on product. Same functionality as group 303 above.
354 = The group for dial-up noninteractive surcharging for the BASE24-atm self service banking (SSB) add-on product. This group must be merged with groups 0, 12, and 305.
355 = The group for dial-up noninteractive deposit surcharge processing and logging canceled transactions to the journal for the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, and 315.
361 = Optional receipts for the Device Handler Extension of the BASE24-atm SSB add-on product. Same functionality as group 311 above.

362 = Optional receipts for the Device Handler Extension of the BASE24-atm SSB add-on product. Same functionality as group 312 above.
363 = Optional receipts for the Device Handler Extension of the BASE24-atm SSB add-on product. Same functionality as group 313 above.
364 = The group for dial-up noninteractive surcharge processing with split deposit support for the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, 52, 305, and 354.
365 = The group for dial-up noninteractive surcharge processing with split deposit support and logging of canceled transactions for the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, 52, 315, and 355.
400 = The Europay, MasterCard, Visa (EMV) base load group. This group must be loaded after group 0.
404 = The EMV surcharging group. This group must be loaded after groups 0, 304, and 400.
405 = The EMV fast cash group. This group must be loaded after groups 0, 5, and 400.
406 = The EMV Non–Currency Dispense group. This group must be loaded after groups 0, 400, and 500.
424 = The EMV PIN change group. This group must be loaded after groups 0, 24, and 400.
425 = The EMV PIN unblock load group. This group must be loaded after groups 0 and 400.
450 = The Wincor EMV base load group. This group must be loaded after group 0.
454 = The Wincor EMV statement print group. This group must be loaded after groups 0, 4, and 450.
455 = The Wincor EMV fast cash group. This group must be loaded after groups 0, 5, and 450.
456 = The Wincor EMV PIN change group. This group must be loaded after groups 0, 24, and 450.
457 = The Wincor EMV surcharging group. This group must be loaded after groups 0, 304, and 450.
458 = The Wincor EMV Non-Currency Dispense group. This group must be loaded after groups 0, 500, and 450.
500 = Non–Currency Dispense group.
503 = The group for Non–Currency Dispense with Mobile Top-Up processing. This group must be merged with groups 0 and 500.
504 = The group for Non–Currency Dispense with Mobile Top-Up processing and logging canceled transactions to the TLF. This group must be merged with groups 0, 500, and 503.
550 = The group for Non–Currency Dispense with Mobile Top-Up definitions-based processing. This group must be merged with groups 0, 500, 503, and 504.

600 = The group for the Merchant Banking Center Device Handler extension.
612 = The group for the Merchant Banking Center Device Handler extension plus the Device Handler extension of the BASE24-atm SSB add-on product. This group must be loaded after groups 0, 6, and 12.
712 = Adds states and screens necessary to support the BNA (Bulk Note Acceptor) device, which is part of the Device Handler Extension of the BASE24-atm self-service banking (SSB) add-on product. This group must be merged with groups 0 and 12.
800 = Shared Bunch Note Acceptor (BNA) cash deposit transaction load group for support of different currencies and different denominations in a single deposit transaction. Must be loaded on top of groups 0, 12, and 712.
801 = Shared CPM check processing support load group. Must be loaded on top of groups 0 and 106.
812 = Adds states and screens necessary to support the Enhanced Note Acceptor (ENA) device. This group must be loaded on top of groups 0, 12, and 712.
901 = The group for the Multiple Currency Dispense add-on feature.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000

Clear Screen — Specifies whether the ATM screen is cleared before this screen is displayed. Valid values are as follows:

Y = Yes, clear the ATM screen before displaying the current screen.
N = No, do not clear the ATM screen before displaying the current screen.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: Y

Overlays — Specifies a screen number or numbers to use as an overlay for this screen. Any screen referenced in this field should have its Clear Screen flag set to a value of N. Up to five screens can be overlaid. Valid values are 000 through 511, 900, 950, and 999 for STP/DX-based processors. Valid values are 000 through 009 (reserved for use by Diebold), 010 through 899, 900 through 950 (reserved for use by Diebold), and 999 for PC-based processors (BTP, ATP, and CTP).

Field Length: 3 numeric characters
Occurs: Up to 5 times
Required Field: No
Default Value: No default value

<text entry> — This portion of the screen is used to create the ATM screen. There can be up to 40 characters of screen text on each of the 20 lines on the screen. The following two special characters, which are unique to the Diebold ATM, can be entered when using the InterBold i Series/MDS Display screen:

@ = Indicates where to position the cursor on an information entry screen. The at sign (@) character is not displayed on the Diebold screen, but must be filled in during entry. The at sign (@) character must be entered on a line by itself. Only one at sign (@) character can be defined on a given screen. In low-level mode, this character shows up as a trailing "set cursor," or the low-level code SI, *row.column*.

^ = Indicates where to display Track 1 data on the screen. (Track 1 usually contains the cardholder's name.) It need not be the only character on a line, but must be the last character on the line. Multiple ^ characters can be defined on a screen, but the Diebold entry program issues a warning before allowing you to continue. In low-level mode, this character shows up as the low-level codes SI, *row.column*, HT.

Function Keys

The use of two function keys on the ATM display screens varies from the standard function keys explained earlier in this section. The use of these function keys is explained below.

Key	Description
F1	**Low —** Displays the InterBold i Series/MDS Display screen. This function key acts as a toggle between the high-level display screen and the low-level display screen.
Shift-F2	**Read-Overlay —** Displays the current screen plus any overlays associated with the screen currently being accessed.

Warning Messages

The following warning messages are specific to the InterBold i Series/MDS Display screen and are generated when something that is not recommended has been done. For institution-specific reasons, however, certain programming practices may cause these warning messages to be displayed. You can override these messages by pressing the **F1** key.

`Cursor Positioning is Normally Done by "Calling" Screen`

Cause: The screen currently displayed is an overlay and is set to position the cursor.

Remedy: Screens that call overlays normally control cursor positioning. If the screen that calls this overlay is set up otherwise, the warning message can be overridden safely.

`Typically a Screen Would Use the Same Overlay Only Once`

Cause: The same overlay has been used more than once with the same screen. Unless the low-level mode has been used to set the cursor before calling each overlay, this is unnecessary since each occurrence occupies the same space on the screen.

Remedy: Verify that a single overlay is needed multiple times. If it is not needed, remove the duplicate reference.

`Typically a Screen Would not be Its Own Overlay`

Cause: You have indicated that the screen currently displayed should overlay itself. Usually, this situation results in an infinite loop.

Remedy: Remove the use of this screen as its own overlay.

`Note: Low Level Access is Recommended for Updates` *`reason`*

Where: *reason* = Imbedded Cir Screen, No Trailing Set Cursor, Trailing Set Cursor, or Imbedded Overlays Used

Cause: One of several warnings was overridden from the low-level mode and you are now attempting to read the screen associated with the override in the high-level mode.

Remedy: While it is possible to update screens from the high-level mode when this message is displayed, the screen display may be different than it appears. It is recommended that the screen be updated from the low-level mode when this message is displayed.

InterBold Low-Level Display Screen

The InterBold Low-Level Display screen allows codes to be embedded in the screen. These codes provide detailed control of how the screen is presented to the cardholder.

The InterBold Low-Level Display screen is shown below, followed by descriptions of its fields.

```
 INTERBOLD DISPLAY SCREEN, LOW-LEVEL    Screen 000   Lang Bank 000  Group 0000
   ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+.
 1
 2
 3
 4
 5
 6
 7
 8
 9
10                        <screen control code>
11
12
13
14
15
16
17
18
19
20

   F1=HI  F2=RD  SF2=HELP  F3=ADD  F4=DEL  F5=UPDT  F9=NXT  F10=PRNT  F16=EXIT
```

Screen — A number that is associated with the desired screen. Valid values are 000 through 511, 900, 950, and 999 for STP/DX-based processors. Valid values are 000 through 009 (reserved for use by Diebold), 010 through 899, 900 through 950 (reserved for use by Diebold), and 999 for PC-based processors (BTP, ATP, and CTP).

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Lang Bank — A number associated with the language in which the screen is displayed. This key field allows the same screen number to be displayed in multiple languages. Valid values are 0 through 7.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

Group — The group number used to identify a particular load group such as a collection of states, screens, etc., that is downloaded to the ATM. Each group provides certain functionality such as, Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic "InterBold i Series/MDS Display Screen" for a complete list of groups.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000

<screen control code> — A variable number of codes that control how the screen is displayed to the cardholder. Place a comma between multiple codes on the same line. There can be up to 76 characters of codes on each of the 20 lines on the screen. A complete list of codes supported by the BASE24-atm product are provided in the topic, "Screen Control Codes."

Function Keys

The use of two function keys on the InterBold Low-Level Display screen varies from the standard function keys explained earlier in this section. The use of these function keys is explained below.

Key	Description
F1	**High —** Displays the InterBold i Series/MDS Display screen. This function key acts as a toggle between the high-level display screen and the low-level display screen.
Shift-F2	**Help —** Accesses Help screens explaining the characters used on the InterBold Low-Level Display screen. Press the **F1** key to move to the next Help screen, and press the **F2** key to move to the previous Help screen. Press the **F16** key to return to the low-level screen.

Screen Control Codes

The following Diebold screen control codes are supported by the BASE24-atm product. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information on the screen control codes.

Code	Description
"*text*"	Displays text in ASCII format. The text must be enclosed in double quotation marks, and embedded quotes must be doubled.
hh	Displays a character in hexadecimal format from the current character set. The valid values of *hh* are 20 through 7E hexadecimal.
BS	Moves the cursor to the left by one column in the current row on the screen.
CR	Returns the cursor to the first column of the present row on the screen. The CR code is also used as a line terminator for scrolled text (defined using the ESC-K code).

Code	Description
DC4	Moves the cursor to the right one column in the current row on the screen.
ESC-;,*t*,*m*,*d*	Controls the voice response by the terminal for Voice Phase I and Voice Phase II. The value *t* specifies the termination flag, with valid values of 0 through 2. The value *m* specifies the message number, with valid values of 0 through 999. The value *d* specifies the delay time in ticks, with valid values of 0 through 255.
ESC-;,*t*, *”message”*,*d*	Controls the voice response by the terminal for Voice Phase III. The value *t* specifies the termination flag, with valid values of 0 through 2. The value *message* specifies the message to be used and must be enclosed in quotation marks. This is either a file name with the .WAV or .TXT extension, a text string, or the character sequence ESC-ZK,*b*,*ff*. The ESC-ZK,*b*,*ff* character sequence is described later in this table. The value *d* specifies the delay time in ticks, with valid values of 0 through 255. Note that the voice control codes must precede all other screen data.
ESC-:,*t*	Displays the current date and time at the cursor position on the screen. The value *t* specifies the format template screen, with valid values of 10 to 511.
ESC-=,*r*	Sets the number of rows displayed on the screen. The value *r* specifies the row size, with valid values of 10 through 16.
ESC->,*d*	Controls fade-in transitions. The value *d* specifies the DAC setup number, with valid values of 0 through 255.
ESC-<	Indicates that a fade-out transition is used.
ESC-?,*s*	Indicates that sprites are used for animation, and controls their use. The value *s* specifies the sprite file number, with valid values of 0 through 127.

Code	Description
ESC-A,*i,s,r*	Indicates that an animation sequence using icons is displayed, and provides controls of the sequence. The icons are sequentially numbered. Refer to the vendor manuals for illustrations of the available icons. The value *i* specifies the first icon number, with valid values of 0 through 999. The value *s* specifies the number of sequential icons to display, with valid values of 0 to 9. The value *r* specifies the animation rate, with valid values of 000 to 999.
ESC-C,*b,f*	Sets the color for displaying text and icons on the screen. The value *b* specifies the background color, with valid values of 0 through 9 and A through F. The value *f* specifies the foreground color, with valid values of 0 through 9 and A through F.
ESC-D,*t*	Creates a delay before the next action is taken. The value *t* specifies the number of ticks before the next action, with valid values of 4 through 255.
ESC-E,*b*	Selects a bank as the current source of text and icons. The value *b* specifies the bank number, with valid values of 0 through 6.
ESC-F,*f,d*	Selects a font for the text. The value *f* specifies the color graphic character set, with valid values of 0 through 7, E through G, and at sign (@). The value *d* specifies the display mode, with valid values of 0 through 3. This code is the same as ESC-G for color graphics and VGA+ mode.
ESC-G,*s,m*	Specifies a text page and display mode. If you are using color graphics, the value *s* specifies the color graphic character set, with valid values of 0 through 7, E through G, and at sign (@). If you are using Enhanced Monochrome Graphics (EMG), the value *s* selects a specified page of the current bank as the source of the text, with valid values of 0 through 7. The value *m* specifies the display mode, with valid values of 0 through 3. This code is the same as ESC-F for color graphics and VGA+ mode.
ESC-H,*c*	Selects the foreground color. The value *c* specifies the color number, with valid values of 0 through 255.

Code	Description
ESC-I,*n*	Inserts a screen into the current screen. This code is the equivalent to the SO code. The value *n* specifies the number of the screen to be inserted.
ESC-J,*d,r,c1*[*,c2, c3,…c15*],ESC	Indicates that the color wheel is used, and controls the resulting colors. The value *d* specifies the DAC setup number, with valid values of 0 through 255. The value *r* specifies the change rate, with valid values of 0 through 999. The values *c1* through *c15* specify the wheel setup numbers, with valid values of 0 through 511. Setup numbers *c2* through *c15* are optional, and the code must be terminated with ESC.
ESC-K,*d,s,r1,r2,t*	Specifies a window through which text may scroll. The value *d* specifies the scrolling direction, with valid values of D, L, R, and U. The value *s* specifies the screen number to roll, with valid values of 10 through 511. The value *r1* specifies the top row of the scrolling window with valid values of positions @ through 3 in Diebold's graphic format, representing rows 1 through 20. The value *r2* specifies the bottom row of the scrolling window with valid values of positions @ through 3 in Diebold's graphic format, representing rows 1 through 20. The value *t* specifies the scrolling speed, with valid values of 0 through 999.
ESC-M,*r*	Sets the resolution of the screen. The value *r* specifies the resolution of the screen, with valid values of 0 through 4.
ESC-N,*p*	Selects the palette registers. The value *p* specifies the palette register number, with valid values of 0 through 255.
ESC-O,*d*	Selects the DAC setup. The value *d* specifies the DAC setup number, with valid values of 0 through 999.
ESC-P,*i,d*	Displays an icon at the cursor position on the screen. The value *i* specifies the icon number, with valid values of 0 through 999. The value *d* specifies the display mode, with valid values of 0 through 5.
ESC-Q	Displays a check image with its upper-left corner at the current cursor position.

Code	Description
ESC-R,*r,b,f,c*	Repeats a particular character starting at the current cursor position on the screen. This code creates a border using a single character. The value *r* specifies the number of times to repeat the character, with valid values of 0 through 800. The value *b* specifies the background color of the character, with valid values 0 through 9 or A through F. The value *f* specifies the foreground color of the character, with valid values 0 through 9 or A through F. The hexadecimal value *c* specifies the cell of the current character set to repeat.
ESC-T,*b,f,c*	Repeats a background color or pattern in a tile pattern outside the addressable area on the screen. The addressable area is a 32 by 16 character display grid. The value *b* specifies the background color of the character, with valid values of 0 through 9 or A through F. The value *f* specifies the foreground color of the character, with valid values of 0 through 9 or A through F. The hexadecimal value *c* specifies the cell of the current character set to repeat.
ESC-U,*r1,c1,r2, c2,b,f,c*	Creates a border around a specified rectangular area by repeating a character on the screen. The value *r1* specifies the upper row with valid values of positions @ to 3 in Diebold's graphic format, representing rows 1 through 20. The value *c1* specifies the left column with valid values of positions @ to W in Diebold's graphic format, representing columns 1 through 40. The value *r2* specifies the bottom row with valid values of positions @ to 3 in Diebold's graphic format, representing rows 1 through 20. The value *c2* specifies the right column with valid values of positions @ to W in Diebold's graphic format, representing columns 1 through 40. The value *b* specifies the background color of the character, with valid values of 0 through 9 or A through F. The value *f* specifies the foreground color of the character, with valid values of 0 through 9 or A through F. The hexadecimal value *c* specifies the cell of the current character set to repeat.

Code	Description
ESC-V,*b,f*	Displays the buffer with a format template. The value *b* specifies the buffer ID, with valid values of A through Z, or the at sign (@). The value *f* specifies the format template ID, with valid values of 00 through 99.
ESC-W,*r1,c1,r2, c2,b*	Indicates that blinking is used within a specified rectangular area and controls the blinking. The value *r1* specifies the top row to blink with valid values of positions @ through 3 in Diebold’s graphic format, representing rows 1 through 20. The value *c1* specifies the left column to blink with valid values of positions @ through W in Diebold’s graphic format, representing columns 1 through 40. The value *r2* specifies the bottom row to blink with valid values of positions @ through 3 in Diebold’s graphic format, representing rows 1 through 20. The value *c2* specifies the right column to blink with valid values of positions @ through W in Diebold’s graphic format, representing columns 1 through 40. The value *b* specifies the blink rate, with valid values of 0 through 999.
ESC-X,*x,y*	Sets the cursor to a particular pixel position on the screen. The value *x* specifies the X coordinate for the cursor, with valid values of 0 through 9999. The value *y* specifies the Y coordinate for the cursor, with valid values of 0 through 9999.
ESC-Y, *o, t*	Selects a special effects time in milliseconds. The value *o* specifies the escape sequence to be adjusted. Valid values are >, <, and P. The value *t* specifies the number of milliseconds. Valid values are 0 through 9999.
ESC-ZC,*t*	Specifies which touch template to use. The value *t* specifies the touch template number, with valid values of 0 through 998.

Code	Description
ESC-ZD,*t,k,x1, x2,y1,y2,i*	Specifies a touch template definition. The value *t* specifies the touch template number, with valid values 0 through 998. The value *k* specifies the key ID, with valid values of 0 through 9, A through M, or a period. The value *x1* specifies the low position for the X coordinate, with valid values of 0 through 9999. The value *x2* specifies the high position for the X coordinate, with valid values of 0 through 9999. The value *y1* specifies the low position for the Y coordinate, with valid values of 0 through 9999. The value *y2* specifies the high position for the Y coordinate, with valid values of 0 through 9999. The value *i* specifies the icon number, with valid values of 0 through 999.
ESC-ZF, "*filename.ext*", *c, n, x, y*	Displays animated graphic files in the high resolution modes. The *filename* value is the name of the animated graphic file. The *ext* value is the file extension FLC or FLI. The file name and extension must be enclosed in quotation marks. The variable *c* specifies whether to display the graphic file once or repeatedly with valid values R and W. The value *n* specifies the animation number. This value must be 0. The value *x* is the X coordinate of the upper left corner of the animation's coordinates, with valid values 0 through 9999. The value *y* is the Y coordinate of the upper left corner of the animation's coordinates, with valid values 0 through 9999.

Code	Description
ESC-ZH, *k, c, x,* "*message*"	Defines a keypad feedback template. A keypad feedback template is used to map a key that can be pressed to a sound wave file. The value *k* specifies the key type. This can be the text NUM or the text ESC. The value *c* contains the ASCII code for the character key entered. The value is stored in hexadecimal format. (For example, the value A is stored as 0041 and the value # is stored as 0023.) The value *x* is a reserved field. It is required and must be set to 000. The "message" variable specifies the name of the sound wave file to be associated with the key. This field is not required, but if it is used, the file name must contain 1–8 characters followed by the .WAV or .TXT extension or a text string. Note that the quotation marks (" ") are required regardless of whether you enter a file name. **Note:** If the value for *k* is NUM and the value for *c* is the symbol # (stored value 0023 hexadecimal), TCS treats the numeric keys as a group and sets them to user-defined default messages. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information about this TCS mapping.
ESC-ZI, *x, y, n*	Enables you to display touch screen buttons from the In Button database. The value *x* specifies the X coordinate for the cursor, with valid values of 0 through 9999. The value *y* specifies the Y coordinate for the cursor, with valid values of 0 through 9999. The value *n* specifies, the number of the button, with values 1 through 999.
ESC-ZK, *b, ff*	Interprets a specified buffer for Voice Phase III. This sequence can be used only with the ESC-; sequence. The value *b* specifies the buffer containing the data to be formatted, with valid values at sign (@) and A through Z. The value *ff* is the template number, with valid values 00 through 99.
ESC-ZO, *x, y, n*	Enables you to display touch screen buttons from the Out Button database. The value *x* specifies the X coordinate for the cursor, with valid values of 0 through 9999. The value *y* specifies the Y coordinate for the cursor, with valid values of 0 through 9999. The value *n* specifies the number of the button, with values 1 through 999.

Code	Description
ESC-ZU, "MPEG-PLAY", 1, *t, "filename", x, y, h, w, n, o, p,* or ESC-ZU, "MEG-PLAY", 1, *t, svvv*	Defines the screen windows that show video sequences complete with sound from MPEG-formatted files stored on a CD-ROM or hard drive. The value *t* indicates the task switch with valid values 0 through 5. If the value of *t* is 0 or 1, the values for *x, y, h, w, n, o,* and *p* are required. If the value of *t* is 4 the value *svv* is required. If the value of *t* is 2, 3, or 5, no other values are required. The value *filename* specifies the name of the MPEG file to be played. It must be contained in quotation marks and have an extension .MPG or .CHN. The value *x* specifies the X coordinate of the upper-left corner of the MPEG display window, with valid values 0000 through 9999. The value y specifies the Y coordinate of the upper-left corner of the MPEG display window, with valid values 0000 through 9999. The value *h* specifies the height of the MPEG display window relative to the Y coordinate, with valid values of 0000 through 9999. The value *w* specifies the width of the MPEG display window relative to the X coordinate, with valid values 0000 through 9999. The value *n* is the color key feature state, with valid values 0 through 2. The value *o* specifies the color palette index for the color key, with valid values 000 through 255. The value *p* specifies the play mode designation, with valid values C, R, and W. The value *s* is the sign + or – and the value *vvv* specifies the volume adjustment in the range of 0 through 100.

Code	Description
ESC-ZV, *c,* "*id*", *zz,* "*url*", *xxxx, yyyy, wwww, hhhh, ttt, bbb*	Enables a web browser to create and manage one or more browser windows that persist across screen changes on the consumer display. The value *c* is the command identifier with the following valid values: C = Create. Open a new browser window D = Destroy. Close an existing browser window. H = Hide. Make an existing browser window invisible. N = Navigate. Change the web page of an existing browser window. R = Restore. Redisplay an existing, hidden browser window. S = Size/position. Change the size or position (or both) of an existing browser window. "*id*" is a brief text string (one or more characters) that identifies which browser window the command acts upon. The value must be enclosed in quotation marks. For the Create command, the value enclosed in the quotation marks becomes the identifier for the browser window being created. You can enter an asterisk (*) as a wild card for the Destroy, Hide, Navigate, or Restore commands. The value *zz* is the visibility priority (z-order index) for the browser window and is used only by the Create command. Valid values are 00–99, with 00 as the lowest priority (least visible) and 99 as the highest priority (most visible). "*url*" a web page or local document to be displayed in the browser window and is used by the Create and Navigate commands. This value is enclosed in quotation marks and must be a valid URL. For the Create command, this variable can be blank. The variables *xxxx* and *yyyy* specify the placement of the upper left corner of a browser window, measured in global coordinates. The variables *wwww* and *hhhh* specify the width and height, respectively, of the browser window in global coordinates. These parameters are used by the Create, Navigate, and Size/position commands. *continued on the following page*

Code	Description
ESC-ZV, *c,* “*id*”, *zz,* “*url*”, *xxxx, yyyy, wwww, hhhh, ttt, bbb* continued	The value *ttt* is the number of seconds to wait for a URL to successfully load. Valid values are 000–999. A value of 000 causes the system to use the Browser Timeout value set in the Agilis Configuration Utility. This parameter is used by Create and Navigate commands that specify a valid URL. The value *bbb* specifies a backup screen number for use in the event that the browser is unable to load the specified URL.
ESC-ZX, *ttt*, *nnn*, *kk*, *ff*	Defines a key map that translates one or more physical keys to key designators (key values) representing function keys. The value *ttt* specifies the map number of the key map that this control sequence defines, with valid values of 001 to AZZ. The value *nnn* indicates the number of key pairs (kk ff) in this control sequence, with valid values of 001 to 100. For each key pair (kk ff), the parameter *kk* specifies the physical key that is to be translated to a specified function key value when the key *kk* is pressed. The valid range for *kk* is 30 through 39 (keys 0 through 9) and 41 through 4D (keys A through M). For each key pair (kk ff), the parameter *ff* specifies which function key value is reported when the key *kk* is pressed. The valid range for *ff* is 41 through 4D (A through M). **Note:** Refer to Diebold’s ***Agilis 91x Terminal Programming Manual*** for more information about this key mapping.

Code	Description
ESC-ZW, *ttt*	Directs the terminal to use a specified key map when displaying the associated screen. The value *ttt* specifies the map number of the key map to be used with the screen in which this control sequence is embedded. The valid range for *ttt* is 001 to AZZ, along with 000. If the value of *ttt* is 000, no key map is selected, including the default key map. Only the keys explicitly enabled by the state or function are enabled. All enabled keys report their original (untranslated) values when pressed. This usage allows the network to suspend key mapping for specific instances. **Note:** Refer to Diebold's ***Agilis 91x Terminal Programming Manual*** for more information about this key mapping.
FF	Clears the screen and places the cursor in the home position. This should be the first code in any screen definition, except overlay screens and screens with voice control and touch templates. The ESC-;, ESC-ZD, and ESC-ZC codes must precede this code.
HT	Displays the name found on Track 1 of the card at the current cursor location.
LF	Moves the cursor to the next row in the current column on the screen.
SI,*rc*	Sets the cursor to a particular position on the screen. The value *r* specifies the row number of the cursor with valid values of positions @ to 3 in Diebold's graphic format, representing rows 1 through 20. The value *c* specifies the column number of the cursor with valid values of positions @ to W in Diebold's graphic format, representing columns 1 through 40.
SO,*n*	Overlays the current screen with another screen. Typically this code is preceded by (and usually followed by) a set cursor (SI) code, unless the referenced screen does its own cursor positioning. The value *n* specifies the number of the screen to be inserted.

Code	Description
US	Moves the cursor to the previous row in the current column on the screen.
VT, *hh*	Displays an alternate character. The value *hh* is the hexadecimal format of the character, with valid values of 20 through 7E (hexadecimal).

Warning Messages

The following warning messages are specific to the InterBold Low-Level Display screen and are generated when something that is not recommended has been done. For institution-specific reasons, however, certain programming practices may cause these warning messages to be displayed. You can override these messages by pressing the **F1** key.

`Clear Screen (FF) at Column` *`XX`* `Should be First Command`

Where: *XX* is the number of the column

Cause: The FF code is embedded in the message. Thus, all preceding data in the message is erased.

Remedy: The FF code should be in the first column of the message.

`Track1 Name (HT) at Column` *`XX`* `May Cause Wraparound`
`on Display`

Where: *XX* is the number of the column

Cause: The entry program does not know where the cursor is located. (This can occur if an embedded overlay exists.)

Remedy: You should return to the High-Level Standard Display screen after the update is complete to verify that the display looks correct. If the display does not look correct, return to the Low-Level Standard Display screen and move the HT character so that wraparound does not occur.

`Text at Column` *`XX`* `May/Will Cause Wraparound on Display`

Where: *XX* is the number of the column

Cause: The entry program does not know where the cursor is located. (This can occur if an embedded overlay exists.)

Remedy: You should return to the High-Level Standard Display screen after the update is complete to verify that the display looks correct. If the display does not look correct, return to the Low-Level Standard Display screen and move the text at column *XX* so that wraparound does not occur.

`Typically the Last Display CMD Should "Home" the Cursor`

Cause: The last code does not explicitly set the cursor and, thus, is left at the last line of display text.

Remedy: Typically, the cursor is set to return to the home position (SI,@@), unless data entry is expected.

`Typically an Overlay Will Not Use Set Cursor as Last CMD`

Cause: The screen currently displayed is an overlay and is set to position the cursor.

Remedy: Screens that call overlays normally control cursor positioning. If the screen that calls this overlay is set up otherwise, the warning message can be overridden safely.

State Tables

The InterBold State Table screen can be accessed by entering function code 2 for State Tables on the ENTIBOLD or ENTIBLD2 Diebold Entry Program Menu or by from the Shared State Select Menu. The InterBold State Table screen allows you to reconfigure state tables and conditions that are associated with various ATM transaction states and groups.

Transaction processing at the ATM is controlled by the configuration of state tables. These state tables comprise a number of sequential instructions that provide the ATM with the information to perform financial and administrative transactions. Most states include a screen number and a next state number as part of the table entry.

When a terminal is in service and a transaction is requested, the state tables are executed. Generally, as each new state is entered, a screen is displayed at the ATM; however, some states (for example, the Clear Keys State) do not have screens associated with them. The screen specifies the action that is required before the transaction can continue. When the specified action is taken, the transaction flow continues to the next state.

Shared State Select Menu

The Shared State Select Menu can be accessed by entering the function code 2 on the ENTWBOLD Diebold Entry Program Menu and pressing the **F1** key. The Shared State Select Menu enables you to access the InterBold State Table screen to reconfigure Wincor Shared states e, @r, and @q. After you enter one of these values in the State Type field, along with state number and group information, press the **F2** key to display the Interbold State Table screen. Refer to the topic “InterBold State Table Screen” for more information about the State Table screen.

The Shared State Table Menu is shown below, followed by a description of its fields.

```
                              SHARED STATE SELECT MENU



                        State Type

                        State Number  000   Group  0000



                 Select the State Type From the Following List



              e  (Wincor EMV Chip Card)
              @r (Wincor Enhanced CheckIn State)
              @q (Wincor Enhanced CashIn State)




================================================================================

 F1=CHANGE STATE TYPE  F2=READ  F9=NEXT      F10=PRINT  F16=EXIT
```

State Type — The state type associated with the number in the State Number field. The BASE24-atm product supports the state types that Wincor has defined. When a state type defined by Wincor is entered in this field, the necessary fields for that state type are listed on the screen. You can define additional state types, but the Diebold entry program does not check the data entered in those states to ensure that they meet the prescribed parameters. These additional state types are also not listed on the cross-reference listings of states and screens, and states and states.

Field Length:	2 alphanumeric characters, followed by a system-protected text description
Required Field:	Yes
Default Value:	e

The values defined by Wincor are listed below.

State Type Designation	Description
e	Wincor EMV Chip Card state
@r	Wincor Enhanced Cash In state
@q	Wincor Enhanced Check In state

For more information about the Wincor EMV Chip Card state, refer to the ***BASE24-atm EMV Support Manual.*** For more information about the Enhanced Check In state, refer to the ***Wincor-Nixdorf NDC/Diebold D91x Message Format Extension for Cheque In, June 13, 2007, Release 1.34.*** For more information about the Enhanced Cash In State, refer to the ***Wincor-Nixdorf NDC_Diebold D91x Message Format Extension for CashIn V1.44.***

State Number — A number that is associated with the desired state. Valid values are 000 through 254, 256 through ZZZ.

Field Length: 3 alphanumeric characters
Required Field: Yes
Default Value: 000

Group — The number used to identify a particular load group (that is, a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic "InterBold i Series/MDS Display Screen" for a complete list of groups.

InterBold State Table Screen

The InterBold State Table screen is shown below, followed by descriptions of its key fields. The items displayed in the highlighted area vary according to the state. For more information on the state tables, refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual***.

```
                              INTERBOLD STATE TABLE
                           Type A  Card Read State
                          State Number  000   Group  0000



                                <state-specific data>











================================================================================
 F1=CHANGE STATE TYPE    F2=READ     F3=ADD      F4=DELETE   F5=UPDATE
                         F9=NEXT    F10=PRINT                F16=EXIT
```

Type — The state type associated with the value in the State Number field. The BASE24-atm product supports the state types that Diebold has defined. When a state type defined by Diebold is entered in this field, the necessary fields for that state type are listed on the screen. You can define additional state types, but the Diebold entry program does not check the data entered in those states to ensure that they meet the prescribed parameters. These additional state types are also not listed on the cross-reference listings of states and screens, and states and states.

Field Length:	3 alphanumeric characters, followed by a system-protected text description
Required Field:	Yes
Default Value:	A

The values defined by Diebold are listed below.

State Type Designation	Description
A	Card Read state
B	Pin Entry state
C	Deposit/Unlock Depository Door state
D	Clear Keys state
E	Select Function state
F	Dollar Entry state
G	Cent Check state
H	Information Entry state
I	Transaction Request state
J	Close state
K	Indirect Next state
L	Card Write state
M	PIN Entry, Track 3 Write state
N	Camera Control state
O	Vandal Shield state
R	Set Dollar Buffer state
T	Card Read (Non-state 000) state
Z	Check Track Buffer state
;	Voice Toggle state
=	Copy Buffer state
?	Withdrawal Area Sensors state
[	Set Language Bank state

State Type Designation	Description
\	Buffer Arithmetic state
AU	Text to Speech state
CI	Enhanced Currency Acceptor state
CM	Camera Enhanced state
CR	Card Reader Enhanced state
DP	Depositor Enhanced state
DR	Dispenser Door Enhanced state
IC	Image Character Recognition State
LT	Lead Through Indicators Enhanced state
NC	Enhanced Network Control state
PR	Consumer/Journal Printer Enhanced state
SC	Extended Smart Card state
SD	Envelope Dispenser state
SP	Statement Printer Enhanced state
SV	Enhanced Service Code Check state
VN	Vandal Shield Enhanced state
@B	Check FIT Enhanced state
@C	Check PIN Enhanced state
@D	Information Entry Enhanced state
@E	Enhanced Buffer Compare state
@F	Set Buffer Enhanced state
@G	Buffer Copy Enhanced state
@H	Check Buffer Enhanced state

State Type Designation	Description
@I	Transaction Request Enhanced state
@K	Indirect Next State Enhanced state
@L	Create Buffer Enhanced state
@M	Track Buffer Compare Enhanced state
@N	Buffer Shift state
@P	Copy Buffer Data state
@Q	Buffer Overlay Enhanced state
@S	Time Delay Enhanced state
@X	Multi-tasking Enhanced state
@Y	Enhanced Set Language Bank state
@Z	Close Enhanced state
@r	Wincor Enhanced Cash In state
@q	Wincor Check In state

State Number — A number that is associated with the desired state. Valid values are 000 through 254, 256 through ZZZ.

Field Length: 3 alphanumeric characters
Required Field: Yes
Default Value: 000

Group — The number used to identify a particular load group (that is, a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic “InterBold i Series/MDS Display Screen” for a complete list of groups.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	0000

<state-specific data> — Data pertaining to the specified state. This field can contain multiple data entry fields. The number of fields, the field labels, and what the fields configure depend upon the Type field. Refer to Diebold’s ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for information about state-specific data.

Field Length:	3 alphanumeric characters
Occurs:	Up to eight times. Extension states add additional occurrences.
Required Fields:	Yes
Default Value:	000

Miscellaneous Data

The Miscellaneous Data screen can be accessed by entering function code 3 on the Diebold Entry Program Menu. The screen is used to enter the following types of data for each group:

- Camera control parameters.
- Logical unit number (LUNO).
- Values for Miscellaneous Features fields that indicate whether features such as the following are allowed: remote balancing, host-controlled solenoid-activated lock, and decimal format entry.
- Up to 16 timers for cardholder transactions. The exact number of timers that should be entered is dependent on the ATM model.

A group number is entered on this screen to read, add, change, or delete the miscellaneous data for the group indicated.

The Miscellaneous Data screen is shown below, followed by descriptions of its fields.

```
                            MISCELLANEOUS DATA    Group 0000

PARAMETERS: Camera Control          LUNO
  Misc Features 1        Misc Features 2        Misc Features 3

TIMERS (in seconds):
00: Customer Keyboard Response         01: Timeout Screen Display ...
02: Close State Screen Display         03: Communications Response ..
04: Envelope Deposit .........         05: Currency Removal .........
06: Poll Interval ............         07: Withdrawal Door Open .....
08: Securomatic Deposit ......         09: Card Removal .............
10: Take Cash Screen Display .         11: Vandal Shield Close ......
15: Card Reader Enable .......         18: Swipe Reader Error Display
20: IDM Deposit ..............         21: Statement Printer Capture



================================================================================
            F2=READ         F3=ADD          F4=DELETE       F5=UPDATE
            F9=NEXT        F10=PRINT                       F16=EXIT
```

Group — The group number used to identify a particular load group such as a collection of states, screens, and so on., that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic "InterBold i Series/MDS Display Screen" for a complete list of groups.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000

Camera Control — Determines whether pictures of the cardholder are taken automatically. Valid values are as follows:

0 or 1 = Automatic
2 or 3 = Controlled

With an automatic picture-taking setting, Diebold ATMs do not take a picture after a bad card read operation.

Field Length: 1 numeric character
Required Field: Yes
Default Value: No default value

Logical Unit Number (LUNO) — The logical unit number of the ATM. This number identifies a specific terminal to the BASE24-atm system.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: No default value

Misc Features 1 — Indicates how the following parameters are set: decimal style, deposit cancel status bit, dollar decimal places, dollar buffer size, and separate ready response. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information on these parameters and the values to enter in this field.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: No default value

Misc Features 2 — Indicates how the following parameters are set: reject reason code to network, maintenance mode log to network, change MMD mispick timer from 2 minutes to 30 seconds, and expanded hardware configuration status. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information on these parameters and the values to enter in this field.

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	No default value

Misc Features 3 — Indicates how the following parameters are set: send all status, extended range for message coordination number, extended timer 03 feature, double expanded hardware configuration status, OK and correction keys, and last transaction status buffer. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information on these parameters and the values to enter in this field.

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	No default value

Timers

The following fields indicate the timeout values, in seconds, for the various ATM timers listed below. The two-character numeric field before the timer name is the timer number. Refer to the appropriate Diebold vendor documentation for timer descriptions.

00: Customer Keyboard Response	01: Timeout Screen Display
02: Close State Screen Display	03: Communications Response
04: Envelope Deposit	05: Currency Removal
06: Poll Interval	07: Withdrawal Door Open
08: Securomatic Deposit	09: Card Removal
10: Take Cash Screen Display	11: Vandal Shield Close
15: Card Reader Enable	18: Swipe Reader Error Display
20: IDM Deposit	21: Statement Printer Capture

Field Length: 3 numeric characters

Occurs:	16 times
Required Field:	Yes
Default Value:	No default value

Financial Institution Table

The Financial Institution Table screen can be accessed by entering function code 4 on the Diebold Entry Program Menu. This screen is used to maintain financial institution tables (FITs).

The screen can be used to add a record to the Diebold Configuration File (CNFGIX) for each financial institution allowed to use the ATM (up to a maximum of 509 FIT records).

This screen also contains the FIT parameters that must be modified to support local PIN verification. Refer to the topic "Local PIN Verification" that follows this screen and field descriptions for a description of these FIT parameters and how to set them.

The Financial Institution Table screen is shown below, followed by descriptions of its fields.

```
                     FINANCIAL INSTITUTION TABLE

                 FIT Entry Number 000   Group 0000

       PIDDX:  Offset to FIID on Card ..................
       PFIID:  FIID Prefix ............................
       PSTDX:  Offset to Indirect Next State ...........
       PAGDX:  Offset to Algorithm Number on Card ......
       PMXPN:  Maximum PIN Entry Length ................
       PCKLN:  Number of Digits used for PIN Check .....
       PINPD:  PIN field Pad Character .................
       PANDX:  Offset to PIN Base Data on Card .........
       PANLN:  PIN Base Data Field Length ..............
       PANPD:  PIN Base Data Pad Character .............
       PRCNT:  Offset to PIN Retry Count on Card .......
       POFDX:  Offset to Offset Field on Card ..........
       PDCTB:  Decimalization Field for DES Algorithm ..
       PEKEY:  Encrypted PIN Key for DES Algorithm .....
       PINDX:  Information for Index Entries in FIT ....


================================================================================
                 F2=READ     F3=ADD      F4=DELETE   F5=UPDATE
                 F9=NEXT    F10=PRINT               F16=EXIT
```

FIT Entry Number — The number associated with the Financial Institution Table (FIT) entry group to be impacted.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

Group — The group number used to identify a particular load group such as a collection of states, screens, and so on, that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic "InterBold i Series/MDS Display Screen" for a complete list of groups.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000

Local PIN Verification

FIT parameters must be modified to support local PIN verification. Refer to the appropriate Diebold vendor documentation for complete information on setting up the FIT data to support local PIN verification.

The FIT data specifies parameters to be used depending on the terminal PIN verification process to be used. The following field values are important to support local PIN verification. These fields are described following the FIT screen.

Field	Value
PINPD	Must be set to 000.
PRCNT	The value is 255 indicating the PIN retry count is not on the ATM card, but in the PIN Entry State (B) table.
PDCTB	The institution is responsible for the values for this field.
PEKEY	The institution is responsible for the values for this field.
PINDX	The fifth PINDX digit is reserved and must be hexadecimal 0.

<table>
<tr><th>Field</th><th colspan="2">Value</th></tr>
<tr><td rowspan="6">PCKLN</td><td colspan="2">The valid values for the entire PCKLN byte, excluding Identikey, are shown below.</td></tr>
<tr><th>Terminal Verification</th><th>PCKLN Byte Value</th></tr>
<tr><td>None*</td><td>000</td></tr>
<tr><td>DES</td><td>004–016</td></tr>
<tr><td>Diebold</td><td>064–068</td></tr>
<tr><td>VISA</td><td>032–036</td></tr>
</table>

* The terminal does not perform local PIN verification.

Rear Balance Function Definition Data

The Rear Balance Function Definition screen defines special information needed for a rear balance administrative transaction. This screen defines such things as which transactions are initiated, the key data that is sent, and what, if any, additional data is required at the time of the transaction in order to complete the transaction.

Note: ATM customers are strongly discouraged from entering PINs into rear keypads, based on Payment Card Industry (PCI) security restrictions. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information on configuring the ATM-REAR-ADMN-TRAN-PIN-ALWD param to decline rear balancing administrative transaction requests containing PINs.

This screen can be accessed by entering function code 5 for Rear Balance Function Definition Data on the Diebold Entry Program Menu.

Each function that can be set up using this screen defines the characteristic of an administrative transaction. Multiple functions can be associated with a single record.

For information on the fields on this screen, refer to the vendor documentation.

The Rear Balance Function Definition screen is shown below, followed by descriptions of its fields.

```
                      REAR BALANCE FUNCTION DEFINITION
                                 Group 0000

                          Function Index Number

                   Autoexecute/Chain Flag ................. 0
                   Track Data Flag ........................ 2

                   Op Key Data Flag ....................... Y
                   Op Key Data .......

                   PIN Flag ............................... Y
                   Servicer Input Flag .................... 0
                   Input Screen Number .................. 000

                   Expected Good Next State ............. 000
                   Immediate Exit on Next State Mismatch .. N

           Additional Data Flag .. 0
           Additional Data .

================================================================================
 F2=READ REC   F3=ADD REC   F4=DEL REC   F5=UPDT REC   F9=NXT REC  F10=PRNT REC
SF2=READ FNCT SF3=ADD FNCT SF4=DEL FNCT SF5=UPDT FNCT SF9=NXT FNCT F16=EXIT
```

Group — The number used to identify a particular load group (that is, a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field on ATD Diebold 10XX/478X screen 7. Refer to the topic “InterBold i Series/MDS Display Screen” for a complete list of valid load groups.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	0000

Function Keys

The use of many of the function keys on the Rear Balance Function Definition screen varies from the standard function keys explained earlier in this section. The use of these function keys is explained below.

Key	Description
F2	**Read Record —** Displays the current record from the CNFGIX.
F3	**Add Record —** Adds a record to the CNFGIX.
F4	**Delete Record —** Deletes a record from the CNFGIX.
F5	**Update Record —** Updates the current record to the CNFGIX.
F9	**Next Record —** Displays the next record from the CNFGIX.
F10	**Print Record —** Sends the record currently being displayed to a specified spooler location. This location is specified on the Diebold Entry Program Menu in the Output Device for PRINT Function field, which is described earlier in this section.
Shift-F2	**Read Function —** Displays the current function from the CNFGIX.
Shift-F3	**Add Function —** Adds a function to the CNFGIX.
Shift-F4	**Delete Function —** Deletes a function from the CNFGIX.

Key	Description
Shift-F5	**Update Function —** Updates the current function to the CNFGIX.
Shift-F9	**Next Function —** Displays the next function from the CNFGIX.

Consumer Printer Configured Text Definition

The Consumer Printer Configured Text screen is used to define special codes that define how the receipts printed on the consumer printer look. These codes are then downloaded to the terminal. This screen offers printer control in addition to those defined in the Receipt File (RCPT). Refer to section 6 for more information on printer control in the RCPT. This screen is accessed by entering function code 6 for Consumer Printer Configured Text Definition on the Diebold Entry Program Menu.

The special codes are described after the screen description. The Consumer Printer Configured Text screen is shown below, followed by descriptions of its fields.

```
CONSUMER PRINTER CONFIGURED TEXT SCREEN  Screen 000  Lang Bank 000  Group 0000
  ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+.
 1
 2
 3
 4
 5
 6
 7
 8
 9
10                         <printer control code>
11
12
13
14
15
16
17
18
19
20

     F2=READ  SF2=HELP  F3=ADD  F4=DEL  F5=UPDT  F9=NEXT  F10=PRNT  F16=EXIT
```

Screen — A number that is associated with the desired screen. Valid values are 000 through 511, 900, 950, and 999 for STP/DX-based processors. Valid values are 000 through 009 (reserved for use by Diebold), 010 through 899, 900 through 950 (reserved for use by Diebold), and 999 for PC-based processors (BTP, ATP, and CTP).

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Lang Bank — A number associated with the language in which the screen is displayed. This key field allows the same screen number to be displayed in multiple languages. Valid values are 0 through 7. The languages associated with these values are user-defined.

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Group — The group number used to identify a particular load group such as a collection of states, screens, etc., that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic "InterBold i Series/MDS Display Screen" for a complete list of groups.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	0000

<printer control code> — A variable number of codes that control how the receipt is printed. Place a comma between multiple codes on the same line. A complete list of codes supported by the BASE24-atm product are provided in the topic "Consumer Printer Control Codes."

Field Length:	76 alphanumeric characters
Occurs:	20 times
Required Field:	Yes
Default Value:	No default value

Function Keys

The use of one function key on the Consumer Printer Configured Text screen varies from the standard function keys explained earlier in this section. The use of this function key is explained below.

Key	Description
Shift-F2	**Help —** Accesses Help screens explaining the characters used on the low-level screen. Press the **F1** key to move to the next Help screen, and press the **F2** key to move to the previous Help screen.

Consumer Printer Control Codes

The following Diebold consumer printer control codes are supported by the BASE24-atm product. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information on the consumer printer control codes.

Code	Description
"*text*"	Prints text in ASCII format. The text must be enclosed in double quotation marks, and embedded quotes must be doubled.
hh	Prints a character from the current character set. The valid values of *hh* are 20 through 7E hexadecimal.
+	Prints graphics without completing a line feed. After printing, the code exits the graphic mode and performs a carriage return.
-	Prints graphics and then completes a 1/12 inch line feed. After printing, the code exits graphic mode and performs a carriage return.
/	Prints graphics and then completes a 1/12 inch line feed. After printing, the code stays in the graphic mode and performs a carriage return.
CR	Ends the current row of printing, and moves the cursor to the beginning of the next line of the receipt.

Code	Description
ESC-=	Selects the graphic mode.
ESC-@	Resets the printer by selecting the standard character set, setting single-width printing, and reverting to the default horizontal and vertical pitches.
ESC-$	Sets the printer to 6 lines per inch vertical pitch. This is the default vertical pitch for Diebold terminals.
ESC-%	Sets the printer to 8 lines per inch vertical pitch.
ESC-&	Sets the printer to 10 characters per inch horizontal pitch.
ESC-’	Sets the printer to 12 characters per inch horizontal pitch. This is the default horizontal pitch for Diebold terminals.
ESC-(	Sets the printer to 16 characters per inch horizontal pitch.
ESC-)	Sets the printer to 10 characters per inch horizontal pitch, with double-density printing.
ESC-4	Selects the alternate character set.
ESC-5	Selects the standard character set.
ESC-6	Selects the standard or alternate character set that was previously selected.
ESC-7	Selects the second alternate character set.
ESC-C	Cuts and delivers the receipt form.
ESC-D, *filename.ext*;	Prints a graphic, logo, or plain text files. The value *filename* specifies the 1–8 character name of the file to be printed. The value *ext* specifies the file extension. This code is supported only on the 48-column thermal consumer printer.
ESC-E, *o*	Enables or disables emphasized printing, where *o* specifies whether emphasized printing is used. Valid values of *o* are 0, which means to disable emphasized printing, and 1, which means to enable emphasized printing. This code is supported only on the 48-column thermal consumer printer.
ESC-I,*s*	Inserts a configured text screen, where *s* specifies the configured text screen number to insert into the receipt.

Code	Description
ESC-L, *o*	Enables or disables landscape printing, where *o* specifies whether landscape printing is used. Valid values of *o* are 0, which means to disable landscape printing, and 1, which means to enable landscape printing. This code is supported only on the 48-column thermal consumer printer.
ESC-P	Advances the receipt to the top of the next form.
ESC-Q	Enables the printing of a scanned check amount to the printer.
ESC-R, *nnn*	Enables an entire scanned check image to be printed, where *nnn* is the length of the check image in millimeters. Valid values are 000–106.
ESC-S, *filename.ext;*	Enables the selecting of a specific character set for printing. The value *filename* specifies the 1–8 character name of the character set map file to select. The value *ext* specifies the file extension. This code is supported only on the 48-column thermal consumer printer.
ESC-U, *o*	Enables or disables underline printing, where o specifies whether underline printing is used. Valid values of *o* are 0, which means to disable underline printing, and 1, which means to enable underline printing. This code is supported only on the 48-column thermal consumer printer.
ESC-W,*o*	Enables or disables double-width printing, where *o* specifies whether double-width printing is used. Valid values of *o* are 0, which means to disable double-width printing, and 1, which means to enable double-width printing.
ESC-Z,*o*	Enables or disables compressed margin printing, where *o* specifies whether compressed margin printing is used. Valid values of *o* are 0, which means to disable compressed margin printing, and 1, which means to enable compressed margin printing.
FF	Causes the receipt paper to be ejected, and the next receipt readied for the next transaction.
LF	Moves the cursor to the beginning of the next line, leaving the current line blank.

Code	Description
SO,*s*	Prints spaces on the receipt, where *s* specifies the number of spaces, of which there can be up to 15 specified. Valid values of *s* are 1 through 9, and the following special characters, which represent the values 10 through 15: 10 = : (colon) 11 = ; (semicolon) 12 = < (less than sign) 13 = = (equals sign) 14 = > (greater than sign) 15 = ? (question mark)
VT,*hh*	Prints a character from an alternate character set. The valid values of *hh* are 20 through 7E hexadecimal.

Statement Printer Configured Text Definition

The Statement Printer Configured Text screen defines special codes that control how the receipts printed on the statement printer look. These codes are then downloaded to the terminal. This screen offers printer control in addition to the control defined in the Receipt File (RCPT). Refer to section 6 for more information on print control in the RCPT. Note that the statement printer is only supported on Diebold *ix*-series terminals. This screen is accessed by entering function code 7 for Statement Printer Configured Text Definition on the Diebold Entry Program Menu.

The special codes are described after the screen description. The Statement Printer Configured Text screen is shown below, followed by descriptions of its fields.

```
STATEMENT PRINTER CONFIGURED TEXT SCREEN Screen 000  Lang Bank 000  Group 0000
  ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+.
 1
 2
 3
 4
 5
 6
 7
 8
 9
10                         <printer control code>
11
12
13
14
15
16
17
18
19
20

     F2=READ  SF2=HELP  F3=ADD  F4=DEL  F5=UPDT  F9=NEXT  F10=PRNT  F16=EXIT
```

Screen — A number that is associated with the desired screen. Valid values are 000 through 511, 900, 950, and 999 for STP/DX-based processors. Valid values are 000 through 009 (reserved for use by Diebold), 010 through 899, 900 through 950 (reserved for use by Diebold), and 999 for PC-based processors (BTP, ATP, and CTP).

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Lang Bank — A number associated with the language in which the screen is displayed. This key field allows the same screen number to be displayed in multiple languages. Valid values are 0 through 7. The languages associated with these values are user-defined.

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000

Group — The group number used to identify a particular load group such as a collection of states, screens, and so on, that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support. The group number of each ATM is assigned in the TERMINAL GROUP field on ATD Diebold 10XX/478X screen 7. Refer to the topic "InterBold i Series/MDS Display Screen" for a complete list of groups.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	0000

<printer control code> — A variable number of codes that control how the receipt is printed. Place a comma between multiple codes on the same line. A complete list of codes supported by the BASE24-atm product is provided in the topic "Statement Printer Control Codes."

Field Length:	76 alphanumeric characters
Occurs:	20 times
Required Field:	Yes
Default Value:	No default value

Function Keys

The use of one function key on the Statement Printer Configured Text screen varies from the standard function keys explained earlier in this section. The use of this function key is explained below.

Key	Description
Shift-F2	**Help —** Accesses Help screens explaining the characters used on the low-level screen. Press the **F1** key to move to the next Help screen, and press the **F2** key to move to the previous Help screen.

Statement Printer Control Codes

The following Diebold statement printer control codes are supported by the BASE24-atm product. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for more information on the statement printer control codes.

Code	Description
"*text*"	Prints text in ASCII format. The text must be enclosed in double quotation marks, and embedded quotes must be doubled.
hh	Prints a character from the current character set. The valid values of *hh* are 20 through 7E hexadecimal.
CR	Moves the print position to the left margin, and prints all data in the buffer. You should have a CR code at the end of each line of text or graphics.
ESC-@	Resets the printer to its default settings.
ESC--,*o*	Selects underlining under all characters and spaces, where *o* specifies whether underlining is used. Valid values for *o* are 0 to disable underlining and 1 to enable underlining. Note that this code is ESC followed by two hyphens.
ESC-+,*n*	Selects *n*/360-inch line spacing, where *n* specifies the fraction of an inch for the line spacing. Valid values for *n* are 20 through 7F hexadecimal.

Code	Description
ESC-%,*p*	Selects the character set, where *p* specifies which character set to use. Valid values of *p* are 0 and 1.
ESC-0	Selects 1/8-inch line spacing between lines. Note that this code is ESC- followed by a zero, not the letter O.
ESC-2	Selects 1/6-inch line spacing between lines.
ESC-3,*n*	Selects *n*/180-inch line spacing, where *n* specifies the fraction of an inch for the line spacing. Valid values of *n* are 20 through 7F hexadecimal.
ESC-4	Selects the italic font.
ESC-5	Cancels the italic font that was set with the ESC-4 code.
ESC-E	Selects the bold font.
ESC-F	Cancels the bold font that was set with the ESC-E code.
ESC-G	Sets the printer to 15 characters per inch horizontal pitch.
ESC-K	Selects the letter-quality prestige typeface.
ESC-M	Sets the printer to 12 characters per inch horizontal pitch.
ESC-P	Sets the printer to 10 characters per inch horizontal pitch.
ESC-T,*t*	Selects a character table where *t* specifies which character table to use. Valid values of *t* are 0 through 3.
ESC-W,*o*	Enables or disables double-width printing, where *o* specifies whether double-width printing is used. Valid values of *o* are 0, which means to disable double-width printing, and 1, which means to enable double-width printing.
FF	Causes the receipt paper to be ejected, moves the print position to the left margin at the top of the next page, and prints all data in the buffer. You should have a CR code before the FF code.
LF	Moves the print position to the left margin of the next line, and prints all data in the buffer. You should have a CR code before the LF code.

Merchant Banking Center Data

The Merchant Banking Center Data screen defines the positions (x and y coordinates) for placing buttons and amounts on the Merchant Banking Center change mix touch screen at the ATM. This screen is used with the Diebold Merchant Banking Center product only.

The Merchant Banking Center Data screen is shown below followed by descriptions of its fields.

```
            Merchant Banking Center Data       Group 0000

                      Fee Amount Coordinates  00000000
                    Total Amount Coordinates  00000000
 Key A Button Coordinates    00000000    Key F Button Coordinates    00000000
 Key A Quantity Coordinates  00000000    Key F Quantity Coordinates  00000000
 Key A Amount Coordinates    00000000    Key F Amount Coordinates    00000000

 Key B Button Coordinates    00000000    Key G Button Coordinates    00000000
 Key B Quantity Coordinates  00000000    Key G Quantity Coordinates  00000000
 Key B Amount Coordinates    00000000    Key G Amount Coordinates    00000000

 Key C Button Coordinates    00000000    Key H Button Coordinates    00000000
 Key C Quantity Coordinates  00000000    Key H Quantity Coordinates  00000000
 Key C Amount Coordinates    00000000    Key H Amount Coordinates    00000000

 Key D Button Coordinates    00000000    Key I Button Coordinates    00000000
 Key D Quantity Coordinates  00000000    Key I Quantity Coordinates  00000000
 Key D Amount Coordinates    00000000    Key I Amount Coordinates    00000000


================================================================================
                F2=READ     F3=ADD      F4=DELETE   F5=UPDATE
                F9=NEXT    F10=PRINT               F16=EXIT
```

Group — The group number used to identify a particular load group such as a collection of states, screens, and so on, that is downloaded to the ATM. Each group provides certain functionality. For example, group 0 is the basic group, and group 600 provides Merchant Banking Center support. The group number of each ATM is assigned in the TERMINAL GROUP field on ATD Diebold 10XX/478X screen 7. Refer to the topic “InterBold i Series/MDS Display Screen” for a complete list of groups.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000

Fee Amount Coordinates — The position coordinates (xxxxyyyy) of the fee amount on the Change Mix touch screen.

Field Length: 8 numeric characters
Required Field: Yes
Default Value: 00000000

Total Amount Coordinates — The position coordinates (xxxxyyyy) of the total amount on the Change Mix touch screen at the ATM.

Field Length: 8 numeric characters
Required Field: Yes
Default Value: 00000000

Key Button Coordinates — The position coordinates (xxxxyyyy) of the ATM buttons (A–I) on the Change Mix touch screen at the ATM.

Field Length: 8 numeric characters
Occurs: 8 times
Required Field: Yes
Default Value: 00000000

Key Quantity Coordinates — The position coordinates (xxxxyyyy) of the quantity of bills and coins associated with the corresponding key on the Change Mix touch screen.

Field Length: 8 numeric characters
Occurs: 8 times
Required Field: Yes
Default Value: 00000000

Key Amount Coordinates — The position coordinates (xxxxyyyy) of the amount of bills and coins associated with the corresponding key on the Change Mix touch screen.

Field Length: 8 numeric characters
Occurs: 8 times
Required Field: Yes
Default Value: 00000000

ACI Worldwide, Inc.

5: Host-Controlled Solenoid-Activated Lock Feature

This section discusses the BASE24 operations required to support the host-controlled solenoid-activated lock feature. This feature is available with the Diebold 10XX/478X 2.1 firmware and provides additional security for the ATM chest door. The terminal must be equipped with a solenoid lock to support the host-controlled solenoid-activated lock feature. This section also describes the host-controlled solenoid-activated lock feature and the tasks necessary to set up the BASE24-atm Operator Identification File (OIF).

Overview

The host-controlled solenoid-activated lock is an optional Diebold 10XX/478X terminal feature available with 2.1 firmware. This feature supports additional controls for access to the terminal chest door. Before an operator can gain access to the ATM chest door, the Device Handler process must approve or deny an operator's request to unlock the solenoid lock on the door.

This feature is a terminal configuration option that requires modification of the Miscellaneous Features 2 field of the Write Command III message. This modification allows the BASE24-atm Diebold 10XX/478X Device Handler process to receive the additional information needed to process a host-controlled solenoid-activated lock request.

When this feature is enabled, a service request message contains the operator's group number and individual operator identification number. Both of these numbers are verified by the terminal. The operator's group number, identification number, and the terminal ID are then verified by the BASE24-atm Diebold 10XX/478X Device Handler process. The Device Handler process verifies the terminal ID by checking the BASE24-atm Terminal Data files (ATD). The Device Handler process verifies the operator group number and operator identification number to determine whether the operator has access to the ATM chest door. Operator group numbers, identification numbers, and access information are stored in the Operator Identification File (OIF).

For more information on the host-controlled solenoid-activated lock feature, refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** or the ***Diebold i Series, ix Series, and MDS Maintenance Manual***. The OIF is described on the following pages.

Operator Identification File (OIF)

The BASE24-atm Operator Identification File (OIF) enables the BASE24-atm Diebold 10XX/478X Device Handler process to control access to the solenoid lock of the ATM chest door.

Each operator servicing the ATM is assigned an operator group number and an individual operator identification number. When a service request is initiated at the ATM, the Device Handler process verifies the terminal ID for the ATM, the operator's group number, the operator's identification number, and whether the operator has access to the ATM chest door. The terminal ID, group number, identification number, and access information are stored in the OIF.

With the settings in the OIF, service operators within a specific group are given, or denied, chest door access. Individual operators in a single group can have different access authority. In addition, operators can have access to service the ATMs of multiple institutions.

Prerequisites for Setting Up the OIF Records

The Diebold 10XX/478X 2.1 firmware must be installed at each device and the Diebold Configuration File (CNFGIX) must be modified and downloaded to each terminal before an institution can support the use of the host-controlled solenoid-activated lock feature and its associated Operator Identification File (OIF). Changes to the CNFGIX are made using the Diebold entry program through the Miscellaneous Data screen, Miscellaneous Features, field 2. Refer to section 4 for a description of changing the CNFGIX using the Diebold entry program and the Miscellaneous Data screen.

The Logical Network Configuration File (LCONF) must be updated to support the host-controlled solenoid-activated lock feature. This feature requires that the LCONF assign ATM-DH-OIF1000 identify the fully qualified name of the Operator Identification File and that the ATM-DH-OIF1000-CHK param be set to Y to indicate that the BASE24-atm Diebold 10XX/478X Device Handler process checks OIF to determine access to the ATM chest door. The LCONF is described in section 2.

File Access

Access to OIF records is dependent on the standard BASE24 security restrictions used for other BASE24 files. The BASE24 security system is used to restrict screen access and specify the functions each CRT operator can use.

A CRT operator performing file maintenance functions has access to Operator ID File records and functions as specified in his or her security records. For example, depending on the specified security access, an operator may not be able to access all of the OIF records for a particular operator group and operator number combination.

Operator group, operator number, and terminal ID are the keys to the OIF. A CRT operator can only access OIF records for terminals belonging to institutions (FIIDs) for which he or she is authorized access. For example, if a logical network consists of Institution A and Institution B and a CRT operator has screen access only to all of Institution B’s terminals, the operator can only access OIF records for the terminals belonging to Institution B.

OIF file maintenance functions are performed after successfully passing two checks. The first check is the ATD. The terminal ID entered in the OIF must have a corresponding terminal ID in the ATD. The second check is the operator’s security file. If the terminal ID check is successful, the FIID, region, and branch specified in the ATD are compared with the FIID, region, and branch specified in the operator’s security file to determine whether the operator has access to that institution’s terminals. When both of these checks are successful, the OIF file can be accessed.

For complete information about BASE24 security, refer to the ***BASE24 User Security Manual***.

Operator ID File Screen

The Operator Identification File (OIF) contains a record for each operator performing service on Diebold 10XX/478X ATMs. A maximum of six groups can be defined. Each group can have eight operator identifiers. Therefore, forty-eight operators can be specified for each terminal.

Multiple records are displayed on an OIF screen. That is, each entry displayed on the screen is a separate record. Records can be added, deleted, or updated.

The keys to the file are the OPERATOR GROUP, OPERATOR NUMBER, and TERM ID fields.

To access the Operator ID File screen, enter OIF in the FILE DESTINATION field at the bottom of this screen. From a menu screen, press the **F1** key. From a file screen, press the **F16** key. The OIF screen is displayed.

The OIF screen is shown below, followed by descriptions of its fields.

```
BASE24-ATM   OPERATOR ID FILE         LLLL        YY/MM/DD  HH:MM  01 OF 01
              DIEBOLD 1000 OPERATOR IDENTIFICATION FILE
             OPERATOR GROUP:            OPERATOR NUMBER:
        TERM ID    ACCESS     TERM ID    ACCESS     TERM ID    ACCESS



















*********************************** BASE24 ***********************************
                  FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                          F12-HELP
```

OPERATOR GROUP — The group number of the operator servicing the ATM. The operator group is defined at the ATM. Valid values are 00 through 05.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: No default value
Data Name: OIF.PRIKEY.GRP

OPERATOR NUMBER — The operator identification number of the operator servicing the ATM. The operator identification number is defined at the ATM. Valid values are 1 through 8.

Field Length: 1 numeric character
Required Field: Yes
Default Value: No default value
Data Name: OIF.PRIKEY.NUM

TERM ID — An identifier (terminal ID) for the terminal associated with this record.

The TERM ID field must be a valid terminal ID number defined in the ATD. The identifier entered in this field must also match a symbolic station name defined in the XPNET network.

Field Length: 1–16 alphanumeric characters
Required Field: Yes
Default Value: No default value
Data Name: OIF.PRIKEY.TERM-ID

ACCESS — Indicates if the operator has access to the ATM chest door. Valid values are as follows:

Y = Yes, the operator does have access to the ATM chest door.
N = No, the operator does not have access to the ATM chest door.

Field Length: 1 alphanumeric character
Required Field: Yes
Default Value: No default value
Data Name: OIF.MENU-ITEM

Function Keys

The use of three function keys on the OIF screen varies from the standard BASE24 function keys explained in the topic “Configuration File Function Keys” in section 4. The use of these function keys is explained below.

Key	Description
F6	**Read Next (Group Number) —** Reads the next operator group and operator number. This function key allows an operator to read OIF records sequentially by operator identification number for each operator group number.
F8	**Clear Screen —** Clears all data from the screen.
F9	**Next Page —** Reads the next page of records that exists for a particular operator group and operator identification number.

Adding OIF Records

Records can be added to the OIF in a variety of ways depending on the operational requirements of the institution. Service access is specified by entering a terminal ID for every terminal and by entering a corresponding access value indicating whether the service operator has access to the terminal chest door. Asterisks are entered for the terminal ID to specify all terminals. To add a record with asterisks for the terminal ID, the operator adding the record must have security access to all terminals for all institutions (FIIDs).

A combination of entries can be made to specify service operator access to all but a few of the network terminals. For example, if an institution wants to allow service access to all but three of the network terminals, four records can be defined. One record is established by entering asterisks for the terminal ID and a value of Y for the access code. The three additional records, which are the exceptions, are established by entering the three terminal IDs and a value of N for the access code for each of the three terminals.

To add OIF records, perform the following steps:

1. Enter the appropriate values in the following fields:

   OPERATOR GROUP
   OPERATOR NUMBER
   TERM ID
   ACCESS

   Multiple records can be added on an OIF screen.

2. After entering values for each of the records to be added, press the **F3** key to add all records listed on the screen to the file.

If any of the entries are incorrect, none of the entries entered on that screen are added to the file. For example, if one of the entries contains an invalid terminal ID or the operator does not have security access to the terminal, none of the entries on that screen are added to the file. Incorrect entries are highlighted on the screen and appropriate error messages are displayed at the bottom of the screen. Incorrect entries must be corrected or deleted before any of the entries on the screen can be added.

After correcting these entries and pressing the **F3** key to add the records, another check is performed for duplicate records. If duplicate records exist for any of the entries, those entries are highlighted on the screen. All entries that do not have duplicate records are added to the OIF.

Deleting OIF Records

To delete OIF records, perform the following steps:

1. Enter the appropriate values in the following fields:

   OPERATOR GROUP
   OPERATOR NUMBER

2. Press the **F2** key to read the records for the OPERATOR GROUP and OPERATOR NUMBER entered.
3. Use the **space bar** to blank out all records that are **not** to be deleted.
4. Press the **F4** key to delete the records listed on the screen.

If the TERM ID to be deleted is known, it is not necessary to read all of the records. In this case, enter the OPERATOR GROUP, OPERATOR NUMBER, and TERM ID at step 2; read the record by pressing the **F2** key; then press the **F4** key to delete that record.

If an incorrect entry is made while deleting records, none of the entries on the screen are deleted. No deletions are made until all screen entries are correct. Incorrect entries are highlighted on the screen and appropriate error messages are displayed at the bottom of the screen.

Updating OIF Records

To update OIF records, perform the following steps:

1. Enter the appropriate values in the following fields:

   OPERATOR GROUP
   OPERATOR NUMBER

2. Press the **F2** key to read the records for the OPERATOR GROUP and OPERATOR NUMBER entered.

3. Update the appropriate ACCESS field and press the **F5** key.

   Only those records with changes to the ACCESS field are updated. If the ACCESS field is left blank, it defaults to N.

If the TERM ID to be updated is known, it is not necessary to read all of the records. In this case, enter the OPERATOR GROUP, OPERATOR NUMBER, and TERM ID at step 2; read the record by pressing the **F2** key; update the ACCESS field; then press the **F5** key to update the record.

If an incorrect entry is made while updating records, none of the entries on the screen are updated. No updates are made until all entries on the screen are correct. Incorrect entries are highlighted on the screen and appropriate error messages are displayed at the bottom of the screen.

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6: ATM Printer Support

Standard default receipt formats based on Diebold guidelines are provided with the BASE24-atm product. You can customize these default receipt formats. This section provides examples of default financial and administrative receipt formats as well as instructions on how to configure customized receipts. Refer to Diebold's ***91x Terminal Control Software (TCS), TCS Plus, and 91x TCS CSP Terminal Programming Manual*** for information on specifications for the physical properties and standard print formats supported by Diebold.

The BASE24-atm self-service banking (SSB) add-on product modifies some of the receipts shown here, and provides additional receipts. Refer to the ***BASE24-atm Diebold 10XX/478X SSB Manual*** for information on these modifications.

BASE24-atm Receipts

The BASE24-atm product can produce a printed receipt for the cardholder, which provides a tangible result of the electronic transaction that just took place. Receipts are also printed on the journal printer, so that the ATM owner has a record of the transaction.

Types of Printed Receipts

The BASE24-atm product can print receipts for the following transaction types:

- Financial transactions
- Statement print transactions
- Administrative transactions

Financial Transactions

Each financial transaction has its own type of receipt, so that the information provided to the cardholder is tailored to the requirements of the transaction. A denied transaction also has its own type of receipt. Whether a receipt is printed can be determined by you at the terminal level, or you can allow the cardholders to decide if they want receipts.

Statement Print Transactions

Statement print transactions also produce receipts given to the cardholder. However, this manual does not group these transactions with financial transactions because the BASE24-atm system does not control the formatting of the statement receipt; the host computer is responsible for the formatting of the statement print transaction receipt.

The MAXIMUM STMT LINES/RCPT field on BASE24-atm Terminal Data files (ATD) Diebold 10XX/478X screen 10 allows the statement printer to have a different number of maximum lines than the consumer printer (if you are using a Diebold *ix*-series terminal that has a statement printer installed) and is described in appendix A. Statement print transactions must be printed; cardholders should not be given the option to continue without a receipt.

The BASE24-atm product provides print compression of statement print transactions, which reduces the message traffic required. The statement print data compression feature is described in appendix A.

Administrative Transactions

Administrative transactions are performed by institution personnel, to perform such tasks as replenishing currency. You can configure whether administrative receipts are printed in the ATD.

Configurable Receipts

Configurable receipts provide you with the ability to establish ATM receipt formats that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, and multiple language support. The configurable receipt option allows for configuring approved and denied financial transaction receipts as well as administrative receipts.

The total number of characters of printer data for all the blocks combined in one Function Command Message (that is, transaction response from the BASE24-atm system) cannot exceed 580. If this number is exceeded, the Device Handler process generates an event message and no receipt is printed.

The ATM device can be installed with one of three consumer printer types. The 40-column consumer printer can print lines up to 2.75 inches wide. The 80-column consumer printer can print up to 6.125 inches wide. The 48-column thermal consumer printer can print lines up to 2.83 inches wide. The statement printer is an additional option on Diebold *ix*-series terminals. The number of characters that can be printed on a single line is determined by the CPI (characters per inch) setting. These restrictions must be considered when modifying receipt templates.

Terminal Receipt File (TRF)

The Terminal Receipt File (TRF) defines the receipt requirements for both approved and denied transactions. The transactions are defined using *ISO processing codes*. These codes consist of a two-character transaction code, a two-character *from* account type, and a two-character *to* account type.

The key to the TRF is the terminal receipt profile, message category (the second digit of the message type code), and the ISO processing code. The terminal receipt profile is specified in the BASE24-atm Terminal Data files (ATD) and can be assigned to a single terminal or a group of terminals. The Device Handler process calls a utility to map the internal BASE24 transaction code to the corresponding ISO processing code before reading the TRF extended memory table.

The Device Handler process reads the TRF extended memory table using the terminal receipt profile, the message category code from the message type, and the ISO processing code for the transaction.

Using the terminal receipt profile, message category, and ISO processing code, the Device Handler process steps through the records in the TRF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection:

Preference	Terminal Receipt Profile	Message Category	Processing Code
1	Exact Match	Exact Match	Exact Match
2	Exact Match	*	Exact Match
3	****************	Exact Match	Exact Match
4	****************	*	Exact Match

If none of the searches return a match, the Device Handler process uses the default value of 0 (no printed receipt, no audit receipt) for a declined transaction and 1 (print receipt only, mandatory) for an approved transaction.

Note: Every time the TRF is modified, the TRF extended memory table file must be rebuilt using the Base Extended Memory Table Build utility. After the TRF extended memory table file is rebuilt, the Device Handler process must reallocate its internal table. You can reallocate the TRF extended memory table by using the ALTERATTRIBUTE command from the EMT Control Commands screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. Note that when you enter the command from the DCT, all processes that use the TRF extended memory table are warmbooted. If you use the network control facility to issue the command, you can specify a specific process for the TRF extended memory table. Refer to the ***BASE24 Text Command Reference Manual*** for more information about the ALTERATTRIBUTE command. Refer to the ***BASE24-atm Transaction***

Processing Manual for information about the Base Extended Memory Table Build utility. Refer to the ***BASE24 Device Control Manual*** for more information on the EMT Control Commands screen.

Receipt File (RCPT)

The Receipt File (RCPT) is the primary file used to configure receipts. It contains receipt templates that define the actual text and format to use for the various types of receipts.

The Device Handler process retrieves receipt templates in the RCPT from the RCPT extended memory table and then uses that information for creating receipts.

Initially, the RCPT contains default templates for each of the receipt types. You can use these default templates as they are, or you can modify the templates to print the information you want on the receipt.

Note: Every time the RCPT is modified, the RCPT extended memory table file must be rebuilt using the Base Extended Memory Table Build utility. After the RCPT extended memory table file is rebuilt, the Device Handler process must reallocate the table. You can reallocate the RCPT extended memory table by using the ALTERATTRIBUTE command from the EMT Control Commands screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process Note that when you enter the command from the DCT, all processes that use the RCPT extended memory table are warmbooted. If you use the network control facility to issue the command, you can specify a specific process for the RCPT extended memory table. Refer to the ***BASE24 Text Command Reference Manual*** for more information about the ALTERATTRIBUTE command. Refer to the ***BASE24-atm Transaction Processing Manual*** for information about the Base Extended Memory Table Build utility. Refer to the ***BASE24 Device Control Manual*** for more information on the EMT Control Commands screen.

Space Compression

The BASE24-atm system supports Diebold’s space compression on all types of transaction receipts that it sends to the ATM, with the exception of the statement print transaction if the terminal has an operational statement printer. To do so, the Device Handler process fully formats the print buffer. The process then searches the buffer for occurrences of more than 3 spaces and replaces these occurrences with the SO control character. The number directly following the SO character identifies the number of spaces that the ATM should insert in the message at that location.

Receipt Templates

Receipt templates are made up of a series of field markers and field type codes that define the information to be included on the receipt and the format of the receipt. Refer to the topics "Field Markers" and "Field Type Codes" later in this subsection for a description of field markers and field type codes that can be used in the receipt templates.

Individual receipt templates can be created for the various types of approved and denied financial transactions, and approved and denied administrative transactions. These same receipt templates can also be created in different languages.

To determine the text and format to use for a receipt, the Device Handler process scans a template to identify the field markers and field type codes and their positions in the template. Then, based on the field type codes, the Device Handler process retrieves the required information and formats the receipt information based on the field markers used and their positions in the template. Examples of the default templates provided in the RCPT extended memory table for various transactions and examples of actual receipts created from these default templates are included later in this section.

Receipt templates are divided into three general categories, receipt header templates, receipt body templates, and receipt trailer templates, which are described in the following paragraphs:

- Receipt header templates control the information printed at the top of the receipt for all types of transactions.
- Receipt body templates control the information printed on the remainder of the receipt; body templates are specific to the type of transaction (for example, withdrawal or deposit, approved or denied, financial or administrative).
- Receipt trailer templates allow additional marketing information to be placed at the end of each financial transaction. Trailers can be multiple lines; however, the length of potential receipts with header, body, and trailer should be taken into consideration so that the number of lines does not exceed the maximum. Trailers are only allowed on financial transactions.

To correctly format receipt information for a transaction, a Device Handler process must be able to retrieve the appropriate templates from its RCPT information.

Receipt Template Keys

Each receipt template is one record in the RCPT, and the key fields for RCPT records are as follows: device type, receipt type, receipt group, language code indicator, FIID, *from* account type, *to* account type, and printer type. This group of key fields, described below, provides you with a great deal of flexibility in defining templates. For a comprehensive list of values that can be used in each of these key fields, refer to the ***BASE24-atm Files Maintenance Manual***.

Key Field	Description
DEVICE TYPE	Groups the RCPT records by type of device, allowing the Device Handler process to locate its records at initialization.
RECEIPT TYPE	Identifies the type of receipt template. Valid type codes have been defined by the BASE24-atm product.
RECEIPT GROUP	Allows the use of a single receipt template for a group of ATMs. Receipt groups are specified for each ATM on ATD screen 8. When selecting template data, the Device Handler process determines the group number assigned to an ATM in the ATD, and then searches for templates with the corresponding group. If an exact match cannot be found for the ATM group, the Device Handler process uses template records with asterisks in their group fields.
LANG CODE MSG/ID	Allows you to define the same receipt templates for more than one language. That is, a separate receipt template can be created in each of eight languages supported by the ATM. For example, you could configure one receipt template in English and one in Spanish in the RCPT. A language indicator is sent to the BASE24-atm system from the ATM after the cardholder selects a preferred language for the transaction. The Device Handler process then uses this language indicator to select the templates to use.
FIID	Allows a receipt to be formatted differently depending on who issued the card or owns the terminal. For instance, you may want different information printed for an on-us transaction than a not-on-us transaction, or you may want an abbreviated journal receipt for a terminal that you own.

Key Field	Description
FROM ACCOUNT	Allows a receipt to be formatted differently depending on the type of *from* account. For example, you can truncate savings account numbers differently than credit card numbers.
TO ACCOUNT	Allows a receipt to be formatted differently depending on the type of *to* account. For example, you can truncate savings account numbers differently than credit card numbers.
PRINTER TYPE	Allows a receipt to be formatted differently depending on the type of printer. For instance, you can truncate the account numbers on a financial receipt to help prevent fraud if the receipt falls into the wrong hands. However, you want the entire account number to print on the journal printer so that you have a complete record of the transaction.

Default Receipt Templates

Default receipt templates are provided in the RCPT for each type of transaction. These default receipt templates can be edited by you to customize specific types of information printed on the ATM receipts. Examples of the various default templates are described later in this section.

Multiple Languages

Receipt templates can be configured in more than one language. That is, a separate receipt template can be created in each of the eight languages supported by each Diebold terminal for a withdrawal transaction. The language indicator is sent to the BASE24-atm system from the ATM after the cardholder selects a preferred language for the transaction. For example, by configuring one receipt template in English and one in Spanish and matching each receipt template to the appropriate language indicator in the RCPT, cardholder receipts can be printed in English or in Spanish.

The Device Handler process uses the eighth byte in the operations code buffer to determine the language to use in screen display and receipt printing.

The Device Handler process uses the language indicator in the RCPT extended memory table, along with the terminal receipt group identifier, receipt type, and device type, to determine the correct receipt template to use.

Components of the Receipt Template

A number of components make up a receipt template. They are as follows:

- Field markers
- Field type codes
- Static text
- Directives

Field Markers

Field markers are special characters that identify the type of data to be used on a receipt or specify an action such as a line feed. The field markers that can be used with financial and administrative configurable receipt templates in the RCPT are described on the following pages.

The following table describes the standard field markers that can be used in receipt templates to define receipt text and format.

Code	Description
#	Indicates that a variable data field is to be inserted. The number symbol (#) is immediately followed by a field type code which indicates the type of data to be inserted. The possible field type codes are discussed in the topic “Field Type Code Descriptions” later in this section.
>	Indicates that a line feed character (hexadecimal OA) is to be inserted.
@	Indicates that a literal data string follows.
?	Indicates that a directive statement follows. A directive gives a command other than printing. The possible directives are discussed below in the topic “Directives.”
^	Indicates that an end-of-text character (hexadecimal 03) is to be inserted.

Field Type Codes

Field type codes correspond to predetermined pieces of fixed-length data which the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the ATD, or the C100NAMS File. The field type codes that can be used in configurable financial transaction receipt templates in the RCPT are described in the topic "Field Type Code Descriptions," later in this section. The field type codes are divided as follows:

- Standard field type codes. These codes are used with all receipt types.
- Cartridge field type codes. These codes are used with administrative receipts only.
- Terminal totals field type codes. These codes are used with administrative receipts only.

Static Text

You can enter any text in the receipt template that you want to print on the receipt. Static text must be preceded by the at (@) symbol in the template. You must ensure that the length of the static text plus the length of any field type codes used does not exceed the maximum number of characters that can be printed on the receipt.

Special characters that are reserved for use with field markers (#, >, @, and ^) can be printed in the text of the receipt by preceding them with the number (#) symbol. For instance, to print the greater than character (>), you enter #> in the template.

Directives

A directive indicates a command other than printing. The IF and ENDIF directives allow one template to print different text or data fields depending upon a condition that is determined outside of the RCPT. The IF directive marks the beginning of a block of receipt template that is printed if the condition is true. On the same line as the IF directive is the variable name of the condition to be tested. The ENDIF directive marks the end of the block. If the condition is not true, the block is skipped, and the receipt continues printing with the receipt line following the ENDIF directive.

Currently, the only valid condition values are as follows:

- FEES and NOFEES
- ISSFEES and NOISSFEES
- NCD and NONCD
- CASH, CHECK, and ENVLP

The FEES and NOFEES condition values can only be used in financial receipt templates. When the FEES condition is on the same line as the IF directive, the following block is printed if there is a surcharge fee assessed on the transaction. When the NOFEES condition is on the same line as the IF directive, the following block is printed if there is no fee for the transaction (that is, surcharging is not configured, or surcharging is configured, but not applicable for this particular transaction).

The ISSFEES and NOISSFEES condition values can be used only in financial receipt templates. When the ISSFEES is on the same line as the IF directive, the following block is printed if there is an issuer fee or rebate for the transaction. When the NOISSFEES condition is on the same line as the IF directive, the following block is printed if there is no issuer fee or rebate for the transaction.

The NCD and NONCD condition values enable noncurrency data to be displayed on withdrawal transaction receipts. When the NCD condition is on the same line as the IF directive, the following block is printed if the transaction is a noncurrency transaction (that is, if the Non-Currency Dispense token is present). When the NONCD condition is on the same line as the IF directive, the following block is printed if there are no noncurrency items in the transaction (that is, the Non–Currency Dispense add-on product is not configured, or the Non–Currency Dispense add-on product is configured but not applicable for this particular transaction).

The CASH, CHECK, and ENVLP condition values enable the type of deposit to be printed on receipts. When the CASH condition is on the same line as the IF directive, the block that follows is printed if a cash deposit transaction was made using the Diebold Bulk Note Acceptor or Enhanced Note Acceptor. When the CHECK condition is on the same line as the IF directive, the following block is printed if a check deposit transaction was made. When the ENVLP condition is on the same line as the IF directive, the following block is printed if an envelope deposit transaction was made.

An example of directives is described in the topics “Withdrawal from Checking, No Surcharge” and “Withdrawal from Checking, with Surcharge,” later in this section.

Financial Transaction Receipts

The BASE24-atm product supports both approved and denied transaction receipts. The following pages describe the default receipt templates that are used to configure approved financial transaction receipts as well as denied transaction receipts.

Default receipt templates are provided by ACI in English only, regardless of the language indicator associated with the template. You must then modify the appropriate template if you want another language. For example, default receipt templates for language indicators 0 and 1 are configured in English. You must modify the templates to correspond to the languages supported, or add templates for additional languages you want to support. This section shows one example for each default receipt template in the RCPT.

The Device Handler process raises EMS events if receipt formatting errors occur when BASE24-atm transaction processing takes place. The Device Handler process also raises EMS events if the maximum number of lines per receipt or the print buffer is exceeded.

The ATM waits until each transaction is completed before printing the receipt. Multiple transactions can be printed on a receipt depending on the transaction receipt option selected in the ATD. How many transactions can be printed on one receipt depends on several factors. If the device installation table (DIT) is set for marked paper, the maximum number of transactions per receipt is two; if the DIT is set for unmarked paper, the maximum number of transactions per receipt is four. The MAX LINES PER RCPT field on ATD Diebold 10XX/478X screen 8 can also determine the number of transactions per receipt.

Balance inquiry information requested by the cardholder can be printed, displayed on the ATM screen, or both, depending on how the ATD and IDF are configured.

Note: If the BASE24-atm Multiple Currency Dispense add-on product has been installed, the balance amount that appears on the receipt and on the ATM screen is in the Account currency. Because of the multiple currency conversions performed on the account balances, the balance on the receipt and on the screen may not be exactly the same as the balance amount as held by the account issuer. For more information about the BASE24-atm Multiple Currency Dispense add-on product, refer to appendix D.

Optional Receipts

You can determine whether receipts are printed for each transaction by several methods. The first method allows you to control receipt and journal printing by transaction type, and by whether the transaction was approved or denied. You enter this information using the Terminal Receipt File (TRF) screens. See the ***BASE24-atm Files Maintenance Manual*** for information on this file.

The second method allows the cardholder to make the final decision as to whether or not a receipt should be printed. If the cardholder chooses not to have a receipt printed, the journal receipt is still printed if the settings indicate that it should print. Note that the cardholder's choice with the second method overrides the settings in the TRF of the first method.

Configuring Optional Receipts

The optional receipts function is determined by settings on ATD Diebold 10XX/478X screen 7, which is also a device-specific screen. The OPTIONAL RECEIPTS field on ATD Diebold 10XX/478X screen 7 has four values, as follows:

0 = Optional receipts are not supported.
1 = The cardholder selects optional receipts if the printer fails. If the printer is working, the settings from the TRF extended memory table are used.
2 = The cardholder selects optional receipts on all approved transactions. If the transaction is denied, the settings from the TRF extended memory table are used.
3 = The cardholder selects optional receipts on all approved and denied transactions.

One exception to the optional receipt selection, no matter which setting you choose, is that statement print transactions must have receipts printed. If the printer fails, the cardholder cannot complete a statement print transaction.

Three additional fields on ATD Diebold 10XX/478X screen 7 determine how often and where in the flow the cardholder is asked about optional receipts. These are as follows:

Terminal Group	This group contains the states and screens that are downloaded with group 0 during an initial load. The states and screens should be ones that are appropriate when the printer is operable.

Printer Operable Group	This group contains the states and screens that are downloaded when the terminal sends a status indicating that the printer has become operable. Downloading this group restores any states and screens from the TERMINAL GROUP that were replaced when the printer became inoperable.
Printer Inoperable Group	This group contains the states and screens that are downloaded when the terminal sends a status indicating that the printer is not working. Downloading this group replaces states and screens from the TERMINAL GROUP, as applicable, with ones that are appropriate when the printer is unavailable.

One of the following options is used, depending on the values that you enter in the two group fields:

- The cardholder selects once at the beginning of the session whether the receipt is printed (decision applies to all chained transactions).
- The cardholder selects during each transaction whether the receipt is printed (decision is for that transaction only).

The groups that ACI provides in a sample configuration file for the optional receipts function are as follows (the options are set in the OPTIONAL RECEIPTS field as described above):

Group 300 = The basic group for the optional receipts function. This group should be specified as the TERMINAL GROUP on ATD Diebold 10XX/478X screen 7. The states and screens from the appropriate PRINTER OPERABLE GROUP below must be manually copied into group 300 so that the desired functionality is present on an initial load.

Group 301 = The printer currently is operational. The cardholder selects optional receipts once for all chained transactions. This group should be specified as the PRINTER OPERABLE GROUP on ATD Diebold 10XX/478X screen 7.

Group 302 = The printer currently is inoperable. The cardholder selects optional receipts once for all chained transactions (option 1, 2, or 3). This group should be specified as PRINTER INOPERABLE GROUP on ATD Diebold 10XX/478X screen 7.

Group 303 = The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, only if the printer becomes inoperable (option 1). This group should be specified as the PRINTER OPERABLE GROUP on ATD Diebold 10XX/478X screen 7.

Group 311 = The printer currently is operational. The cardholder selects optional receipts before each transaction. This group should be specified as the PRINTER OPERABLE GROUP on ATD Diebold 10XX/478X screen 7.

Group 312 = The printer currently is inoperable. The cardholder selects optional receipts before each transaction (option 1, 2, or 3). This group should be specified as the PRINTER INOPERABLE GROUP on ATD Diebold 10XX/478X screen 7.

Group 313 = The printer currently is operational. The cardholder selects optional receipts before each transaction, only if the printer becomes inoperable (option 1). This group should be specified as the PRINTER OPERABLE GROUP on ATD Diebold 10XX/478X screen 7.

You can use the states, FITs, and screens to create subsets of cardholders that use various settings of the optional receipts functionality. However, the sample configuration file provided by ACI only shows settings for all cardholders. Refer to section 4 for a description of groups, states, FITs, and screens.

Transaction Log File

The Transaction Log File (TLF) contains a flag indicating whether a receipt was requested. Valid values for the receipt option flag are as follows:

0 = The cardholder was not asked whether a receipt was required.
1 = A receipt was requested.
2 = A receipt was not printed because the printer has faulted.
3 = A receipt was not printed because the cardholder requested that one not be printed.

You can view this flag on the TLF. Refer to the ***BASE24-atm Files Maintenance Manual*** for information on how to peruse the TLF. However, this flag is not on any of the printed BASE24 reports.

Samples of Optional Receipt Settings

The following topics show how the different settings of the ATD work together to provide six different scenarios for optional receipts. These settings determine when the optional receipts function is used, and how often the cardholders are asked about whether they want to complete a transaction without a printed receipt.

Scenario 1—Ask Once per Session Only if Printer Fails

To ask the cardholder only once per session whether a receipt is printed if the printer fails, enter the following values in the following ATD Diebold 10XX/478X screen 7 fields:

OPTIONAL RECEIPTS = 1
PRINTER OPERABLE GROUP = 0303
PRINTER INOPERABLE GROUP = 0302

If the printer is working, the settings in the TRF extended memory table are used. If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked at the beginning of the session, “Continue without a receipt?” If the cardholder selects yes, all chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 2—Ask Each Transaction Only if Printer Fails

To ask the cardholder during each transaction whether a receipt is printed if the printer fails, enter the following values in the following ATD Diebold 10XX/478X screen 7 fields:

OPTIONAL RECEIPTS = 1
PRINTER OPERABLE GROUP = 0313
PRINTER INOPERABLE GROUP = 0312

If the printer is working, the settings in the TRF extended memory table are used. If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked during each transaction, “Continue without a receipt?” If the cardholder selects yes, the current transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 3—Ask Once per Session on Approved Transactions

To ask the cardholder once per session whether a receipt is printed on all approved transactions, enter the following values in the following ATD Diebold 10XX/478X screen 7 fields:

OPTIONAL RECEIPTS	=	2
PRINTER OPERABLE GROUP	=	0301
PRINTER INOPERABLE GROUP	=	0302

When the printer is working, the cardholder is asked once at the beginning of the session, “Do you want a receipt?” If the cardholder selects yes, the receipts for all approved chained transactions are printed. There is one exception to the yes selection; if the transaction is denied, the settings in the TRF extended memory table are used.

If the cardholder selects no, receipts are not printed for all approved chained transactions. There are two exceptions to the no selection:

- If the transaction can only complete a partial dispense, a receipt always is printed.
- If the transaction is denied, the settings in the TRF extended memory table are used.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked at the beginning of the session, “Continue without a receipt?” If the cardholder selects yes, all chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 4—Ask Once per Session on Approved and Denied Transactions

To ask the cardholder once per session whether a receipt is printed on all approved and denied transactions, enter the following values in the following ATD Diebold 10XX/478X screen 7 fields:

OPTIONAL RECEIPTS	=	3
PRINTER OPERABLE GROUP	=	0301
PRINTER INOPERABLE GROUP	=	0302

When the printer is working, the cardholder is asked once at the beginning of the session, “Do you want a receipt?” If the cardholder selects yes, the receipts for all approved and denied chained transactions are printed. If the cardholder selects no, receipts are not printed for all approved or denied chained transactions. There is one exception to the no selection; if the transaction can only complete a partial dispense, a receipt is always printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked at the beginning of the session, “Continue without a receipt?” If the cardholder selects yes, all approved and denied chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 5—Ask Each Transaction on Approved Transactions

To ask the cardholder during each transaction whether a receipt is printed on all approved transactions, enter the following values in the following ATD Diebold 10XX/478X screen 7 fields:

OPTIONAL RECEIPTS	=	2
PRINTER OPERABLE GROUP	=	0311
PRINTER INOPERABLE GROUP	=	0312

When the printer is working, the cardholder is asked during each transaction, “Do you want a receipt?” If the cardholder selects yes, the receipt for the approved transaction is printed. There is one exception to the yes selection; if the transaction is denied, the settings in the TRF extended memory table are used.

If the cardholder selects no, a receipt is not printed for the approved transaction. There are two exceptions to the no selection:

- If the transaction can only complete a partial dispense, a receipt always is printed.
- If the transaction is denied, the settings in the TRF extended memory table are used.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked during each transaction, “Continue without a receipt?” If the cardholder selects yes, the approved transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 6—Ask Each Transaction on Approved and Denied Transactions

To ask the cardholder during each transaction whether a receipt is printed on all approved and denied transactions, enter the following values in the following ATD Diebold 10XX/478X screen 7 fields:

OPTIONAL RECEIPTS	=	3
PRINTER OPERABLE GROUP	=	0311
PRINTER INOPERABLE GROUP	=	0312

When the printer is working, the cardholder is asked during each transaction, “Do you want a receipt?” If the cardholder selects yes, the receipt for the approved or denied transaction is printed. If the cardholder selects no, a receipt is not printed for the approved or denied transaction. There is one exception to the no selection; if the transaction can only complete a partial dispense, a receipt is always printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked during each transaction, “Continue without a receipt?” If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Sample Financial Transaction Receipt

The Device Handler process sends to the terminal all of the information needed to print a financial transaction receipt.

The table below describes the standard lines used on Diebold ATM default financial receipts. Because receipts are configurable, however, the information on each receipt line could be different.

Line	Description
1–2	A space reserved for the name and address of the institution along with space for an institution’s free-format marketing message. The Diebold ATM printer can utilize preprinted receipt paper or can print these lines.
3	DATE TIME STATION ID
4	The date of the transaction, the time the transaction occurred, and the first 8 digits of the 16-digit terminal ID of the Diebold ATM.
5–6	Blank
7	CARD NUMBER
8	The card number that was used to initiate the transaction.
9	Blank
10	LOCATION
11	ATM location; obtained from the ATD.
12–13	Blank
14	A label of RECORD NO. and the terminal sequence number assigned to this transaction.
15	The transaction type that has been performed and the amount of the transaction.

Line	Description
16	The account number of the affected account. If funds are being withdrawn from the affected account the account number is preceded by the label FR. If funds are being deposited to the affected account the account number is preceded by the label TO. If a two-sided transaction such as transfer, inquiry, or payment has been performed, the affected *from* account number is printed on this line, preceded by the label FR.
17	A space reserved for the LEDGER BALANCE field, if needed. The Diebold ATM prints the ledger balance of the affected account immediately following the LEDGER BALANCE field for deposit, withdrawal, or inquiry transactions. If a two-sided transaction such as transfer or payment has been performed, the affected *to* account number is printed on this line, preceded by the label TO.
18	A space reserved for the AVAILABLE BALANCE field, if needed. The Diebold ATM prints the available balance of the affected account immediately following the AVAILABLE BALANCE field for withdrawal, deposit, or inquiry transactions.

The following is a sample receipt that demonstrates the standard lines for a Diebold receipt.

```
 1       CUSTOMER LOGO SCREEN
 2
 3     DATE    TIME   STATION ID
 4   DD/MM/YY  HH:MM   S1AC9100
 5
 6
 7   CARD NUMBER
 8    37666600000000001-000
 9
10   LOCATION
11   330 S. 108TH AVE.
12
13
14   RECORD NO.            83
15   DEPOSIT           $20.00
16   TO 5555566666777778888
17   LEDGER BAL       $123.45
18   AVAIL BAL        $678.90
19

              1         2
    1234567890123456789012345678
```

Field Type Code Descriptions

The following table describes the standard field type codes that are used in receipt templates. Each field type code corresponds to a predetermined piece of fixed-length data that the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the ATD, or the C100NAMS File. Data names are included in parentheses after each field type code description to indicate the specific data field from which the information is taken. This list includes field type codes that are not used in the default receipt templates, but which you can use to customize the receipt templates.

Note: Additional field type codes and descriptions associated with the BASE24-atm self service banking (SSB) add-on product can be found in the ***BASE24-atm Diebold 10XX/478X SSB Manual.***

For further information on the STM data fields, refer to the ***BASE24-atm Transaction Processing Manual***. For further information on the ATD data names, refer to the ***BASE24-atm Files Maintenance Manual*** (using the data name index) or refer to the DDLs.

Code	Description
A1	The transaction amount (STM.RQST.AMT-1), for instance, the withdrawal amount or deposit amount. In the case of a withdrawal with a surcharge fee, the A1 code contains the withdrawal amount plus the surcharge (which is the amount that is deducted from or added to the cardholder's account). In the case of a deposit with a surcharge fee, the A1 code contains the deposit amount minus the surcharge (i.e., the amount deducted from or added to the cardholder's account).
A2	The transaction amount (STM.RQST.AMT-2). This amount is the ledger balance for most transactions. The exceptions include the cash-back amount for a deposit with cash-back transaction, and the second deposit amount for a split deposit.
A3	The transaction amount (STM.RQST.AMT-3). This amount is the available balance for most transactions. The exceptions include the cash received in a partial deposit with cash-back transaction.
AB	The available balance (STM.RQST.AMT-3). This code does not print the text, "AVAIL BAL" (available balance), on the receipt. Use code XA below to print the text on the receipt. If this code is used on a balance inquiry receipt, the available balance is only printed if a response code 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields on the Institution Definition File (IDF) screen 13 and the CUSTOMER BALANCE INFORMATION field on ATD Diebold 10XX/478X screen 8 indicate that Amount 3 (STM.RQST.AMT-3) is to be displayed.
AI	The application identifier (DF-NAME field of the EMV Discretionary Data token). This field type code is used with the BASE24-atm EMV add-on product. Refer to the ***BASE24-atm EMV Support Manual*** for more information about the EMV Discretionary Data token.

Code	Description
AL	The Application Label that the ATM sent in the card authentication method (CAM) data on the request. This field type code is used with the BASE24-atm EMV add-on product. Refer to the ***BASE24-atm EMV Support Manual*** for more information about the EMV Request Data token.
AN	Indicates that the receipts shows all denominations along with the number of rolls dispensed for a money exchange transaction, even if the number of rolls dispensed is zero for the corresponding denomination. This code is used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension only.
AS	The application PAN sequence number (APPL-PAN-SEQ-NUM field of the EMV Status token). This field type code is used with the BASE24-atm EMV add-on product. Refer to the ***BASE24-atm EMV Support Manual*** for more information about the EMV Status token.
AV	Indicates that the receipt shows all denominations along with the value dispensed for a money exchange transaction, even if the value dispensed is zero for the corresponding denomination. This code is used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension only.
AX	Indicates the amount of money exchange transactions since the last balance period. This code is used for administrative receipts for the Diebold Merchant Banking Center product.
CA	Indicates the amount for the particular currency deposited after it has been converted to the terminal currency (AMT-INFO.CONV-AMT of the BNA Multiple Currency token).
CN	The card number (ATM.D1000.CRD-NUM) and member number (STM.RQST.MBR-NUM) from the ATM card. To print the card number only, see the CX field type code below.
CV	The equivalent converted amount for the transaction. The Device Handler process retrieves this amount from the Multiple Currency token. The Device Handler process uses the amount specified in the currency of the account, if available. If it is not available, the Device Handler process uses the amount in the currency of the terminal. This code is used with the Multiple Currency Dispense add-on feature only.

Code	Description
CX*l*,*r*,*c*	The card number from the ATM card (ATM.D1000.CRD-NUM). The member number is not printed. The three variables (*l*, *r*, and *c*) are used to control the truncation of the card number. If the variables are not present, the card number is printed in its entirety. Refer to the topic "Account Number Truncation" later in this section for additional information about these variables.
DA	The transaction date. (STM.TRAN-DAT—MM/DD/YY or DD/MM/YY). The date displayed is dependent on the value specified in the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF).
DC	The text "CONVERTED AMOUNT" that is printed on the receipt. This text is configured in the mult_crncy_tbl_g table in the C100NAMS File. This code does not print the converted amount. Use code CA (above) to print the converted amount.
DE	The text "EXCHANGE RATE" that is printed on the receipt. This text is configured in the mult_crncy_tbl_g table in the C100NAMS File. This code does not print the value of the exchange rate. Use code EA (below) to print the value of the exchange rate.
DO	The text "ORIGINAL AMOUNT" that is printed on the receipt. This text is configured in the mult_crncy_tbl_g table in the C100NAMS File. This code does not print the original amount. Use code OA (below) to print the original amount.
DS	The last statement print date. This code uses the date display format param (DATE-DISPLAY-TYPE) and STM.STMT-INFO.STMT.LAST-STMT-DAT.
EA	The rate used to convert the original amount into the currency used by the terminal (ATM-INFO.CONV-RATE of the BNA Multiple Currency token).
EC*xc*	Controls two-color printing. The value *x* specifies whether to disable (the value 0) or enable (the value 1) two-color printing. If two color printing is enabled, the variable *c* indicates the intensity or shade. Valid values are 0–9, ;, :, <, =, >, or ? (30–3F hex). This is used on terminal receipt or statement printers only.

Code	Description
ED @ *filename;*	Indicates that a graphic, logo, or plain text file be printed. The value *filename* is the valid DOS file name including an extension. The PRNTR TYPE field on RCPT screen 1 must be set to a value of C when using this code.
EI	Inserts configured text. The three characters that immediately follow are the screen number that contains the configured text. (The screen number is obtained from the Diebold Configuration File.)
EL*x*	Controls landscape printing. The value *x* specifies whether to disable (the value 0) or enable (the value 1) landscape printing. The PRNTR TYPE field on RCPT screen 1 must be set to a value of C when using this code.
EQ	Prints a scanned check amount to a printer.
ER *nnn*	Prints a scanned check image to a printer, where *nnn* is the length of the check image in millimeters. Valid values are 000–106.
ERV *nnn*	Prints a scanned check image (reverse side) to a printer, where *nnn* is the length of the check image in millimeters. Valid values are 000–106. This function requires an Intelligent Depository Module (IDM) equipped with dual scanners.
ES @ *filename;*	Indicates the selection of a character set for printing. The value *filename* is the valid DOS file name including an extension of the character set map. The PRNTR TYPE field on RCPT screen 1 must be set to a value of C when using this code.
EV	The exchange rate value. The Device Handler process retrieves the exchange rate from the Multiple Currency token. The Device Handler process uses the exchange rate for the account currency, if available. If it is not available, the Device Handler process uses the terminal currency rate. This code is used with the Multiple Currency Dispense add-on feature only.
E&	Sets 10 characters per inch (CPI) horizontal pitch.
E'	Sets 12 characters per inch (CPI) horizontal pitch.
E(	Sets 16 characters per inch (CPI) horizontal pitch.

Code	Description
E @	Resets the printer to the standard character set, single-width printing, and the default horizontal and vertical pitches.
FA	The *from* account type (e.g., CK, SV, CC). The account type printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the acct_typ_tbl_g table in the C100NAMS File.
FD	The *from* account description (e.g., CHECKING, SAVINGS, CREDIT CARD). The account description printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process retrieves the account description from the appl_tbl_g table in the C100NAMS File.
FR*l*,*r*,*c*	The *from* account number (STM.RQST.FROM-ACCT). If the STM.RQST.FROM-ACCT is all blanks or zeros, the Device Handler process uses the account description from the appl_tbl_g table in the C100NAMS File. The value of STM.RQST.FROM-ACCT-TYP and the language indicator determine which table entry is used. The three variables (*l*, *r*, and *c*) are used to control the truncation of the account number. If the variables are not present, the account number is printed in its entirety. Refer to the topic "Account Number Truncation" later in this section for additional information about these variables.
ID	Indicates the bag ID for a bag deposit transaction. This code is used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension only.
LB	The ledger balance (STM.RQST.AMT-2). This code does not print the text, "LEDGER BAL" (ledger balance), on the receipt. Use code XL below to print the text on the receipt. If this code is used on a balance inquiry receipt, the ledger balance is printed only if a response code of 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields on IDF and the CUSTOMER BALANCE INFORMATION field on ATD Diebold 10XX/478X screen 8 indicate that Amount 2 (STM.RQST.AMT-2) is to be displayed.
MB	Cardholder member number (STM.RQST.MBR.NUM).

Code	Description
MX	Indicates the number of money exchange transactions performed since the last balance period. This code is used for administrative receipts for the Diebold Merchant Banking Center product.
N2	Indicates the total amount of currency inserted for a money exchange from cash transaction. This code is used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension only.
OA	The amount of the deposit in the original currency (AMT-INFO. ORIG-AMT of the BNA Multiple Currency token).
OL	The number of the log-only transaction selected (for example, Log Only #1.) Four log-only transactions are supported. If the *from* account is 01, then a 1 is printed on receipt; if the *from* account is 02, a 2; if the *from* account is 03, a 3; and if the *from* account is 04, a 4.
RC	The response code (STM.RQST.RESP).
RN	Indicates that the receipt shows only the rolled coin denominations dispensed along with the number dispensed for a money exchange transaction (i.e., no information is displayed if no rolls of the corresponding denomination are dispensed). This code is used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension only.
RT	The response text. This code is used with denied transaction receipts. The printed receipt text depends on the value of STM. RQST.RESP. This response code is mapped to an internal response code specific to this Device Handler process. The Device Handler process searches the response_text_g table in the C100NAMS File to find a match on the internal response code. The corresponding text is then moved to the print buffer. The response_text_g table is configurable; refer to the topic "Response Code Text Table for Financial Receipts" later in this section.
RV	Indicates that the receipt shows only the rolled coin denominations dispensed along with the value dispensed for a money exchange transaction (i.e., no information is displayed if no rolls of the corresponding denomination are dispensed). This code is used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension only.

Code	Description
S1	The amount to be dispensed or deposited, without including the surcharge amount (used in a withdrawal or deposit transaction with a surcharge fee). (STM.RQST.AMT-1 – TRAN-FEE)
S2	In the case of a transaction with a surcharge fee, this code is the amount of the surcharge (TRAN-FEE, from the Surcharge Data token).
S3	In the case of a withdrawal with an issuer fee or rebate, this code is the amount of the issuer fee or rebate (TRAN-FEE, from the Issuer Fee/Rebate token).
SQ	The four-digit BASE24 sequence number assigned to the transaction. (ATDD1.[atdd1-dev-info-diebold-10xx].TRAN-SEQ-NUM)
T1	The time the transaction took place—24 Hour clock: hh:mm. (STM.TRAN-TIM)
T2*xx/yy*	The time the transaction took place (STM.TRAN-TIM). Time is displayed in a 12-hour format. If the time is less than 1200, the time appears as hh:mm*xx*. If the time is greater than or equal to 1200, the time appears as hh:mm*yy* where hh has valid values of 1 through 12. *xx* and *yy* are the configurable variable length symbols for a.m. and p.m., respectively. The format of this field type code is T2*xx/yy*. You can optionally insert spaces before *xx* and *yy* if you do not want the text to directly follow the time. The value *yy* is terminated by either the next field marker code (#, >, @, or ?), or by the end of the template.
TA	The *to* account type (e.g., CK, SV, CC). The account type printed on the receipt depends on the language indicator and the value of the STM.TO-ACCT-TYP field. The Device Handler process matches account type numbers to a two-character account type description in the acct_typ_tbl_g table in the C100NAMS File.
TC	The city where the terminal is located. (STM.RQST.TERM-CITY)
TD	The *to* account description (e.g., CHECKING, SAVINGS, CREDIT CARD). The account description printed on the receipt depends on the language indicator and the value of the STM.TO-ACCT-TYP field. The Device Handler process retrieves the account description in the appl_tbl_g table in the C100NAMS File.

Code	Description
TE	All 16 bytes of the terminal ID (STM.TERM-ID).
TL	The terminal location (STM.RQST.TERM-NAME-LOC).
TM	The first 8 bytes of the terminal ID (STM.TERM-ID).
TN	The name of the terminal owner (STM.RQST.TERM-OWNER-NAME).
TO*l*,*r*,*c*	The *to* account number (STM.RQST.TO-ACCT). If the STM.RQST.TO-ACCT field is all blanks or zeros, the Device Handler process uses the account description from the appl_tbl_g table in the C100NAMS File. The value of STM.RQST.TO-ACCT-TYP and the language indicator determine which table entry is used. The three variables (*l*, *r*, and *c*) are used to control the truncation of the account number. If the variables are not present, the account number is printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
TS	The state where the terminal is located. (STM.RQST.TERM-ST-X)
TT	The transaction type located in C100NAMS tran_tbl_g or T100NAMS mtu_tbl_g). The transaction type printed on the receipt depends on the value of STM.TRAN-CDE. The Device Handler process searches the tran_tbl_g table in the C100NAMS or the mtu_tbl_g table in the T100NAMS to find a match on the transaction code (for a standard transaction), then the corresponding text is moved to the print buffer. If the transaction is supported by the Diebold Merchant Banking Center Device Handler extension, the Device Handler process searches the X100NAMS File for the default transaction code.
V0	The activation telephone number used to activate a mobile top-up voucher (Pre-Pay Top-Up token). If the token contains blanks, blank spaces are printed on the receipt. If the token cannot be retrieved, no activation information is printed. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.

Code	Description
V1	The stock number (Inventory Voucher token) of a voucher. If the Inventory Voucher token is present, the Device Handler process uses the stock number in the token. If the token is not present, no stock number information is printed on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V2	The expiration date (Pre-Pay Voucher token) of a voucher. If the Pre-Pay Voucher token is present, the Device Handler process uses the shelf date in the token. If the Pre-Pay Voucher token is not present, the Device Handler process uses the value in the stock expiration date field in the Inventory Voucher token. If the Inventory Voucher token is not present, no print expiration date is included on the receipt. The Device Handler process uses the date format specified in the DATE-DISPLAY TYPE param in the LCONF. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V3	The telephone number used in a mobile top-up transaction (Pre-pay Top-Up token). If the Pre-Pay Top-Up token is present, the Device Handler process uses the telephone number in the token. If the Pre-Pay Top-Up token is not present, no telephone number information is included in the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V4	The unique transaction number (Pre-Pay Top-Up token) for a mobile top-up transaction. If the Pre-Pay Top-Up token is present, the Device Handler process uses the reference number in the token. If the field contains a value, that value is printed on the receipt. If the field contains blanks or the token is not present, the unique transaction number is not printed on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V5	A telecommunications company marketing message (Pre-Pay Receipt token). If the Pre-Pay Receipt token is present, the Device Handler process retrieves the operator message and the corresponding length field. If it cannot retrieve the token, no marketing message is printed on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.

Code	Description
V6	The telecommunications company name (Pre-Pay Response token). If the token is not present, no telecommunications company information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V7	The carrier telephone number (Pre-Pay Response token). The Device Handler process uses the telephone number from the Pre-Pay Response token. If the token is not present, the carrier telephone number is not included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VA	The new minutes balance on the cardholder's mobile telephone (Pre-Pay Top-Up token). The Device Handler process uses the service balance from the Pre-Pay Top-Up token. If the token is not present, no new minutes balance information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VB	The batch ID (Pre-Pay Voucher Receipt token). The Device Handler process uses the batch ID from the Pre-Pay Voucher Receipt token. If the token is not present, no batch ID information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VC	Indicates the amount of cash vaulted by the currency acceptor. This code is used for administrative receipts for the Diebold Merchant Banking Center product.
VE	The number of expiration days (Pre-Pay Voucher Receipt token). The Device Handler process uses the expiration days from the Pre-Pay Voucher Receipt token. If the token is not present, no expiration days information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VI	The control ID number (Pre-Pay Voucher Receipt token). The Device Handler process uses the control number from the Pre-Pay Voucher Receipt token. If the token is not present, control ID information is not included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.

Code	Description
VN	The voucher number (Pre-Pay Voucher Receipt token). The Device Handler process uses the value of the NUM field in the Pre-Pay Voucher Receipt token. If the token is not present, no voucher number is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VS	The online expiration date (Pre-Pay Online token). The Device Handler process uses the expiration date from the Pre-Pay Online Receipt token. If the token is not present, no online expiration date is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VT	The tax amount (Pre-Pay Response token). The Device Handler process uses the tax amount from the Pre-Pay Response token. If the token is not present, no tax information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VX	The amount of the mobile top-up (STM.RQST.AMT-1 minus the amount value of the tax in the Pre-Pay Response token). This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
XA	The text "AVAILABLE BAL" (available balance) that is printed on the receipt. This text is configured in the ledg_avail_tbl_g table in the C100NAMS File. This code does not print the amount of the ledger balance. Use code AB (above) to print the available balance amount.
XC	The converted currency description text. The Device Handler process retrieves the converted amount description text from the diebold_1000_mult_crncy_tbl table in the C100NAMS File. This code is used with the Multiple Currency Dispense add-on feature only.
XE	The exchange rate description text. The Device Handler process retrieves the exchange rate text from the diebold_1000_mult_ crncy_tbl table in the C100NAMS File. This code is used with the Multiple Currency Dispense add-on feature only.

Code	Description
XL	The text "LEDGER BAL" (ledger balance) that is printed on the receipt. This text is configured in the ledg_avail_tbl_g table in the C100NAMS File. This code does not print the amount of the ledger balance. Use code LB (above) to print the ledger balance amount.
XS	The selected currency description text. The Device Handler process retrieves the selected amount description text from the diebold_1000_mult_crncy_tbl table in the C100NAMS File. This field is used with the Multiple Currency Dispense add-on feature only.

Account Number Truncation

Three of the field type codes listed above have additional variables that allow the account number or card number to be truncated. This feature can help prevent fraud by making it more difficult for an unscrupulous person to use a carelessly discarded receipt to gain access to an account.

The three field type codes are CX (card number without member number), FR (*from* account number), and TO (*to* account number). The three variables that can be appended to these field type codes are *l*, a two-character numeric code that indicates the number of the left-most digits that are printed; *r*, a two-character numeric code that indicates the number of the right-most digits that are printed; and *c,* a one-character alphanumeric code that indicates the pad character, or the character used to fill the remaining digits. If these three variables are not present, the card or account number is printed in its entirety.

Note that on the pad character that you can use a blank space as a pad. To use a reserved special character, (#, @, ?, >, and ^), precede it with the number symbol (#) in the receipt template.

You must configure another receipt template with a different printer type if you want the card or account number to print in its entirety on the journal printer. Also note that if you truncate one or more of the numbers, this does not affect how the number is stored internally; the complete number is shown on the TLF and reports.

The following template examples use a 19-digit primary account number (PAN) value of 1123581321345589144.

Field Type Code in Receipt Template	PAN as Printed on Receipt
`#CX00,04,%`	%%%%%%%%%%%%%%%9144
`#CX06,04,#@`	112358@@@@@@@@@9144
`#CX06,00,X`	112358XXXXXXXXXXXXX

Currency Symbols

The amount field type codes (A1, A2, A3, AB, LB, S1, and S2) can have a monetary symbol (such as $, •, or £) that prints on the receipt. You can enter the monetary symbol either manually or have it determined automatically by the Device Handler process. To enter the symbol manually, type the appropriate symbol after each occurrence of the field type code. Any text after the code is considered part of the monetary symbol until the next field marker is encountered.

The monetary symbol that is used can be determined automatically by the use of a table in the C100NAMS File. For the symbol to be determined automatically, you must enter CC$ after the field type code. The symbol is determined by the currency code table (RCPT_MONEY_SYM_TBL_G). This table contains an entry for each monetary system for which you want a special symbol to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbol(s) to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is encountered that is not in the table, the 3-character ISO currency code is used (for example, USD for US dollars). If neither the country code nor the ISO currency code exist in the table, no symbol is printed.

Transaction Receipt Header

The receipt header is printed at the top of the receipt for all financial transactions. The Device Handler process uses the header receipt template (the HDR receipt type in the RCPT extended memory table) to format the receipt header for all transactions. As described previously in this section, the process uses the device type, receipt group, transaction type, language code, card issuer FIID, *from* and *to* account type, and printer type from the RCPT extended memory table to find the corresponding header receipt template.

Once this step is completed, the Device Handler process reads the first character of the header receipt template and uses the field markers and field type codes described previously in this section to scan the receipt template to determine the receipt text.

The following is a representation of the default entry in the header receipt template (receipt type HDR) in the RCPT extended memory table for the Device Handler process.

Transaction Type	Receipt Type	Default RCPT Extended Memory Table Template Data

General Header	HDR	`#EI490>>`
		`#E@ DATE   TIME   STATION ID>`
		`#DA #T1 #TM>>>`
		`@CARD NUMBER>`
		` #CN>>`
		`@LOCATION>`
		`#TL>>>`

Transaction Receipt Trailer

A trailer or footer can also be added to a receipt to contain marketing information, or other information you want to print at the bottom of all financial receipts. You can choose not to use a trailer by not having a TRLR receipt type.

So that the trailer is not printed multiple times on one receipt, the trailer is ignored if the cardholder is chaining transactions and you have the TRANSACTIONS ON RECEIPT field on ATD Diebold 10XX/478X screen 8 set to 0, for multiple transactions per receipt.

An example of the trailer receipt (receipt type TRLR) is not given, because it is specific to each institution and no default template is included in the RCPT.

Samples of a Standard Receipt

The following pages describe the default receipt templates provided in the RCPT. Default templates are provided in English only regardless of the language indicator associated with the template. For example, default receipt templates for language indicators 0 and 1 are configured in English. You can either create a new template with a language indicator of 2 through 7 if you want another language or modify the appropriate template.

The following pages show default financial receipts for the following transactions:

- Withdrawal from checking, no surcharge
- Withdrawal from checking, with surcharge
- Deposit to savings
- Transfer from checking to savings
- Checking account inquiry
- Credit payment from checking
- Payment enclosed
- Message enclosed
- Log only
- PIN change
- Denied financial transactions

Withdrawal from Checking, No Surcharge

The receipt type WD is used for withdrawal transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a withdrawal from checking account transaction that does not have an added surcharge fee. Note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

```
@RECORD NO.       #SQ>
?IF NOFEES
@WITHDRAWAL#A1$>
?ENDIF
?IF FEES
@WITHDRAWAL#S1$>
@FEE       #S2$>
@TOTAL     #A1$>
?ENDIF
@FR  #FR>>
#XL#LB$>
#XA#AB$>>
```

```
     CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 3766660000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.             570
WITHDRAWAL          $20.00
FR     013333

LEDGER BAL        $3412.00
AVAIL BAL         $1760.00




         1         2
1234567890123456789012345678
```

Withdrawal from Checking, with Surcharge

The following is an example of the Diebold default receipt format for a withdrawal from checking account transaction that has an added surcharge fee. Note that this receipt is produced using the same template as the example on the previous page. Also note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

```
     CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 37666600000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.              571
WITHDRAWAL           $20.00
FEE                   $1.00
TOTAL                $21.00
FR     013333

LEDGER BAL         $3412.00
AVAIL BAL          $1760.00


         1         2
1234567890123456789012345678
```

Deposit to Savings Account

The receipt type DP is used for deposit transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a deposit to savings account transaction. The account type can be printed on the receipt, instead of the cardholder account number, depending on the authorization method used and whether the account number is available to the BASE24-atm system.

```
#E(@RECORD NO.            #SQ>
#TT    #A1$>>
@TO  #TO>
#XL#LB$>
#XA#AB$>>
```

```
      CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 37666600000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.            83
DEPOSIT           $20.00
TO 55555666667777778888
LEDGER BAL       $123.45
AVAIL BAL        $678.90

         1         2
1234567890123456789012345678
```

Transfer from Checking to Savings

The receipt type TR is used for transfer transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a transfer from checking to savings transaction. Cardholder account numbers can be printed on the receipt, instead of the account type, depending on the authorization method used and whether the account number is available to the BASE24-atm system.

```
#E(@RECORD NO.            #SQ>
@TRANSFER  #A1$>
@FR  #FR>
@TO  #TO>>
```

```
    CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 3766660000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.                840
TRANSFER              $240.00
FR CHECKING
TO SAVINGS

         1         2
1234567890123456789012345678
```

Checking Account Inquiry

The receipt type INQ is used for inquiry transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a balance inquiry from checking transaction. The cardholder account number can be printed on the receipt, instead of the account type, depending on the authorization level used and whether the account number is available to the BASE24-atm system.

```
#E(@RECORD NO.            #SQ>
@INQUIRY>
@FR  #FR>>
#XL#LB$>
#XA#AB$>>
```

```
   CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 37666600000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.              846
INQUIRY
FR CHECKING

LEDGER BAL          $123.45
AVAIL BAL           $678.90

         1         2
1234567890123456789012345678
```

Credit Account Payment from Checking Account

The receipt type PY is used for payment transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a credit account payment made from a checking account. The cardholder account number can be printed on the receipt, instead of the account type, depending on the authorization method used and whether the account number is available to the BASE24-atm system.

```
#E(@RECORD NO.           #SQ>
@PAYMENT   #A1$>
@FR  #FR>
@TO  #TO>>
```

```
    CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 3766660000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.                 843
PAYMENT                 $47.00
FR CHECKING
TO CREDIT CARD

         1         2
1234567890123456789012345678
```

Payment Enclosed

The receipt type PYEN is used for payment-enclosed transactions. The Diebold default receipt template is shown below, followed by an example of the payment enclosed transaction.

```
#E(@RECORD NO.            #SQ>
@PAYMT ENCL#A1$>>
```

```
 1       CUSTOMER LOGO SCREEN
 2
 3     DATE    TIME   STATION ID
 4   DD/MM/YY  HH:MM   S1AC9100
 5
 6
 7   CARD NUMBER
 8    37666600000000001-000
 9
10   LOCATION
11   330 S. 108TH AVE.
12
13
14   RECORD NO.                 843
15   PAYMT ENCL              $47.00
16

              1         2
     1234567890123456789012345678
```

Message Enclosed

The receipt type MGEN is used for message-enclosed transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a message from the cardholder to the institution placed in the envelope depository.

```
#E(@RECORD NO.            #SQ>
@MSG ENCL>>
```

```

    CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 37666600000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.                843
MSG ENCL




          1         2
1234567890123456789012345678
```

Log Only

The receipt type OL is used for log-only transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a log-only transaction.

```
#E(@RECORD NO.            #SQ>
@TYPE #OL @LOG ONLY MSG>>
```

```
      CUSTOMER LOGO SCREEN

    DATE    TIME   STATION ID
  DD/MM/YY  HH:MM   S1AC9100


  CARD NUMBER
   376666000000000001-000

  LOCATION
  330 S. 108TH AVE.


  RECORD NO.                843
  TYPE 1 LOG ONLY MSG
```

```
          1         2
1234567890123456789012345678
```

PIN Change

The receipt type PC is used for PIN change transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a PIN change transaction.

```
#E(@RECORD NO.             #SQ>
@PIN CHANGE>>
```

```
 1      CUSTOMER LOGO SCREEN
 2
 3    DATE    TIME   STATION ID
 4  DD/MM/YY  HH:MM   S1AC9100
 5
 6
 7  CARD NUMBER
 8   37666600000000001-000
 9
10  LOCATION
11  330 S. 108TH AVE.
12
13
14  RECORD NO.              843
15  PIN CHANGE
16
17
18
19

             1         2
    1234567890123456789012345678
```

Denied Financial Transactions

The Device Handler process converts receipt templates for denied transactions in the same manner as approved transactions with one exception. In addition to the receipt templates, a response code text table is used to associate one line of receipt text for every ATM response code. Refer to the topic "Response Code Text Table for Financial Receipts" later in this section for a description of the response code text table.

The receipt type DENY is used for denied financial transactions. The Diebold default receipt template is shown below, followed by an example of the default receipt for a denied withdrawal transaction.

```
#E(>@RECORD NO.           #SQ>
#TT>>
@TRANSACTION DECLINED>
#RC>
#RT>>
```

```
    CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100


CARD NUMBER
 3766660000000001-000

LOCATION
330 S. 108TH AVE.


RECORD NO.             580
WITHDRAWAL
TRANSACTION DECLINED
51
EXPIRED CARD

         1         2
1234567890123456789012345678
```

Response Code Text Table for Financial Receipts

The response code text table is a separate table in the C100NAMS File and is used in conjunction with the #RT (Response Text) data field when that particular data field is used in a receipt template for denied transactions.

The response code text table (response_text_g) provides additional receipt text for denied transaction receipts. This text gives the cardholder an explanation for the denied transaction and is in addition to the text provided in the receipt template.

The Device Handler process maps any BASE24 response code numbers that it receives into device-specific internal response code numbers. These internal response code numbers are in the C100NAMS File. Refer to the ***BASE24-atm Transaction Processing Manual*** for complete descriptions of the BASE24-atm response codes.

Default Entries for Denied Transactions

Default entries for the response code text table are provided in the C100NAMS File.

All of the default entries are in English. The default entries have a prefix to make it easier to distinguish the languages. For example, the first response code text entry has eight variations, listed in the Default Text column in the table below. Each variation corresponds to a language indicator, and has a prefix that makes it easier to distinguish the languages.

Default Text	Language Indicator	Prefix
INVALID PIN	0	none
*INVALID PIN	1	*
!INVALID PIN	2	!
@INVALID PIN	3	@

Default Text	Language Indicator	Prefix
#INVALID PIN	4	#
%INVALID PIN	5	%
&INVALID PIN	6	&
+INVALID PIN	7	+

The following table lists all of the default response code text messages in English. You can tailor all messages to meet your needs.

Device-specific Response Code	Default Response Code Text
1	INVALID PIN
2	PIN TRIES EXCEEDED
3	ENTER LESSER AMOUNT
9	TRANSACTION NOT ALLOWED
10	UNABLE TO PROCESS
11	INVALID TRAN OR ACCOUNT
12	UNAUTHORIZED CARD USAGE
13	EXPIRED CARD
14	INVALID CARD
16	UNABLE TO PROCESS AMT
17	INVALID CREDIT CARD AMT
18	USES LIMIT EXCEEDED
19	INSUFFICIENT FUNDS
21	NO MATCHING SHARING GRP
22	CONTACT BANK

Device-specific Response Code	Default Response Code Text
23	INSUFFICIENT FUNDS
34	UNABLE TO DISPENSE
35	UNABLE TO DEPOSIT
37	DAILY WD LIMIT REACHED
38	TRANSACTION TIMED OUT
41	UNABLE TO DISPENSE COINS
42	INVALID CASH BCK AMOUNT
44	NO CHECK ENTERED
45	UNABLE TO ACCEPT CHECKS
62	UNABLE TO DEPOSIT

Mapping Response Codes to Receipt Text

The response code text table includes one line of receipt text for each response code. Each line of text can be configured in up to 8 languages. Text for each language can be defined in groups of up to 24 characters each. The line of text from the table is printed on the denied transaction receipt at the location of the #RT field type code in the receipt template.

To locate the correct information in the response code text table, the Device Handler process performs the following steps:

1. Locates the applicable response code entry in the response code text table.
2. When a match is made on the response code, the Device Handler process determines which portion of the entry to send to the ATM as receipt print data by using the language indicator. If the language indicator is 0, the Device Handler process uses the first 24 characters of the response code entry as receipt print data. If the language indicator is 1, the Device Handler process scans past the first 24 characters of the response code entry and uses the second group of 24 characters of the entry as receipt print data. Each additional language skips another 24 characters.

Statement Print Transaction Receipts

Cardholders have the option of requesting statement print transactions at Diebold ATMs. The information for the body of this receipt is determined by the host computer linked to the ATM. The MAXIMUM STMT LINES/RCPT field allows the statement printer to have a different number of maximum lines than the consumer printer (if you are using a Diebold *ix*-series terminal that has a statement printer installed), and is described in appendix A. The BASE24-atm product provides methods for print compression of statement print transactions, which reduces the message traffic required. The statement print data compression feature is also described in appendix A.

Statement Print Journal Entry

The statement headers can be printed by either the journal or receipt printer. The printer flag preceding the specific transaction data determines which printer receives that data. When formatting the journal header, the Device Handler process uses the STMT template with the value A in the PRNTR TYPE field. When formatting the receipt header, the Device Handler process uses the STMT template with the value asterisk (*) in the PRNTR TYPE field. Examples of these templates are shown below. The format of the body of a statement print receipt is determined by the host computer linked to the ATM.

Statement Print Header Templates

The following illustrations represent template entries in the statement print header templates for both the journal and receipt printers. The journal print header template shown below controls printing of statement header information for the journal printer only when a statement is requested.

Statement Print Header	RECEIPT TYPE	PRNTR TYPE
#DA #T1 #TM> #CN> #TL> @RECORD NO. #SQ> @LAST STMT DATE #DS>	STMT	A

The template shown below controls the printing of header information for cardholder statements that are printed on receipt paper. Cardholder statement information printed after the first three header lines (two data lines plus one blank line specified by the template below) is supplied by the host computer.

Statement Print Header	**RECEIPT TYPE**	**PRNTR TYPE**

`@  DATE   TIME  STATION ID>`
`#DA #T1  #TM>>>`
`@CARD NUMBER>`
`#CN>>`
`@LOCATION>`
`#TL>>`
`@RECORD NO.       #SQ>`
`@LAST STMT DATE #DS>>`
	STMT	*

Sample Statement Print Receipt

The following is an example of a statement print receipt generated at a Diebold ATM. Note that the exact format of this receipt can differ because the host formats this receipt.

```
RECORD NO.          860
LAST STMT DATE 30/06/01


                L A S T   N A T I O N A L  B A N K
                _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
                   INTERIM FINANCIAL STATEMENT

JOHN R. CUSTOMER                          DATE:  22/07/01
131 ANYSTREET                             TIME:  14:00 PM
ANYTOWN ANYSTATE, 99999                   LOCATION:   076
ACCOUNT NUMBER: 123456789

  DATE      AMOUNT                  TYPE
_ _ _ _   _ _ _ _ _ _   _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _


7-06-01        100.00+   TELLER DEPOSIT
7-07-01         20.00+   ATM DEPOSIT #076
7-12-01        200.00+   TELLER DEPOSIT #076
7-15-01         75.00-   ATM WITHDRAWAL #134


_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

                          CHECK DETAIL
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

  DATE    CHECK NO.    AMOUNT     DATE     CHECK NO.    AMOUNT
_ _ _ _  _ _ _ _ _    _ _ _ _ _  _ _ _ _  _ _ _ _ _    _ _ _ _
7-12-01     1116        339.12   7-19-01     1122        64.50
7-15-01     1117        350.00   7-19-01     1123       114.50
7-15-01     1118        247.00   7-20-01     1124       100.00
7-16-01     1119        202.00   7-20-01     1125         9.00


_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

                         INTERIM SUMMARY
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _


PREVIOUS BALANCE:                                     2,032.76
DEPOSITS:                                               320.00
```

Administrative Transaction Receipts

The BASE24-atm Terminal Data Dynamic File—general data (ATDD1) accumulates transaction totals as transactions are completed for each ATM. These totals can then be printed on an ATM administrative receipt. These totals are used in balancing the ATM and monitoring the cash supply. They can also be printed at the ATM using terminal balancing and terminal subtotaling administrative transactions. All balancing totals are accumulated from the last time the ATM was balanced and the totals were reset to zero.

Administrative Receipt Descriptions

Administrative transaction receipts are printed on the same type of receipt paper forms as those used for financial receipts.

The position of the information printed on the receipt relative to the right side and the bottom of the receipt is dependent on the width of the paper used and whether the paper used is marked or unmarked. The totals are printed in hopper ascending order. That is, the totals for the hopper with the lowest denomination are printed first, while the totals for the highest denomination hopper are printed last.

Formatting and contents of administrative receipts is configurable according to your needs. Note that when establishing receipt templates, printer buffers have a maximum of 580 bytes. If additional information needs to be sent to the ATM, an additional message is sent.

Field Type Code Descriptions

The following tables describe the field type codes that are used in receipt templates for administrative receipts. Each field type code corresponds to a segment of fixed-length data. The data is retrieved from the ATD. Data names are shown in parentheses following each field type code description.

Refer to the ***BASE24-atm Files Maintenance Manual*** for further explanation of ATD fields.

Hopper Field Type Code Descriptions

The following field type codes are used in that portion of the administrative receipt that includes hopper totals. Where applicable, ATD data names are included in parentheses after each field type code description.

Code	Description
BV	Bill value. The denomination of the bills in the hopper (ATDD1.ATDD1-CORE.HOPR[*i*].BILL-VAL, where *i* refers to hoppers 0 through 3 for Diebold ATMs).
CD	Cash decreased. The number of bills removed from the hopper using midpoint adjustment during the current balancing period multiplied by the denomination of the bills in the hopper (ATDD1.ATDD1-CORE.HOPR[*i*].CASH-DECR × ATDD1.ATDD1-CORE.HOPR*i*.BILL-VAL, where *i* refers to hoppers 0 through 3 for Diebold ATMs). This field type code has a 10-byte length.
CE	End cash. The current number of bills in the hopper multiplied by the denomination of the bills in the hopper (ATDD1.ATDD1-CORE.HOPR[*i*].CASH-END × ATDD1.ATDD1-CORE.HOPR[*i*].BILL-VAL where *i* refers to hoppers 0 through 3 for Diebold ATMs). This field type code has a 10-byte length.
CI	Cash increased. The number of bills added to the hopper using midpoint adjustment during the current balancing period multiplied by the denomination of the bills in the hopper (ATDD1.ATDD1-CORE.HOPR[*i*].CASH-INCR × ATDD1.ATDD1-CORE.HOPR[*i*].BILL-VAL where *i* refers to hoppers 0 through 3 for Diebold ATMs). This field type code has a 10-byte length.
CO	Cash out. The number of bills dispensed from the hopper or removed from the hopper for cardholder withdrawals during the current balancing period multiplied by the denomination of the bills in the hopper (ATDD1.ATDD1-CORE.HOPR[*i*].CASH-OUT × ATDD1.ATDD1-CORE.HOPR[*i*].BILL-VAL where *i* refers to hoppers 0 through 3 for Diebold ATMs). This field type code has a 10-byte length.

Code	Description
CS	Start cash. The number of bills in the hopper at the beginning of the current balancing period multiplied by the denomination of the bills in the cassette. (ATDD1.ATDD1-CORE.HOPR[*i*].BEG-CASH × ATDD1.ATDD1-CORE.HOPR[*i*].BILL-VAL where *i* refers to hoppers 0 through 3 for Diebold ATMs). This field type code has a 10-byte length.
HN	The hopper number.

Each hopper field type code can be followed with a variable length monetary symbol in the receipt template to identify the monetary unit used for the transaction.

Totals for a specific hopper are printed only if the bill value for that hopper has been defined in the ATD.

Currency Symbols

The amount field type codes (BV, CD, CE, CI, CO and CS) can have a monetary symbol (such as $, •, or £) that prints on the receipt. You can enter the monetary symbol either manually or have it determined automatically by the Device Handler process. To enter the symbol manually, type the appropriate symbol after each occurrence of the field type code. Any text after the code is considered part of the monetary symbol until the next field marker is encountered.

The monetary symbol that is used can be determined automatically by the use of a table in the C100NAMS File. For the symbol to be determined automatically, you must enter CC$ after the field type code. The symbol is determined by the currency code table (RCPT_MONEY_SYM_TBL_G). This table contains an entry for each monetary system for which you want a special symbol to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbol(s) to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is encountered that is not in the table, the 3-character ISO currency code is used (for example, USD for US dollars). If neither the country code nor the ISO currency code exist in the table, no symbol is printed.

Depository Field Type Code Descriptions

The following field type codes are used in the portion of the administrative receipt that includes depository totals.

Code	Description
AC	The total unverified amount of commercial deposits accepted since the terminal was last balanced (ATDD1. ATDD1-CORE.AMT-CMRCL-DEP).
AD	The unverified amount of deposits accepted since the terminal was last balanced (ATDD1.ATDD1-CORE.AMT-DEP).
AK	The unverified amount of checks cashed or deposited and received since the terminal was last balanced (ATDD1. ATDD1-CORE.AMT-CHK).
AP	The unverified amount of payments enclosed received since the terminal was last balanced (ATDD1.ATDD1-CORE. AMT-PAY).
AT	Administrative type (for example, 'SUB' for subtotals).
CK	The total number of checks cashed or deposited since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-CHK).
CR	The total number of cards retained since the terminal was last balanced (ATDD1.ATDD1-CORE.CRDS-RET).
D0	The number of deposits received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-DEP).
D1	The number of commercial deposits received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-CMRCL-DEP).
EN	The sum of the following: number of deposits, number of commercial deposits, number of message envelopes, number of payment envelopes, and number of checks cashed or deposited since the terminal was last balanced.

Code	Description
MG	The total number of “message to financial institution” envelopes received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-MSG).
PY	The total number of payment envelopes received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-PAY).
RA	Response text for denied administrative transactions. The text table is in the C100NAMS File.

Note: Each depository amount field type code can be followed with a variable-length currency symbol, which identifies the monetary unit used for the transaction. See the “Currency Symbols” discussion in this section for additional information about the use of currency symbols.

Administrative Templates

The administrative header does not have a default receipt template in the RCPT extended memory table. If you create a receipt type ADHD, this receipt type is used only for administrative transactions. If the ADHD receipt type does not exist, the HDR receipt type is used for financial and administrative transactions.

Two administrative receipt templates are used for approved administrative transactions: ADHO for hopper information and ADTO for administrative totals. In addition, ADDN is a template used for denied administrative transactions.

The following are representations of the default entries in the RCPT extended memory table for administrative receipts.

Transaction Type	Receipt Type	Default RCPT Extended Memory Table Template Data
ADMIN BIN INFO (Administrative Bin Information)	ADBI	`#E(@BIN #BN @TOT CHKS #BC>` `@TOT AMT  #BA$>>`
Administrative transaction denied	ADDN	`@ADMIN TRANSACTION DENIED>` `#RA>>`

Transaction Type	Receipt Type	Default RCPT Extended Memory Table Template Data

Cassette Totals	ADHO	`#E(@C#HN @STR#CS$>`
		`@C#HN @INC#CI$@  C#HN @DEC#CD$>`
		`@C#HN @OUT#CO$@  C#HN @END#CE$>`

Depository Totals
(for terminal subtotaling, terminal balancing, and midpoint adjustment)
	ADTO	`#E(@DEP0#D0  @DEP1#D1  @CHK #CK>`
		`@MSG #MG  @PAY #PY>`
		`@TOT #EN  @CRD #CR  #AT>`

Sample Administrative Receipt

The following is an example of a Diebold ATM default receipt format for an administrative subtotal transaction. Note that this same format is used for all administrative transactions.

```
  CUSTOMER LOGO SCREEN

DATE      TIME   STATION ID
YY/MM/DD  17:06    S1A10035

CARD NUMBER
 56666600000001-000

LOCATION
330 SOUTH 108TH STREET


CO STR $10000
CO INC     $0 CO DEC     $0
CO OUT     $0 CO END $10000
C1 STR $10000
C1 INC     $0 C1 DEC     $0
C1 OUT     $0 C1 END $10000
DEPO  00 DEP1 00 CHK     00
MSG   00  PAY 00
TOT   00  CRD 00 SUB

         1         2
1234567890123456789012345678
```

Response Code Text Table for Administrative Receipts

The denied administrative transaction response code text table is a separate table in the C100NAMS File and is used in conjunction with the #RA (Response Text Administrative) field type code when that particular field type code is used in an administrative receipt template for denied transactions.

The response code text table for denied administrative transactions (admin_resp_text_g) provides additional receipt text for denied transactions. This text gives the service person an explanation for the denied transaction and is in addition to the text provided in the receipt template.

When the Device Handler process encounters a #RA field type code in a receipt template, the process uses the response code that it generated or that it received from the Authorization process, maps that code to an internal response code, moves to the administrative response code text table, and scans for a match on the internal response code. From this point on, the Device Handler process uses the administrative response code text table in the same manner as the response code text table for financial transactions. This processing is described previously in this section. The following entries are the default entries in the response code text table for denied administrative transactions. Note that response codes 26, 28, 30, and 43 are for transactions provided by the BASE24-atm self-service banking (SSB) add-on product. All of the default entries are in English; however, up to eight languages can be defined for the Diebold terminals.

Response Code	Initial Message Text
16	UNABLE TO PROCESS AMT
26	INVALID ADMIN SELECTION
28	COUNTS ALREADY OBTAINED
29	INVALID AMT FOR HOPPER
30	COUNTS NEEDED
33	ATM ALREADY BALANCED
39	ADMIN TRAN CANCELLED
43	REQUESTED BIN IS EMPTY

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7: Diebold Merchant Banking Center Support

This section describes BASE24-atm operations used to support the Diebold Merchant Banking Center. Diebold created Merchant Banking Center hardware to enable institutions to provide automated services to their merchant customers, such as purchasing cash, making change, and performing off-hour bag deposits. The Merchant Banking Center eliminates the need for financial institutions to have branch locations, and the associated overhead expenses, in high-priced retail locations.

The Diebold 10XX/478X Merchant Banking Center Device Handler extension supports money exchange and bag deposit transactions and provides receipts and settlement and reporting information.

Overview

The Diebold Merchant Banking Center terminal is a combination of the Cash Acceptor, Rolled Coin Dispenser, and (optionally) an After Hours Depository attached to a Diebold 10XX/478X device. This hardware configuration enables customers to get change and make bag deposits at an ATM. The BASE24-atm Diebold 10XX/478X Merchant Banking Center Device Handler extension enables institutions to perform these transactions and provide receipts and detailed reports on terminal activity to the financial institution.

Note: The Merchant Banking Center transactions cannot be used with the BASE24-atm Multiple Currency Dispense add-on feature or surcharging with response notification.

Supported Transactions

Merchant Banking Center functions include processing three Merchant Banking Center-specific transactions: money exchange from account, money exchange from cash, and bag deposit. These transactions are withdrawal, log-only, and deposit transactions, respectively, and are flagged with tokens as shown below:

Merchant Banking Center Transaction	BASE24-atm Transaction	Token
Money exchange from account	Withdrawal	Money Exchange Transaction token
Money exchange from cash	Log only	Money Exchange Transaction token
Bag deposit	Deposit	Bag Deposit token

Refer to the ***BASE24 Tokens Manual*** for more information about the layout of these tokens. Refer to the ***BASE24-atm Transaction Processing Manual*** for more information about how withdrawal, log only, and deposit transactions are processed.

Money Exchange Transactions

Cardholders can use the money exchange transactions to purchase change by accessing an online funding account (money exchange from account) or by inserting cash into the ATM currency acceptor (money exchange from cash). A merchant obtains multiple copies of receipts for money exchange transactions.

This subsection describes the following topics:

- Transactions allowed
- Coin and bill dispensing
- Surcharging
- Administrative Support

Transactions Allowed

Money exchange from account and money exchange from cash transactions are withdrawal and log-only transactions, respectively. You can make these transactions available to the entire customer base or you can limit these transactions strictly to merchant customers by adding the card prefixes to the Financial Institution Table (FIT). Configuring the FIT ensures that the screens for the Merchant Banking Center transactions are presented to the cardholder.

Note: The money exchange from cash transaction requires a card to be entered to initiate a transaction (though not a PIN).

Coin and Bill Dispensing

The money exchange transactions enable a cardholder to make change. The change cash for these transactions is provided in bills or rolled coins. The 50 bill per transaction limit imposed by Diebold 10XX/478X ATMs applies to Merchant Banking Center transactions as well. There also is a similar limit for the number of rolls of coins that can be dispensed in a transaction.

Surcharging

The money exchange from account transaction supports surcharging (request notification processing only) using the Surcharge File (SURF). Surcharging for money exchange from cash transactions, however, are supported using a state flow in the Diebold Configuration File (CNFGIX). The surcharge fee is maintained only for Merchant Banking Center transactions. The institution also can use the FIT to further control surcharging for Merchant Banking Center transactions.

Administrative Support

The following administrative transactions support the money exchange functions of the Merchant Banking Center:

- Rolled coin increase/decrease
- Rolled coin subtotals
- Balance to standard cash
- Balance to current cash

These transactions are allowed from the ATM only (that is, they cannot be entered from the Device Control Terminal (DCT)). The balancing transactions are applied to all hoppers, including rolled coins. For more information about administrative transactions, refer to the ***BASE24-atm Transaction Processing Manual.***

Bag Deposit Transactions

The Diebold 10XX/478X Merchant Banking Center Device Handler extension enables a merchant to obtain multiple copies of receipts for bag deposits. The cardholder can enter a deposit bag ID as part of the transaction, and this ID can be included on receipt copies. If the cardholder does not enter a deposit bag ID, the transaction is treated as a standard commercial deposit. Bag Deposit transactions are for on-us customers only.

Bag transaction totals are added to the commercial deposit totals.

Files Maintenance Considerations

The following Diebold and BASE24 files contain fields used with the Diebold 10XX/478X Merchant Banking Center Device Handler extension:

- Diebold Configuration File (CNFGIX)
- BASE24-atm Terminal Data files (ATD)
- Receipt File (RCPT)

Diebold Configuration File

The Diebold Configuration File (CNFGIX) enables the institution to maintain ATM screen displays, state table entries, and other configuration parameters. This information is identified by a load group number. The load group number for the Diebold 10XX/478X Merchant Banking Center Device Handler extension is 600.

States for the Merchant Banking Center Device Handler extension include the following:

- A (Card read state)
- B (PIN entry state)
- D (Clear keys state)
- E (Select function state)
- I (Transaction request state)
- @L (Create buffer enhanced state)

You can access the CNFGIX by using the Diebold Entry program In addition to defining these states, you can use the Diebold Entry program to configure the screen positions for buttons and cash amounts on the ATM screen. This is done by selecting the value 9 on the Diebold Entry Program Menu and pressing the **F1** key to display the Merchant Banking Center Data screen. Refer to section 4 for more information about using the Diebold Entry program and the Merchant Banking Data Center screen.

BASE24-atm Terminal Data Files

The BASE24-atm Terminal Data files (ATD) contain configuration information for each terminal in the logical network. The fields described below have specific Merchant Banking Center requirements. These ATD fields are described in more detail in the ***BASE24-atm Files Maintenance Manual.***

ATD Diebold 10XX/478X Screen 7

ATD Diebold 10XX/478X screen 7 is a device-specific screen. The screen includes the MERCHANT BANKING CENTER field on ATD screen 7 which indicates whether the terminal supports the Diebold Merchant Banking Center. This field must be set to a value of Y when the Merchant Banking Center Device Handler extension is used.

ATD Diebold 10XX/478X Screens 14 and 15

ATD Diebold 10XX/478X screens 14 and 15 are device-specific screens that define the rolled coin hopper contents. Fields on these screens define the contents of each hopper that contains rolled coins. Screen 14 defines hoppers 1 through 3, and screen 15 defines hoppers 4 through 6. If a hopper is used for rolled coins, the corresponding CONTENTS field must be set to the value 11. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about setting the fields on these screens.

ATD Diebold 10XX/478X Screen 16

ATD Diebold 10XX/478X screen 16 is a device-specific screen that contains configuration and usage information for Merchant Banking Center transactions. Included on this screen is the RECEIPT GROUP field, which identifies the set of receipt templates used for Merchant Banking Center transactions. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about this screen.

Receipt File

You can add receipt templates to the Receipt File (RCPT) for money exchange from account, money exchange from cash, and bag deposit transactions. While these transactions are processed as standard withdrawal, log-only, or deposit transactions, respectively, you can customize the receipts using the following field type codes: AN, AV, AX, ID, N2, MX, RN, RV, and VC. Refer to section 6 for descriptions of these field type codes.

Money Exchange from Account Transactions

The receipt type WD (withdrawal) is used for a money exchange from account transaction. The receipt group is configured in the RECEIPT GROUP field on ATD screen 16. A sample receipt template from the RCPT is shown below.

```
@RECORD NO.   #SQ>
 ?IF NOFEES
 #TT  #A1>
 ?ENDIF
 ?IF FEES
 #TT  #S1>
 @FEE        #S2$>@TOTAL    A1$>
 ?ENDIF
 @  DENOMINATION COUNT>
 #XL#LB$>
 #XA#AB$>>
```

The field type code TT in the receipt template causes the Device Handler process to retrieve the default description stored in the X100NAMS File. This can be replaced with text.

The above template creates the following receipt for a money exchange from account transaction:

```
 1      A C I   D E M O
 2
 3     DATE      TIME  TERM
 4     MM/DD/YY  HH:MM S1AC9100
 5
 6      123456700000003-003
 7
 8     330 SOUTH 108TH AVE
 9
10     RECORD NO.              570
11     MONEY XCHG           $33.00
12     FEE                   $1.00
13     TOTAL                $34.00
14     DENOMINATION    COUNT
15          0.01         2
16          0.05         1
17          0.25         2
18          10.00        1
19
20     LEDGER BAL     $3412.00
21     AVAIL BAL      $1760.00
22
23
24

                1         2
       1234567890123456789012345678
```

Money Exchange from Cash Transactions

The receipt type OL (log only transaction) is used for a money exchange from account transaction. The receipt group is configured in the RECEIPT GROUP field on ATD screen 16. A sample receipt template from the RCPT is shown below.

```
@RECORD NO.   #SQ>
?IF NOFEES
#TT  #A1$>
?ENDIF
?IF FEES
#TT        #S1$>@TOTAL    A1$>
?ENDIF
@    DENOMINATION   AMOUNT    #A1$
#RV>
@NOTES INSERTED #N2
```

The field type code TT in the receipt template causes the Device Handler process to retrieve the default description stored in the X100NAMS File. This can be replaced with text. The RV field type code displays only the denominations and value of the cash dispensed. The N2 field type code displays the total amount of notes inserted.

The above template creates the following receipt for a money exchange from cash transaction:

```
     A C I   D E M O

DATE      TIME  TERM
MM/DD/YY  HH:MM S1AC9100

 123456700000003-003

330 SOUTH 108TH AVE

RECORD NO.              570
MX FR CASH           $60.00
    DENOMINATION     AMOUNT
      0.05             4.00
      0.10            10.00
      0.25            40.00
      1.00             1.00
      5.00             5.00

NOTES INSERTED        60.00
```

Bag Deposit Transactions

The receipt type DP (deposit) is used for a bag deposit transaction. The receipt group is configured in the RECEIPT GROUP field on ATD screen 16. A sample receipt template from the RCPT is shown below.

```
@RECORD NO.   #SQ>
#TT
@TO    #TO00,06,X>>
@BAG ID  #ID
@TOTAL AMOUNT    #A1$>>
#XL#LB$>
#XA#AB$>>
```

The field type code TT in the receipt template causes the Device Handler process to retrieve the default description stored in the X100NAMS File. This can be replaced with text. The ID field type code displays the bag ID.

The above template creates the following receipt for a bag deposit transaction:

```
    A C I   D E M O

DATE      TIME  TERM
MM/DD/YY  HH:MM S1AC9100

 123456700000003-003

330 SOUTH 108TH AVE

RECORD NO.             570
DEPOSIT
TO   XXXX223455

BAG ID  12345678
TOTAL AMOUNT       $5620.35
LEDGER BAL        $3412.00
AVAIL BAL         $1760.00


         1         2
1234567890123456789012345678
```

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A: Statement Print Data Compression

This appendix describes the statement print data compression feature. This feature is available for use with statement print transactions initiated at Diebold ATMs that are supported by the BASE24-atm Diebold 10XX/478X Device Handler process.

This appendix includes the following information:

- A description of data compression and its purpose
- A description of BASE24-atm support of data compression
- An analysis of the impacts of data compression
- BASE24-atm data compression standards
- Statement print message specifications
- Configuration requirements for the statement print data compression feature
- A description of fields in the BASE24-atm Terminal Data files (ATD) that are specific to the statement print data compression feature and statement print processing in general

Data Compression on Statement Print Transactions

Data compression involves reducing, or compressing, the amount of transaction data that is carried throughout the BASE24 system. Reducing the amount of transmitted data results in a more efficient use of processing resources. Data compression is accomplished by reducing the number of characters and spaces transmitted within an individual transaction.

The greatest benefits of data compression can be achieved by reducing the amount of data carried in a statement print transaction. An ATM statement print transaction provides the cardholder with a receipt containing a detailed listing of account transactions for a specified period of time. The number of items listed on the receipt depends on the amount of transaction activity that has taken place in the account and on the host's capability to store and send such data. In the case of an active account, the receipt can contain many lines of account information. This type of transaction contains more information than other types of ATM transactions.

The additional amount of information contained in a statement print transaction requires that a series of messages be sent back and forth between the BASE24-atm system and the host. These messages consist of request messages from the BASE24-atm system to the host and response messages from the host to the BASE24-atm system. The number of request and response messages sent between the BASE24 system and the host depends on the amount of transaction activity in the account and on the host's capacity for sending statement information. As a result, compared to other types of transactions, statement print transactions take longer to send, make greater use of communications lines, and place a heavier burden on transaction processing efficiencies.

Forms of Data Compression

Two techniques are used to compress the data in statement print transactions. These techniques are referred to as *space compression* and *text compression.* The benefits of data compression can be achieved by using either space compression or text compression or by using a combination of both techniques. Implementing both techniques results in the greatest efficiency. These two techniques are discussed below.

Space Compression

Space compression reduces the number of blank spaces or repetitive characters transmitted in a statement print transaction message. Space compression uses special reserved symbols or characters to represent multiple spaces, blank lines, and repetitive characters contained in the statement print transaction message as it is carried through the ATM transaction processing system.

For example, in a statement print transaction, the first line of information may contain ten blank spaces between two items of information. When the statement print transaction message is sent, a space compression symbol is substituted for the ten blank spaces. Using space compression symbols to represent blank spaces reduces the amount of data sent in a statement print message.

Text Compression

Text compression reduces the amount of descriptive information (for example, headings and labels) transmitted in a statement print transaction message. Text compression uses special reserved symbols or characters to represent the descriptive text contained in the statement print message.

For example, one line of text in a statement print transaction might be displayed as AVAILABLE BALANCE. When the statement print transaction message is sent, a text compression symbol is substituted for the words AVAILABLE BALANCE. One text compression symbol replaces two words in the message. Using text compression symbols to represent descriptive text reduces the amount of data sent in a statement print message.

BASE24-atm Statement Print Data Compression Support

The BASE24-atm product establishes data compression standards that allow compressed data to be carried in a statement print transaction message. These data compression standards consist of a set of reserved symbols or characters used for space and descriptive text compression. These reserved symbols or characters are referred to as *format codes.*

BASE24-atm standardized data compression format codes are used to carry statement print messages from the host to the BASE24-atm system. That is, the statement print message format sent from the host to the BASE24-atm system contains the set of reserved symbols or characters established by the BASE24-atm product to achieve space and text compression.

Diebold terminals do not support the BASE24-atm data compression message format. Therefore, before the compressed statement print message can be sent to an ATM, the BASE24-atm system converts the BASE24-atm data compression format codes into the expanded device-specific message format supported by the ATM.

Data Compression Impacts

One of the primary objectives of implementing the statement print data compression feature is to decrease the number of messages required for statement print transactions. Decreasing the number of statement print messages can result in significant savings to system resources. The amount of savings to a financial institution is specific to the operations of that institution and cannot be generalized.

The performance gains that can be achieved by implementing the statement print data compression feature are directly related to the extent that space and text compression standards are used by the host, and the number of statement print transactions an institution typically processes.

Reducing the number of messages involved in processing a statement print transaction can impact the following areas:

- Host processing. Fewer messages mean fewer times a host needs to handle the same statement print transaction.
- Internal BASE24-atm processing. Fewer messages mean less interprocess traffic for statement print transactions.
- Communications lines. Fewer messages mean less traffic on communications lines to the host.

Again, the extent to which any of the above factors is affected is based on the number of statement print transactions the institution typically handles as well as the extent to which data compression is implemented.

C100NAMS File

The C100NAMS File is a Hewlett-Packard (HP) NonStop edit file that contains a series of tables used by the BASE24-atm Device Handler process. You can edit these tables to allow customization of specific types of information displayed on ATM receipts or screens. At the time the Device Handler process is compiled, the information in the C100NAMS File is incorporated into the Device Handler object file. The Device Handler process must be recompiled every time the C100NAMS File is modified.

The C100NAMS File contains a table with a series of identifiers that cross-reference BASE24-atm data compression symbols. The Device Handler process uses these identifiers as a key to convert BASE24-atm compression symbols into a format the ATM can use to print statement print receipts.

Statement Print Transaction Description Table

The statement print transaction description table in the C100NAMS File supports the statement print data compression feature. The BASE24-atm Device Handler process uses information in this table to convert compressed descriptive text in the statement print transaction message from the BASE24-atm data compression format into a format that the ATM can use to print statement print receipts.

ASCII Hexadecimal Index

The statement print transaction description table is built using an ASCII hexadecimal index with valid codes ranging from 00 to FF. Each of the codes used in the index must correspond to a BASE24-atm text compression symbol used by the host. For example, if index code 00 is established and defined in the statement print transaction description table, the corresponding BASE24-atm text compression symbol [00 can be used by the host. The compressed descriptive text in the message, which is represented by the symbol [00, is converted into the value defined for index code 00 in the statement print transaction description table.

You establish the index codes used in the ASCII hexadecimal index for the primary, secondary and tertiary languages and determine the value for each of the index codes. This is done by editing the statement print transaction description table in the C100NAMS File and recompiling the Device Handler process. The values defined for each of the index codes determine the text that is printed on statement print receipts.

Using the ASCII Hexadecimal Index

The following illustrates how the ASCII hexadecimal index is used. In this example, the value of index code 00 in the statement print transaction description table has been defined as CURRENT BALANCE. Whenever the BASE24-atm Device Handler process receives the text compression symbol [00 in a statement print message from the host, the Device Handler process finds the corresponding index value in the statement print transaction description table. In this case, since 00 has been defined as CURRENT BALANCE, the Device Handler process translates the text compression symbol [00 into the text CURRENT BALANCE before sending the statement print message to the ATM. The ATM then prints CURRENT BALANCE on the statement print receipt.

The following table illustrates how ASCII hexadecimal index values can be defined in a statement print transaction description table.

Index Value	Receipt Text
00	CURRENT BALANCE
01	BEGINNING BALANCE
02	AVAILABLE BALANCE
FF	RETURNED CHECK-NSF

Setting Up the ASCII Hexadecimal Index

You can establish up to 256 index codes in the statement print transaction description table. The length of the text for each of the index values is predefined within the table and cannot be altered.

When setting up values for index codes, it is important to list entries by frequency of use. That is, the most frequently used text should be included in the entries at the beginning of the table. If CURRENT BALANCE is the text displayed most often on statement print receipts, the first index code defined in the statement print transaction description table should be CURRENT BALANCE. In this way, the BASE24-atm Device Handler process can scan the statement print transaction description table more efficiently when converting BASE24-atm text compression symbols.

Text Compression Conversion

Since the Diebold devices do not support text compression, the BASE24-atm Device Handler process must convert BASE24-atm text compression symbols used in the statement print response message before sending the message to the ATM. This conversion is accomplished by using BASE24-atm data compression symbols called table lookup characters.

Table Lookup Characters

The standard BASE24-atm data compression format codes carried in the statement print message from the host to the Device Handler process include two text compression symbols called the table lookup character and the suppress table lookup character. These lookup-character symbols are used to compress text as the statement print message is carried from the host to the Device Handler process. The ATM does not support the use of these table lookup-character symbols; therefore, the Device Handler process must convert these symbols before sending the message on to the ATM.

When the Device Handler process receives the statement print transaction message from the host, it matches the lookup-character symbols in the message to the index codes in the ASCII hexadecimal index in the statement print transaction description table in the C100NAMS File. The Device Handler process converts the table lookup characters into the wording defined for each of the index codes before sending the statement print message to the ATM.

Displaying or Suppressing Blank Characters

The at symbol (@) can be used in conjunction with the table lookup character ([) or the suppress table lookup character (]) to determine whether blank spaces are displayed or suppressed when pulling information from the table. The at symbol (@) can be placed after a specific table entry in the statement print transaction description table.

When the table lookup character is used to access a table entry, at symbols (@) placed after a table entry are converted to blank spaces on the statement receipt. The at symbols (@) are converted to blank spaces up to the maximum table length.

When the suppress table lookup character is used to access a table entry, blank spaces are suppressed beginning with the first at symbol (@) placed after the table entry. Suppressing the blanks enables the host to eliminate the blank spaces after the text in the table entry and insert data on the receipt immediately after the entry. Data to be inserted is included in the statement data response message sent from the host.

Using the table lookup characters in combination with the at symbols (@) in a table entry makes it possible to display one table entry in the statement print transaction description table in two different ways. The following table is an example of how the same table entry can be displayed on the statement receipt depending on the table lookup character used to access the entry in the statement print transaction description table and whether at symbols (@) are used in the entry.

Text Compression Symbol Used		**Table Entry**	**Receipt Shows**
Lookup character	[	"RET CK #@@@@@@@@"	RET CK # 5600
Suppress lookup character	]	"RET CK #@@@@@@@@"	RET CK #5600
Lookup character	[	"RET CK # "	RET CK # 5600
Suppress lookup character	]	"RET CK # "	RET CK # 5600

Example of Descriptive Text Compression Conversion

The following is an example of how the BASE24-atm Device Handler process converts text compression symbols into receipt text using the statement print transaction description table in the C100NAMS File.

Statement Print Transaction Description Table

You can edit the statement print transaction description table in the C100NAMS File and compile the Device Handler process to establish ASCII hexadecimal index codes and specify the values for each of the index codes in the table.

Sample entries used in a statement print transaction description table are as follows:

Index Value	Receipt Text
00	CURRENT BALANCE
01	BEGINNING BALANCE
02	AVAILABLE BALANCE
FF	RETURNED CHECK-NSF

The Device Handler process uses the values in the statement print transaction description table to convert text compression symbols received in the statement print message from the host.

Table Lookup Character

The Device Handler process receives table lookup-character symbols in the statement print message received from the host.

The BASE24-atm table lookup character consists of a bracket character ([) followed by two digits from the ASCII hexadecimal index. The lookup character is represented as [*hh*, where *hh* is a two-digit hexadecimal value. The table lookup character is a cross-reference to the ASCII hexadecimal index in the statement print transaction description table in the C100NAMS File.

For example, the table lookup character might take the following forms: [01, [99, or [FF.

Using the index entries shown in the table on the previous page, the Device Handler process converts the table lookup-character symbol [01 as follows:

[01 = BEGINNING BALANCE

Conversion

Using the index entries shown in the sample table described previously in the topic "Statement Print Transaction Description Table," the following illustrates how the Device Handler process converts compressed descriptive text data received in the statement print message from the host.

Data Received from the Host	Data Sent to the ATM
[00	CURRENT BALANCE
[01	BEGINNING BALANCE
[02	AVAILABLE BALANCE
[FF	RETURNED CHECK-NSF

Data Compression Format Codes

The BASE24-atm product uses a standardized set of format codes that allows a host to send a compressed statement print message to the BASE24-atm system.

These codes can be placed in the host's statement print response messages to the BASE24-atm system. The host does not have to use all of the data compression format codes. These format codes can be used to achieve space compression or text compression or both. However, the greatest savings are attained when both forms of compression are used.

The following table identifies and describes the BASE24-atm data compression format codes used for space and text compression.

Format Code	Definition
{	**Line Feed —** A begin brace ({) indicates the end of a line of statement data. This symbol must be placed at a column position less than or equal to the value of the maximum number of characters per line. (This message is described later in this appendix.)
}	**Page Feed —** An end brace (}) indicates the last line of a paged receipt. This symbol must be at a line position less than or equal to the value specified in the LINES/PAGE field in the BASE24-atm Terminal Data files (ATD).
^*x*	**Compression Character —** The space compression character is a caret (^) followed by the variable *x*. The variable *x* indicates the number of spaces that have been compressed. Valid values for the variable *x* are 1 through 9 and A through Z for a maximum of 35 spaces. The values A through Z indicate the number of spaces according to alphabetic rank (for example, A equals 10, B equals 11, etc.).
>*znn*	**Repeat Character —** The repeat character is a greater than sign (>) followed by the variables *znn*, where *z* indicates the character to be repeated and *nn* is the number of times it is to be repeated.

Format Code	Definition
[*hh*	**Table Lookup Character —** The table lookup character is a begin square bracket ([) followed by the variable *hh,* where *hh* is the two-digit cross-reference into the alpha hexadecimal index in the statement print transaction description table in the C100NAMS File. Using the table lookup character [, all at symbols (@) used in a specific table entry are changed to blanks up to the maximum table length.
]*hh*	**Suppress Table Lookup Character —** The suppress table lookup character is an end square bracket (]) followed by the variable *hh,* where *hh* is the two-digit cross-reference into the alpha hexadecimal index in the statement print transaction description table in the C100NAMS File. Using the suppress table lookup character] indicates that blanks are suppressed at the first at symbol (@) found in a specific table entry.
nn	**Increment Date Character —** The increment date character, or date calculation, is a backslash (\\) followed by the variable *nn*. This symbol tells the BASE24-atm Device Handler process to calculate a date using the STMT.LAST-STMT-DAT field in the STM. The format of this date is based upon the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF). The variable *nn* indicates the number of days by which the date in the STMT. LAST-STMT-DAT field should be increased.
?*nnn*	**Ignore Text Character —** The ignore text character is a question mark followed by a three-character variable *nnn*. The variable *nnn* indicates the number of characters the Device Handler process is to ignore in the statement print data from the host. This symbol, if used, must be included in the first position of the statement data to indicate that the specified number of characters in the statement print data are not to be used as statement data, but simply echoed back to the host as is. The purpose of this character is to allow the host to set aside a certain number of characters in the statement data for tracking purposes. If this command is used, the IGNORE TEXT field on Diebold screen 10 of the ATD must be set to 1 (ignore text code is used). (Refer to the topic “Configuration Requirements” later in this appendix.)

Application of Data Compression

The following pages provide examples of how data compression standards work when applied to statement print transaction data. This is illustrated by showing a sample statement print receipt and applying data compression standards to information in the receipt. The receipt is used as a representation of the effect data compression has on the information contained in the statement print message.

The examples show how data compression affects the statement print message format from the host to the BASE24-atm Device Handler process. Also included are descriptions of the data compression symbols used for selected lines of the statement print receipt.

Sample Statement Print Receipt

An example of an ATM statement print receipt appears below. The information on this receipt is used in later examples to show how data compression techniques are applied.

```
                  ACCOUNT UPDATE SINCE MOST RECENT STATEMENT
                INFORMATION FROM mm/dd/yy TO mm/dd/yy hh:mm xM.

    ACCOUNT NUMBER - 3333331111123456

      DATE      TRANSACTION                         DEBIT          CREDIT

      mm/dd/yy  BEGINNING BALANCE                                 1,703.70

      mm/dd/yy  RETURNED CHECK-NSF                                   22.97
      mm/dd/yy  RETURNED CHECK-NSF                                   56.41
      mm/dd/yy  CHECK WITHDRAWAL #5654             60.90
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           20.00
      mm/dd/yy  DEPOSIT BIG BUCKS ATM                               840.76
      mm/dd/yy  DEPOSIT BIG BUCKS ATM                               840.76
      mm/dd/yy  DEPOSIT                                           1,000.00
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           10.00
      mm/dd/yy  CHECK WITHDRAWAL #5653             20.00
      mm/dd/yy  CHECK WITHDRAWAL #5660            247.82
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           40.00
      mm/dd/yy  STOP PAYMENT CHARGE                10.00
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           50.00
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           30.00
      mm/dd/yy  DEPOSIT BIG BUCKS ATM                                 7.62
      mm/dd/yy  CHECK WITHDRAWAL #5658             71.19
      mm/dd/yy  CHECK WITHDRAWAL #5659              3.00
      mm/dd/yy  CHECK WITHDRAWAL #5661             31.99
      mm/dd/yy  CHECK WITHDRAWAL #5662          1,012.50
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           20.00
      mm/dd/yy  CHECK WITHDRAWAL #5665             32.70
      mm/dd/yy  CHECK WITHDRAWAL #5664             69.96
      mm/dd/yy  DEPOSIT                                           1,222.33
      mm/dd/yy  CHECK WITHDRAWAL #5663             61.50
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           75.00
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM            5.00
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           40.00
      mm/dd/yy  CHECK WITHDRAWAL #5667             78.29
      mm/dd/yy  CHECK WITHDRAWAL #5668             89.69
      mm/dd/yy  CHECK WITHDRAWAL #5669             21.50
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM           50.00
      mm/dd/yy  WITHDRAWAL BIG BUCKS ATM            5.00
      mm/dd/yy  ACCOUNT UPDATE FEE                  1.00
******************************************************************************

      mm/dd/yy  CURRENT BALANCE                                   1,815.71

      mm/dd/yy  AVAILABLE BALANCE                                 1,915.71

```

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
```

Sample Statement Print Transaction Description Table

The following is the statement print transaction description table used for the statement print transaction represented by the statement shown on the previous page.

ASCII Hexadecimal Index — **Table Description**

```
                       1         2
          1234567890123456789012345
  01     "BEGINNING BALANCE        "
  02     "AVAILABLE BALANCE        "
  03     "CURRENT BALANCE          "
  04     "ACCOUNT UPDATE FEE       "
  05     "WITHDRAWAL BIG BUCKS ATM "
  06     "CHECK WITHDRAWAL @@@@@@@@"
  07     "STOP PAYMENT CHARGE      "
  08     "DEPOSIT BIG BUCKS ATM    "
  09     "DEPOSIT                  "
  0A     "RETURNED CHECK-NSF       "
  94     "DEBIT          CREDIT@@@@"
  95     "DATE      TRANSACTION@@@@"
  96     "ACCOUNT NUMBER - @@@@@@@@"
  97     "INFORMATION FROM @@@@@@@@"
  98     " RECENT STATEMENT@@@@@@@@"
  99     "ACCOUNT UPDATE SINCE MOST"
```

Note: Entries between ASCII hexadecimal index values 0A and 94 are not shown in this table because they are not used in the statement illustration.

Message Data—Host to the BASE24-atm Device Handler Process

The example below illustrates how the statement print transaction data looks when it is transmitted from the host to the BASE24-atm Device Handler process. This example applies BASE24-atm space and text compression standards to the statement print transaction data contained in the statement print receipt shown in the topic "Sample Statement Print Receipt" presented previously in this appendix.

Stmt Line	Host Compressed Data Format	Line Length	Position	Message Number
1	^G[99]98{	9	1-9	1
2	^E]97\00 TO \23 hh:mm xM.{	26	10-35	
3	{	1	36	
4	^4]963333331111123456{	22	37-58	
5	{	1	59	
6	^6]95^O]94{	11	60-70	
7	{	1	71	
8	^6\nn [01^O1,703.70{	21	27-92	
9	{	1	93	
10	^6\nn [0A^R22.97{	18	94-111	
11	^6\nn [0A^R56.41{	18	112-129	
12	^6\nn]06#5654^D60.90{	23	130-152	
13	^6\nn [05^B20.00{	18	153-170	
14	^6\nn [08^Q840.76{	19	171-189	
15	^6\nn [08^Q840.76{	19	190-208	
16	^6\nn [09^O1,000.00{	21	209-229	
17	^6\nn [05^B10.00{	18	230-247	
18	^6\nn]06#5653^D20.00{	23	248-270	
19	^6\nn]06#5660^C247.82{	24	271-294	
20	^6\nn [05^B40.00{	18	295-312	
21	^6\nn [07^B10.00{	18	313-330	
22	^6\nn [05^B50.00{	18	331-348	
23	^6\nn [05^B30.00{	18	1-18	2
24	^6\nn [08^S7.62{	17	19-35	
25	^6\nn]06#5658^D71.19{	23	36-58	
26	^6\nn]06#5659^E3.00{	22	59-80	
27	^6\nn]06#5661^D31.99{	23	81-103	
28	^6\nn]06#5662^A1,012.50{	26	104-129	
29	^6\nn [05^B20.00{	18	130-147	
30	^6\nn]06#5665^D32.70{	23	148-170	
31	^6\nn]06#5664^D69.96{	23	171-193	
32	^6\nn [09^O1,222.33{	21	194-214	
33	^6\nn]06#5663^D61.50{	23	215-237	
34	^6\nn [05^B75.00{	18	238-255	
35	^6\nn [05^C5.00{	17	256-272	
36	^6\nn [05^B40.00{	18	273-290	

Stmt Line	Host Compressed Data Format	Line Length	Position	Message Number
37	^6\nn]06#5667^D78.29{	23	291-313	
38	^6\nn]06#5668^D89.69{	23	314-336	
39	^6\nn]06#5669^D21.50{	23	337-359	
40	^6\nn [05^B50.00{	18	1-18	3
41	^6\nn [05^C5.00{	17	19-35	
42	^6\nn [04^C1.00{	17	36-52	
43	>*78{	5	53-57	
44	{	1	58	
45	^6\nn [03^O1,815.71{	21	59-79	
46	{	1	80	
47	^6\nn [02^O1,915.71}	21	81-101	

Applying Data Compression

Using the sample statement print receipt in the topic "Sample Statement Print Receipt" and the sample statement print transaction description table in the topic "Sample Statement Print Transaction Description Table," the following examples describe how data compression symbols are used to compress the data in the statement print receipt.

The following pages show the data compression symbols sent in the statement print message for lines 1, 2, 3, 4, 6, 7, 8, 9, 10, 12 and 43 of the sample statement print receipt.

LINE 1

ACCOUNT UPDATE SINCE MOST RECENT STATEMENT eol

{ - End of Line Indicator

]98 – Suppress Table Lookup Entry - 98

[99 – Table Lookup entry - 99

^G – 16 blanks

LINES 2 and 3

INFORMATION FROM mm/dd/yy to mm/dd/yy hh:mm xM. eol eol

{ - End of Line Indicator

{ - End of Line Indicator

hh:mm xM.

\nn – Date Calculation

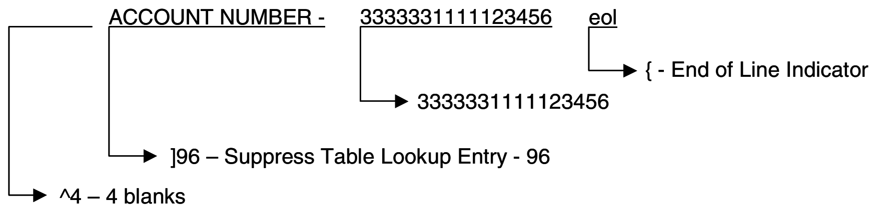
TO

\nn – Date Calculation

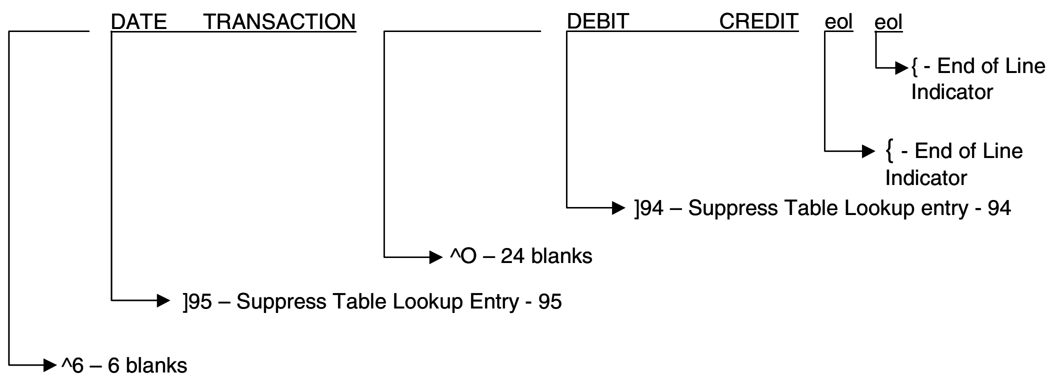
]97 – Suppress Table Lookup Entry - 97

^E 14 blanks

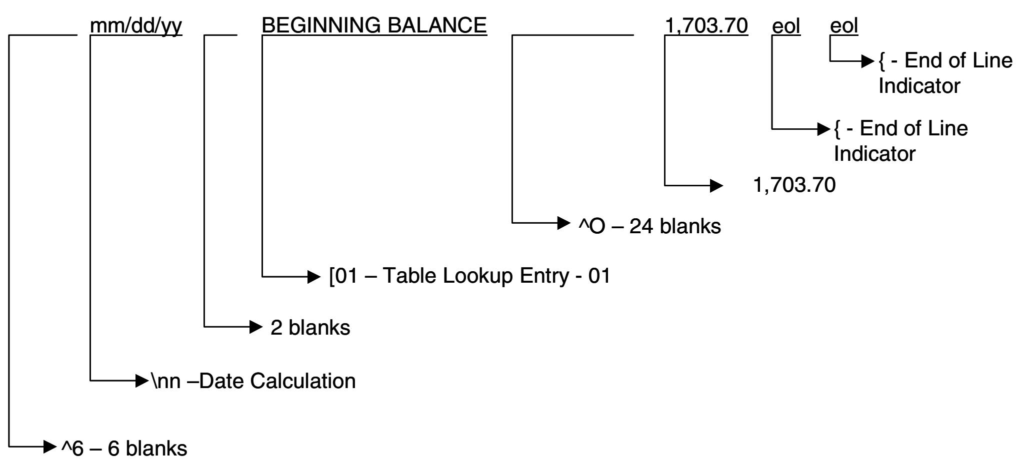
LINE 4



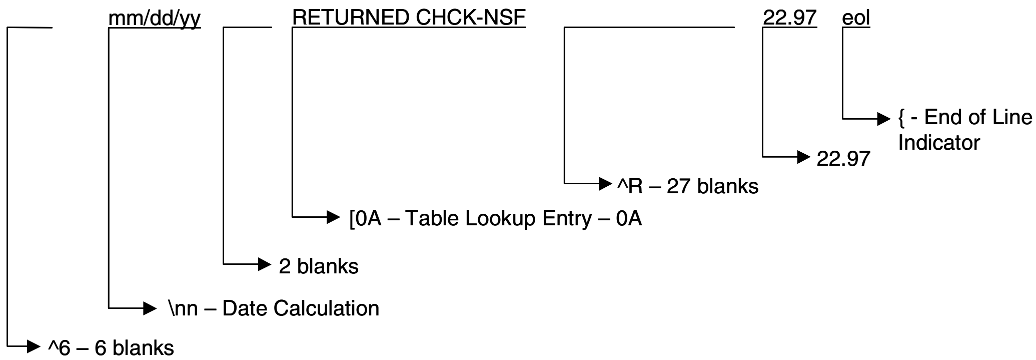
LINES 6 and 7



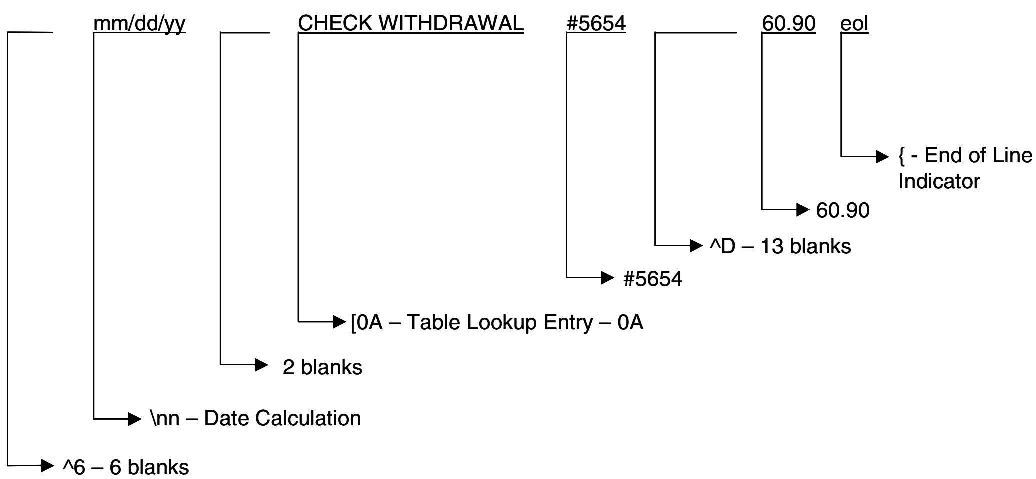
LINES 8 and 9



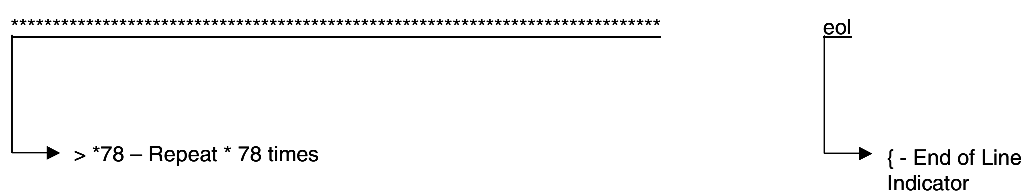
LINE 10



LINE 12



LINE 43



Statement Print Message Specifications

The following pages provide information on statement print message specifications for hosts using the statement print data compression feature.

The BASE24-atm product supports two external message formats to exchange information between the BASE24-atm system and the host. These two message formats—American National Standards Institute (ANSI) and International Organization for Standardization (ISO)—both support the statement print data compression feature.

The following pages cover host specifications for ISO and ANSI statement print requests and responses.

Statement Print Messages

Statement print transaction messages routed through the BASE24-atm system to the host do so in a series of transaction messages. These messages consist of request messages to the host and response messages from the host. Because of the amount of information carried in the statement print transaction message, several request and response messages can be sent between the BASE24-atm system and the host.

BASE24-atm Statement Print Message Types

The transaction types used by the BASE24-atm product for statement print messages are shown below:

Type	Description
0200	**Financial Transaction Request —** A request to authorize a transaction.
0205	**Statement Print Transaction Request —** A request for additional statement information for a transaction in progress.
0215	**Statement Print Transaction Response —** A response carrying the statement information for printing.

BASE24-atm Statement Print Transaction Processing

The following steps describe the processing for statement print transactions sent from the BASE24-atm system to the host.

1. The BASE24-atm system receives a request for statement information that is sent to the host as a 0200 request message. (The ANSI Host Interface process converts the 0200 statement print request to a 0205 message.)
2. The host processes the 0200 message and returns a 0215 response message to the BASE24-atm system. The 0215 message carries the statement information.
3. If additional pages of statement information are required, the BASE24-atm system sends a 0205 request message to the host requesting additional information.

4. The host processes the 0205 message and returns a 0215 message carrying additional statement information.

This processing continues until all of the statement information has been sent.

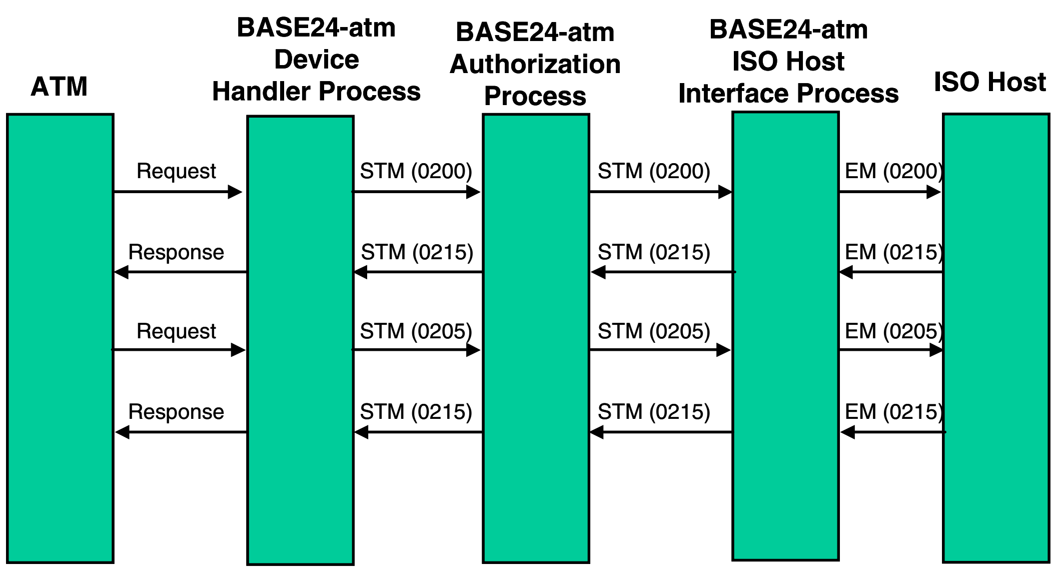
ISO Statement Print Messages

The BASE24 external message is based on the standard external message developed by the International Organization for Standardization (ISO) and is used to route messages between a host and the BASE24-atm system. The BASE24-atm Standard Internal Message (STM) is used to route messages between BASE24-atm processes. The BASE24-atm ISO Host Interface process provides the message interface between the BASE24-atm system and the ISO host.

The data elements used to carry data compression information are described in this appendix.

For detailed information on the BASE24 external message (identified in the following diagram by the code EM), refer to the ***BASE24 External Message Manual***. For information on the BASE24-atm Standard Internal Message (STM) and the BASE24-atm ISO Host Interface process, refer to the ***BASE24-atm Transaction Processing Manual*** and the ***BASE24 ISO Host Interface Manual***.

The following diagram illustrates the message flow when the BASE24-atm system is sending a statement print transaction to an ISO host.



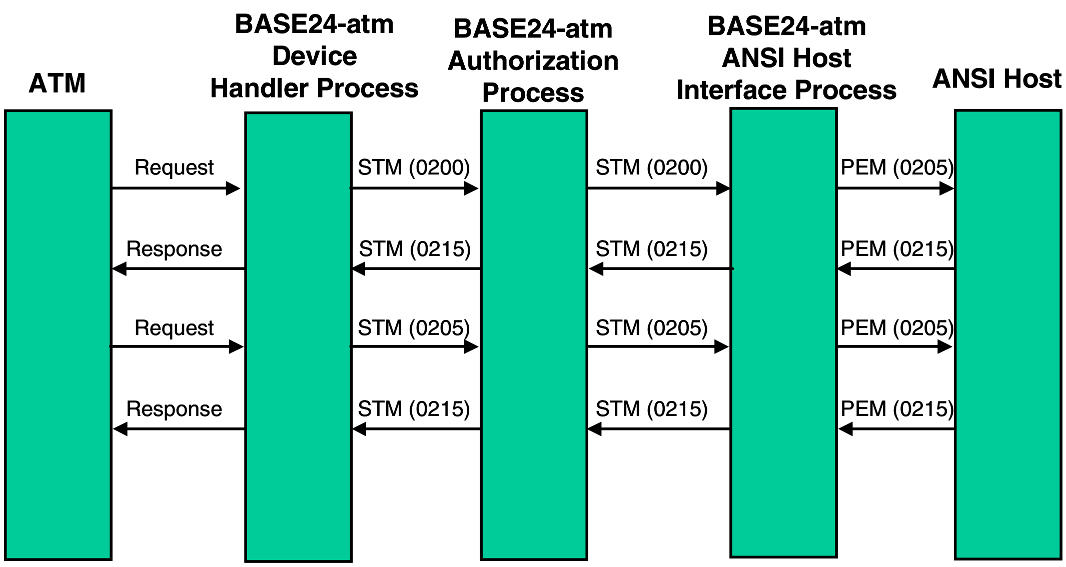
ANSI Statement Print Messages

The Print External Message (PEM) is used for exchanging messages with hosts that have yet to upgrade their communications with the BASE24-atm system to the ISO-based external message. The BASE24-atm Standard Internal Message (STM) is used to route messages between BASE24-atm processes. The BASE24-atm ANSI Host Interface process provides the message interface between the BASE24-atm system and the host.

The PEM format is used to route statement print data messages (0215 and 0205) between a host and the BASE24-atm system. For detailed information on the BASE24-atm ANSI Standard External Message and the BASE24-atm ANSI Host Interface process, refer to the ***BASE24 ANSI Host Interface Manual.***

A summary of the PEM is included in this appendix.

The following diagram illustrates the message flow when the BASE24-atm system sends a statement print transaction to an ANSI host.



ISO Statement Print Data Elements

Statement print data is carried in two data elements in the BASE24 external message. Data element S-126 (BASE24-atm Additional Data) carries the request information needed for the statement print transaction request messages (0200 and 0205) from the BASE24-atm system to the host. Data element S-125 (BASE24-atm Account Indicator/Statement Print Data) carries the response data needed from the host to the BASE24-atm system.

Data Element S-126

The BASE24-atm system uses data element S-126 (BASE24-atm Additional Data) to carry BASE24 message tokens. The Statement Print token (token ID 02) must be added to data element S-126 to include the statement print information in the external message to the host. For outgoing messages, the tokens included in the data element are specified in the Token File (TKN). For information on configuring tokens to be included in outgoing external messages, refer to the ***BASE24 Tokens Manual***. For information on the structure of data element S-126, refer to the ***BASE24 External Message Manual***.

Data Element S-125

Data element S-125 (BASE24-atm Account Indicator/Statement Print Data) carries the statement print transaction response (0215) data from the host to the BASE24-atm system. Data element S-126 must be echoed back with the statement print transaction response data from the host. Data element S-125 is returned to the host by the BASE24-atm system in subsequent requests (0205) if the statement print transaction requires more than one request to the host. For information on the structure of Data element S-125, refer to the ***BASE24 External Message Manual***.

Data Compression Fields

Data element S-126 contains the FRMT (format) field and the PRNT-SIZ (printer size) field. The FRMT field indicates to the host whether the BASE24-atm Device Handler process supports the statement print data compression feature. Information for the FRMT field is taken from the BASE24-atm Terminal Data files (ATD). This field is displayed on ATD Diebold 10XX/478X screen 10. The PRNT-SIZ field is not used with the Diebold 10XX/478X Device Handler process.

The FRMT field is loaded into the BASE24 external message from the BASE24-atm Standard Internal Message (STM).

BASE24-atm ANSI Print External Message (PEM)

The BASE24-atm ANSI Print External Message (PEM) is used to route the statement print data messages (0205 and 0215) between the host and the BASE24-atm system.

Data Compression Fields

Two fields in the PEM support data compression. These fields are FRMT (format) and PRNT-SIZ (printer size). The FRMT field indicates to the host whether the BASE24-atm Device Handler process supports the statement print data compression feature. The FRMT field is displayed on ATD Diebold 10XX/478X screen 10. The PRNT-SIZ field is not used by the Diebold 10XX/478X Device Handler process.

These fields replace the first 4 bytes of the USER-FIELD in the STATEMENTS structure of the PEM.

BASE24-atm PEM Field Summary

The PEM supporting the statement print data compression feature is defined as follows. The fields related to data compression are highlighted.

Position	Level	Field Name	Data Type
	02	ROUTE	
1-2	04	STAT	PIC 9(2)
	04	PATH	
3-4	05	PRODUCT-IND	PIC X(2)
5-6	05	RELEASE-NUM	PIC X(2)
7-10	05	DPC-NUMBER	PIC 9(4)
11-14	05	FILLER	PIC X(4)
15	05	ORIGI-TYPE	PIC X(1)
16-24	05	FILLER	PIC X(9)
25-28	02	TYP	PIC 9(4)
29-44	02	BYTE-MAP	PIC X(16)
	02	CARD-ACCT	
45-46	03	LEN	PIC 9(2)
47-65	03	NUM	PIC X(19)
	02	TRAN-CDE	
66-67	03	TYP	PIC X(2)
68-69	03	FROM-ACCT-TYP	PIC X(2)
70-71	03	TO-ACCT-TYP	PIC X(2)
72-77	02	TRACE-NO	PIC 9(6)
78-81	02	TRAN-DAT	PIC X(4)
82-85	02	POST-DAT	PIC X(4)
86-89	02	TRANSMISSION-DAT	PIC X(4)
	02	TRAN-TIM	
90-91	04	HH	PIC 9(2)
92-93	04	MINUTE	PIC 9(2)
94-95	04	SS	PIC 9(2)
96-105	02	RTTN	PIC 9(10)
106-115	02	ISSUER-ROUTING-NUMBER	PIC 9(10)
116-123	02	RESP-CDE-R	PIC X(8)
	02	RESP-CDE REDEFINES RESP-CDE-R	
116-120	04	FILLER	PIC X(5)
121-123	04	RESP	PIC X(3)
124-138	02	TERM-CITY-ST	PIC X(15)
139-154	02	PIN	PIC X(16)
	02	FROM-ACCT	
155-156	04	LTH	PIC X(2)
157-175	04	NUM	PIC X(19)

Position	Level and Field Name		Data Type
	02	TERM	
176-179	04	OWNER	PIC X(4)
180-204	04	TERM-NAME-LOC	PIC X(25)
205-229	04	TERM-OWNER-NAME	PIC X(25)
230-245	04	ATM-ID	PIC X(16)
246-249	04	BRANCH-ID	PIC X(4)
250-253	04	REGION-ID	PIC X(4)
254-259	04	REC-NUM	PIC 9(6)
260-261	04	TYP	PIC 9(2)
262-265	04	TIM-OFST	
			STRUCTURE TYPE 15
	02	CARD	
266-267	04	FIID-LEN	PIC 9(2)
268-271	04	CARD-FIID	PIC X(4)
272-287	04	PIN-OFST	PIC X(16)
	02	MISC	
288-291	04	ICHG-SETT-DAT	PIC X(4)
	02	TRACK2	
292-293	04	LEN	PIC 9(2)
294-330	04	T2DATA	PIC X(37)
	02	STATEMENTS	
331-332	04	PAGE-INDICATOR	PIC X(2)
	04	LAST-STATEMENT-DATE	
333-334	06	YY	PIC X(2)
335-336	06	MM	PIC X(2)
337-338	06	DD	PIC X(2)
339-340	04	HEADER-LINES	PIC X(2)
341-342	04	COLUMN-LINES	PIC X(2)
343-702	04	STMT-FIELD	PIC X(360)
703	**04**	**FRMT**	**PIC X(1)**
704-706	**04**	**PRNT-SIZ**	**PIC X(3)**
707-714	04	USER-FIELD	PIC X(8)

Configuration Requirements

The following pages provide configuration requirements for the statement print data compression feature.

The information assumes knowledge of the HP NonStop edit function and the BASE24 user access system, as well as BASE24-atm and BASE24 files maintenance. For more information about screen access and files maintenance, refer to the ***BASE24 User Security Manual***, the ***BASE24-atm Files Maintenance Manual***, and the ***BASE24 Base Files Maintenance Manual***.

Information is provided for the following:

- Setting up the BASE24-atm Terminal Data files
- Setting up the External Message File
- Editing the C100NAMS File for data compression support
- Configuring token processing

Setting Up the BASE24-atm Terminal Data Files

The BASE24-atm Terminal Data files (ATD) contain one record for each terminal in the logical network. These records define the characteristics and functionality of the terminal.

The ATD must be updated for the terminals that support the BASE24-atm statement print data compression feature. Changes to the ATD are necessary to specify if the terminal supports data compression and if the host uses the ignore text option.

The following fields on ATD Diebold 10XX/478X screen 10 are used for statement print data compression: FRMT, IGNORE TEXT, and MAXIMUM STMT LINES/RCPT. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about ATD Diebold 10XX/478X screen 10 and complete descriptions of these fields.

Setting Up the External Message File (EMF)

External Message File (EMF) records specify which data elements are included in the BASE24 External Message for incoming and outgoing messages. This file is used with the ISO-based external message format.

The following EMF settings are required for data elements S-125 and S-126 to support the statement print data compression feature. These are the default settings for 0200, 0205, and 0215 messages.

For PRODUCT #	For MSG TYPE	Data Element S-126 Set To	Data Element S-125 Set To
01 (ATM)	0200	C	C (Outgoing)
01 (ATM)	0205	C	C
01 (ATM)	0215	C	M

If anything other than these settings is used for data elements S-125 or S-126, the settings must be updated to support the statement print data compression feature.

For more detailed information on updating the EMF, refer to the ***BASE24 Base Files Maintenance Manual.***

Editing the C100NAMS File

The statement print transaction description table in the C100NAMS File must be configured to support the statement print data compression feature. This table allows you to define receipt text for statement print transactions. The BASE24-atm Device Handler process uses this table to convert compressed text data into the actual receipt text displayed at the ATM.

The statement print transaction description table is described previously in this appendix.

To set up the ASCII hexadecimal index and its values in the statement print transaction description table, edit the C100NAMS File. When setting up the statement print transaction description table, it is important to list the entries for each index code by frequency of use. This enables the BASE24-atm Device Handler process to scan the index codes in the table more quickly.

Note: Any changes made to the statement print transaction description table must be communicated to the host programming personnel so that the same information is available to both the BASE24-atm system and to the host.

After the statement print transaction description table has been built or modified, the Diebold 10XX/478X Device Handler process must be recompiled.

Configuring Token Processing

The Token File (TKN) is used to configure token processing. It specifies the following information:

- The tokens that get logged to the BASE24-atm Transaction Log File (TLF) and the Interchange Log Files (ILFs).
- The tokens that get extracted from the TLF by the Super Extract process. Extracting allows the token information to be processed by the host.
- The tokens that are sent to the host with each transaction in the ISO-based external message.

The Statement Print token contains the FRMT field that indicates whether the statement print data compression feature is supported. The Token File (TKN) must be modified to ensure that this token gets sent from the BASE24 system to the ISO Host Interface process or the BASE24 Interchange (BIC ISO) Interface process in the external message and to ensure that this token gets logged to the appropriate transaction log files.

Refer to the ***BASE24 Tokens Manual*** for complete information on the Statement Print token (token ID 02) and on using the TKN to specify tokens to be sent in the external message. Refer to the ***BASE24 Base Files Maintenance Manual*** for detailed field descriptions for the TKN and the HCF.

B: Dispensing Bills and Noncurrency Items

This appendix describes the algorithms the BASE24-atm Diebold 10XX/478X Device Handler process uses to dispense bills and noncurrency items. The methods include the least bills and configurable dispense algorithm methods. A third method, the thresholds method, is used with the BASE24-atm self-service banking (SSB) add-on product, and is described in detail in the ***BASE24-atm Diebold 10XX/478X SSB Manual.***

This appendix also describes the Non–Currency Dispense add-on product.

Bill Dispensing Methods

You can configure the BASE24-atm Diebold 10XX/478X Device Handler process to dispense notes in three different ways:

- Least bills method
- Thresholds method
- Configurable dispense algorithm method

The thresholds method is used with the BASE24-atm self-service banking (SSB) add-on product. Refer to the ***BASE24-atm Diebold 10XX/478X SSB Manual*** for more information about the thresholds method of configurable bill dispensing.

The DISPENSE ALGORITHM field on BASE24-atm Terminal Data files (ATD) Diebold 10XX/478X screen 7 enables you to select the bill dispensing method for your terminal. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about this field.

Least Bills Method

The least bills method is used with the standard BASE24-atm product. Using the least bills method, the Diebold Device Handler process calculates the hopper bill counts from high to low bill denomination for bill dispensing. The Device Handler process dispenses as many bills as it can from the highest denomination. When the amount left to be dispensed is lower than the highest denomination, the Device Handler process calculates the number of bills to be dispensed from the hopper with the next highest denomination. For example, a transaction for $375 (U.S. dollars) at a terminal with hoppers containing $5, $20, $50, and $100 is dispensed as follows:

- Three $100 bills
- One $50 bill
- One $20 bill
- One $5 bill

To configure the least bills method, set the DISPENSE ALGORITHM field on ATD Diebold10XX/478X screen 7 to a value of 0. The least bills method is the default bill dispensing method of the BASE24-atm product. If the ATM cannot dispense the requested amount using one of the other methods, the least bills method is used. The Non–Currency Dispense and Multiple Currency Dispense add-on products can use the least bills method to dispense noncurrency items.

Configurable Dispense Algorithm Method

The configurable dispense algorithm method gives an institution flexibility in its hopper configuration. The algorithm uses start and stop amounts and the maximum number of bills allowed to calculate the correct number of bills to dispense. This method is used when the DISPENSE ALGORITHM field on ATD Diebold 10XX/478X screen 7 is set to a value of 2.

The *start dispense amount* is the minimum amount needed before the ATM starts dispensing bills from the hopper. For example, if a terminal contains $5 (U.S. dollars) bills in hoppers 1 and 2 and $50 bills in hoppers 3 and 4, hopper 4 could be configured with a start amount of $200. This means the device does not dispense bills out of hopper 4 unless the amount to be dispensed is $200 or more. For amounts less than $200, the device could dispense $50 bills from hopper 3.

The *stop dispense amount* is the amount at which the ATM stops dispensing from the hopper. When the amount to be dispensed is less than or equal to this amount, the ATM proceeds to the next hopper. Using the hopper values from the previous example, you could set a $100 stop dispense amount on hopper 4. This means that when the amount to be dispensed is $100 or less, the ATM would stop dispensing bills from hopper 4 and start dispensing bills from hopper 3. For a $200 transaction, the ATM dispenses $100 from hopper 4 and the remaining $100 from the other hoppers.

The maximum number of bills is the number of bills at which the ATM stops dispensing bills from a hopper.

If you have purchased the BASE24-atm Non–Currency Dispense add-on product and one or more hoppers contains nonvalue items, you can use the configurable dispense algorithm *nonvalue dispense amount* to specify, by nonvalue hopper, the amount at which the ATM dispenses a nonvalue item. For example, you could configure one hopper to dispense a coupon for transactions of $50 or more and a second hopper to dispense an advertisement for transactions of $20 or more. Refer to the topic “Non–Currency Dispense Add-on Product” later in this appendix for more information about dispensing cash value and nonvalue items.

If the algorithm fails (that is, the limits set by the start dispense amount, stop dispense amount and maximum number of bills prevent the ATM from dispensing the requested amount), the Device Handler process defaults to the least bills method.

Configuring the BASE24-atm Terminal Data Files (ATD)

Amounts for the configurable dispense algorithm are defined using the BASE24-atm Terminal Data files (ATD).

The DISPENSE ALGORITHM field on ATD Diebold 10XX/478X screen 7 must be set to a value of 2 (start and stop amounts) for the configurable dispense algorithm.

ATD Diebold 10XX/478X screen 13 is a device-specific screen. You can use this screen to define the configurable dispense algorithm.

The START DISPENSE AMT field contains the start dispense amount. Entering a value in this field is optional.

The STOP DISPENSE AMT field contains the stop dispense amount. Entering a value in this field is optional.

The MAX BILLS field indicates the maximum number of bills per hopper. Note that the maximum number of bills dispensed for a transaction cannot exceed 50.

If you have purchased the Non–Currency Dispense add-on product and you plan to use the configurable dispense algorithm, you can use the NON-VALUE DISPENSE field to specify the amount at which a nonvalue item is dispensed. This field is used only when the Non–Currency Dispense add-on product and the configurable dispense algorithm are used in combination. Refer to the topic “Non–Currency Dispense Add-On Product” later in this section for more information about dispensing noncurrency items.

For more information about the ATD, refer to the ***BASE24-atm Files Maintenance Manual.***

Device Handler Processing

The Device Handler process uses the bill value, start dispense amounts, stop dispense amounts, and maximum number of bills to compute the bill mix for a withdrawal transaction. The Device Handler process starts with hopper 4 and works its way down. If the ATM cannot dispense bills from the hopper (that is, the checks described below fail), the Device Handler process resumes calculations with the next hopper. If the Device Handler process runs out of hoppers before the entire transaction amount is dispensed, the process defaults to the least bills method.

The Device Handler process performs the following steps starting with hopper 4:

1. Verifies that the hopper is operational, the ATD hopper contents value is equal to the local contents value, the bill value is greater than 0, and the amount to be dispensed is greater than or equal to the hopper's start dispense amount. If the conditions are not met, the Device Handler process proceeds to the next hopper and performs the same checks until the conditions are met.
    - Once the conditions are met, processing continues with step 2.
    - If the conditions are not met with the final hopper, the Device Handler process defaults to the least bills method.
2. Compares the amount to be dispensed to the stop dispense amount for the hopper.
    - If the amount to be dispensed is less than or equal to the stop dispense amount, the Device Handler process proceeds to the next hopper and starts again with step 1.
    - If the amount to be dispensed is greater than the stop amount, processing continues with step 3.
3. Compares the amount to be dispensed to the bill value.
    - If the amount to be dispensed is less than the bill value, the Device Handler process proceeds to the next hopper and starts again with step 1.
    - If the amount to be dispensed is greater than or equal to the bill value, processing continues with the step 4.
4. Compares the number of bills dispensed from the hopper to the hopper's maximum number of bills allowed.
    - If the number of bills dispensed is greater than or equal to the hopper's maximum, the Device Handler process proceeds to the next hopper and starts again with step 1.

- If the number of bills dispensed is less than the hopper's maximum number of bills, processing continues with step 5.

5. Compares the number of bills to be dispensed to the maximum number of bills allowed for the device.
   - If the number of bills to be dispensed is greater than or equal to the device's maximum, the Device Handler process proceeds to the next hopper and starts again with step 1.
   - If the number of bills to be dispensed is less than the device's maximum, the Device Handler process continues with step 6
6. The Device Handler process calculates that the hopper can dispense a bill and subtracts the bill value from the transaction amount, giving the new amount to be dispensed. The Device Handler process also updates the bill count for the hopper and the device.
7. Repeats steps 2, 3, and 4 until it reaches the full transaction amount, the stop amount, or maximum number of bills is reached.
   - If the stop amount or maximum bills is reached, the Device Handler process resumes calculations with the next highest hopper, starting with step 1.
   - If there are no more hoppers available, the Device Handler process defaults to the least bills method.

Currency Dispense Algorithm Examples

The following is an example of how the Device Handler process calculates the bill mix using the Configurable Dispense algorithm. The table below shows a sample ATM hopper configuration in U.S. currency.

	Hopper 1	Hopper 2	Hopper 3	Hopper 4
Bill Value	$5	$20	$20	$100
Start dispense amount	0	20	100	300
Stop dispense amount	1	19	89	101
Max bills	6	6	4	6

Using this configuration and the configurable dispense algorithm, the Device Handler process would calculate the number of bills as shown in the table below. Descriptions of the transactions follow the table.

Transaction Amount	Hopper 1	Hopper 2	Hopper 3	Hopper 4
$15	3 bills	0 bills	0 bills	0 bills
$20	0 bills	1 bill	0 bills	0 bills
$50	2 bills	2 bills	0 bills	0 bills
$100	0 bills	4 bills	1 bill	0 bills
$150	2 bills	3 bills	4 bills	0 bills
$250*	2 bills	0 bills	2 bills	2 bills
$300	0 bills	4 bills	1 bill	2 bills
$425	1 bill	1 bill	0 bills	4 bills
$500	0 bills	4 bills	1 bills	4 bills
$750	2 bills	3 bills	4 bills	6 bills

* The configurable dispense algorithm fails, and the Device Handler process defaults to the least bills method. Refer to the transaction description below for more processing information

$15 Transaction. The transaction amount is less than the start dispense amount for hoppers 4, 3, and 2. The Device Handler process calculates 3 bills from hopper 1.

$20 Transaction. The transaction amount is less than the start dispense amount hoppers 4 and 3. The transaction amount is equal to the start dispense amount, greater than the stop dispense amount, and equal to the bill value for hopper 2, so one bill can be dispensed from the hopper.

$50 Transaction. The transaction amount is less than the start dispense amount for hoppers 4 and 3, so the Device Handler process begins calculating the bill mix with hopper 2. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for the hopper, so bills can be dispensed from the hopper. After hopper 2 dispenses $40 (two bills), the amount remaining to be dispensed ($10) is less than the stop dispense amount, so the Device Handler begins calculating the amount to be dispensed from the next hopper. The remaining $10 is dispensed as two bills from hopper 1.

$100 Transaction. The transaction amount is less than the start dispense amount for hopper 4, so the Device Handler process begins calculating the bill mix with hopper 3. The transaction is equal to the start dispense amount and greater than the stop dispense amount and bill value for that hopper. The Device Handler process calculates that $20 (one bill) can be dispensed from hopper 3. After that the amount to be dispensed ($80) is less than the $89 stop dispense amount. The Device Handler process begins calculating the bill mix with hopper 2. Four bills ($80) can be dispensed from hopper 2.

$150 Transaction. The transaction amount is less than the start dispense amount for hopper 4, so the Device Handler process begins calculating the bill mix using hopper 3. The transaction amount is greater than the start dispense amount, stop dispense amount and bill value, and the Device Handler process begins calculating the bill mix from that hopper. The first $80 (four bills) can be dispensed from hopper 3. After that, the amount remaining to be dispensed ($70) is less that the $89 stop dispense amount for that hopper. The Device Handler process begins calculating using hopper 2. The next $60 (three bills) can be dispensed from hopper 2. After that, the amount remaining to be dispensed ($10) is less than the stop dispense amount for that hopper. The Device Handler process begins calculating using hopper 1. The remaining amount can be dispensed with 2 bills from hopper 1.

$250 Transaction. The transaction amount is less than the start dispense amount for hopper 4, so the Device Handler process begins calculating the bill mix with hopper 3. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 3, so the Device Handler process begins calculating the bill mix from that hopper. The Device Handler process calculates the first $80 can be dispensed from hopper 3. Although the remaining amount ($170) is greater than the stop dispense amount, the maximum bills limit has been reached. The Device Handler process resumes calculations with hopper 2. The next $120 (six bills) can be dispensed from hopper 2. The maximum number of bills is reached, so the Device Handler process must resume calculations with hopper 1. The $50 dollars remaining to be dispensed cannot be dispensed from hopper 1 because there is a six bill maximum. The algorithm fails, and the Device Handler defaults to the least bills method.

$300 Transaction. The transaction amount is equal to the start dispense amount and greater than the stop dispense amount and bill value for hopper 4, so the Device Handler process begins calculating the bill mix. The first $200 can be dispensed from hopper 4 (two bills). After this, the amount remaining to be dispensed ($100) is less than the $101 stop dispense amount. The Device Handler process resumes calculations with hopper 3. The amount remaining to be dispensed is equal to the start dispense amount and greater than the stop dispense amount and the bill value for that hopper. Hopper 3 can dispense $20 (1 bill), after this the amount to be dispensed ($80) is less than the $89 stop dispense amount. The Device Handler process resumes calculations with hopper 2. Hopper 2 can dispense the remaining $80 (4 bills).

$425 Transaction. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 4, so the Device Handler process begins calculating the bill mix with hopper 4. The first $400 dollars can be dispensed from hopper 4. After this, the amount remaining to be dispensed ($25) is less than the $101 stop dispense amount. The Device Handler process resumes calculations with hopper 3. The amount to be dispensed is less than the start dispense amount ($100) for hopper 3, so no bills can be dispensed from hopper 3. The Device Handler process resumes calculations with hopper 2. The amount to be dispensed is greater than the start dispense amount, stop dispense amount, and bill value for hopper 2. One bill ($20) can be dispensed from hopper 2. The amount remaining ($5) is less than the stop dispense amount of hopper 2, so the Device Handler process resumes calculations with hopper 1. The remaining $5 can be dispensed with one bill from hopper 1.

$500 Transaction. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 4, so the Device Handler process begins calculating the bill mix. The first $400 dollars (four bills) can be dispensed from hopper 4. After this, the amount remaining to be dispensed ($100) is less than the $101 stop dispense amount. The Device Handler process resumes calculations with hopper 3. $20 (1 bill) can be dispensed from hopper 3. After this, the amount remaining to be dispensed ($80) is less than the $89 stop dispense amount for hopper 3. The Device Handler process resumes calculations with hopper 2. Hopper 2 can dispense the remaining $80 (4 bills).

$750 Transaction. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 4, so the Device Handler process begins calculating the bill mix. The first $600 (six bills) can be dispensed from hopper 4. This is the maximum that can be dispensed from the hopper. The Device Handler process resumes calculations with hopper 3. $80 dollars (4 bills) can be dispensed from hopper 3. After this, the amount remaining to be dispensed ($70) is less than the $89 stop dispense amount. The Device Handler process resumes calculations with hopper 2. $60 can be dispensed from hopper 2. After this, the amount remaining to be dispensed ($10) is less than the stop dispense amount for the hopper. The Device Handler process resumes calculations with hopper 1. The remaining $10 can be dispensed from hopper 1 with two bills.

Non–Currency Dispense Add-On Product

The BASE24-atm Non–Currency Dispense add-on product enables an institution to dispense noncurrency items through its ATMs. Noncurrency items can include cash value items, such as travelers checks, telephone cards, postage stamps, and transportation tickets, or nonvalue items, such as coupons or advertisements.

An institution can configure hoppers for up to four noncurrency items at an ATM. This is the maximum number of note hoppers for a Diebold 10XX/478X ATM.

The Diebold 10XX/478X Device Handler process can support the Non–Currency Dispense add-on product when operating in 911 or 912 mode.

Processing Non–Currency Dispense Transactions

The BASE24-atm system treats a Non–Currency Dispense transaction as it does a cash withdrawal or cash advance transaction. When the Diebold 10XX/478X Device Handler process receives a Non–Currency Dispense transaction, it adds the Non–Currency Dispense token to the end of the Standard Internal Message (STM). The token carries the information necessary for the transaction to be processed by the BASE24-atm system.

The Diebold 10XX/478X Device Handler process performs the following functions on a Non–Currency Dispense transaction:

- Adds the Non–Currency Dispense token to the STM.
- Computes the total cost of the purchase, and places the value in the STM. This value is later displayed to the cardholder on the ATM screen. This is done for quantity-based transactions (for example, purchasing postage stamps) but not amount-based transactions (that is, traveler's checks).
- Checks thresholds configured for dispensing nonvalue items.

The Device Handler process searches the internal operations code table when it receives a request message. If the operations code indicates that the cardholder has requested a noncurrency item, the Device Handler process moves the value in the NCD ROUTING GROUP field on ATD screen 1 to the RQST.RTE-GRP field of the STM instead of the value in the ALT ROUTING GROUP field on ATD screen 1.

Refer to the ***BASE24-atm Transaction Processing Manual*** for more information about processing cash withdrawal and cash advance transactions.

Non–Currency Dispense Configuration

The Non–Currency Dispense add-on product can be used with the BASE24-atm Diebold 10XX/478X Device Handler process. The following Diebold and BASE24 files contain fields needed for the Non–Currency Dispense add-on product:

- C100NAMS
- Acquirer Processing Code File (APCF)
- BASE24-atm Terminal Data files (ATD)
- Card Prefix File (CPF)
- Cardholder Authorization File (CAF)
- External Message File (EMF)
- Institution Definition File (IDF)
- Logical Network Configuration File (LCONF)
- Receipt File (RCPT)
- Surcharge File (SURF)
- Terminal Receipt File (TRF)
- Usage Accumulation File (UAF)

Modifying the C100NAMS

The C100NAMS is an edit file that contains a series of tables used by the BASE24-atm Diebold 10XX/478X Device Handler process. You can edit these tables to customize specific information displayed on ATM receipts.

The valid denomination table in the C100NAMS (diebold_1000_curr_tbl) supports the Non–Currency Dispense enhancement. The Device Handler process uses information in this table to map noncurrency items to denominations.

Modifying the Acquirer Processing Code File

The Acquirer Processing Code File (APCF) contains one record for each combination of acquirer transaction profile, message category, and ISO processing code supported by the BASE24-atm system. Each acquirer transaction profile defines a set of transactions supported by a terminal or group of terminals. Each

APCF record contains a code indicating if and how the transaction is allowed at the terminal. APCF screen 2 contains fields that enable you to specify the types of Non–Currency Dispense transactions that can be performed at your ATMs. For more information about the APCF screens, refer to the ***BASE24 Base Files Maintenance Manual.***

Modifying the BASE24-atm Terminal Data Files

The BASE24-atm Terminal Data files (ATD) contain configuration information for each individual terminal in the logical network. The fields described below have specific Non–Currency Dispense requirements. These ATD fields are described in detail in the ***BASE24-atm Files Maintenance Manual.***

ATD Screen 1

ATD screen 1 contains information about the Device Handler process and Authorization process associated with the terminal and sharing information. The NCD ROUTING GROUP field specifies the routing group number used to route Non–Currency Dispense transactions initiated by cardholders who are outside the logical network.

ATD Screen 2

You can use ATD screen 2 to specify the acquirer transaction profile for a Non–Currency Dispense transaction available to not-on-us cardholders at a particular terminal.

ATD Screens 4 and 5

ATD screens 4 and 5 enable you to define hopper contents. You can define up to four Non–Currency Dispense hoppers on a Diebold 10XX/478X ATM. (This is the maximum number of note hoppers for these ATMs.)

Hoppers containing cash are configured first, followed by noncurrency hoppers. The CONTENTS field for noncurrency hoppers is set to a value of 02 through 10. Values 02 through 10 are defined in the COBNAMES file, with the value 02 reserved for travelers checks. Within a contents type, the value in the BILL VALUE field is entered from lowest to highest. For example, if your ATMs carry

$20 travelers checks and $50 travelers checks, the hopper with the $20 travelers checks would be configured first. When configuring the BILL VALUE field for nonvalue items, use the value 0.

ATD Diebold 10XX/478X Screen 7

ATD Diebold 10XX/478X screen 7 is a device-specific screen. The screen includes the NON-VALUE ACCTS fields used to indicate the type or types of accounts for which a nonvalue item is dispensed. A value of Y in the CK, SV, or CC field indicates that a nonvalue item can be dispensed for a checking account transaction, savings account transaction, or credit account transaction, respectively.

Note that this screen also includes the DISPENSE ALGORITHM field. You can use any algorithm with the Non–Currency Dispense add-on product.

ATD Diebold 10XX/478X Screen 8

ATD Diebold 10XX/478X screen 8 is a device-specific screen. You can use this screen to specify the terminal receipt profile for Non–Currency Dispense transactions.

ATD Diebold 10XX/478X Screen 13

ATD Diebold 10XX/478X screen 13 is a device-specific screen. You can use this screen to define nonvalue item dispense amounts.

The NON-VALUE DISPENSE field contains the amount at which nonvalue items are dispensed when using the configurable dispense algorithm. Using this field and the configurable dispense algorithm enables the institution to set a nonvalue dispense amount for each hopper that contains a nonvalue item.

Setting Card Prefix File and Cardholder Authorization File Usage Limits

Non–Currency Dispense transactions are processed in the same manner as cash withdrawal transactions, with the Authorization process checking limits in the Card Prefix File (CPF), Cardholder Authorization File (CAF), and the Usage

Accumulation File (UAF). Your institution can include Non–Currency Dispense transactions in the limits set for ATM transactions, or it can set limits specifically for Non–Currency Dispense transactions.

You can set Non–Currency Dispense limits in the CPF and CAF for general noncurrency items and two specific noncurrency items. Hopper content codes specify the two noncurrency items for which to set limits. The general noncurrency items include any noncurrency item dispensed by an ATM other than those specified by content codes. For example, you could define a set of limits for the amount of general noncurrency items, a set of limits specifically for the amount of travelers checks, and a third set of limits for the amount of event tickets. Within the content categories, you can specify limits for cash withdrawals and credit withdrawals. Within the cash withdrawal category, you can set limits for the total amount of transactions and the total amount of offline transactions.

CPF screen 5 contains Non–Currency Dispense limits set at the card prefix level. CAF screen 9 contains Non–Currency Dispense limits and activity for individual cardholders. Fields on these screens are grouped by hopper contents. During authorization processing, the Authorization process checks the HOPR-CONTENTS field of the Non-Currency Dispense token. If the value in the HOPR-CONTENTS field is equal to one of the content codes, the Authorization process uses limits specific to that content code field. If the value of the HOPR-CONTENTS field is not equal to one of the content codes, the Authorization process uses the set of limits defined for all other noncurrency items. Refer to the ***BASE24 Tokens Manual*** for more information about the Non-Currency Dispense token.

Fields are arranged in a similar manner on UAF screen 5. This screen contains system-protected fields showing activity for the usage period.

For more information about the CPF, CAF, and UAF, refer to the ***BASE24 Base Files Maintenance Manual.***

Modifying the External Message File

When configuring the External Message File (EMF) you can change the default value of data element S-116 (BASE24-from host maintenance CAF Non-Currency Dispense) from a value of C (conditional) to a value of M (mandatory) if your institution requires that this element be present in messages to and from the From Host Maintenance process. If the element is not included in the message, you can set the value to a blank.

Modifying the Institution Definition File

The ACQUIRER TRANSACTION PROFILE field on Institution Definition File (IDF) screen 9 enables you to specify an acquirer transaction profile for Non–Currency Dispense transactions. The processing codes for the profile are defined in the Acquirer Processing Code File (APCF). For more information about IDF screen 9, refer to the ***BASE24 Base Files Maintenance Manual.***

Modifying the Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains one param for the Non–Currency Dispense add-on product.

The ATM-DH-NON-VALUE-THRESHOLD param should be present at initialization if an ATM is configured to dispense nonvalue items. This param sets a minimum whole dollar withdrawal amount for the entire network for determining whether a nonvalue item should be dispensed along with the requested cash or noncurrency item. This param is used when the DISPENSE ALGORITHM field on ATD Diebold 10XX/478X screen 7 is set to a value of 0 or 1. If the DISPENSE ALGORITHM field is set to a value of 2, this param is ignored.

Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information about the ATM-DH-NON-VALUE-THRESHOLD param.

Modifying the Receipt File

You can add receipt templates to the Receipt File (RCPT) for noncurrency dispense transactions. Receipt condition codes NCD and NONCD that are used with the IF and ENDIF directives allow noncurrency data to be displayed on withdrawal transaction receipts. The field type code NC displays the noncurrency description text defined in the C100NAMS.

See section 5 for more information about condition values and field type codes.

The receipt type WD (withdrawal) is used for a Non–Currency Dispense transaction. A sample receipt template as defined in the RCPT is shown below.

```
@RECORD NO.   #SQ>
?IF NONCD
@WITHDRAWAL#A1$>
?ENDIF
?IF NCD
```

```
@PURCHASE#A1$>
#NC>>
?ENDIF
@FR #FR>>
#XL#LB$>
#XA#AB$>>
```

The above template creates the following receipt for a cash withdrawal transaction:

```
 1             A C I   D E M O
 2
 3        DATE      TIME  TERM
 4        MM/DD/YY  HH:MM S1AN5070
 5
 6         123456700000003-003
 7
 8        330 SOUTH 108TH AVE
 9
10        RECORD NO.             570
11        WITHDRAWAL          $20.00
12        FR     013333
13
14        LEDGER BAL        $3412.00
15        AVAIL BAL         $1760.00

                   1         2
          123456789012345678901234567
```

The same template creates the following receipt for a Non–Currency Dispense purchase transaction (in this case, traveler's checks).

```
1             A C I   D E M O
2
3        DATE      TIME  TERM
4        MM/DD/YY  HH:MM S1AN5070
5
6         123456700000003-003
7
8        330 SOUTH 108TH AVE
9
10       RECORD NO.             577
11       PURCHASE            $50.00
12       TRAVELERS CHECKS
13
14       FR     013333
15
15       LEDGER BAL        $3362.00
16       AVAIL BAL         $1710.00

                  1         2
         123456789012345678901234567
```

Modifying the Surcharge File

The Surcharge File (SURF) contains information regarding surcharging. If you want to assess a surcharge for Non–Currency Dispense transactions, set the NON-CURRENCY DISPENSE field on SURF screen 3 to a value of 1. This enables the Authorization process to assess the surcharge as configured in the SURF. If you do not want to assess a surcharge for Non–Currency Dispense transactions, set the value in this field to 0.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about configuring the SURF.

Modifying the Terminal Receipt File

Terminal Receipt File (TRF) screen 2 contains fields that enable you to specify the receipt options for approved and denied Non–Currency Dispense transactions at your ATMs. Each terminal receipt profile defines a set of receipt options supported by a terminal or a group of terminals. Each TRF record contains a code indicating if and how the transaction is allowed at the terminal. For more information about the TRF screens, refer to the ***BASE24-atm Files Maintenance Manual.***

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C: Diebold 10XX/478X Dial-Up Device Handler Extension

With the saturation of ATMs, the financial industry demands cost-effective methods of distributing ATMs to new locations. Because a dial-up ATM runs on a standard business line, it is a less expensive alternative to a leased line, especially in areas that do not see much consumer traffic.

An institution can support the use of dial-up ATMs using the BASE24-atm Diebold 10XX/478X Dial-up Device Handler extension.

The Diebold 10XX/478X Dial-Up Device Handler extension can be used with transaction surcharging and the BASE24-atm Non–Currency Dispense add-on product.

This appendix describes the Diebold 10XX/478X Dial-Up Device Handler extension. It includes the following topics:

- Supported protocols
- Dial-up message header
- Device Handler processing
- Files maintenance considerations
- Network control commands

Supported Protocols

Dial-up ATMs are physically connected to the BASE24 system using standard communication protocols (refer to section 1 for more information on the protocols supported by the Diebold 10XX/478X Device Handler process) in conjunction with the Visa II protocol.

If the dial-up device uses the Async Visa II protocol, the TYPE attribute of the corresponding XPNET process DEVICE object must be set to a value of 255. This suppresses the automatic disconnect function for connection complete messages. A value other than 255 retains the automatic disconnect function.

Dial-Up Message Header

The request message from a dial-up ATM is distinguished from other request messages by the dial-up header, which precedes the standard message header. The format of the dial-up header is shown below, followed by descriptions of its fields:

Position	Field Length	Field Name
1–2	2 numeric characters	Device Type
3–9	7 numeric characters	BIN
10–21	12 alphanumeric characters	Terminal ID
22	1 hexadecimal character	Field Separator
23–24	2 alphabetic characters	Message Type

Device Type — A code indicating the Visa II device. This field is always set to a value of 50.

BIN — Communication and routing information. This information is used in routing the message to the BASE24 system. The value of this field is .777777. The Device Handler process does not use this information.

Terminal ID — A value that identifies the Diebold 10XX/478X terminal, entered locally at the ATM. When a message comes into the BASE24 system, the Device Handler process reads the BASE24-atm Terminal Data Static File—general data (ATDS1) record using this terminal ID instead of the BASE24 station name from the XPNET process.

Field Separator — A period (.) representing a field separator character (hexadecimal 1C).

Message Type — A two-character code representing the type of inbound message. Valid values are as follows:

CC = Connect continuation
CP = Communication complete
RQ = Transaction request message

Device Handler Processing

The following pages discuss request and response message processing. They also describe how the Device Handler process handles transaction surcharges.

Request Message Processing

When the Device Handler process receives a message, it checks for the dial-up header. If the message does not start with the dial-up header, the Device Handler process reads the BASE24-atm Terminal Data Static File—general data (ATDS1) using the BASE24 station name from the XPNET process. If the message starts with the dial-up header, the Device Handler process reads the ATDS1 with the terminal ID from the header but saves the BASE24 station name, so that the response can be sent back to the correct terminal.

The ATM can send the following types of messages:

- Request messages (e.g., cardholder transaction requests)
- Communication complete message indicating that a transaction is complete or that a disconnect command was sent
- Connect continuation message used to maintain a connection with the BASE24 system when the current sequence of messages is not complete (e.g., downloading the ATM)

In addition, the ATM can connect when its internal *I'm Alive* timer has expired (indicating that no messages have been sent for a period of time). In the case of an event, such as an indicator from the tamper sensor or a card reader fault, the ATM connects to the BASE24 system and sends an unsolicited status message.

Response Message Processing

When responding to the ATM, the Device Handler process uses the BASE24 station name from the XPNET process to send a message back to the correct ATM.

When the Device Handler is done processing a request and there are no messages to be sent to the ATM, it sends a disconnect command to the ATM. The format of this command is the normal terminal command format with a command code of F. If the ATM has another message to send to the BASE24 system, the message is sent. If the ATM does not have any other messages when it receives the disconnect command, it sends a request message with the Message Type field of the dial-up header set to a value of CP.

Failed Messages

In the case of a message failure, the Device Handler process reads the ATDS1 with the terminal ID from the XPNET process. For message failure type 106 with the message flag turned off (i.e., the MSG FAIL field on ATD Diebold 10XX/478X screen 7 is set to a value of 0), the Device Handler process generates an event message stating that the transaction was not reversed. If the MSG FAIL field on ATD Diebold 10XX/478X screen 7 is set to a value of 1, the Device Handler process reverses all failed messages.

Surcharge Processing

The Diebold 10XX/478X Dial-Up Device Handler extension supports ATM surcharging. An institution using the surcharge application in a dial-up environment can choose, for each ATM, the interactive message sequence of the standard BASE24-atm processing or a noninteractive message sequence.

Interactive Surcharge Processing

Standard surcharge processing uses an interactive message sequence. In this environment, the cardholder selects a withdrawal or deposit transaction, and it is sent to the Authorization process, which calculates the surcharge. A message containing the surcharge amount is presented to the cardholder to allow him or her to agree to the charge or terminate the transaction. When the cardholder agrees to the surcharge, the transaction is sent to the authorization process again for approval.

The Diebold 10XX/478X Dial-Up Device Handler extension can support interactive response notification processing.

Noninteractive Surcharge Processing

Using the surcharging interactive message sequence can result in long dial-up times for the institution. To decrease dial-up time, an institution can choose a noninteractive message sequence. The Diebold 10XX/478X Dial-Up Device

Handler extension enables an institution to define a maximum surcharge amount that can be charged for a transaction at the terminal. This maximum amount is displayed to the cardholder during the initial transaction sequence. The actual amount the cardholder is charged might be less than the maximum amount shown during transaction selection. The maximum surcharge amount is sent to the ATM during any download that contains the surcharge notification screen. Because there is only one communication sequence, the dial-up time is shorter.

When the Device Handler process receives the surcharge notification message (9901) from the Authorization process, it determines whether noninteractive surcharging is in use (that is, the DOWNLOAD SURCHARGE AMT field on ATD Diebold 10XX/478X screen 7 contains a nonzero value, and the value has been downloaded to the ATM). If noninteractive surcharging is in use, and the actual surcharge is less than or equal to the maximum amount, the request is sent to the Authorization process without presenting a surcharge notification screen to the cardholder.

Refer to the ***BASE24-atm Transaction Processing Manual*** for a message flow diagram of a noninteractive surcharge transaction.

Institutions using the surcharging noninteractive message sequence must use load group 305 for withdrawal transactions and load group 306 for fast cash transactions.

Files Maintenance Considerations

The following Diebold and BASE24 files contain fields used with the Diebold 10XX/478X Dial-up Device Handler extension:

- Logical Network Configuration File (LCONF)
- BASE24-atm Terminal Data files (ATD)
- Diebold Configuration File (CNFGIX)
- Visa II Dial Application File (CONFIG.DAT)
- Application Dial Interface File (ADI.DAT)

In addition, this subsection describes system configuration using text commands issued from a network control facility.

Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains one assign and two params for the Diebold 10XX/478X Dial-up Device Handler extension.

The ATM-ADCF assign contains the fully qualified file name of the ATM Dial-Up Command File (ADCF). The ADCF queues commands to a dial-up ATM until it initiates a connection with the BASE24 system.

The ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP param specifies the number of seconds the Device Handler process waits for the next leg of a multiple-leg transaction before the dynamic data for a terminal connected to a dial-up line can be removed from the dynamic data table. A value of 0 disables this processing.

The ATM-DH-MAX-QUEUED-CMD param specifies the maximum number of commands that can be queued to a terminal This param can be any integer value greater than zero.

The ATM-DH-MAX-LOAD-CMD param specifies the number of ADCF LOAD commands that can be processed by the Device Handler process at one time.

Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information about the assign and params.

BASE24-atm Terminal Data Files

BASE24-atm Terminal Data files (ATD) Diebold 10XX/478X screen 7 is a device-specific screen. This screen contains the following fields used with the Diebold 10XX/478X Dial-Up Device Handler extension:

- The DIAL-UP field indicates whether the terminal is a dial-up ATM.
- The MSG FAIL field indicates failed-message processing.
- If surcharging is supported at the terminal, the DOWNLOAD SURCHARGE AMT field specifies the surcharge amount to be downloaded to the device. This amount is displayed to the cardholder during noninteractive surcharge processing. Refer to the topic, "Surcharging" in this appendix for more information on noninteractive surcharge processing.

  When you update the DOWNLOAD SURCHARGE AMT field, an ATD flag is set to indicate that the field has been changed and must be downloaded to the ATM.

  Note: It is important that the DOWNLOAD SURCHARGE AMT field is set with the same rules as the load groups. For example, if you are using load groups for interactive surcharge processing, and the DOWNLOAD SURCHARGE AMT field contains a nonzero value, a surcharge is applied to the transaction but the cardholder is not presented with a surcharge notification screen. Likewise, if you are using load groups for noninteractive surcharge processing and the DOWNLOAD SURCHARGE AMOUNT field is set to a value of zero, the transaction is processed as if interactive surcharging is in effect, except that the cardholder is presented with both the noninteractive surcharge screen and the interactive surcharge screen.

In addition, the INACTIVITY TIME-OUT (WORKDAY), INACTIVITY TIME-OUT (WEEKEND/HOLIDAY), and NON-PEAK INACTIVITY TIME-OUT fields on this screen are used to identify ATMs that are not active because of communication problems. They should be set to values greater than the longest anticipated disconnect period for a dial-up ATM.

Refer to the ***BASE24-atm Files Maintenance Manual*** for more information on ATD Diebold 10XX/478X screen 7.

Diebold Configuration File

The Diebold Configuration File (CNFGIX) enables the institution to maintain ATM screen displays, state table entries, FIT table data, configuration parameters, and timers. The following load groups are used for dial-up processing with surcharging. Load groups are assigned in the TERMINAL GROUP field on ATD Diebold 10XX/478X screen 7.

305 = The load group for dial-up noninteractive surcharge processing for withdrawal and deposit transactions. Canceled transactions are not logged to the journal.

306 = The load group for dial-up noninteractive surcharge processing for fast cash transactions.

310 = The load group for dial-up surcharging with surcharge interactive response notification processing for withdrawal and deposit transactions.

315 = The load group for dial-up noninteractive surcharge processing for withdrawal and deposit transactions. Canceled transactions are logged to the journal.

354 = The load group for dial-up noninteractive surcharge processing for BASE24-atm self-service banking (SSB) deposit with cash back transactions. Canceled transactions are not logged to the journal. This group must be merged with groups 0, 12, and 305.

355 = The load group for dial-up noninteractive surcharge processing for BASE24-atm SSB deposit with cash back transactions. Canceled transactions are logged to the journal. This group must be merged with groups 0, 12, and 315.

364 = The load group for dial-up noninteractive surcharge processing for BASE24-atm SSB split deposit transactions. Canceled transactions are not logged to the journal. This group must be merged with groups 0, 12, 52, 305, and 354.

365 = The load group for dial-up noninteractive surcharge processing for BASE24-atm SSB split deposit transactions. Canceled transactions are logged to the journal. This group must be merged with groups 0, 12, 52, 315, and 355.

Refer to section 4 for more information about the CNFGIX.

The following table summarizes the load groups and values in the DOWNLOAD SURCHARGE AMT field on ATD Diebold 10XX/478X screen 7 that an institution can use for surcharging functionality. Refer to section 4 for descriptions of load groups 0, 5, 12 52, 304, and 314.

Surcharge Processing	**Load Groups**										**ATD DOWNLOAD SURCHARGE AMT**
	0	**5**	**12**	**52**	**304 or 314**	**305 or 315**	**306**	**310**	**354 or 355**	**364 or 365**	
Interactive surcharging for withdrawals and deposits	✓	✓			✓						0
Noninteractive surcharging for withdrawals and deposits	✓	✓				✓	✓				Nonzero
Noninteractive surcharging for deposit with cash back transactions	✓		✓			✓			✓		Nonzero
Noninteractive surcharging for split deposit transactions	✓		✓	✓		✓			✓	✓	Nonzero
Interactive response notification for withdrawals and deposits	✓	✓						✓			0

To use surcharge interactive response notification processing with a dial-up ATM (i.e., using load group 310), you must set the ATM-FEE-INTRACTV-RESP-NTFY param to a value of Y. If you are not using surcharge interactive response notification processing (i.e, using load groups other than 310) and the ATM-FEE-INTRACTV-RESP-NTFY param is present, it must be set to a value of N.

Visa II Dial Application File

The Visa II Dial Application File (CONFIG.DAT) contains general configuration data for the Visa II Dial Application for the Diebold 10XX/478X device. For more information about the CONFIG.DAT refer to the Diebold publication ***Visa II Dial Protocol Manual*** (TP-820321-001A).

Application Dial Interface File

The Application Dial Interface File (ADI.DAT) contains data used by the Application Dial Interface for the Diebold 10XX/478X device. For more information about the ADI.DAT, refer to the Diebold publication ***Visa II Dial Protocol Manual*** (TP-820321-001A).

Network Control Commands

Because the Device Handler process cannot initiate a call to a dial-up ATM to execute network control commands, it queues and stores commands in the ATM Dial-Up Command File (ADCF) until the ATM connects to the BASE24 system.

When a request initiating a connection is received, the Device Handler process checks the ADCF. If there are no stored commands, the request is processed. Otherwise, the transaction request is declined while the ATM closes and the ACDF commands are processed. If the ATM is already connected to the BASE24 system and a command is received, the command is queued to the ADCF and processed after all preceding commands in the file are processed. The commands stored in the ADCF are executed using first in, first out (FIFO) processing, except when priority commands are generated programmatically (for example, initial loads started by the Device Handler process as a result of receiving ATM unsolicited power failure status messages).

The maximum number of commands that can be queued to a terminal is specified in the ATM-DH-MAX-QUEUED-CMD param. When the number of commands in the queue surpasses the limit specified by the param, the new command is dropped and an event message is generated indicating the command was dropped.

The maximum number of simultaneous downloads to multiple terminals that can be queued by the Device Handler process at one time is specified in the ATM-DH-MAX-LOAD-CMD param. Once this limit is reached, the Device Handler process retains the load command in the ACDF. A counter is increased any time a queued load message is sent from the Device Handler process and decreased when the command has been processed.

You can list the commands in the ADCF by entering the LISTQ command from a network control facility. This command causes the Device Handler process to list all records from the ADCF for the specified terminal ID. Each command is listed in an event message on the network log.

To clear the queue for a terminal you can enter the CLEARQ command from a network control facility. This command causes the Device Handler process to delete all records from the ADCF for a specified terminal ID. Each command is listed in event messages on the network log before it is deleted. The number of commands pending for that terminal is then set to a value of 0.

Neither the LISTQ or CLEARQ commands can be entered from the Device Control Terminal (DCT).

Refer to the ***BASE24 Text Command Reference Manual*** for more information about the LISTQ and CLEARQ commands.

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D: Multiple Currency Dispense Add-On Feature

This appendix describes the BASE24-atm Multiple Currency Dispense add-on feature. This feature enables Diebold 10XX/478X ATMs running in 912 emulation mode to dispense cash in multiple currencies. This add-on product and the BASE24-atm Multiple Currency authorization add-on product are independent of one another. The Multiple Currency authorization add-on product is not required to process Multiple Currency dispense transactions if accounts are not supported in multiple currencies, and the Multiple Currency Dispense add-on product is not required if cash is not dispensed in more than one currency.

This appendix documents the cash dispensing portion of BASE24-atm Multiple Currency functions only. Refer to the ***BASE24-atm Multiple Currency Support Manual*** for information about authorization and currency conversion processing of multiple currency transactions.

Overview

The BASE24-atm Multiple Currency Dispense add-on feature enables an institution to dispense cash in a single currency or up to four different currencies from a single ATM.

Note: This function is for dispensing cash only. The BASE24-atm Multiple Currency add-on product for authorization, which supports the authorization of multiple currencies, can work with this product but is not required. Refer to the ***BASE24-atm Multiple Currency Support Manual*** for more information about the BASE24-atm Multiple Currency add-on product for authorization.

Currency Types

BASE24-atm supports transaction processing in multiple currencies by defining currency types and allowing these to be used in data files and financial messages within BASE24. Some of these currency types are briefly defined in the following list. The ***BASE24-atm Multiple Currency Support Manual*** describes these currency types in more detail.

Transaction Currency — The currency in which the transaction is delivered to BASE24 from a device, a host, or an interchange. This is the key currency for multiple currency processing.

Acquirer Institution Currency — The default currency for the institution that acquires the transaction.

Issuer Institution Currency — The default currency for the institution that is the issuer for the transaction.

Account Currency — The designated currency for the account or accounts to which the transaction applies.

Base Currency — The currency through which all currency conversion takes place. All exchange rates are defined in the Base currency. This currency is selected at installation and it is unlikely to change.

Original Currency — The currency used at the ATM at which the transaction originates. If the transaction is acquired by BASE24, Original currency is the same as the Transaction currency. However, if the transaction enters BASE24 through an interchange, the Original currency can be different from any other currency known to BASE24.

Terminal Currency — The currency for the ATM at which the transaction is performed.

Logical Network Currency — Logical network currency is maintained in the BASE24-atm Multiple Currency add-on product primarily to provide compatibility with standard BASE24 processing for a number of subsidiary functions.

Supported Transactions

The BASE24-atm Multiple Currency Dispense add-on feature supports cash withdrawals from checking, savings, and credit accounts.

Support with Other BASE24 Products

The BASE24-atm Multiple Currency Dispense add-on feature is supported by the Surcharge option. The BASE24-atm Non–Currency Dispense add-on product is compatible with the BASE24-atm Multiple Currency Dispense add-on feature (i.e., they can exist in the same environment), but BASE24-atm Non–Currency Dispense transactions cannot occur in more than a single currency. Only cash withdrawals can be done in more than a single currency.

Support is not provided for the Fraud Control option or the BASE24-atm self-service banking application or the Merchant Banking Center.

Account Balance Information

When the Authorization process processes a Multiple Currency Dispense transaction, it converts the account currency to the Transaction currency. To display the account balance on the receipt and on the screen, the Device Handler process converts the amount given in the Transaction currency to an amount in the Account currency. Because of the multiple currency conversions, the account balance that appears on the receipt and on the ATM screen may differ slightly from the actual account balance.

Exchange Rate Notification

Institutions can optionally include an exchange rate notification step in multiple currency dispense transactions. If the value of the ATM-DH-XCHG-RATE-NTFY param is set to a value of Y, the cardholder is presented the rate of exchange and is prompted to confirm or cancel the transaction. If the cardholder cancels the transaction, the BASE24-atm Diebold 10XX/478X Device Handler process sends a reversal message to the BASE24-atm Authorization process. Refer to the topic "Processing" for more information about exchange rate notification and the topic "Configuring the Multiple Currency Dispense Add-On Feature" for more information about the ATM-DH-XCHG-RATE-NTFY param.

Exchange rate notification can exist in the same environment as response notification surcharging; however, only one interactive response message can be presented to the cardholder during a transaction. If both types of interactive response scenarios occur in the same transaction, the surcharge response notification takes precedence, and the exchange rate screen is not displayed.

Processing

Any process performing multiple currency processing converts the Transaction currency into the various currencies specified by the currency type using the Currency Conversion utilities. Currency conversion uses exchange rates held against the user-selected Base currency and follows appropriate conversion rules. Currency conversion processing uses the triangulation method, in which conversion between two currencies uses exchange rates held against the Base currency.

Currency conversion processing is discussed in the ***BASE24-atm Multiple Currency Support Manual.***

Exchange rates are configured at the network level in the Exchange Rate File (ERF). The ERF contains one record for each currency (except the Base currency) that is used by the BASE24 network. The Base currency in multiple currency processing is the single currency against which exchange rates for all other currencies are expressed. All exchange rates held in the ERF are in respect to the Base currency to a specific currency rate. That is, the value by which the amount must be multiplied or divided to obtain the equivalent.

The exchange rate information is stored in and retrieved from the Multiple Currency token. Refer to the ***BASE24-atm Tokens Manual*** for a detailed description of this token.

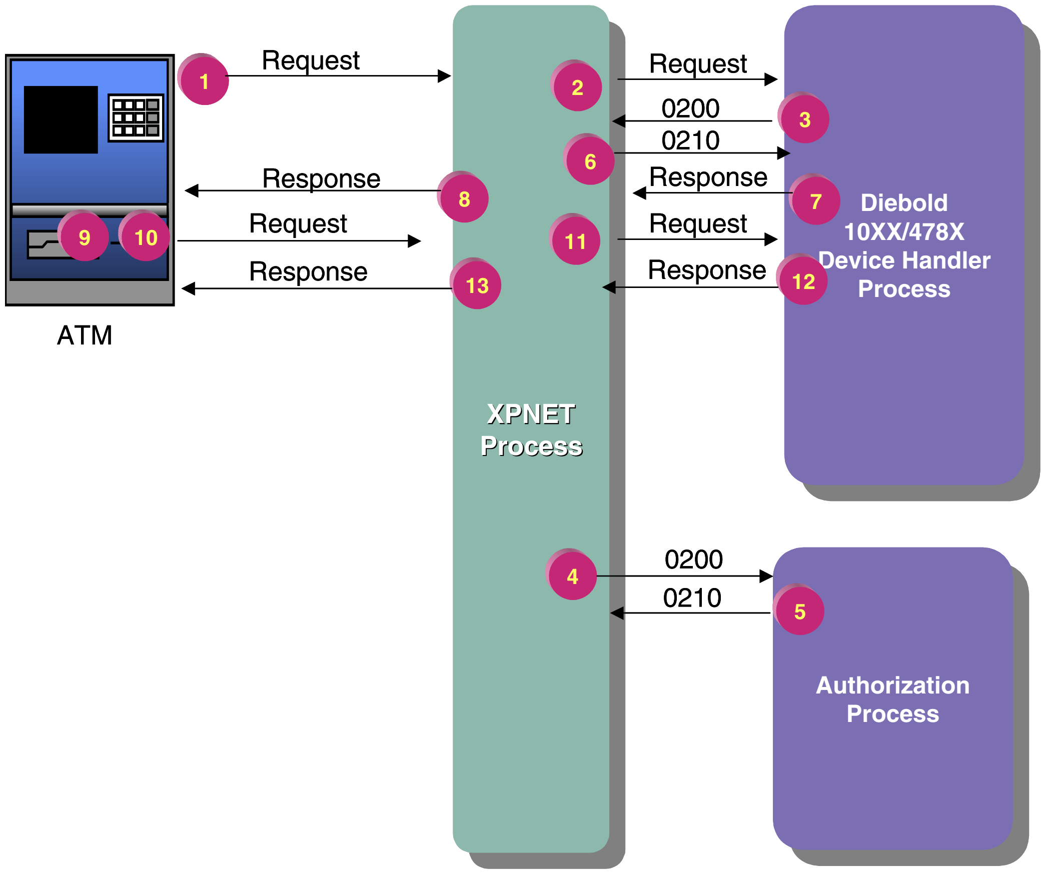
Message Flows

The following message flow diagrams describe the processing that takes place when the institution uses exchange rate notification (that is, the ATM-DH-XCHG-RATE-NTFY param is present and set to a value of Y). With exchange rate notification, the exchange rate is displayed to the cardholder, and the cardholder can confirm or cancel the transaction.

Note: The message flow for a multiple currency dispense transaction without exchange rate notification is the same as the message flow for a withdrawal transaction. Refer to the ***BASE24-atm Transaction Processing Manual*** for message flows for withdrawal transactions. Message flows for multiple currency dispense transactions that include multiple account selection or surcharges are similar to those in the ***BASE24-atm Transaction Processing Manual,*** with the exchange rate notification steps described here occurring after the multiple account selection and surcharging steps.

Withdrawal Transaction, Exchange Rate Notification, Cardholder Confirms Exchange Rate

The following diagram shows the message flow of a multiple currency dispense transaction with the cardholder confirming the exchange rate. Descriptions of each step follow the illustration.

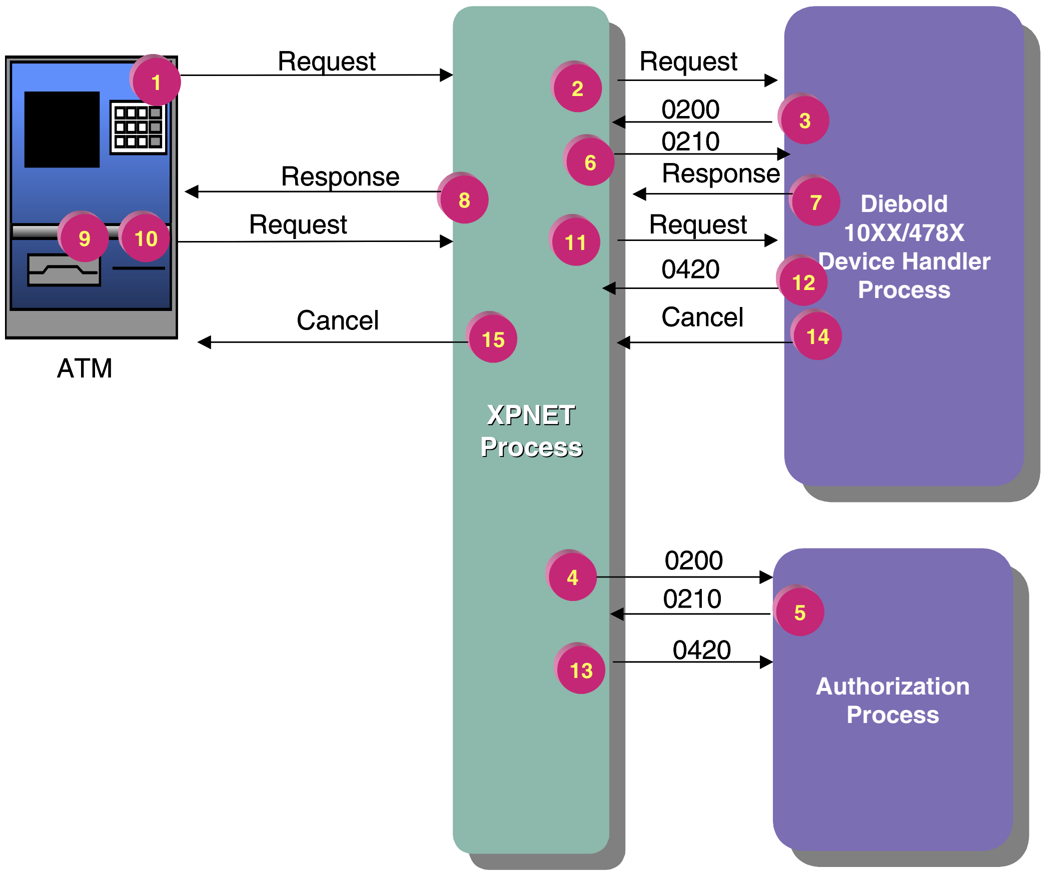


Step	Description
1, 2	The cardholder selects a transaction in a specific currency, the account to be used, and the transaction amount. The ATM sends a native-mode request message to the BASE24-atm Diebold 10XX/478X Device Handler process.
3, 4	The Device Handler process formats a 0200 request message with the multiple currency token and sends it to the BASE24-atm Authorization process.

Step	Description
5, 6	Once the transaction is authorized (by the Authorization process itself or by the host or an interchange), the Authorization process formats a 0210 response message and sends it to the Device Handler process.
7, 8	The Device Handler process formats a native mode response message instructing the ATM to display the exchange rate and prompt the customer for confirmation.
9	The ATM displays the exchange rate and prompts the customer for confirmation. The customer elects to confirm the transaction.
10, 11	The ATM formats a native-mode request and sends it to the Device Handler process.
12, 13	The Device Handler process sends the dispense and print information to the ATM in a native response message.

Withdrawal Transaction, Exchange Rate Notification, Cardholder Cancels Transaction

The following message flow diagram illustrates a transaction in which the cardholder is presented with the exchange rate information and chooses to cancel the transaction. A description of the steps follows the diagram.



Step	Description
1, 2	The cardholder selects a transaction in a specific currency, the account to be used, and the transaction amount. The ATM sends a native-mode request message to the BASE24-atm Diebold 10XX/ 478X Device Handler process.
3, 4	The Device Handler process formats a 0200 request message with the multiple currency token and sends it to the BASE24-atm Authorization process.

Step	Description
5, 6	Once the transaction is authorized (by the Authorization process itself or by the host or an interchange), the Authorization process formats a 0210 response message and sends it to the Device Handler process.
7, 8	The Device Handler process formats a native mode response message instructing the ATM to display the exchange rate and prompt the customer for confirmation.
9	The ATM displays the exchange rate and prompts the customer for confirmation. The customer elects to cancel the transaction.
10, 11	The ATM formats a native-mode request and sends it to the Device Handler process.
12, 13	The Device Handler process formats a 0420 reversal message which includes the Multiple Currency token and sends it to the Authorization process.
14, 15	The Device Handler process formats a native-mode message to cancel the transaction and sends it to the ATM.

Receipt Processing

The BASE24-atm Diebold 10XX/478X Device Handler process uses the following field type codes to process receipts for multiple currency dispense transactions:

Field Type Code	Description
CV	Equivalent converted amount
EV	Exchange rate value
XC	Converted currency description
XE	Exchange rate description
XS	Selected currency description

The following is an example of a receipt for a transaction in which the currency selected by the cardholder is different than the Account or Terminal currency. The selected amount, exchange rate, and converted amount were formatted from the XS, EV, and CV field type codes, respectively.

```
     A C I   D E M O

DATE      TIME  TERM
MM/DD/YY  HH:MM S1AC9100

 123456700000003-003

330 SOUTH 108TH AVE

RECORD NO.             570
WITHDRAWAL
 SELECTED AMOUNT    $20.00
 EXCHANGE RATE       .1000
 CONVERTED AMOUNT  p200.00
FR     1111222233334444

LEDGER BAL     $3412.00
AVAIL BAL      $1760.00

         1         2
1234567890123456789012345678
```

Note that although the balance information is in the Account currency, the figure may differ slightly from the actual account balance because of multiple currency conversions that have taken place.

Refer to section 6 for detailed information about these field type codes and configuring receipts.

Configuring the Multiple Currency Dispense Add-On Feature

The Multiple Currency Dispense add-on feature can be used with the BASE24-atm Diebold 10XX/478X Device Handler process. The following Diebold and BASE24 files contain fields needed for the Multiple Currency Dispense add-on feature.

- C100NAMS File
- Diebold Configuration File (CNFGIX)
- Logical Network Configuration File (LCONF)
- BASE24-atm Terminal Data files (ATD)
- Receipt File (RCPT)
- Exchange Rate File (ERF)

C100NAMS File

The C100NAMS File is an edit file that contains a series of tables used by the BASE24-atm Diebold 10XX/478X Device Handler process. You can edit these tables to customize specific information displayed on ATM receipts. The diebold_1000_mult_crncy_tbl table in the C100NAMS File supports the BASE24-atm Multiple Currency Dispense add-on feature. This table allows the receipt information specific to the BASE24-atm Multiple Currency Dispense add-on feature to be displayed in multiple languages.

Diebold Configuration File (CNFGIX)

The Diebold Configuration File (CNFGIX) enables institutions to maintain ATM screen displays, state table entries, FIT table data, configuration parameters, and timers, which are then downloaded to the ATM. These screens, states, etc., are identified by a load group number. The load group number for the BASE24-atm Multiple Currency Dispense add-on feature is 901.

States for the BASE24-atm Multiple Currency Dispense add-on feature include the following:

State Type	State Number
D—Clear Keys State	138
D—Clear Keys State	163
D—Clear Keys State	166
E—Select Function State	029
E—Select Function State	137
F—Dollar Entry State	041
H—Information Entry State	037
I—Transaction Request State	139

Refer to section 4 for more information about configuring the CNFGIX.

Logical Network Configuration File (LCONF)

The Logical Network Configuration File (LCONF) links physical file names to the files names as they are referenced in the BASE24 code and contains processing values used throughout the logical network. The LCONF contains one assign and three params that are related to the BASE24-atm Multiple Currency Dispense add-on feature.

The ERFEMT assign, which is not specific to the BASE24-atm Multiple Currency Dispense add-on feature (but is used by the BASE24-atm and BASE24-pos Multiple Currency products), defines the location of the Exchange Rate File (ERF) extended memory table to be used by this logical network.

The BASE-CURRENCY-CODE param contains the ISO currency code of the Base currency to be used by this logical network. This code must match the Base currency code as set in the COBNAMES File at installation. This param is used by the BASE24-atm Multiple Currency Dispense add-on feature and the BASE24-atm Multiple Currency add-on product.

The ATM-DH-XCHG-RATE-NTFY param indicates whether to prompt the cardholder to confirm the exchange rate. This param is specific to the BASE24-atm Multiple Currency Dispense add-on feature.

The CURRENCY-CODE param contains the ISO currency code of the primary type of currency the logical network uses. The value in this param must match a value in the COBNAMES File. This param is not specific to the BASE24-atm Multiple Currency Dispense add-on feature.

BASE24-atm Terminal Data Files (ATD)

The BASE24-atm Terminal Data files (ATD) contain configuration information for each terminal in the logical network. The CURRENCIES CONFIGURED FOR SELECTION BY THE CARDHOLDER fields on ATD screen 4 have specific requirements for the BASE24-atm Multiple Currency Dispense add-on feature. These fields correspond to the currencies displayed to the cardholder at the ATM and must contain valid ISO currency codes from the COBNAMES File.

These ATD fields are described in detail in the ***BASE24-atm Files Maintenance Manual.***

Receipt File (RCPT)

The Receipt File (RCPT) contains receipt templates that define the actual text and formats for the various types of receipts. You can configure receipts for the BASE24-atm Multiple Currency Dispense add-on feature using the field type codes described in the topic “Receipt Processing” provided earlier in this section.

Refer to section 6 for more information about configuring the RCPT.

Exchange Rate File (ERF)

The Exchange Rate File (ERF) contains one record for each currency used in the BASE24 system. When using the BASE24-atm Multiple Currency Dispense add-on feature, ensure that all currencies used are configured in the ERF. Refer to the ***BASE24-atm Multiple Currency Support Manual*** for more information about the ERF.

ACI Worldwide, Inc.

E: Distributing Master Keys and Transaction Keys

This appendix describes add-on products that manage master keys and transaction keys. The BASE24-atm uses transaction keys, also known as working keys, to perform specific cryptographic operations, such as encrypting a message authentication code (MAC) or a PIN. A *master key* is used to encrypt the transaction key.

The Automated Key Distribution System add-on product enables you to remotely load a master key to the ATM. The Dynamic Key Management add-on product enables you to manage transaction keys.

Note that the Automated Key Distribution System add-on product manages master keys only, and the Dynamic Key Management add-on product manages transaction keys only.

Automated Key Distribution System Add-On Product

The BASE24-atm Automated Key Distribution System add-on product enables an institution to remotely load a unique master key to an ATM. In the past, master keys, which are used to encrypt all other keys, were loaded manually at ATMs. Using the BASE24-atm Automated Key Distribution System add-on product, you can load the master key to the ATM from the BASE24-atm system. In addition, the key is randomly generated, ensuring a unique key for all ATMs in the network.

This topic describes the BASE24-atm Automated Key Distribution System add-on product, its configuration, and its processing.

Overview

The ATM master key (also known as the key exchange key or A-key) is the cryptographic key used to encrypt all other keys (e.g., PIN encryption key, MAC key) that are exchanged between the BASE24-atm system and an ATM.

Master keys are by nature long term, and, once in place, are usually left unchanged unless they are compromised. Without the BASE24-atm Automated Key Distribution System add-on product, a new key must be manually loaded into the ATM if the key is ever compromised.

The BASE24-atm Automated Key Distribution System add-on product enables an institution to remotely load a unique and secure master key to an ATM using the Rivest-Shamir-Adleman (RSA) public key scheme.

Before the master key is downloaded, the BASE24-atm system and the Encrypting PIN Pad (EPP) on the ATM must authenticate each other and exchange their public keys. The EPP is a tamper-proof component of the ATM consisting of a PIN pad, a security module, data encryption algorithms, and the cryptographic keys. The Transaction Security Services process provides an enhanced interface between the BASE24-atm system and the host security module.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more information about the role of the Transaction Security Services process in automated key distribution.

Configuration Considerations

The following Hardware and Logical Network Configuration File (LCONF) configuration considerations must be addressed for the use of the BASE24-atm Automated Key Distribution System add-on product.

In addition, you must address the Transaction Security Services process and hardware security module implementation requirements. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more information about these requirements.

Diebold Software and Hardware Requirements

Diebold's signature-based processing: The NCR 5XXX ATM must be running Agilis 91x XV 2.1 or later terminal application software.

Diebold's certificate-based processing: The Diebold 10XX/478X ATM must be running Terminal Control Software (TCS) level 1.3 (or greater) software. The ATM also must be equipped with a Diebold encrypting PIN pad (EPP).

Wincor-Nixdorf's shared specification signature- and certificate-based processing: The ATM (whether it is a Wincor, NCR, Diebold, or another vendor's device) must be running Wincor-Nixdorf's ProCash/NDC firmware version 13.00 (or greater) or firmware that is compatible with Wincor-Nixdorf's shared RKL specification, version 1.5.

Hardware Security Module Requirements

The Automated Key Distribution System add-on product supports Atalla Key Block (AKB) and the Thales e-Security (Racal) security modules.

Logical Network Configuration File

The following Logical Network Configuration File (LCONF) assigns and params must be configured to use the Automated Key Distribution System add-on product. For full descriptions of these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual.***

Assigns. The following assigns must be configured in the LCONF to use the Automated Key Distribution System add-on product.

SECURE-DEV1	The PPD name of the primary Transacation Security Services process, rather than a hardware security module. This assign is not specific to the Automated Key Distribution System add-on product.
SECURE-DEV2	The PPD name of the secondary Transaction Security Services process, rather than a hardware security module. This assign is not specific to the Automated Key Distribution System add-on product.
SECURE-DEV-*xx*	The PPD name of one of 20 Transaction Security Services processes accessed on a rotating basis, where *xx* is any two characters. This assign is not specific to the Automated Key Distribution System add-on product.

Params. The following params must be configured in the LCONF to use the Automated Key Distribution System add-on product.

ATM-DH-AKDS-TSS-TIMER	The number of seconds the Diebold 10XX/478X Device Handler process waits for a reply from the Transaction Security Services process for certain Automated Key Distribution System commands. If the Device Handler process has not received a reply before the timer expires, it aborts the load. Valid values are 10 through 600 seconds. This param is specific to the Automated Key Distribution System add-on product.
ATM-DH-1000-AKDS-SERIAL-NUM-VRFY	A flag indicating whether to verify the EPP serial number against a list of approved numbers maintained in the Transaction Security Services (TSS) database. This verification occurs during the authentication and public key exchange sequence of Automated Key Distribution System processing. Valid values are Y (yes) and N (no).

SECURE-DEV-TYP	Specifies the security device or process the BASE24 security utilities communicate with directly. This param must be set to the value TSS to indicate the use of the Transaction Security Services process instead of a hardware security module. This param is not specific to the Automated Key Distribution System add-on product.

Terminology

The following terminology is used in describing the Automated Key Distribution add-on product:

Base64 Encoding — A method of encoding data used by Diebold to convert binary data into encoded characters. A base64 algorithm can produce a 4 to 3 ratio of encoded bytes to binary bytes.

Certificate Authority (CA) — An organization that creates signatures and certificates. A certificate authority might be an ATM manufacturer, a financial institution, or a third-party certificate authority.

Encrypting PIN Pad (EPP) — A tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, data encryption algorithms (e.g., DES, Triple DES, MAC, RSA), and the cryptographic keys. All cryptographic keys are stored in the EPP. All cryptographic functions are performed by the EPP.

Host Security Module (HSM) — A security device that acts as a peripheral to the host computer. It performs cryptographic processing in a physically secure environment on behalf of the host.

Key — A string of bits widely used in cryptography, enabling users to encrypt and decrypt data. A key can be used to perform other mathematical operations, as well. Given a cipher, a key determines the mapping of the plaintext to the ciphertext. Also known as the A-Key, Key Exchange Key (KEK), and Terminal Master Key (TMK).

Master key — A key used to encrypt and decrypt the working keys that are downloaded from the host to the ATM. Also known as the A-Key, Key Exchange Key (KEK), and Terminal Master Key (TMK).

Nonce — A randomly generated value used to defeat *playback* attacks in communication protocols. One party randomly generates a nonce and sends it to the other party. The receiver encrypts it using an agreed-upon secret key and returns it to the sender. Because the sender randomly generated the nonce, this defeats playback attacks; the receiver cannot know in advance which nonce the sender will generate. The receiver denies connections that do not have a properly encrypted nonce.

Out-of-Band — Describes a function that is performed offline, rather than via online transaction processing. For example, manually loading keys at an ATM is a function that is performed out-of-band.

PKCS #7 cryptographic message — A message standard for data that has cryptography applied to it, such as digital certificates. The Diebold EPP transports the certificates and other data to and from the BASE24 system using this message.

Private key — Synonymous with secret key.

Public key — One half of a cryptographic key pair used in the RSA public key algorithm (the other half being the secret key). The public key is uniquely associated with an entity and is available publicly. It is used to encrypt data that can only be decrypted using the matched secret key. It also is used to verify signatures that were created with the matched secret key. There is a mathematical relationship between the public key and the secret key, but it is virtually impossible to derive one from the other. In the Automated Key Distribution System add-on product, the BASE24 system, the EPP, and the Diebold certificate authority have public keys.

Public Key Cryptography Standards (PKCS) — A series of cryptographic standards dealing with public key issues, published by RSA laboratories.

Public Key Infrastructure (PKI) — A security infrastructure whose services are implemented and delivered using public key concepts and techniques.

Remote Key Loading (RKL) — Wincor-Nixdorf's keying process using RSA or public cryptography.

RSA — A public key algorithm based on the factoring problem. RSA stands for Rivest-Shamir-Adleman, the developers of the RSA public key algorithm.

Secret key — One half of a cryptographic key pair used in the RSA public key algorithm (the other half being the public key). The secret key is uniquely associated with an entity and is not available publicly. It is used to decrypt data that was encrypted using the matched public key. It also is used to create signatures that only can be verified using the matched public key. There is a mathematical relationship between the secret key and the public key, but it is virtually impossible to derive one from the other. In the Automated Key Distribution System add-on product, the BASE24 system, the EPP, and the Diebold certificate authority have secret keys.

Signature — A method of preventing data from being changed during transmission and of preventing the sender from being impersonated. A secure hashing algorithm is applied to the data and then the result is encrypted. The signature is generated using the secret key. Only the holder of the secret key can generate the signature but anybody can verify it using the matched public key.

Working key — A category of keys used to perform specific cryptographic operations. This key is encrypted under the Master Key prior to being downloaded from the Host to the ATM. PIN encryption keys, MAC keys, and PIN verification keys are all examples of Working Keys. Also known as the B-Key.

Signature-Based Processing

The following topics describe the processing steps the BASE24-atm Diebold 10XX/478X Device Handler process takes in automated key distribution using signatures. Automated key distribution is performed in phases as follows:

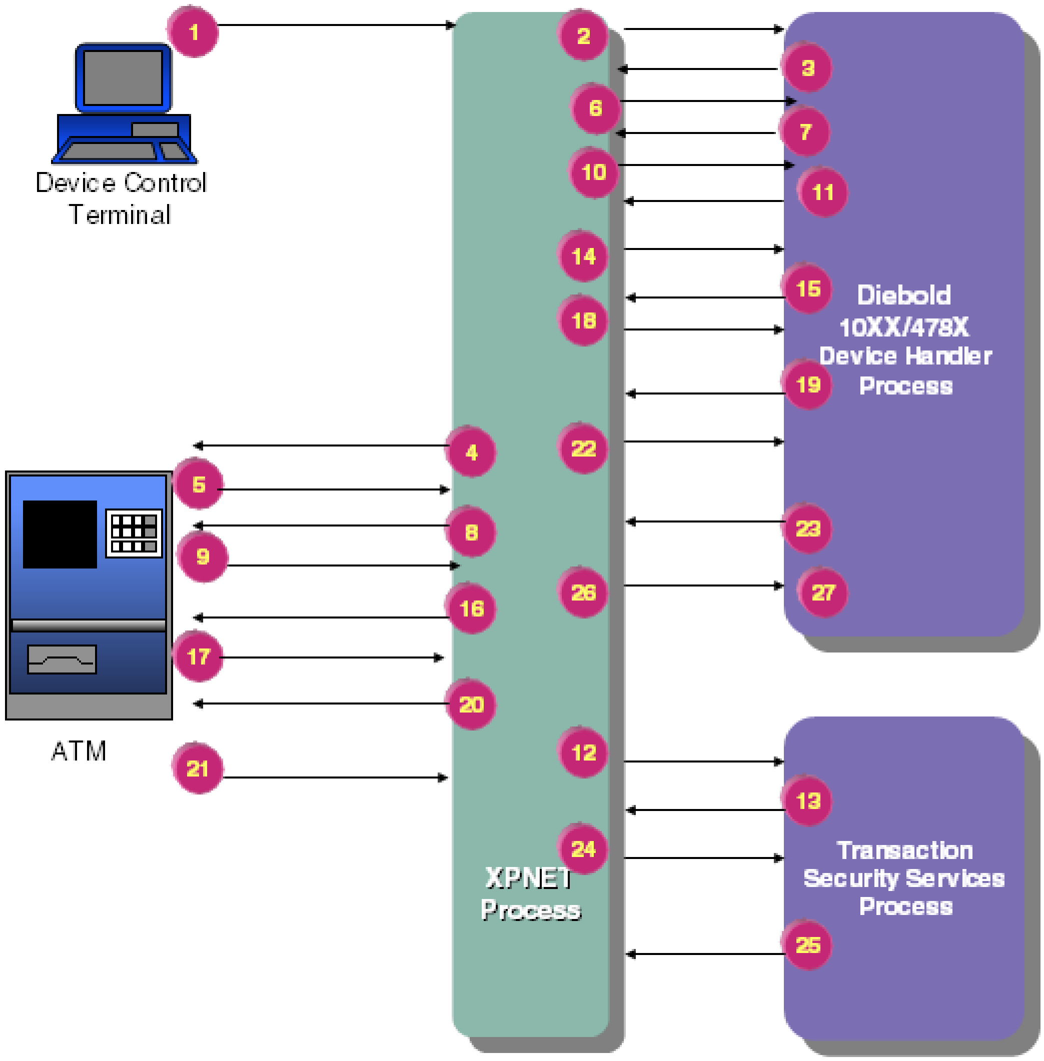
Phase	Description
1	Generation of the NCR RSA key pair.
2	Manufacture of the NCR Encrypting PIN Pad (EPP).
3	Generation of the BASE24 RSA key pair.
4	Authentication and public key exchange.
5	Master key download.

The processing described in this appendix is for phases 4 and 5. Phases 1 through 3 are described in the ***BASE24 Transaction Security Services Processing Guide.***

Authentication and Public Key Exchange Using Signatures

The following message flow illustrates the processing that takes place during authentication and public key exchange (phase 4 of automated key distribution processing) which utilizes signatures. At this stage in the exchange, the BASE24 system and the EPP have established the keys and signatures required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for the message flow for establishing keys and signatures.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Device Control Terminal
ATM
XPNET Process
Diebold 10XX/478X Device Handler Process
Transaction Security Services Process
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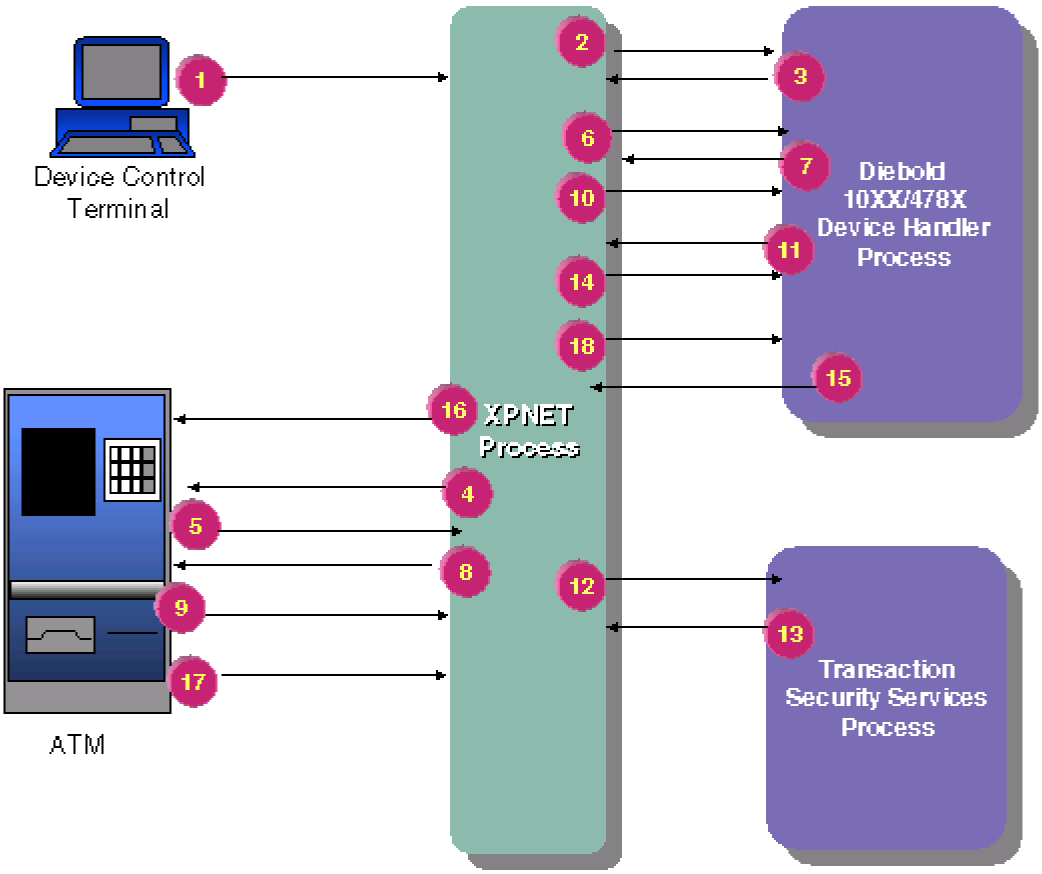
Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends an operational command message with a command code indicating "Send EPP's Serial Number and Signature" to the ATM.
9, 10	The ATM responds with a solicited status message, as long as there is AKDS capable firmware running at the ATM. Otherwise, a command reject will be received.
11, 12	The Device Handler process decodes the base94-encoded signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.
13, 14	The Transaction Security Services process verifies the signature using the NCR public key and stores the serial number of the EPP in its database. The Transaction Security Services process returns a response to the Device Handler process. Included in the response is the BASE24 public key and the signature of the public key.
15, 16	The Device Handler process encodes the BASE24 public key and its signature using the base94 algorithm. The Device Handler process downloads the encoded BASE24 public key and signature to the ATM EPP by sending an encryption key change (EKC) message.
17, 18	The ATM EPP verifies the signature using the NCR public key. The BASE24 public key is stored in the EPP and then the ATM responds with a solicited status message.

Steps	Processing
19, 20	The Device Handler process requests the EPP public key by sending an operational command message, “Send EPP’s Public Key and Signature” to the ATM.
21, 22	The ATM responds to the Device Handler process with a solicited status message.
23, 24	The Device Handler process decodes the base94-encoded EPP public key and signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.
25, 26	The Transaction Security Services process verifies the signature using the NCR public key and stores the EPP public key in its database. It also generates a message authentication code (MAC) of the EPP public key and stores it in its database. The Transaction Security Services process sends a response to the Device Handler process.
27	The Device Handler process receives the response. At this point the host and EPP have successfully authenticated each other and exchanged their public keys.

Downloading the Master Key Using Signatures

The following message flow illustrates the processing that takes place during the master key download phase (phase 5). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

Steps	Processing
7, 8	The Device Handler process sends an operational command message to the ATM to “Initiate Remote Key Transport.”
9, 10	The ATM responds with a solicited status message, as long as there is AKDS capable firmware running at the ATM. Otherwise, a command reject will be received.
11, 12	The Device Handler process sends a nowaited request to the TSS to generate a master key.
13, 14	The Transaction Security Services process generates the master key, which is encrypted under the EPP public key, and a signature for the master key. It stores the master key in its database and returns a response to the Device Handler process.
15, 16	The Device Handler process encodes the master key and the signature using the base94 algorithm. The Device Handler process downloads the encoded master key and signature to the EPP by sending a Write Command VII message.
17, 18	The EPP verifies the signature using the BASE24 public key and decrypts the master key using the EPP secret key. The master key is stored in the EPP. Finally, the ATM responds to the Device Handler process with a solicited status message.

The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Using Signatures

The Diebold 10XX/478X Device Handler process enables you to download all keys (i.e., public key, master key, MAC encryption key, and PIN encryption key) with one command.

After an operator initiates a request to download all keys, the BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown, as described in the previous message flows. The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

The Device Handler process sends an operational command message, “Send EPP’s Serial Number and Signature” to the ATM.

Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.

The Device Handler process decodes the signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.

The Transaction Security Services process verifies the signature using the NCR public key, stores the EPP public key in its database, and returns a nowaited response to the Device Handler.

The Device Handler process encodes the BASE24 public key and its signature using the base94 algorithm, and then downloads the encoded BASE24 public key and signature to the ATM EPP by sending a Write Command VII message.

The EPP verifies the signature using the BASE24 public key. The public key is stored in the EPP. Then, the ATM responds to the Device Handler process with a solicited status message.

The Device Handler process requests the EPP public key by sending a terminal command message with a command code indicating, “Send EPP’s Public Key and Signature” to the ATM.

Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.

The Device Handler process decodes the base94-encoded signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.

The Transaction Security Services process verifies the signature using the NCR public key, stores the EPP public key in its database, and returns a nowaited response to the Device Handler.

The Device Handler process sends an operational command to the ATM indicating "initiate remote key transport."

Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.

The Device Handler process sends a nowaited request to the Transaction Security Services process to generate a master key.

The Transaction Security Services process generates the master key, along with a signature for the master key and random number, and stores it in its database. Then, the Transaction Security Services process sends a nowaited response.

The Device Handler process encodes the master key and the signature using the base94 algorithm. The Device Handler process downloads the encoded master key and signature to the EPP by sending a Write Command VII message.

The EPP verifies the signature using the BASE24 public key and decrypts the master key using the EPP secret key. The master key is stored in the EPP. Finally, the ATM responds to the Device Handler process with a solicited status message.

Certificate-Based Processing

The following topics describe the processing steps the BASE24-atm Diebold 10XX/478X Device Handler process takes in automated key distribution using certificates.

Automated key distribution is performed in phases as follows:

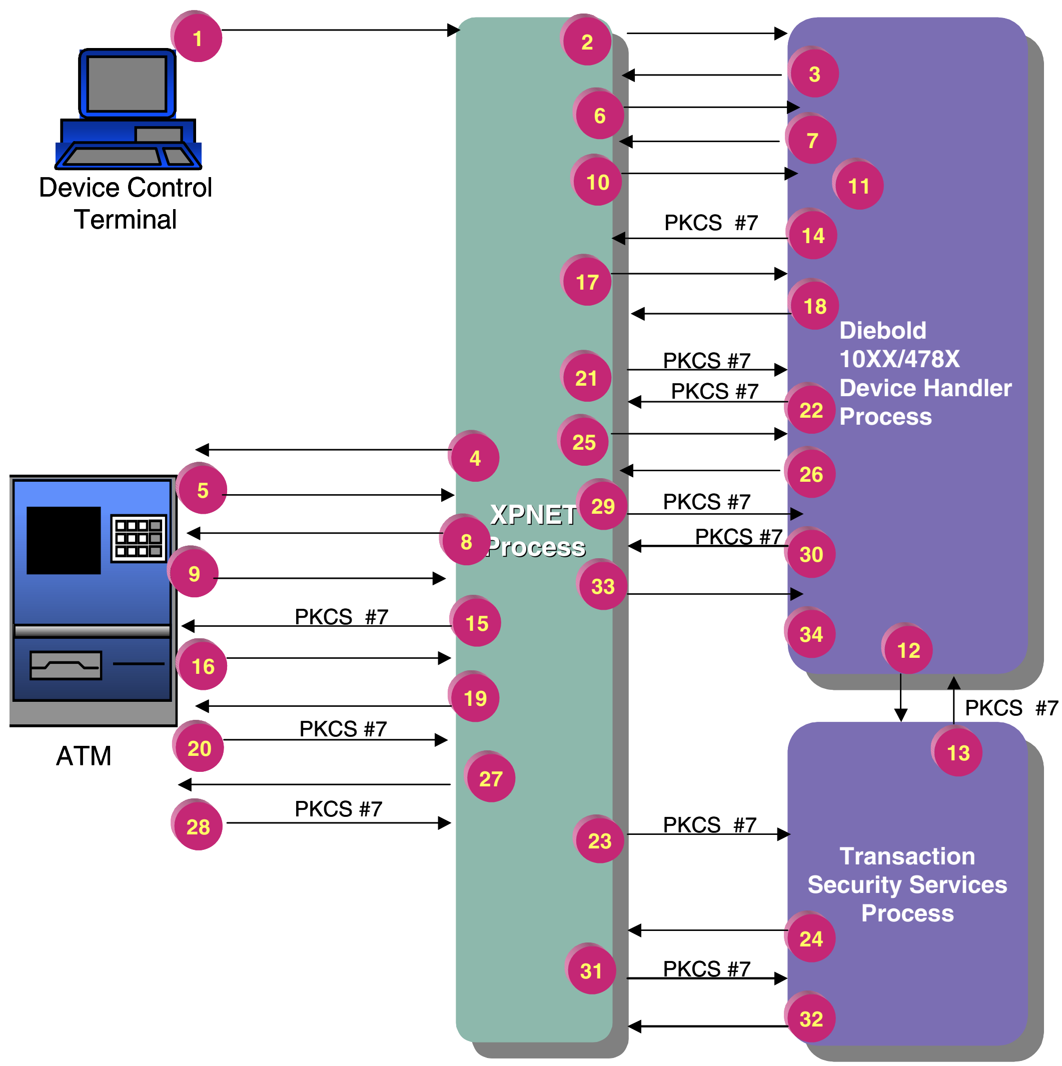
Phase	Description
1	Initial registration and certification for the Diebold Encrypting PIN Pad (EPP).
2	Initial registration and certification for the BASE24 system.
3	Initial certificate exchange.
4	Remote key transport.

The processing described in this appendix is for phases 3 and 4. Phases 1 and 2 are described in the ***BASE24 Transaction Security Services Processing Guide.***

Initial Certificate Exchange Using Certificates

The following message flow illustrates the processing that takes place during the initial certificate exchange (phase 3 of automated key distribution processing). At this stage in the exchange, the BASE24 system and the EPP have established the keys and certificates required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for a message flow example for establishing keys and certificates.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request also can be made from a network control facility or from the ATM itself. If the request comes from the ATM, refer to the topic “Initial Certificate Exchange Initiated from the ATM” following these procedures for information about this step.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.

Steps	Processing
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request message to the ATM.
9, 10	The ATM sends a configuration response to the Device Handler process.
11	The Device Handler process searches the Additional Hardware Configuration Status area of the configuration response message for the value EP0104RK, which indicates the ATM has an AKDS-capable EPP installed. If the value is not found, this phase is not performed and the ATM is restarted. If the value is found, the Device Handler process continues with step 12.
12	The Device Handler process sends a message to the Transaction Security Services to request the BASE24 primary certificate containing the BASE24 public signature verification key.
13	The Transaction Security Services process responds to the Device Handler process with the length of the Public Key Cryptography Standards (PKCS) #7 message and the PKCS #7 message that contains the BASE24 primary certificate containing the BASE24 public signature verification key.
14, 15	The Device Handler process base64-encodes the PKCS #7 message and downloads it to the ATM EPP.
16, 17	The ATM EPP verifies the BASE24 primary certificate using the certificate authority's primary public signature verification key. The BASE24 public signature verification key is stored in the EPP. The ATM responds to the Device Handler process with a solicited status message indicating the key was loaded successfully.
18, 19	The Device Handler process sends an operational command message to the ATM requesting the EPP public signature verification key.

Steps	Processing
20, 21	The ATM responds with a solicited status message. The solicited status message contains a base64-encoded PKCS #7 cryptographic message, which contains the EPP primary certificate which in turn contains the EPP public signature verification key.
22, 23	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
24, 25	The Transaction Security Services process verifies the EPP primary certificate using the certificate authority's primary public signature verification key and stores the EPP public signature verification key in its database. The Transaction Security Services process sends a nowaited response to the Device Handler process.
26, 27	The Device Handler process receives the nowaited response message. After processing the response, the Device Handler process sends an operational command message to the ATM requesting the EPP public enciphering key.
28, 29	The ATM responds with a solicited status message. The solicited status message contains a base64-encoded PKCS #7 cryptographic message, which contains the EPP primary certificate which in turn contains the EPP public enciphering key.
30, 31	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
32, 33	The Transaction Security Services process verifies the EPP primary certificate using the certificate authority's primary public signature verification key and stores the EPP public enciphering key in its database. The Transaction Security Services process sends a nowaited response message to the Device Handler process.
34	The Device Handler process receives the message and processes the response. At this point the BASE24 system and EPP have successfully authenticated each other and exchanged their public keys.

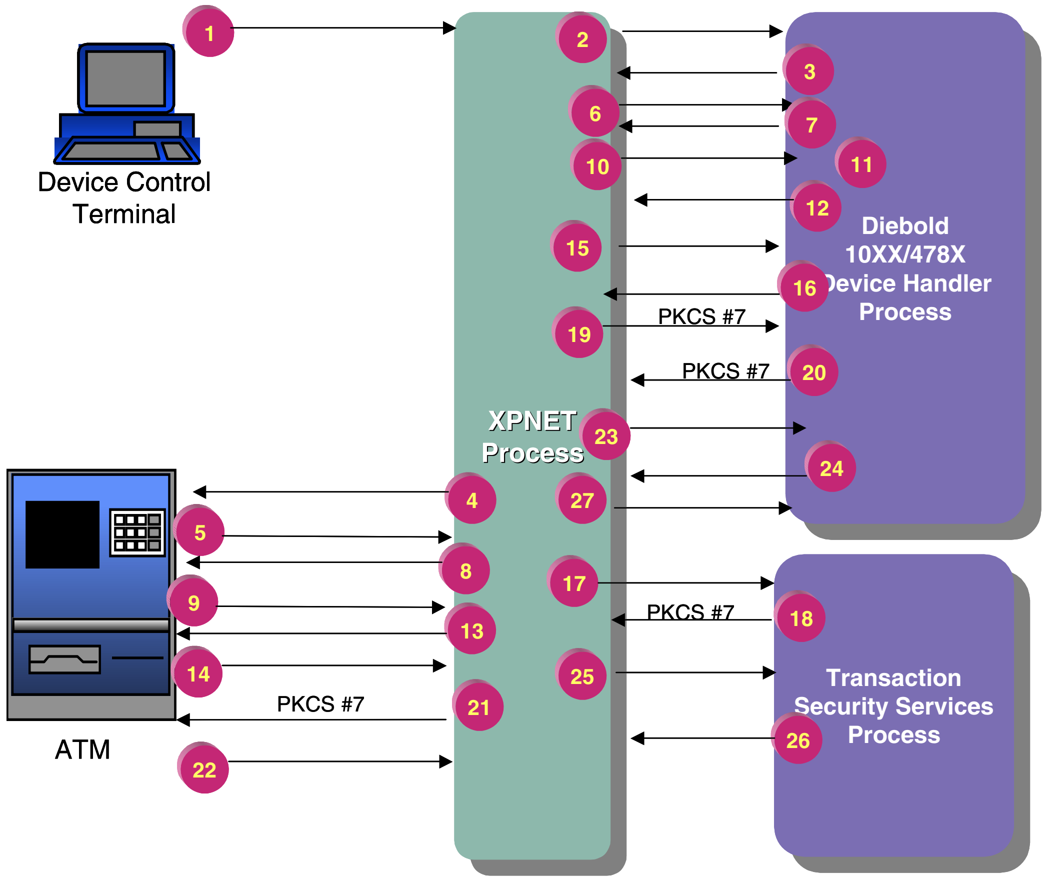
Initial Certificate Exchange Initiated from the ATM

Operators can enter a request for the initial certificate exchange (phase 3) directly from the ATM. When this occurs, the ATM sends an unsolicited status message to the Device Handler process requesting the BASE24 public signature verification key. Once the Device Handler process receives this message, processing continues with step 8, above.

Remote Key Transport

The following message flow illustrates the processing that takes place during the remote key transport phase (phase 4). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but the command also can be made from a network control facility or from the ATM itself. If the request comes from the ATM, refer to the topic “Remote Key Transport Initiated from the ATM” following these procedures for more information about this step.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.

Steps	Processing
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request message to the ATM.
9, 10	The ATM sends a configuration response to the Device Handler process.
11	The Device Handler process searches the Additional Hardware Configuration Status area of the configuration response message for the value EP0104RK, which indicates the ATM has an AKDS-capable EPP installed. If the value is not found, this phase is not performed and the ATM is restarted. If the value is found, the Device Handler process continues with step 12.
12, 13	The Device Handler process sends an operational command to the ATM indicating “initiate remote key transport.”
14, 15	The ATM responds with a solicited status message containing the EPP’s random nonce.
16, 17	The Device Handler process sends a nowaited request to the Transaction Security Services process to generate a master key.
18, 19	The Transaction Security Services process generates the master key and stores it in its database. It then constructs a key block containing BASE24’s distinguishing identifier, the master key, and the key identifier and encrypts the key block using the EPP public enciphering key. The Transaction Security Services process then constructs a key token containing a data block of BASE24’s random nonce, the EPP’s random nonce, the EPP’s distinguishing identifier and the encrypted key block. The Transaction Security Service process then signs the data block using BASE24’s private signature generation key. Finally, the Transaction Security Services process puts the entire key token in the PKCS #7 message and returns it to the Device Handler process.

Steps	Processing
20, 21	The Device Handler process base64-encodes the PKCS #7 message containing the enciphered and signed key block. The Device Handler process downloads the encoded PKCS #7 message to the EPP.
22, 23	The EPP verifies the signature on the key token using the BASE24 public signature verification key. It also checks the EPP's distinguishing identifier in the key token, and verifies that the EPP's random nonce in the key token is the same as the one sent in step 9. The EPP decrypts the key block using the EPP private deciphering key. The EPP also checks BASE24's distinguishing identifier. If all checks are successful, the master key is stored in the EPP according to the key identifier. The EPP then constructs a key token containing a data block of BASE24's random nonce, the EPP's random nonce, and BASE24's distinguishing identifier. The EPP then signs the data block using the EPP private signature verification key. Finally, the EPP puts the entire key token in the PKCS #7 cryptographic message and sends it in a solicited status message to the Device Handler process.
24, 25	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the key token.
26, 27	The Transaction Security Services process verifies the signature on the key token using the EPP public signature verification key. It also checks BASE24's distinguishing identifier and checks that the received BASE24 random nonce agrees with the one sent in step 13. In addition it checks that the received EPP random nonce is the same as the one received in step 12. The Transaction Security Services process returns a nowaited response to the Device Handler process.

The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Remote Key Transport Initiated from the ATM

Operators can enter a request for the remote key transport (phase 4) directly from the ATM. When this occurs, the ATM sends an unsolicited status message to the Device Handler process requesting a remote key transport and containing the EPP's random nonce. Once the Device Handler process receives this message, processing continues with step 12, above.

Load All Keys Using Certificates

The Diebold 10XX/478X Device Handler process enables you to download all keys (i.e., public key, master key, MAC encryption key, and PIN encryption key) with one command.

After an operator initiates a request to download all keys, the BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown, as described in the previous message flows. The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready. The Device Handler process sends a configuration request to the ATM, and the ATM responds with a configuration response.

The Device Handler process searches the Additional Hardware Configuration Status area of the configuration response message for the value EP0104RK, which indicates the ATM has an AKDS-capable EPP installed. If the value is found, the Device Handler process downloads the following RSA and DES keys in this order:

1. Public key
2. Master key
3. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
4. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the EP0104RK value is not found in the configuration response message, the Device Handler process downloads only the following DES keys (in this order):

1. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
2. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

The public key and master key are loaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are loaded as described in the ***BASE24 Transaction Security Manual.***

Note that this command is supported only in the Automated Key Distribution System extension of the Diebold 10XX/478X Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Wincor-Nixdorf's Shared Specification Certificate-Based Processing

The following topics describe the processing steps the BASE24-atm Diebold 10XX/478X Device Handler process takes in automated key distribution using certificates for Diebold terminals running firmware that is compatible with Wincor-Nixdorf's RKL specification running in Diebold 912 emulation mode.

Automated key distribution is performed in phases as follows:

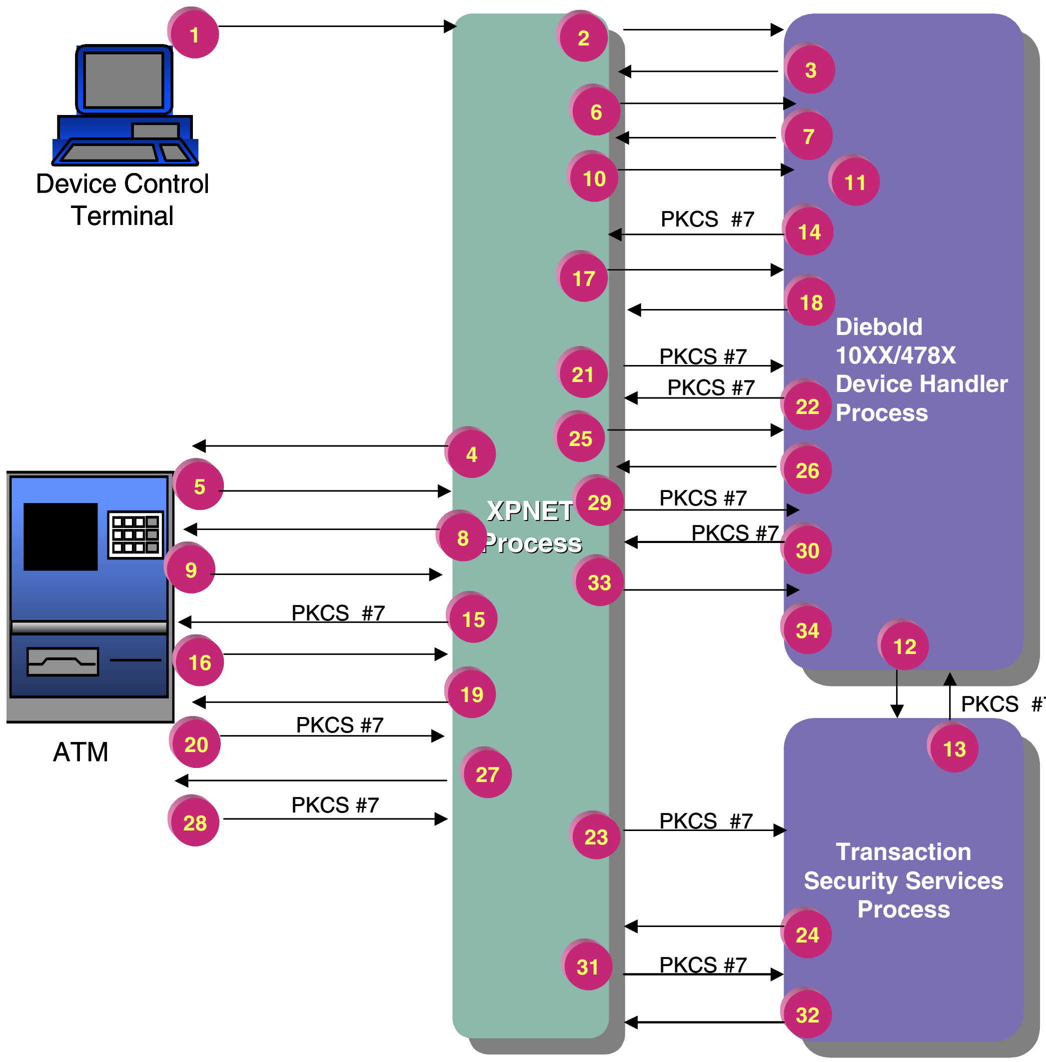
Phase	Description
1	Initial registration and certification for the Diebold Encrypting PIN Pad (EPP).
2	Initial registration and certification for the BASE24 system.
3	Initial certificate exchange.
4	Remote key transport.

The processing described in this appendix is for phases 3 and 4. Phases 1 and 2 are described in the ***BASE24 Transaction Security Services Processing Guide.***

Initial Certificate Exchange Using Certificates

The following message flow illustrates the processing that takes place during the initial certificate exchange (phase 3 of automated key distribution processing). At this stage in the exchange, the BASE24 system and the EPP have established the keys and certificates required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for a message flow example for establishing keys and certificates.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request also can be made from a network control facility.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends an operational command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.
11	The Device Handler process searches the signature/certificate indicator for the value 2 (certificate). A value of 2 indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is discontinued and a startup message is sent to the ATM. If the value is not found (a value of 0 is present), this phase is not performed and the ATM is restarted. If the value is found, the Device Handler process continues with step 12.
12	The Device Handler process sends a message to the Transaction Security Services to request the BASE24 primary certificate containing the BASE24 public signature verification key.
13	The Transaction Security Services process responds to the Device Handler process with the length of the Public Key Cryptography Standards (PKCS) #7 message and the PKCS #7 message that contains the BASE24 primary certificate containing the BASE24 public signature verification key.
14, 15	The Device Handler process base64-encodes the PKCS #7 message and downloads it to the ATM EPP.

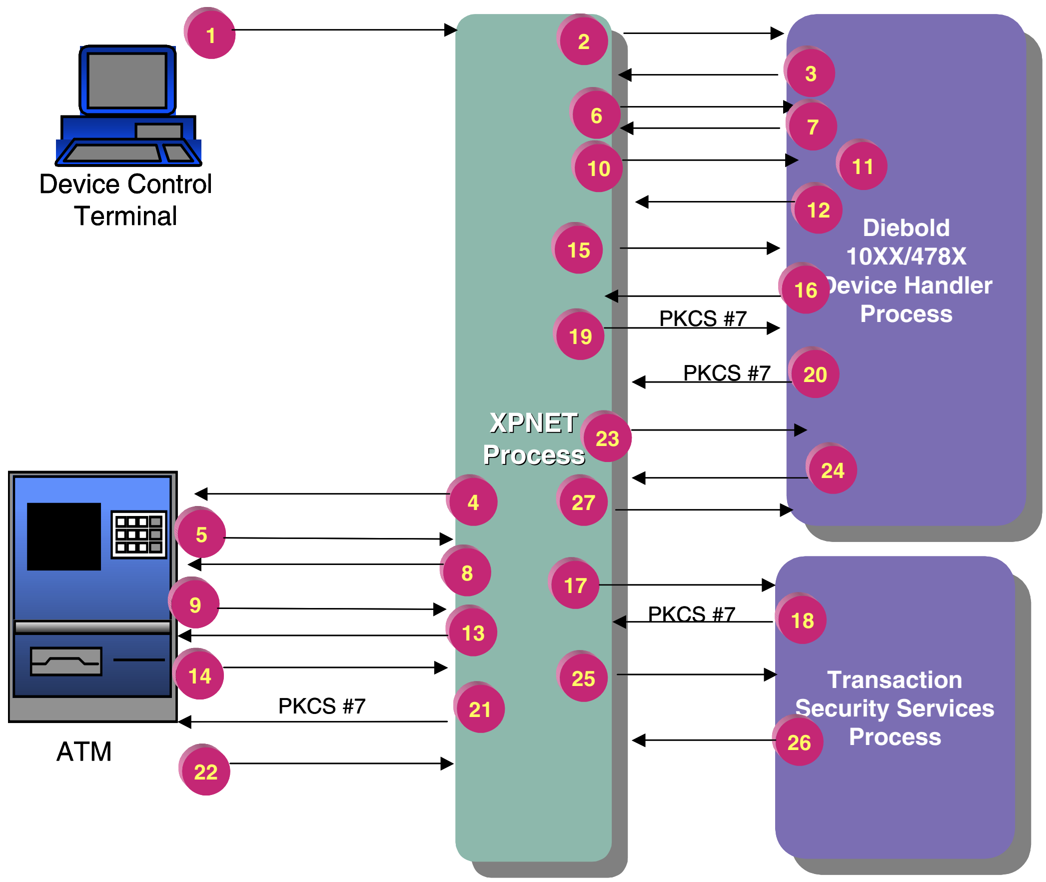
Steps	Processing
16, 17	The ATM EPP verifies the BASE24 primary certificate using the certificate authority's primary public signature verification key. The BASE24 public signature verification key is stored in the EPP. The ATM responds to the Device Handler process with a solicited status message indicating the key was loaded successfully.
18, 19	The Device Handler process requests the EPP public encryption/decryption key by sending an operational command message, "Send RKL information, EPP encryption/decryption Public Key certificate" to the ATM.
20, 21	The ATM responds with a solicited status message. The solicited status message contains a base64-encoded PKCS #7 cryptographic message, which contains the EPP primary certificate which in turn contains the EPP public enciphering key.
22, 23	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
24, 25	The Transaction Security Services process verifies the EPP primary certificate using the certificate authority's primary public signature verification key and stores the EPP public enciphering key in its database. The Transaction Security Services process sends a nowaited response to the Device Handler process.
26, 27	The Device Handler process receives the nowaited response message. After processing the response, the Device Handler process requests the EPP public signature verification key by sending an operational command message, "Send RKL information, EPP signature/verification Public Key certificate" to the ATM.
28, 29	The ATM responds with a solicited status message. The solicited status message contains a base64-encoded PKCS #7 cryptographic message, which contains the EPP primary certificate which, in turn, contains the public signature verification key.

Steps	Processing
30, 31	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
32, 33	The Transaction Security Services process verifies the EPP primary certificate using the certificate authority's primary public signature verification key and stores the EPP public signature verification key in its database. The Transaction Security Services process sends a nowaited response message to the Device Handler process.
34	The Device Handler process receives the message and processes the response. At this point the BASE24 system and EPP have successfully authenticated each other and exchanged their public keys.

Remote Key Transport

The following message flow illustrates the processing that takes place during the remote key transport phase (phase 4). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	**Processing**
1, 2	An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but the command also can be made from a network control facility.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

Steps	Processing
7, 8	The Device Handler process sends an operational command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.
11	The Device Handler process searches the signature/certificate indicator for the value 2 (certificate). A value of 2 indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is discontinued and a startup message is sent to the ATM. If the value is not found (a value of 0 is present), this phase is not performed and the ATM is restarted. If the value is found, the Device Handler process continues with step 12.
12, 13	The Device Handler process sends an operational command message to the ATM indicating “Send RKL information, Start key exchange, create random number.”
14, 15	The ATM responds with a solicited status message containing the EPP’s random nonce.
16, 17	The Device Handler process sends a nowaited request to the Transaction Security Services process to generate a master key.
18, 19	The Transaction Security Services process generates the master key and stores it in its database. It then constructs a key block containing BASE24’s distinguishing identifier, the master key, and the key identifier and encrypts the key block using the EPP public enciphering key. The Transaction Security Services process then constructs a key token containing a data block of BASE24’s random nonce, the EPP’s random nonce, the EPP’s distinguishing identifier and the encrypted key block. The Transaction Security Service process then signs the data block using BASE24’s private signature generation key. Finally, the Transaction Security Services process puts the entire key token in the PKCS #7 message and returns it to the Device Handler process.

Steps	Processing
20, 21	The Device Handler process base64-encodes the PKCS #7 message containing the enciphered and signed key block. The Device Handler process downloads the encoded PKCS #7 message to the EPP.
22, 23	The EPP verifies the signature on the key token using the BASE24 public signature verification key. It also checks the EPP's distinguishing identifier in the key token, and verifies that the EPP's random nonce in the key token is the same as the one sent in step 9. The EPP decrypts the key block using the EPP private deciphering key. The EPP also checks BASE24's distinguishing identifier. If all checks are successful, the master key is stored in the EPP according to the key identifier. The EPP then constructs a key token containing a data block of BASE24's random nonce, the EPP's random nonce, and BASE24's distinguishing identifier. The EPP then signs the data block using the EPP private signature verification key. Finally, the EPP puts the entire key token in the PKCS #7 cryptographic message and sends it in a solicited status message to the Device Handler process.
24, 25	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the key token.
26, 27	The Transaction Security Services process verifies the signature on the key token using the EPP public signature verification key. It also checks BASE24's distinguishing identifier and checks that the received BASE24 random nonce agrees with the one sent in step 13. In addition it checks that the received EPP random nonce is the same as the one received in step 12. The Transaction Security Services process returns a nowaited response to the Device Handler process.

The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Using Certificates

The Diebold 10XX/478X Device Handler process enables you to download all keys (i.e., public key, master key, MAC encryption key, and PIN encryption key) with one command.

After an operator initiates a request to download all keys, the BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown, as described in the previous message flows. The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

The Device Handler process sends an operational command message, "Send RKL information, EPP capabilities" to the ATM.

Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.

The Device Handler process searches the signature/certificate indicator for the value 2 (certificate). A value of 2 indicates that the ATM supports the Automated Key Distribution System add-on product.

If the ATM supports the Automated Key Distribution System add-on product, the Device Handler process downloads the following RSA and DES keys (in this order):

1. Public key
2. Master key
3. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
4. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the ATM does not support the Automated Key Distribution System add-on product, the Device Handler process downloads only the following DES keys (in this order):

1. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
2. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

The public key and master key are loaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are loaded as described in the ***BASE24 Transaction Security Manual.***

Note that this command is supported only in the Automated Key Distribution System extension of the Diebold 10XX/478X Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Wincor-Nixdorf's Shared Specification Signature-Based Processing

The following topics describe the processing steps the BASE24-atm Diebold 10XX/478X Device Handler process takes in automated key distribution for Wincor-Nixdorf terminals running in Diebold 912 emulation mode, or other vendor's terminals running firmware that is compatible with Wincor-Nixdorf's RKL specification running in Diebold 912 emulation mode. Automated key distribution is performed in phases as follows:

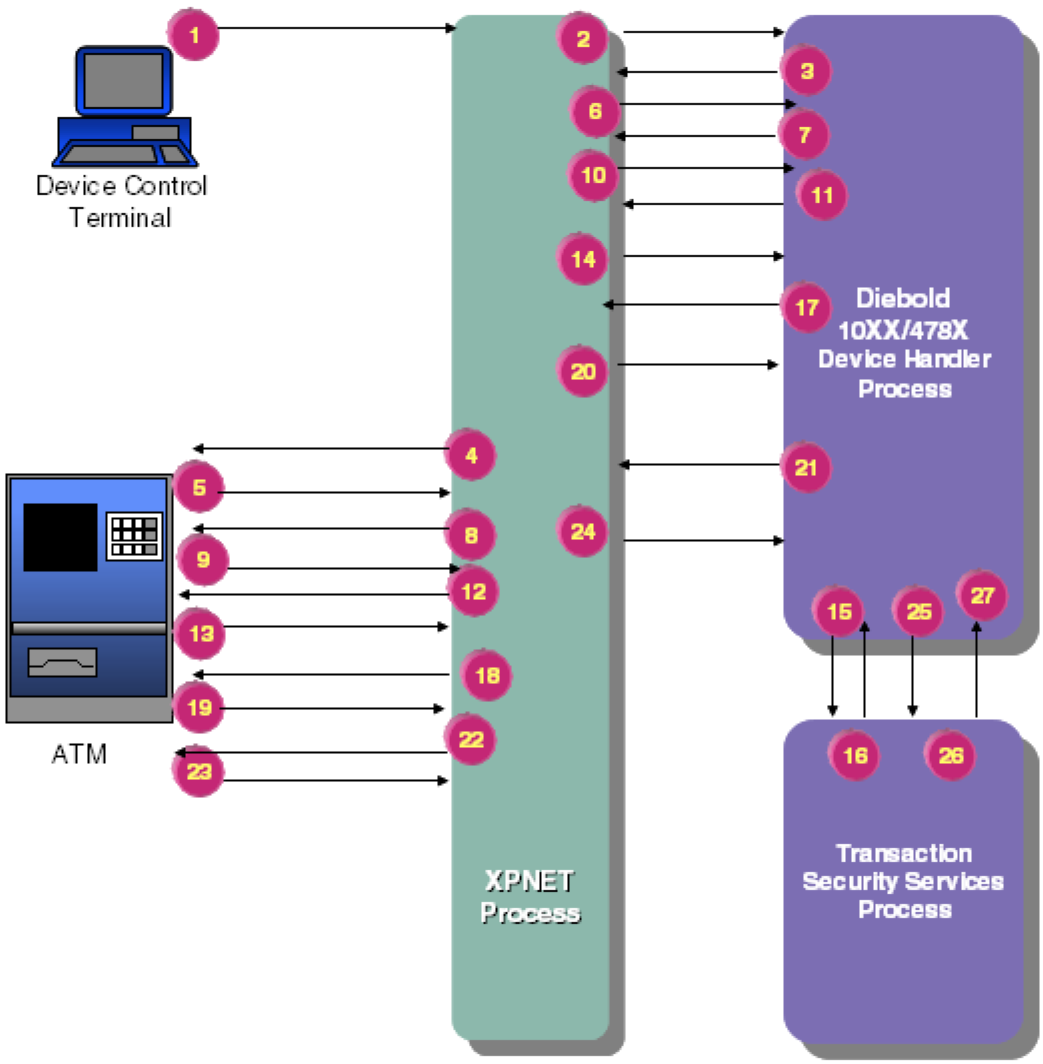
Phase	Description
1	Generation of the Wincor-Nixdorf RSA key pair.
2	Manufacture of the vendor's encrypting pin pad (EPP).
3	Generation of the BASE24 RSA key pair.
4	Authentication and public key exchange.
5	Download of the master key.

The processing described in this appendix is for phases 4 and 5. Phases 1 through 3 are described in the ***BASE24 Transaction Security Services Processing Guide***.

Authentication and Public Key Exchange for Wincor-Nixdorf

The following message flow illustrates the processing that takes place during authentication and public key exchange (phase 4 of automated key distribution processing). At this stage in the exchange, the BASE24 system and the EPP have established the keys and signatures required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for the message flow for establishing keys and signatures.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



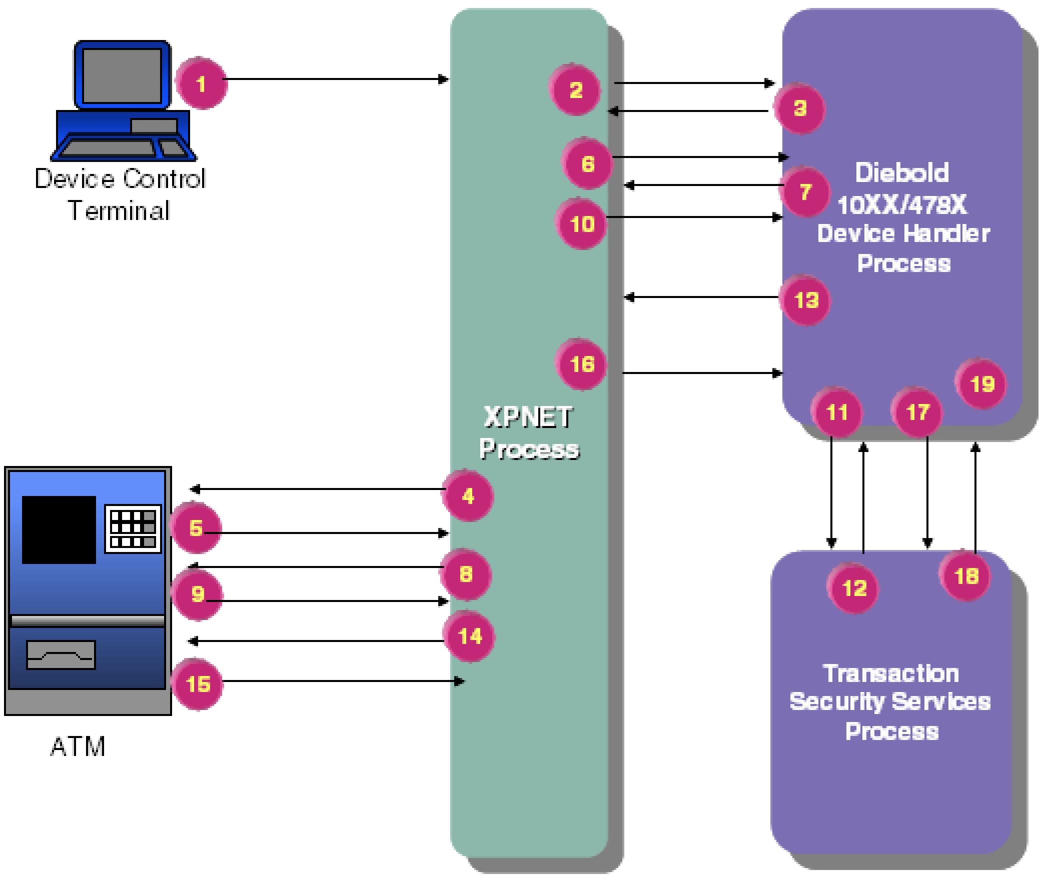
Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends an operational command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.
11, 12	The Device Handler process searches the signature/certificate indicator for the value 1 or 3. A value of 1 or 3 indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is discontinued and a startup message is sent to the ATM. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process requests the encrypting PIN pad (EPP) serial number and signature using an operational command message, “Send RKL information, EPP serial number and signature” to the ATM.
13, 14	The ATM responds to the Device Handler process with a solicited status message.
15	The Device Handler process decodes the base64-encoded signature and sends a request to the Transaction Security Services process to verify the decoded signature.

Steps	Processing
16	The Transaction Security Services process verifies the signature using the Wincor-Nixdorf public key and stores the serial number of the EPP in its database. The Transaction Security Services process returns a response to the Device Handler process. Included in the response is the BASE24 public key and the signature of the public key.
17, 18	The Device Handler process encodes the BASE24 public key and its signature using the base64 algorithm. The Device Handler process downloads the encoded BASE24 public key and signature to the ATM EPP by sending a Write Command VII message.
19, 20	The ATM EPP verifies the signature using the Wincor-Nixdorf public key. The BASE24 public key is stored in the EPP and then the ATM responds with a solicited status message.
21, 22	The Device Handler process requests the EPP public key by sending an operational command message, "Send RKL information, EPP Public Key and Signature" to the ATM.
23, 24	The ATM responds to the Device Handler process with a solicited status message.
25	The Device Handler process decodes the base64-encoded EPP public key and signature and sends a request to the Transaction Security Services process to verify the decoded signature.
26	The Transaction Security Services process verifies the signature using the Wincor-Nixdorf public key and stores the EPP public key in its database. Then, the TSS process sends a response to the Device Handler process.
27	The Device Handler process receives the response. At this point the host and EPP have successfully authenticated each other and exchanged their public keys.

Downloading the Master Key for Wincor-Nixdorf

The following message flow illustrates the processing that takes place during the master key download phase (phase 5). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm Diebold 10XX/478X Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

Steps	Processing
7, 8	The Device Handler process sends an operational command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.
11	The Device Handler process searches the signature/certificate indicator for the value 1 or 3. A value of 1 or 3 indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is discontinued and a startup message is sent to the ATM. The Device Handler process sends a request to the Transaction Security Services process to generate a master key.
12	The Transaction Security Services process generates the master key, which is encrypted under the EPP public key, and a signature for the master key. It stores the master key in its database and returns a response to the Device Handler process.
13, 14	The Device Handler process encodes the master key and the signature using the base64 algorithm. The Device Handler process downloads the encoded master key and signature to the EPP by sending a Write Command VII message.
15, 16	The EPP verifies the signature using the BASE24 public key and decrypts the master key using the EPP secret key. The master key is stored in the EPP. The ATM responds to the Device Handler process with a solicited status message.
17	The Device Handler process sends a request to the Transaction Security Services process to verify the key verification value returned by the ATM.
18	The Transaction Security Services process verifies the key verification value by comparing it to the master key check digits stored in its database. The Transaction Security Services process returns a response to the Device Handler process.
19	The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Command for Wincor-Nixdorf

When you enter the Load All Keys command (entered as LOADALLKEYS), the Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. This message requests the ATM to send remote key loading (RKL) information and EPP capabilities to the BASE24 system. If the ATM does not have an AKDS-capable EPP installed, it rejects this request. If the ATM responds, the Diebold 10XX/478X Device Handler process checks the Signature/Certificate status in the response. If the status is not 1 (Signature) or 3 (Signature and certificate), the process aborts the load, generates one or more event messages, and reopens the ATM.

If the ATM supports AKDS, the Diebold 10XX/478X Device Handler process sequentially performs the following for the Load All Keys command:

1. Performs phase 4, which is to authenticate and exchange public keys.
2. Performs phase 5, which is to download the master key.
3. Downloads the MAC encryption key if the MAC SUPPORT TYPE field on Diebold 10XX/478X screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 3 for the ATM.
4. Downloads the PIN encryption key if the PIN ENCRYPT TYPE field on Diebold 10XX/478X screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 01, 03, 04, or 05 for the ATM.
5. Reopens the ATM.

If the ATM does not support AKDS, the Diebold 10XX/478X Device Handler process sequentially performs only steps 3 through 5 above for the Load All Keys command.

Load Commands

You can issue a command to download the public key, the master key, or all keys to the ATM using the Device Control Terminal (DCT) or a network control facility.

The Automated Key Distribution System add-on product supports both leased-line and dial-up modes of the Diebold 10XX/478X Device Handler process. When using the Diebold 10XX/478X Dial-Up Device Handler extension, load commands are written to the ATM Dial-Up Command File (ADCF). The ADCF queues commands to a dial-up ATM until the ATM initiates a connection with the BASE24-atm Device Handler process. Refer to appendix C for more information about the Diebold 10XX/478X Dial-Up Device Handler extension.

Command Syntax

The Load Public, Load Master, and Load All Keys commands can be entered from a network control facility. The following paragraphs describe the commands and their syntax. Refer to the ***BASE24 Text Command Reference Manual*** for more information about these commands.

9500LOADPUBLIC

The 9500LOADPUBLIC command initiates an authentication and public key exchange load sequence. The BASE24 system and the named ATM authenticate each other and exchange their public keys.

Syntax

9500LOADPUBLIC t*erminal-name*

terminal-name — The name of the ATM with which to exchange public keys.

9500LOADMASTER

The 9500LOADMASTER command initiates a request to generate a new master key and download it to the encrypted PIN pad of the named ATM.

Syntax

9500LOADMASTER *terminal-name*

terminal-name — The name of the ATM to which the master key is loaded.

9500LOADALLKEYS

The 9500LOADALLKEYS command initiates a request to download new public, master, MAC encryption, and PIN encryption keys and download them to the encrypted PIN pad of the named ATM.

Syntax

9500LOADALLKEYS *terminal-name*

terminal-name — The name of the ATM to which the keys are loaded.

Sending a Command Using the DCT

You also can use the Device Control Terminal (DCT) to initiate a Load Public, Load Master, or Load All Keys command. The commands are entered using the LOAD PUBLIC KEY, LOAD MASTER KEY, or LOAD ALL KEYS field on the DCT Diebold 10XX Commands screen. Refer to the ***BASE24 Device Control Manual*** for more information on these fields and the Diebold 10XX Commands screen.

BASE24-atm Dynamic Key Management Add-On Product

The BASE24-atm Dynamic Key Management add-on product enables an institution to determine when PIN and MAC keys need to be changed. The BASE24-atm Dynamic Key Management add-on product shuts down the ATM and initiates commands to change the keys when predetermined thresholds are exceeded.

This topic describes the BASE24-atm Dynamic Key Management add-on product and its configuration.

Overview

The BASE24-atm system uses transaction keys to secure PINs and perform message authentication in the BASE24 network. Transaction keys should be changed regularly to maintain proper security. The standard BASE24-atm Diebold 10XX/478X Device Handler process monitors transactions that include message authentication codes (MACs) and shuts down an ATM when a predetermined number of consecutive MAC errors is exceeded. In this case the ATM's MAC key must be changed. An operator must then issue a command to load the key and reopen the ATM. The operator also must manually monitor and load PIN keys.

The BASE24-atm Dynamic Key Management add-on product automates the key management process. When the Diebold 10XX/478X Device Handler process determines that the keys must be changed (based on criteria set at the process level), it issues commands to automatically load the keys and reopen the ATM.

Determining When Keys are Changed

The standard BASE24-atm Diebold 10XX/478X Device Handler process uses the ATM-DH-CONS-MAC-ERR-LMT Logical Network Configuration (LCONF) param to determine when a MAC key may have been compromised. This param specifies the number of consecutive number of MAC errors that can occur before the Device Handler process shuts down the ATM.

The BASE24-atm Dynamic Key Management add-on product allows you to set thresholds for PIN and MAC keys at the process level. The following table lists the thresholds that can initiate a transaction key change and the associated LCONF param in which the threshold is set:

Threshold	Param
The number of PIN verifications exceeds a predetermined limit.	ATM-DH-DKM-PIN-TXN-LMT
The number of MAC verifications and MAC generations exceed a predetermined limit.	ATM-DH-DKM-MAC-TXN-LMT
The number of PIN verification failures exceeds a predetermined limit.	ATM-DH-DKM-PIN-ERR-LMT
The number of MAC verification and MAC generation failures exceeds a predetermined limit.	ATM-DH-DKM-MAC-ERR-LMT
The number of consecutive MAC failures exceeds a predetermined limit.	ATM-DH-CONS-MAC-ERR-LMT
A specified time interval has passed since the last MAC and PIN key change.	ATM-DH-DKM-KEY-CHNG-INTRVL

The counters and timestamp that correspond to these thresholds are stored in the BASE24-atm Terminal Data files (ATD). A timestamp for the last key change initiated by the Dynamic Key Management add-on product also is stored in the ATD.

Refer to the ***BASE24 Logical Network Configuration File Manual*** for more information about these params.

Changing Keys

Once any of the thresholds for a key change has been reached, the process to load keys to the ATM is automatically initiated and all of the threshold counters and timer are reset. To minimize the time an ATM is down, both the PIN key and MAC key are loaded at the same time. The Dynamic Key Management key change timestamp also is updated.

Refer to the ***BASE24 Transaction Security Manual*** for information about transmitting transaction keys to the device.

ACI Worldwide, Inc.

F: BASE24-atm Mobile Top-Up Extension

The BASE24-atm Mobile Top-Up Device Handler extension enables cardholders to replenish, or *top up,* their mobile telephone accounts at an ATM. The replenishment transaction is a noncurrency dispense transaction. The institution must purchase both the Non–Currency Dispense add-on product and the Mobile Top-Up Device Handler extension to offer mobile top-up services.

A telecommunications services provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account, and calculating any taxes applied to the transaction. The BASE24-atm product authorizes the funds portion of the transaction. Refer to the topic "Processing Mobile Top-Up Transactions" for more information about routing transactions to the telecommunications services provider.

Cardholders who perform mobile top-up transactions receive either real-time replenishment or voucher replenishment. Real-time replenishment enables the telecommunications services provider or mobile operator to replenish the cardholder's account as soon as the transaction is authorized. Voucher replenishment provides a stock receipt (i.e., voucher) on the ATM receipt for the top-up portion of the transaction. Voucher information includes a voucher number and expiration date. The cardholder must then use his or her mobile telephone to enter this information for the replenishment to take place. Whether a customer receives real-time replenishment or a voucher depends on the configuration of a host system (either the institution's host or the telecommunications services provider host). Typically, if the host recognizes the mobile telephone number and carrier, the cardholder receives real-time replenishment. If the host does not recognize the number and carrier, the cardholder receives a voucher printed on the ATM receipt for the top-up portion of the transaction.

Terminology

The following terms appear in this documentation to describe the BASE24-atm Mobile Top-Up Device Handler extension.

Definitions-based model — A transaction processing method for the BASE24-atm Mobile Top-Up Device Handler extension that uses predefined values for mobile operator-supported allowed amounts and telephone time increments.

Funds authorizer — The entity responsible for authorizing the withdrawal, or *funds*, portion of a mobile top-up transaction (e.g., the BASE24-atm Authorization process).

Message-based model — A transaction processing method for the BASE24-atm Mobile Top-Up Device Handler extension that uses a series of messages between the Transaction Context Manager process, the funds authorizer, and the top-up authorizer to verify mobile telephone numbers, the number of minutes, and other purchase information.

Mobile operator — A company that provides application services for the transaction (e.g., an individual mobile telephone company).

Telecommunication service provider — An entity that provides telecommunication services for individual mobile operators.

Top-up authorizer — The entity responsible for authorizing the top-up portion of a mobile top-up transaction. This entity can verify that the mobile telephone number exists, that the account can be replenished, and that the amount or number of minutes requested is valid or may issue a voucher.

Transaction Context Manager process — A BASE24 module that enables the BASE24 system to receive a single request from an endpoint and split the routing between two or more destinations.

Voucher — Information printed on the ATM receipt that indicates mobile telephone time was purchased and a code to enter to activate the replenishment. An ATM generates two receipts for a mobile top-up transaction: one receipt for the funds portion and one for the top-up portion.

Mobile Top-Up Models

Customers can employ one of two models in processing mobile top-up transactions: the definitions-based model or the message-based model.

Definitions-Based Model

The *definitions-based* model uses locally defined values for mobile telephone operators (i.e., the actual service providers), the amount of time that can be purchased, and other information needed to complete the purchase of mobile telephone time. This information is stored in the Mobile Operator File (MOF), a separately licensable BASE24 module, which is read by the Transaction Context Manager process. Customers that employ the definitions-based model must license the Non–Currency Dispense add-on product, the Mobile Top-Up Device Handler extension, the Mobile Top-Up Infrastructure, the Transaction Context Manager product, and the Mobile Top-Up Transaction Context Manager extension.

Because operator information is locally defined, a single message sequence between BASE24 and the mobile operator contains all the information needed to complete the purchase.

Message-Based Model

The *message-based* model of mobile top-up processing uses a series of messages between BASE24 and a telecommunication service provider to gather the information needed to complete the purchase. These messages can include verifying whether the mobile telephone number supplied in the request can accept a replenishment and whether the number of minutes or the amount of the purchase is valid.

Customers that employ the message-based model must license the Non–Currency Dispense add-on product, the Mobile Top-Up Device Handler extension, and the Transaction Context Manager product.

Processing Mobile Top-Up Transactions

A mobile top-up transaction is initiated by the cardholder at the ATM. The cardholder inserts his or her card and selects a noncurrency dispense (withdrawal) transaction, specifying the purchase item as mobile top-up. In addition, the cardholder specifies an amount measured in monetary units (e.g., $20 US dollars) or time units (e.g., 30 minutes). The cardholder can use the ATM keypad to specify a mobile telephone number. The transaction is processed first by the Diebold 10XX/478X Device Handler process and then the Transaction Context Manager process, which routes the transaction, based on settings in the Split Transaction Routing File (STRF), to the BASE24-atm Authorization process and the mobile operator or telecommunication service provider.

Diebold 10XX/478X Device Handler Process

The Diebold 10XX/478X Device Handler process performs the following functions for a mobile top-up transaction:

- Receives the transaction request from the ATM and appends the necessary tokens to identify the transaction as a mobile top-up transaction
- Forwards customer actions to the Transaction Context Manager process
- Receives information from the Transaction Context Manager process
- Formats and sends native mode messages to the ATM

After the ATM sends the transaction information to the Diebold 10XX/478X Device Handler process, the Device Handler process formats the transaction as a noncurrency dispense transaction and builds the Non-Currency Dispense token (setting the HOPR-CONTENTS field to the value 11), Pre-Pay Top-Up token, and the Transaction Subtype token. Although the mobile top-up transaction is a noncurrency dispense (withdrawal) transaction, the transaction requires additional processing. The Device Handler process uses mtu_op_key_tbl table in the T100NAMS file to fill the TXN-SUBTYP field of the Transaction Subtype token.

A mobile top-up transaction message has two processing requirements: the funds portion and the top-up portion (includes the mobile telephone number and all processing information). If the SPLIT-TXN-RTG-DEST param is present, the Device Handler process forwards the transaction to the Transaction Context Manager process specified in the param, which can route the transaction to multiple authorizing entities. If the param is not present, the transaction is declined.

Note that when the SPLIT-TXN RTG-DEST param is present, the Device Handler process always sends transactions to the Transaction Context Manager process and not the BASE24-atm Authorization process.

After the Device Handler process retrieves the destination name from the SPLIT-TXN-RTG-DEST param it adds the Split Transaction Routing token with the name of the Authorization process in the FUNDS-AUTH-DEST field.

When the BASE24-atm system uses the Transaction Context Manager process, all messages between the Device Handler process and the Authorization process are routed through the Transaction Context Manager process.

The BASE24-atm Mobile Top-Up Device Handler extension also supports receiving a 9909 mobile top-up notification message from the Transaction Context Manager process, requesting additional action from the cardholder at the ATM, such as selecting a mobile operator and amount, selecting a new amount, or approving a tax. The content of the 9909 message is based on the value of the Pre-Pay Response token RESP-IND field.

Refer to the ***BASE24 Tokens Manual*** for more information about the Split Transaction Routing token, Pre-Pay Response token, Transaction Subtype token, Pre-Pay Top-Up token and Non-Currency Dispense token.

Transaction Context Manager Process

The Transaction Context Manager process performs routing to the telecommunication service provider and the issuer financial institution. When the Transaction Context Manager process receives a transaction it searches for the Transaction Subtype token. If the token is not present, the transaction is declined. If the token is present, the Transaction Context Manager retrieves the transaction subtype. Using the transaction subtype as the key, the Transaction Context Manager process reads the STRF extended memory table to find the secondary source and the authorizer (primary or secondary) to which it sends the transaction first. The ROUTING HIERARCHY field on the STRF screen specifies the order in which the Transaction Context Manager process sends the transaction message to the destinations.

Splitting the Transaction Routing

The Transaction Context Manager process splits the transaction routing to different destinations. In the case of mobile top-up processing, mobile top-up with funds transaction routing is split between the top-up authorizer and the funds authorizer.

The Transaction Context Manager process uses the routing hierarchy specified in the STRF to determine the order to send the transaction message to the authorizing entities. The Transaction Context Manager process uses the Split Transaction Routing token to keep track of the transaction routing. The Transaction Context Manager process updates the TXN-RESP-IND, the FUNDS-AUTH-STAT, and the SCND-SVC-STAT fields in the Split Transaction Routing token each time it passes the transaction to either the funds authorizer or the top-up authorizer.

Logging to the Transaction Log File (TLF)

The Transaction Context Manager process logs the secondary service (i.e., top-up) transaction information to the Transaction Log File (TLF). This is done prior to sending the final response for the mobile top-up transaction to the Device Handler process. All approved mobile top-up transactions have two transactions logged to the TLF: one for the funds authorization and one for the top-up authorization. In the definitions-based model of mobile top-up processing, the Transaction Context Manager process logs the top-up transaction with a transaction code of 25 (top-up). In the message-based model of mobile top-up processing, the Transaction Context Manager process logs the top-up transaction with a transaction code of 10 (withdrawal). In both models the Transaction Context Manager process logs the value S in the responder field, indicating a secondary service transaction and enables Transaction-based Pricing and ATM reports to process the secondary service transactions appropriately.

Configuration Considerations

The following files contain fields needed for the mobile top-up product. These files are in used in conjunction with the files used in Non–Currency Dispense processing.

- T100NAMS
- Logical Network Configuration File (LCONF)
- Transaction Code/Subtype Relationship File (TSRF)
- Surcharge File (SURF)
- Receipt File (RCPT)
- Split Transaction Routing File (STRF)
- Mobile Operator File (MOF)

Modifying the T100NAMS

The T100NAMS is an edit file that contains a series of tables used by the BASE24-atm Diebold 10XX/478X Mobile Top-Up Device Handler extension for mobile top-up transactions. Customers can edit these tables to customize specific information displayed on ATM receipts.

The transaction type table in the T100NAMS (mtu_tran_tbl_g) supports the mobile top-up transaction type in receipt printing. The Device Handler process uses information in this table to format receipt descriptions with text specific to mobile top-up transactions, instead of generic withdrawal text.

Also in this file is the operations code table containing a subtype for each mobile top-up authorizer destination.

Modifying the Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains seven params specific to the Mobile Top-Up Device Handler extension.

The SPLIT-TXN-RTG-DEST param specifies the destination of the Transaction Context Manager process. The Transaction Context Manager process is required for split transaction routing (in a mobile top-up transaction, part of the transaction is routed to the third-party telecommunications process and part to the BASE24-atm Authorization process).

The ATM-DH-RCPT-PHONE-NUM-FRMT param specifies whether the telephone number printed on the receipt is formatted in the xxx-xxx-xxxx format.

The CARRIER-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type, specifies a four-digit mobile operator ID. This mobile operator ID matches a mobile operator with an entry in the Mobile Operator File (MOF) and enables the mobile operator to be identified for the transaction.

The CARRIER-AMT-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type, specifies the six predetermined amounts, in monetary units, supported for this mobile operator. Valid values must be formatted with eight amounts and each amount must be separated by semicolons. If the mobile carrier does not support eight amounts, the remaining values in this param must be set to zeros.

The MOBILE-OPERATOR-ID-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type, specifies a mobile operator six-digit IIN, two-digit digit product ID, and the mobile operator name (up to 20 characters) associated with the mobile telephone operator.

The MOBILE-NUM-MIN-LGTH param specifies the minimum length for the mobile telephone number entered by the cardholder.

The MOBILE-NUM-MAX-LGTH param specifies the maximum length for the mobile telephone number entered by the cardholder.

Modifying the Transaction Code/Subtype Relationship File

The Transaction Code Subtype Relationship File (TSRF) contains one record for each transaction subtype supported in the network. You can use the TSRF to set up a transaction subtype and related transaction codes for mobile top-up transactions. The transaction subtype distinguishes the mobile top-up transaction from other withdrawal transactions. You can use the subtype when configuring the Surcharge File (SURF) and the Split Transaction Routing File (STRF).

Modifying the Surcharge File

Some customer contracts with telecommunications services providers prohibit surcharging on mobile top-up transactions at an ATM. If you use the surcharging feature at your ATMs and want to offer mobile top-up transactions without a surcharge, you must configure the Surcharge File (SURF) to prevent surcharges.

Using the transaction subtype defined in the TSRF, you can specify transactions with a subtype to not have a surcharge by setting the MIN TXN, FLAT FEE, and PERCENT FEE to the value 0.00 and the USE MIN/MAX field to a blank.

For more information about the SURF, refer to the ***BASE24 Base Files Maintenance Manual.***

Modifying the Receipt File

The Receipt File (RCPT) contains receipt templates. You can add receipt templates for mobile top-up transactions. ACI supplies the PPTU receipt type template for real-time mobile top-up transactions that do not involve a voucher and the VCHR receipt type template for mobile top-up transactions that include a voucher. The VT00 through VT29 receipt types are used in the definitions-based model to provide a unique voucher receipt record for each mobile operator supported.

Note that the ATM prints two receipts for a mobile top-up transaction. The funds portion of the transaction is printed using the same template as a noncurrency dispense (withdrawal) transaction. The receipt for the top-up portion uses the field type codes and templates described here. The ATM always prints the top-up portion first.

Mobile top-up transaction templates can use the following codes specific to the BASE24-atm Mobile Top-Up add-on product:

- TT (print transaction type)
- V0 (print activation number
- V1 (print stock number)
- V2 (print shelf date, that is the date the minutes expire once activated)
- V3 (print telephone number)
- V4 (print unique transaction number)
- V5 (print telecommunications company marketing message)

- V6 (print telecommunications company name)
- V7 (print carrier telephone number)
- VA (print mobile telephone balance)
- VB (print batch ID)
- VE (print expiration days)
- VI (print control ID number)
- VN (print voucher number)
- VS (print expiration date)
- VT (print tax amount)
- VX (print actual amount added to the mobile telephone account)

Note: Care should be taken when configuring the receipt templates as to avoid exceeding the maximum size of the print buffer in the device message.

Refer to section 6 for detailed definitions of these codes and information about configuring the RCPT.

A sample PPTU template as defined in the RCPT is shown below.

```
@MOBILE TOP-UP SUMMARY>>
#V6>
@MOBILE: #V3>
@TOP-UP VALUE:            #VX>
@TAXES PAID:            #VT>
@UPDATED PREPAID BALANCE: #VA>
@NEW EXPIRATION DATE: #VS>@REF NUM: #V4>>
@CUSOTMER SERVICE: #V7>
#V5
```

The above template creates the following receipt for a mobile top-up transaction that does not require a voucher.

```
    CUSTOMER LOGO SCREEN

  DATE    TIME   STATION ID
DD/MM/YY  HH:MM   S1AC9100

MOBILE TOP-UP SUMMARY

XYZ TELECOMMUNICATIONS
MOBILE:  402-555-5555
TOP-UP VALUE:            $20.00
TAXES PAID:               $1.00
UPDATED PREPAID BALANCE: 30
NEW EXPIRATION DATE: MM/DD/YY
REF NUM: 123456789012345

CUSTOMER SERVICE: 402-390-7600
   USE XYZ TEXT MESSAGING
          AND WIN




         1         2         3
12345678901234567890123456789012
```

A sample VCHR or VT*xx*, where *xx* is 00–29, template as defined in the RCPT is shown below.

```
@MOBILE TOP-UP SUMMARY>>
#V6>
@MOBILE: #V3>@TOP-UP VALUE: #VX>@TAXES PAID:   #VT>
@TOP-UP VOUCHER: #VN>@VOUCHER EXPIRY: #VE @DAYS>
@REF NUM: #V4>>
@CUSTOMER SERVICE: #V7>
#V5>>
```

The above template creates the following receipt for a mobile top-up transaction that includes a voucher.

```
 1       CUSTOMER LOGO SCREEN
 2
 3     DATE    TIME   STATION ID
 4   DD/MM/YY  HH:MM   S1AC9100
 5
 6   MOBILE TOP-UP SUMMARY
 7
 8   XYZ TELECOMMUNICATIONS
 9   MOBILE: 402-555-5555
10   TOP-UP VALUE: $20.00
11   TAXES PAID: $1.00
12   TOP-UP VOUCHER: 123456789012345
13   VOUCHER EXPIRY: 7 DAYS
14   REF NUM: 123456789012345
15
16   CUSTOMER SERVICE: 402-390-7600
17      USE XYZ TEXT MESSAGING
18             AND WIN
19

              1         2         3
    12345678901234567890123456789012
```

Modifying the Split Transaction Routing File

The Split Transaction Routing File (STRF) contains one record for each transaction subtype in the BASE24 network that requires split routing. Once you have set up transaction subtypes for mobile top-up transactions in the TSRF, you must create STRF records for each subtype so that the transactions can be routed properly. This file contains the process name of the top-up authorizer for the subtype.

The ROUTING HIERARCHY field specifies the order to which the Transaction Context Manager process routes the transaction for authorization of the funds and top-up portions of the transaction. For the definitions-based model of mobile top-up processing, you can configure sequential routing, with the transaction going to the funds authorizer first by setting this field to the value B0, or you can configure sequential routing with the transaction going to the top-up authorizer first by setting this field to the value B1. For the message-based model of mobile top-up processing, you can configure multiple messages with the transaction going to the top-up authorizer first by setting the field to the value B2.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about this file.

Modifying the Mobile Operator File

The Mobile Operator File (MOF) contains one record for each mobile operator defined to the BASE24 system. This file is used in definitions-based processing only. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about the MOF.

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